

Provenance 14 — Fermion fable: the complex 16-spinor, its canonical quantization in $4 + 4$ dimensions, its field equations in the primordial gravitational field, and its energy–momentum tensor operator

Effort A of 2026-09-24

2026-09-24

Effort A of 2026-09-24. The classical 16-component spinor field *fable* of 2026-09-16 is refined into a fermion: a complex 16-component Grassmann spinor. It is quantized canonically with the observer’s time x_4 as time, in the $4 + 4$ -dimensional pre-universe. Its field equations are written out component by component in the primordial gravitational field (the author’s canonical frame, with the free function a_4 never given a value). Its energy–momentum tensor operator, energy density, pressures and equations of state are derived, and the canonical spin connection is discussed completely.

This page is complete on its own. Everything a reader needs is restated here. Code, logs, the notebook, and the TeX and PDF files are named only as the places where things are computed, recorded or produced.

How to read the evidence. Every mathematical statement on this page carries one of these marks:

mark	meaning
[A], [proved P §n]	an assertion <code>cfAssert[label, test]</code> of the Mathematica notebook, Part P, Section n. The run log prints <code>PASS label</code> ; the label is quoted verbatim , in typewriter type, exactly as the log prints it
[D], [displayed P §n]	a displayed output of the notebook (a table, a count, a closed form), not an assertion
[P], [prose P §n]	stated only in a <code>Text</code> cell (or a code comment) of the notebook; not checked by an assertion
[derived here]	a short derivation carried out on this page from marked results, every step shown
[file]	read from a named file of the repository (code, CSV, log), which is named

Numbers are quoted from the files and logs named next to them. Nothing is invented. a_4 is never given a value anywhere.

Two kinds of “section”. “Section n” with a capital S is a section of the Mathematica notebook (Sections 1–33; Part VII is Sections 26–29). “section n” in lower case is a section of this page. The mathematics is written in standard notation; where the notebook’s Wolfram names matter they are given in typewriter type. $T_{16}[a]$ (`T16[a]`) are the 16×16 Dirac matrices, σ_{16} (`sigma16`) the spinor metric, $\Psi^\dagger$ the classical conjugate, $\bar{\Psi}$ the Dirac conjugate, ∂_μ a partial derivative, D_μ the spin-covariant derivative, and $\sqrt{g}$ (`Sqrt[g]`) the volume factor $\sqrt{|\det g|}$.

Contents

Where things live	4
0 The task, in the author's words	5
1 The complete set of ideas	6
2 Foundations: the pre-universe, its algebra, the canonical frame, the author's La and eLa, and the classical fable of Part VI	6
2.1 How the evidence is cited	6
2.2 Coordinates, the flat 4+4 metric, and which directions are timelike	7
2.3 SO(4): the self-dual and anti-self-dual blocks	8
2.4 The real 8×8 Clifford generators τ , $\bar{\tau}$, and the spinor metric σ	8
2.5 The 16×16 Dirac matrices, chirality, σ_{16} , and $\mathfrak{so}(4, 4)$	9
2.6 Cartan triality and the split octonions (Section 9): what is proved	10
2.7 The wave function of the un-universe and the author's Lagrangian L_a (Section 11)	11
2.8 The Euler–Lagrange equations eLa (Section 12)	12
2.9 The (z, t) chart, four blocks of four, and the closed-form solutions (Section 13)	13
2.10 Bilinears, the solution branches, and $M_6 = 3$ generations of Einstein–Rosen 2-planes (Section 14)	14
2.11 The canonical frame field (Section 15), restated	15
2.12 The classical field fable of 2026-09-16 (Part VI, Section 24)	18
2.13 Why the classical fable must be refined	24
3 The refinement: why fable must be a complex 16-spinor	27
3.1 The Clifford symmetry table	27
3.2 The Lorentz-invariant bilinears: exactly two, both symmetric	28
3.3 A Grassmann algebra, built and tested	28
3.4 The real options, and why each fails	29
3.5 The complex 16-spinor	31
3.6 Four four-dimensional Dirac fermions	31
3.7 The refined Lagrangian, and the fidelity chain to Part VI, L_a and eLa	32
3.8 The conclusion, with its scope	33
4 The Lagrangian and its Hermiticity	33
4.1 The four candidates	33
4.2 The facts, each asserted on the canonical frame for generic $\Psi(x_0..x_7)$, $\bar{\Psi}(x_0..x_7)$	34
4.3 Why the connection drops out, and why this is not Part VI's mechanism	35
5 The field equations in the primordial gravitational field, written out	35
5.1 The derivation	36
5.2 The 16×16 matrix form on the canonical frame	36
5.3 The split-octonion 8+8 form	37
5.4 All sixteen component equations	37
5.5 All sixteen adjoint equations	40
5.6 The 8+8 form, component by component	42
5.7 The rescaled form: the spin connection disappears	45
5.8 The x_4 -only family and the condition $V''s' = 0$	47
5.9 The operator form of the field equation, and its ordering	47
5.10 The U(1) current and the particle-number current	48
5.11 The only internal symmetry is the vector phase	48

6	Canonical quantization in 4+4 dimensions	49
6.1	Time, slices, and why the slice is not a Cauchy surface	49
6.2	The momenta, the kinetic form G , and the signature $(8, 8)$	49
6.3	The Dirac bracket and the canonical anticommutator	50
6.4	The admissible truncation, V_{hid} , and the rescaled field	51
6.5	What J is, and the Hilbert adjoint	51
6.6	Why this is legitimate for fermions, and why it would fail for bosons	53
6.7	Chirality, and why V depends on s only	53
6.8	The Hermiticity rule and the J table	53
6.9	The symmetry that survives, and what is lost	54
6.10	The admissible sector: a theorem	54
6.11	The Hamiltonian on the canonical frame, and the spin connection	56
6.12	The x_0 direction and the boundary condition at the wall	57
6.13	Krein modes, and why the mode basis must consist of J -eigenvectors	57
6.14	A finite Fock space at one momentum (Jordan–Wigner): the Dirac sea, 8 particles and 8 antiparticles	58
6.15	The particle number	60
6.16	The dynamical-frame field χ	60
7	The energy–momentum tensor operator	60
7.1	The definition, and where it comes from	61
7.2	Symmetric and Hermitian	61
7.3	Conservation on shell, and why the commuting proxy proves it for the fermion	61
7.4	On shell: the Lagrangian, the density, the pressures and the trace	62
7.5	Which components are Hermitian operators: the J -parity table	63
7.6	The real commuting limit	63
8	Density, pressure and equations of state	64
8.1	The operators	64
8.2	Expectation values: the mode sums, and one particle	64
8.3	The vacuum: normal ordering, the sea energy, and the cosmological-constant problem	64
8.4	The homogeneous Fermi sea (the Kohn–Sham ground state) in closed form	65
8.5	The numerically stable forms	67
8.6	The mean field	67
8.7	The classical limit is the non-relativistic limit, and the sign rule	68
8.8	The author’s mass term: w falls from $1/3$ to 0	68
8.9	$w \geq -1$ always: no phantom crossing for the quantized fable	69
8.10	On the pre-universe: the quantum gas cools, the classical field does not	69
9	The canonical spin connection: a complete discussion	70
9.1	Sources, and how the status of each statement is marked	71
9.2	Conventions	72
9.3	The canonical frame (vielbein), the metric, and the volume factor	73
9.4	The generalized Christoffel symbols	74
9.5	The vielbein postulate, and how <code>cfSpinConnection</code> solves it	75
9.6	The canonical spin connection: all its non-zero components	76
9.7	The spinor connection Γ_μ and its exact normalization	77
9.8	The gauge-covariant derivative of the 16-component spinor	78
9.9	The three bridges of Part IV	81
9.10	The fable-5.1 bridge of Part V: the Weitzenböck (teleparallel) connection	84
9.11	Master comparison of the five connections (Section 22)	87

9.12	What the canonical connection does for the complex fermion fable	88
9.13	Status ledger	94
10	How it was verified	95
10.1	Part VII of the notebook: sections, cells, assertions	95
10.2	Every Part VII assertion, verbatim	97
10.3	The same assertions in the runs of Parts I–VIII	103
10.4	The design review, and every correction it made to Effort A	105
10.5	The Fock check	106
10.6	The TeX export	106
11	Every command, in full	107
11.1	Build the notebook from its manifests	107
11.2	Run the notebook end to end, straight out of the <code>.nb</code>	108
11.3	Check the generated notebook without evaluating it	109
11.4	Print the section-to-cell map	110
11.5	Export the field equations, the anticommutator, J and the equations of state to TeX and text	111
11.6	Build the LaTeX twin and its PDF	111
12	Push and repository verification	112

Where things live

what	where (relative to the repository root)
the Mathematica notebook (generated, never edited by hand)	<code>claude-fable/claude-fable_Einstein-Rosen-2-Planes.nb</code>
its cell manifests; Part VII (Sections 26–29) is this effort	<code>claude-fable/cells_part1.wl ... cells_part8.wl;</code> <code>claude-fable/cells_part7.wl</code>
the builder and the run harness	<code>claude-fable/build_tools.py,</code> <code>claude-fable/runner_header.wl</code>
the runner, the checker, the section map	<code>claude-fable/run_from_nb.wls,</code> <code>claude-fable/verify_nb.wls,</code> <code>claude-fable/nb_section_map.wls</code>
Part VII’s own acceptance run, the record of Part VII: the notebook through Part VII (198 Input cells, 528/528, 0 cells with messages)	<code>claude-fable/run_fermion_fable_part7.log</code>
the acceptance run of Part VIII, kept as a historical record (217 Input cells, 642/642: the Rust solver’s CSVs did not exist yet, so Part VIII’s two Rust-vs-Mathematica comparisons were skipped, and three Section 32 labels still carry the superseded index notation; section 10.3)	<code>claude-fable/run_fermion_fable_part8.log</code>
the final run of the whole notebook, Parts I–VIII, in its committed state (217 Input cells, 644/644, 0 cells with messages, 198.352 s, 46 witnesses, 0 identities accepted on numerical evidence alone)	<code>claude-fable/run_fermion_fable_final.log</code>
the checker’s logs; the section map	<code>claude-fable/verify_nb_part7.log,</code> <code>claude-fable/verify_nb_part8.log;</code> <code>claude-fable/nb_section_map.log</code>

what	where (relative to the repository root)
the TeX/text export of the Part VII results	claude-fable/export_fermion_fable_tex.wls → provenance-latex/generated/*.tex, *.txt
the LaTeX twin of this page, its preamble and build scripts	provenance-latex/PROVENANCE-14-FERMION-FABLE- CANONICAL-QUANTIZATION.tex → .pdf; provenance-latex/preamble.tex; provenance-latex/build_all.sh, build_all.ps1
the author's two helper packages, which must sit next to the notebook	claude-fable/ConvertMapleToMathematicaV2.wl, claude-fable/EtoExp.wl
the numerically stable Kohn-Sham integrals and Dirac-sea energy used by the solver	fable- cosmology/rust/fable_fermion/src/kohn_sham.rs
fable_fermion (final at 2bc936d: 34/34 tests)	

<repo> below is wherever the repository is cloned. Every shell block finds it with `git rev-parse --show-toplevel`.

0 The task, in the author's words

On 2026-09-24 the author gave three instructions. Their operative parts, verbatim (the ellipses [...] are cuts in the quotation, not in the instruction's meaning):

A. “Refine the new 16-component fermi spinor field named fable to be a complex 16-spinor, and quantize it using canonical quantization, extended to 4+4 dimensions. Write out the field equations for the fermion fable field in the presence of a primordial Gravitational Field. Create new provenance markdown file, latex, and pdf files, that contain the exact, correct, fermion fable field equations, energy-momentum tensor operator, pressure, density and equations of state for fermion fable field. Include a complete discussion of the canonical spin connection.”

B. “Include fable as a source in the Einstein Gravitational Field Equations for this primordial Gravitational Field, and solve the coupled field equations over a time interval from the beginning of the present universe until the present time. Try to use some of the relevant ideas from Density Functional Theory to obtain the fable ground and first excited states. Create another new provenance markdown file, latex, and pdf files [...] the interacting (fermion fable field, primordial Gravitational Field) system [...] coupled equations, canonical spin connection for the interacting system, energy-momentum tensor operator, pressure, density and equations of state [...] complete discussion of the canonical spin connections. Answer: [1] Does this system provide a physical mechanism for a time-varying dark energy equation of state (w)? [2] [...] time-varying dark matter equation of state (w)?”

C. “Thoroughly discuss the commands that you use to solve the coupled system, and to test, verify, execute, and display your calculations of the coupled field equations. Create another new provenance markdown file, latex, and pdf files that contain your complete solution. Each of your major efforts should have its own section in the markdown file, latex, and pdf files. Do not ever refer the USER to another markdown file for any reason. The full set of 100% complete ideas must be given its own section in each markdown file.” And: push, check and verify the repository. “DO NOT TAKE SHORTCUTS.”

This page is Effort A. Efforts B and C have their own pages. The complete set of ideas of all three efforts is section 1 below, identical on all three pages.

The author’s choice. Instruction A itself names the field: “a complex 16-spinor”. As section 3 shows, that is forced by the author’s own algebra: a real anticommuting 16-spinor with the author’s spinor metric has no dynamics, a Majorana 16-spinor is massless with no potential, and a one-chirality (Weyl or Majorana–Weyl) spinor cannot propagate. When these alternatives were laid out on 2026-09-24, the author confirmed the choice: **a complex 16-spinor**, with Dirac conjugate $\bar{\Psi} = \Psi^\dagger \sigma_{16}$. Every result below is for that field.

The work, in the order it was done. A design (revision 1) was written and then reviewed adversarially by four reviewers — quantization; field equations, energy–momentum tensor and spin connection; Einstein equations and cosmology; density-functional theory and numerics — each followed by an independent skeptic. Every finding was adopted, some only in corrected form (section 10.4). The physics was then implemented as Part VII of the Mathematica notebook (Sections 26–29), where every claim is an assertion. The notebook was run end to end (528/528 assertions, 0 messages). Part VIII (the coupled system of Effort B, Sections 30–33) was added afterwards; the final run of the whole notebook, Parts I–VIII, evaluates 217 Input cells with 644/644 assertions and 0 cells with messages, and its Part VII lines are those of the Part VII run, line for line apart from the timings (section 10.3). The exact field equations, anticommutator, fundamental symmetry and equations of state were exported from the notebook’s own objects to TeX and plain text, and those exports are quoted verbatim on this page.

1 The complete set of ideas

{{COMPLETE-SET-OF-IDEAS}}

2 Foundations: the pre-universe, its algebra, the canonical frame, the author’s L_a and eLa, and the classical fable of Part VI

This section sets out everything that the new work on fermion fable stands on. It restates the author’s pre-universe (a curved $4 + 4$ -dimensional spacetime), the real Clifford algebra and split-octonion spinors built on it, the author’s own wave function, Lagrangian and field equations, the canonical frame field, and the classical field fable of 2026-09-16. It ends with the reason why that classical field has to be refined into a complex 16-component spinor. The reader needs nothing outside this document to follow it.

2.1 How the evidence is cited

All the mathematics lives in one Mathematica notebook, `claude-fable/claude-fable_Einstein-Rosen-2-Planes.nb`. That notebook is never edited by hand. Its Parts I–VI, which this section restates, are generated from the cell manifests `claude-fable/cells_part1.wl ... cells_part6.wl`. In a manifest, Text cells explain and Input cells compute. Every claim the notebook checks is an assertion `cfAssert[label, test]`, which prints `PASS label` or `FAIL label`. The complete run of Parts I–VI is recorded in `claude-fable/run_fable51_part6.log`. It evaluated 172 Input cells in 162.5s with no cell raising a message, and its final tally reads (the witness line is shortened here; in the log it continues with a one-sentence explanation of what a witness is):

```
identities accepted on numerical evidence alone : 0    (stage 3 returned True)
non-vanishing witnesses                        : 27
PASS no identity in this notebook was accepted on numerical evidence alone
Assertions run: 358   passed: 358   failed: 0
```

Each statement below is tagged with one of four kinds of evidence:

- [A] an assertion. Its label is quoted exactly as the log prints it after PASS.

- **[D]** a displayed result: an Output the notebook prints but does not assert. Where a displayed result was re-read for this document from the saved kernel state, that is said.
- **[P]** prose only: a statement that appears in a Text cell and is not checked by any assertion.
- **[VII]** proved in Part VII, the new part of the notebook written for fermion fable (Sections 26–29). Its labels are quoted verbatim from `claude-fable/run_fermion_fable_part7.log`, which reports **Assertions run:** 528 **passed:** 528 **failed:** 0 for Parts I–VII together.

How an identity is certified (restated from Section 15 of the notebook). `cfZeroQ` and `cfZeroArrayQ` try three stages in turn:

1. exact structural zero;
2. **Simplify** under the geometric assumptions, with a 5-second budget;
3. a 30-digit numerical evaluation at three rational probe points, with test functions substituted for the undetermined a_4 (and for the undetermined rapidity of the notebook’s boosted-frame bridge); the result is accepted as zero only if its magnitude is below 10^{-22} .

An identity accepted at stage 3 counts as a *numerical certificate*, and the run reports zero of those: every identity in Parts I–VI is closed symbolically. A claim that something is *not* zero is made with `cfNonZeroWitnessQ`. It passes only if, at some probe point, every entry evaluates to a number and at least one of them has magnitude above 10^{-10} . That is a complete proof of non-vanishing; the run has 27 such witnesses. The probe functions are test inputs only, never definitions.

The session policy is the original notebook’s, unchanged: **Simplify** has a 1-second budget and **FullSimplify** a 3-second budget. $\text{DIM8} = 8$, and the model’s three constants M (a mass), K and H (a Hubble-like constant) are **Protected**, so no cell can assign them (Section 1: a Text cell and the Input cells that set this up; no assertion) [P].

2.2 Coordinates, the flat 4+4 metric, and which directions are timelike

The coordinates are $X = \{x_0, x_1, \dots, x_7\}$. The flat tangent-space metric is

$$\text{eta4488} = \eta = \text{diag}(+1, +1, +1, +1, -1, -1, -1, -1), \quad \text{signature } (4, 4). \quad (1)$$

coordinate	η	name in the original notebook (Section 2)	reading used in this work
x_0	+1	hidden space	the hidden spacelike coordinate
x_1, x_2, x_3	+1	ordinary 3-space	the observed sheet
x_4	−1	time	the observer’s time
x_5, x_6, x_7	−1	superluminal deflating time directions	the hidden timelike sheet

[A] `eta4488 . eta4488 == ID8; eta4488 is symmetric; signature of eta4488 is (4,4).`

Index conventions (restated from Section 2). Latin indices a, b, A, B are flat and run 0..7. Greek indices μ, ν are curved and run 0..7. Wolfram arrays are 1-based, so index A sits at array position $A + 1$. For example, x_4 is entry 5 of X , so Part VI’s energy density $\rho = T_{44}$ is the array element `T[[5,5]]`.

The notebook uses two sets of assumptions:

- `ssX = H > 0 && sX0 && 6 H x0 > 0 && 2 la[x4] > 0 && Cot[6 H x0] > 0 && Sin[6 H x0] > 0, with sX0 = x0 > 0 && x1 > 0 && ... && x7 > 0` (all eight coordinates positive), for the (x_0, x_4) chart;

- `constraintVars = x0 > 0 && x4 > 0 && z > 0 && t > 0`, for the chart $z = 6Hx_0$, $t = Hx_4$.

1a is a free function inherited from the original notebook. It occurs only inside **ssX**. It is never given a value, and nothing here invents one [P, Section 2].

2.3 SO(4): the self-dual and anti-self-dual blocks

The six antisymmetric real 4×4 matrices split under the Hodge star into a self-dual 3-space and an anti-self-dual 3-space, the two commuting $\mathfrak{su}(2)$ factors of $\mathfrak{so}(4)$. The original notebook builds both from two elementary tensors (Section 3):

$$Q_a[h, p, q] = \text{Signature}[\{h, p, q, 4\}], \quad Q_b[h, p, q] = \delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}, \quad (2)$$

$$\mathbf{s4by4}[h] = Q_a - Q_b \text{ (self-dual)}, \quad \mathbf{t4by4}[h] = Q_a + Q_b \text{ (anti-self-dual)}, \quad (3)$$

$$h = 1, 2, 3. \quad (4)$$

Explicitly [D]:

$$s_1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}, \quad s_2 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad s_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix},$$

$$t_1 = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad t_2 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad t_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

with $s_h = \mathbf{s4by4}[h]$ and $t_h = \mathbf{t4by4}[h]$.

[A] `s4by4` is self-dual; `t4by4` is anti-self-dual; `s4by4` and `t4by4` are antisymmetric; the two $\mathfrak{su}(2)$ factors commute; and, for the nine mixed products $\mathbf{st}[J, K] = s_J t_K$, $\mathbf{st}[J, K]$ is symmetric.

2.4 The real 8×8 Clifford generators τ , $\bar{\tau}$, and the spinor metric σ

The $O(4,4)$ spinor metric is

$$\sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix}. \quad (5)$$

[A] `sigma . sigma == ID8`; `sigma` is symmetric; `Tr[sigma] == 0`. The type-1 and type-2 split-octonion spinors both carry this same σ , because $\sigma = \sigma^{-1}$ [P].

The generators (Section 4) are

$$\tau[0] = I_8, \quad (6)$$

$$\tau[h] = \begin{pmatrix} 0 & s_h \\ s_h & 0 \end{pmatrix}, \quad h = 1, 2, 3, \quad (7)$$

$$\tau[3+k] = \begin{pmatrix} 0 & t_{4-k} \\ -t_{4-k} & 0 \end{pmatrix}, \quad k = 1, 2, 3 \quad (t_3, t_2, t_1), \quad (8)$$

$$\tau[7] = \tau[1]\tau[2]\tau[3]\tau[4]\tau[5]\tau[6] = \text{diag}(-I_4, +I_4) \quad [\text{D}], \quad (9)$$

$$\bar{\tau}[A] = \sigma (\sigma \tau[A])^T, \quad A = 0, \dots, 7 \quad (\text{the } \sigma\text{-adjoint}). \quad (10)$$

The order 3, 2, 1 is the original's. It is what makes $\tau[7]$ the product of the first six and $\sigma = \tau[1]\tau[2]\tau[3]$ [P]. The original calls the list of the six blocks `sixAntiSymmetric8by8`, but that name fits only half of it: [A] blocks 1-3 (built from `s4by4`) are antisymmetric; blocks 4-6 (built from `t4by4`) are symmetric.

The defining relation is the split Clifford algebra of the $4 + 4$ quadratic form in its real 8×8 “half-spinor” form:

$$\frac{1}{2}(\tau[A] \bar{\tau}[B] + \tau[B] \bar{\tau}[A]) = \eta_{AB} I_8. \quad (11)$$

[A] (1/2)(tau[A].taubar[B] + tau[B].taubar[A]) == eta4488[[A+1,B+1]] ID8.

The ordering identities [A]:

- sigma == tau[1].tau[2].tau[3]
- sigma == tau[4].tau[5].tau[6].tau[7]
- tau[1]...tau[7] == tau[0] == ID8
- sigma . taubar[A] == Transpose[sigma . tau[A]]
- -eta4488[[A+1,A+1]] tau[A] == Transpose[tau[A]] for A = 1..7

The last one says that $\tau[1..3]$ are antisymmetric and $\tau[4..7]$ are symmetric. The second invariant of the original is $\Omega = \sigma \tau[7]$: [A] Omega == tau[4].tau[5].tau[6].

A complete basis of the 8×8 algebra (Section 7) is the 63 ordered products of 1, 2 or 3 of $\tau[1..7]$, together with I_8 . [A] the 8x8 basis has 64 elements; the 64 elements are distinct; (1/8) Tr[bas64[A].bas64[B]] == eta64[[A,B]]; symmetric 8x8 basis elements number 36; antisymmetric 8x8 basis elements number 28.

2.5 The 16×16 Dirac matrices, chirality, σ_{16} , and $\mathfrak{so}(4, 4)$

Following E. A. Lord's reduced Brauer–Weyl construction (Math. Proc. Camb. Phil. Soc. 64 (1968) 765–778), the 8×8 generators are doubled off-diagonally (Section 5):

$$T_{16}[A] = \begin{pmatrix} 0 & \bar{\tau}[A] \\ \tau[A] & 0 \end{pmatrix} \quad (A = 0, \dots, 7), \quad (12)$$

$$T_{16}[8] = T_{16}[0]T_{16}[1] \cdots T_{16}[7], \quad \sigma_{16} = T_{16}[0]T_{16}[1]T_{16}[2]T_{16}[3]. \quad (13)$$

The $T_{16}[A]$ are real 16×16 matrices, and “ $\gamma[a]$ ” means $T_{16}[a]$. $T_{16}[8]$ is the chirality operator and σ_{16} is the 16×16 spinor metric.

Assertions [A]:

- {T16[A], T16[B]}/2 == eta4488[[A+1,B+1]] ID16
- sigma16 == ArrayFlatten[{{-sigma,0},{0,sigma}}]
- sigma16 . T16[A] is antisymmetric for A = 0..7
- T16[8] == sigma16 . T16[4].T16[5].T16[6].T16[7]
- sigma16 is symmetric and sigma16 . sigma16 == ID16
- sigma16 . covariantDiffMatrix is symmetric, with covariantDiffMatrix = $T_{16}[5]T_{16}[6]T_{16}[7]$

The two chiralities are the two split-octonion spinors. The notebook defines $P_L = \frac{1}{2}(I_{16} - T_{16}[8])$ and $P_R = \frac{1}{2}(I_{16} + T_{16}[8])$. [A] PL + PR == ID16; PL is idempotent; PR is idempotent; PL . PR == PR . PL == 0; PL projects onto the upper (type-1) components, PR onto the lower (type-2) components.

The last assertion is equivalent to

$$T_{16}[8] = \text{diag}(-I_8, +I_8). \quad (14)$$

A 16-component spinor is therefore the direct sum $\Psi = (\psi_1, \psi_2)$:

- the upper eight components are the **type-1** split-octonion spinor, of chirality -1 ;
- the lower eight components are the **type-2** spinor, of chirality $+1$.

Every $T_{16}[A]$ exchanges the two halves, since it is block off-diagonal [P].

The $\mathfrak{so}(4,4)$ generators are

$$S^{AB} = \text{SAB}[[A+1, B+1]] = \frac{1}{4} [T_{16}[A], T_{16}[B]]. \quad (15)$$

Assertions [A]:

- SAB is antisymmetric in its two flat indices
- sigma16 . SAB is antisymmetric
- T16[8] commutes with every SAB: the chiral halves are each Spin(4,4)-invariant
- [SAB, SAB] closes on the $\mathfrak{so}(4,4)$ algebra, which is the relation $[S_{AB}, S_{CD}] = -(\eta_{AC}S_{BD} - \eta_{AD}S_{BC} - \eta_{BC}S_{AD} + \eta_{BD}S_{AC})$
- [SAB, T16] reproduces the vector representation, which is $[S_{AB}, \gamma_C] = -\eta_{CA}\gamma_B + \eta_{CB}\gamma_A$

The spin connection of the frame field enters as $\frac{1}{8} \omega_{\mu ab}[\gamma^a, \gamma^b] = \frac{1}{2} \omega_{\mu ab} S^{ab}$.

The 256-element basis (Section 6) is the ordered products of k distinct generators, $k = 1..8$, followed by I_{16} , taken in the original's order. [A] base16[[93]] is sigma16 (the word $\{0, 1, 2, 3\}$); base16[[255]] is the chirality operator T16[8]; base16[[256]] is the identity; Tr[M.M]/16 is +1 or -1 for every basis element; 136 basis elements have Tr[M.M] > 0; 120 basis elements have Tr[M.M] < 0; symmetric basis elements number $16 \cdot 17/2 = 136$; antisymmetric basis elements number $16 \cdot 15/2 = 120$; Tr[M.M] > 0 exactly when M is symmetric; Tr[M.M] < 0 exactly when M is antisymmetric.

Section 8 also builds the ordinary 4×4 Dirac algebra of the (x_1, x_2, x_3, x_4) block, whose metric is $g_{44} = \text{diag}(1, 1, 1, -1)$. For example [A]: {gamma4[h], gamma4[k]} == 2 g44[[h,k]] ID4 for h,k = 1..4; [S44, S44] closes on the $\mathfrak{so}(3,1)$ algebra. It plays no role below.

2.6 Cartan triality and the split octonions (Section 9): what is proved

The vector, the type-1 spinor and the type-2 spinor are three 8-dimensional representations of Spin(4,4) that triality relates. The original makes the vector-to-spinor bridge concrete with one distinguished spinor, unit. It is row 8 of the eigenvectors of σ , rescaled by $1/\sqrt{2}$. Assertions [A]:

- unit is pinned to the author's spinor $\{1/\text{Sqrt}[2], 0, 0, 0, 1/\text{Sqrt}[2], 0, 0, 0\}$ - the row order of Eigensystem is not relied on silently
- each uEig row is sigma-normalized to +/-1
- unit is sigma-normalized to +1

The bridge and its inverse are

$$\text{triVecToSpin}[[A+1, a+1]] = (\text{unit} \cdot \sigma \cdot \bar{\tau}[A])[[a+1]], \quad (16)$$

$$\text{triSpinToVec} = \text{triVecToSpin}^{-1}, \quad (17)$$

where the row index A of triVecToSpin is a flat vector index and the column index a is a type-1 spinor index. Assertions [A]:

- triVecToSpin . triSpinToVec == ID8
- triSpinToVec . triVecToSpin == ID8
- triSpinToVec == Transpose[eta_AA tau[A].unit] (the original's closed form)
- etaTri is symmetric; etaTri has signature (4,4)

- triVecToSpin transports etaTri back to eta4488
- THE TRIALITY METRIC IS EXACTLY THE SPINOR METRIC: etaTri == sigma

The split-octonion structure constants are read off from the generators through the bridge. In the type-1 spinor basis they are mabc; in the vector basis they are mABC, the usual multiplication table. Assertions [A]:

- eA[1] is a two-sided identity for the vector-basis product
- <eA, eB> composition is compatible with eta (basis-unit norm form)
- m_{ABC} restricted to imaginary directions is totally antisymmetric

The scope of what is proved: the norm is multiplicative *on the basis units*, and the imaginary structure constants are totally antisymmetric. The general composition law $N(xy) = N(x)N(y)$ for arbitrary elements, and alternativity, are **not** asserted.

2.7 The wave function of the un-universe and the author's Lagrangian L_a (Section 11)

The original notebook calls Psi16 the wave function of the “un-universe”. It is sixteen scalar functions of the hidden coordinate x_0 and the time x_4 :

$$\Psi_{16} = (f_0, \dots, f_{15}), \quad f_k \equiv \mathbf{f16}[\mathbf{k}][\mathbf{x0}, \mathbf{x4}], \quad (18)$$

where the upper eight components are the type-1 spinor and the lower eight the type-2 spinor. [A] Psi16 is the direct sum of the type-1 and type-2 spinors. Two re-parametrizations are recorded:

- f16[k] -> Z[k][6 H #1, H #2] &, which is the one used downstream;
- f16[k] -> nZ[k][6 H #1, H #2]/Sqrt[Sin[6 H #1]] &, the author's alternative normalization.

The author's Lagrangian, exactly as defined in Section 11:

```
La[] := ((1/H) Transpose[Psi16] . sigma16 .
  Sum[T16[alpha1 - 1] . D[Psi16, X[[alpha1]]], {alpha1, 1, Length[X]}]
  + 2 (M/H) Transpose[Psi16] . sigma16 . Psi16
) // Simplify[#, constraintVars] &
```

that is,

$$L_a = \frac{1}{H} \Psi_{16}^T \sigma_{16} T_{16}[\alpha] \partial_\alpha \Psi_{16} + \frac{2M}{H} \Psi_{16}^T \sigma_{16} \Psi_{16} \quad (19)$$

$$= \frac{1}{H} \Psi_{16}^T \sigma_{16} (T_{16}[0] \partial_{x_0} + T_{16}[4] \partial_{x_4}) \Psi_{16} + \frac{2M}{H} s, \quad s = \Psi_{16}^T \sigma_{16} \Psi_{16}. \quad (20)$$

The second line holds because Ψ_{16} depends only on x_0 and x_4 . It was checked for this document against the evaluated La[] (a Simplify difference of exactly 0); it is not a notebook assertion. The mass term is $(2M/H)s$: it is $+2M/H$ times the scalar bilinear s .

The evaluated La[] [D] is shown below. Write $f_k = \mathbf{f16}[\mathbf{k}][\mathbf{x0}, \mathbf{x4}]$, $\partial_0 f_k = \text{Derivative}[1, 0][\mathbf{f16}[\mathbf{k}]][\mathbf{x0}, \mathbf{x4}]$ and $\partial_4 f_k = \text{Derivative}[0, 1][\mathbf{f16}[\mathbf{k}]][\mathbf{x0}, \mathbf{x4}]$. Then $H L_a$ is exactly

$$\begin{aligned} H L_a = & -4M (f_0 f_4 + f_1 f_5 + f_2 f_6 + f_3 f_7 - f_8 f_{12} - f_9 f_{13} - f_{10} f_{14} - f_{11} f_{15}) \\ & + f_{12}(\partial_4 f_5 + \partial_0 f_0) + f_{13}(-\partial_4 f_4 + \partial_0 f_1) + f_{14}(-\partial_4 f_7 + \partial_0 f_2) \\ & + f_{15}(\partial_4 f_6 + \partial_0 f_3) + f_8(-\partial_4 f_1 + \partial_0 f_4) + f_9(\partial_4 f_0 + \partial_0 f_5) \\ & + f_{10}(\partial_4 f_3 + \partial_0 f_6) + f_{11}(-\partial_4 f_2 + \partial_0 f_7) + f_4(\partial_4 f_{13} - \partial_0 f_8) \end{aligned}$$

$$\begin{aligned}
& -f_5(\partial_4 f_{12} + \partial_0 f_9) - f_6(\partial_4 f_{15} + \partial_0 f_{10}) + f_7(\partial_4 f_{14} - \partial_0 f_{11}) \\
& -f_0(\partial_4 f_9 + \partial_0 f_{12}) + f_1(\partial_4 f_8 - \partial_0 f_{13}) + f_2(\partial_4 f_{11} - \partial_0 f_{14}) \\
& -f_3(\partial_4 f_{10} + \partial_0 f_{15}).
\end{aligned}$$

The structural identity behind it. $\sigma_{16}T_{16}[A]$ is antisymmetric (Section 5). An antisymmetric form vanishes on the diagonal, so for commuting components $\Psi_{16}^T \sigma_{16} T_{16}[\alpha] \Psi_{16} = 0$ identically. The kinetic term therefore equals minus its transpose, with no boundary term. [A] Psi16 . sigma16 . T16[alpha] . Psi16 == 0 identically (antisymmetric form); the kinetic term equals minus its transpose, exactly; La[] == its conjugate form when the coefficients match. The last one is $L_a = -\frac{1}{H}(\partial_\alpha \Psi_{16})^T \sigma_{16} T_{16}[\alpha] \Psi_{16} + \frac{2M}{H} \Psi_{16}^T \sigma_{16} \Psi_{16}$. The original's scratch cell compared L_a with a form carrying mismatched factors of H and so found a non-zero difference [P].

2.8 The Euler–Lagrange equations eLa (Section 12)

The original's Euler–Lagrange operator, transcribed exactly:

```

eL[Lagrangian_Symbol, detsqrt_] := Module[{L, t},
  L = Lagrangian[];
  t = Table[
    FullSimplify[
      (1/detsqrt) ( D[L, f16[k][x0, x4]]
        - D[D[L, Derivative[1, 0][f16[k]][x0, x4]], x0]
        - D[D[L, Derivative[0, 1][f16[k]][x0, x4]], x4] ),
      constraintVars],
    {k, 0, 15}];
  Return[t /. subsDefects];
  (* subsDefects = {} *)
eLa = eL[La, 1]

```

[A] there are sixteen field equations. $eLa[[k]]$ is the Euler–Lagrange expression for the component f_{k-1} , and the field equations are $eLa[[k]] = 0$. **The sixteen expressions, exactly as the notebook computes them** [D], in the notation above:

$$\begin{aligned}
eLa[[1]] &= -\frac{2}{H}(2Mf_4 + \partial_4 f_9 + \partial_0 f_{12}) & (\text{varied: } f_0) \\
eLa[[2]] &= -\frac{2}{H}(2Mf_5 - \partial_4 f_8 + \partial_0 f_{13}) & (\text{varied: } f_1) \\
eLa[[3]] &= -\frac{2}{H}(2Mf_6 - \partial_4 f_{11} + \partial_0 f_{14}) & (\text{varied: } f_2) \\
eLa[[4]] &= -\frac{2}{H}(2Mf_7 + \partial_4 f_{10} + \partial_0 f_{15}) & (\text{varied: } f_3) \\
eLa[[5]] &= -\frac{2}{H}(2Mf_0 - \partial_4 f_{13} + \partial_0 f_8) & (\text{varied: } f_4) \\
eLa[[6]] &= -\frac{2}{H}(2Mf_1 + \partial_4 f_{12} + \partial_0 f_9) & (\text{varied: } f_5) \\
eLa[[7]] &= -\frac{2}{H}(2Mf_2 + \partial_4 f_{15} + \partial_0 f_{10}) & (\text{varied: } f_6) \\
eLa[[8]] &= -\frac{2}{H}(2Mf_3 - \partial_4 f_{14} + \partial_0 f_{11}) & (\text{varied: } f_7) \\
eLa[[9]] &= \frac{2}{H}(2Mf_{12} - \partial_4 f_1 + \partial_0 f_4) & (\text{varied: } f_8) \\
eLa[[10]] &= \frac{2}{H}(2Mf_{13} + \partial_4 f_0 + \partial_0 f_5) & (\text{varied: } f_9) \\
eLa[[11]] &= \frac{2}{H}(2Mf_{14} + \partial_4 f_3 + \partial_0 f_6) & (\text{varied: } f_{10}) \\
eLa[[12]] &= \frac{2}{H}(2Mf_{15} - \partial_4 f_2 + \partial_0 f_7) & (\text{varied: } f_{11}) \\
eLa[[13]] &= \frac{2}{H}(2Mf_8 + \partial_4 f_5 + \partial_0 f_0) & (\text{varied: } f_{12}) \\
eLa[[14]] &= \frac{2}{H}(2Mf_9 - \partial_4 f_4 + \partial_0 f_1) & (\text{varied: } f_{13}) \\
eLa[[15]] &= \frac{2}{H}(2Mf_{10} - \partial_4 f_7 + \partial_0 f_2) & (\text{varied: } f_{14}) \\
eLa[[16]] &= \frac{2}{H}(2Mf_{11} + \partial_4 f_6 + \partial_0 f_3) & (\text{varied: } f_{15})
\end{aligned}$$

In Wolfram InputForm the first reads, character for character (broken over two lines here),

```
eLa[[1]] = (-2*(2*M*f16[4][x0, x4] + Derivative[0, 1][f16[9]][x0, x4]
+ Derivative[1, 0][f16[12]][x0, x4]))/H
```

and the other fifteen follow the same pattern with the indices and signs above.

These were printed for this document by loading the notebook's own `DumpSave` file `claude-fable/claude-fable_Einstein-Rosen-2-Planes-eLa.mx`. That file is `SameQ` to the author's original `Pre-gravityPre-Big_Bang_M6=3-Generations_of_Einstein-Rosen-2-Planes-eLa.mx` in the repository root; this was re-checked for this document with `SameQ`, which returned `True`. So they are the author's equations, unchanged. In matrix form they read

$$\mathbf{eLa} = \frac{2}{H} \sigma_{16} (T_{16}[0] \partial_{x_0} + T_{16}[4] \partial_{x_4} + 2M) \Psi_{16}, \quad \text{i.e.} \quad (21)$$

$$(T_{16}[0] \partial_{x_0} + T_{16}[4] \partial_{x_4}) \Psi_{16} = -2M \Psi_{16}. \quad (22)$$

This matrix form was checked for this document (a `Simplify` difference of exactly 0); it is not a notebook assertion. In Part VI's language (below) it is the flat-frame Dirac equation $\gamma^\mu \partial_\mu \Psi = H V'(s) \Psi$ with $\gamma^\mu = T_{16}[\mu]$ and $V'(s) = -2M/H$. What Part VI asserts is the fidelity anchor below: the Euler–Lagrange expressions of its fable Lagrangian with $V = -(2M/H)s$, on the flat frame and with no volume factor, are exactly `eLa`.

The coupling pattern. Iterating `MergeSetsStep` to a fixed point merges any two sets of component indices that share an index. The result partitions the sixteen components into [D]

$$\mathbf{eLaCouplings} = \{\{0, 5, 8, 13\}, \{1, 4, 9, 12\}, \{2, 7, 10, 15\}, \{3, 6, 11, 14\}\}. \quad (23)$$

[A] the coupled blocks partition all sixteen components; there are four blocks of four. The notebook writes `eLa` and `eLazt` to `.mx` files beside itself, as the original does.

2.9 The (z, t) chart, four blocks of four, and the closed-form solutions (Section 13)

The chart. The chart is $z = 6Hx_0$, $t = Hx_4$. Substituting `f16[k] → Z[k]` [6 H x0, H x4] and then $x_0 \rightarrow z/(6H)$, $x_4 \rightarrow t/H$, with the original's normalization $1/(2H)$, gives `eLazt` = $\frac{1}{2H}$ `eLa` in the new chart. The first equation, for example, reads [D]

$$\mathbf{eLazt}[[1]] = -\frac{2M Z_4 + H \partial_t Z_9 + 6H \partial_z Z_{12}}{H^2}, \quad Z_k \equiv Z[k][z, t]. \quad (24)$$

Decoupling. Solving $0 = \mathbf{eLazt}$ for the sixteen t -derivatives gives [A] all sixteen t -derivatives are determined. Relabelling each coupling set as a consecutive block, `sZt0yZ` maps Z_k to yZ_j as follows [D]:

- $Z_0, Z_5, Z_8, Z_{13} \rightarrow yZ_0..yZ_3;$
- $Z_1, Z_4, Z_9, Z_{12} \rightarrow yZ_4..yZ_7;$
- $Z_2, Z_7, Z_{10}, Z_{15} \rightarrow yZ_8..yZ_{11};$
- $Z_3, Z_6, Z_{11}, Z_{14} \rightarrow yZ_{12}..yZ_{15}.$

Assertions [A]:

- the relabelling is a bijection of the sixteen components
- the system splits into four blocks of four equations
- each 4x4 block is closed in its own four components

The relabelling is a linear map. [A] the yZ coefficient matrix is the identity; ORTHOGONAL: Transpose[S] . S == ID16; ORTHOGONAL: S . Transpose[S] == ID16; but NOT a O(8,8) transformation: it does not preserve sigma16; and NOT a direct sum: it mixes type-1 with type-2.

Closed forms. DSolve, wrapped in a 60-second TimeConstrained, returns every block unevaluated. The log prints `block 1: returned unevaluated` through `block 4: returned unevaluated`. The author obtained closed forms with Maple and pasted them into the original as strings. The notebook reproduces them verbatim, parses them with the author's ConvertMapleToMathematicaV2, and then substitutes them back into the four blocks. That substitution, not the parser, is what certifies them. [A] all sixteen components have a closed form; there are sixteen pure-function rules; THE MAPLE CLOSED FORMS SOLVE ALL SIXTEEN EQUATIONS.

Block j has one constant C_j and four constants $c_{j1}..c_{j4}$. Every component of block j carries the same exponential

$$\begin{aligned} E_j &= \exp\left(\frac{6(z/6 + t)H^2 C_j^2 - 6(-z/6 + t)M^2}{6C_j H^2}\right) \\ &= \exp\left(C_j\left(t + \frac{z}{6}\right) - \frac{M^2}{C_j H^2}\left(t - \frac{z}{6}\right)\right). \end{aligned} \quad (25)$$

Block 1, verbatim from the Maple string:

$$\begin{aligned} yZ_0 &= E_1 \frac{H^2(c_{11}c_{12} - c_{13}c_{14})C_1^2 - M^2(c_{11}c_{12} + c_{13}c_{14})}{2HC_1M}, & yZ_2 &= c_{13}c_{14} E_1, \\ yZ_1 &= -E_1 \frac{H^2(c_{11}c_{12} - c_{13}c_{14})C_1^2 + M^2(c_{11}c_{12} + c_{13}c_{14})}{2HC_1M}, & yZ_3 &= c_{11}c_{12} E_1. \end{aligned}$$

Blocks 2, 3 and 4 have the same shape with their own constants and signs (Section 13 prints all four strings).

2.10 Bilinears, the solution branches, and $M_6 = 3$ generations of Einstein–Rosen 2-planes (Section 14)

Five bilinears are formed from the solution written in the relabelled basis. `psi1sol` and `psi2sol` are its first and second halves of eight components **in the relabelled yZ basis**, that is, coupling blocks 1–2 and coupling blocks 3–4. They are not the type-1 and type-2 spinors. [A] the yZ split at 8 is NOT the type-1/type-2 split; the first relabelled block holds four type-1 and four type-2 components.

For the full sixteen-component norm, [A]:

- `psi . sigma16 . psi == psi2.sigma.psi2 - psi1.sigma.psi1`
- `sigma16` alone is SYMMETRIC, so `psi . sigma16 . psi` is not forced to vanish
- and for this solution it does NOT vanish: it is not the zero expression

With the light-cone coordinates $ztadv = 6t + z$ and $ztret = 6t - z$, the notebook requires the two block norms to be light-cone constants. [A] the light-cone-constant condition has solutions. Solve returns four branches [D], among them $\{C_2 \rightarrow -C_1, C_4 \rightarrow -C_3\}$, the original's branch. [A] on the branch $C_2 = -C_1, C_4 = -C_3$ the two norms are independent of $ztadv$ and $ztret$.

The displayed values on that branch [D, re-read for this document from the saved kernel state] are sharper than the assertion. There, $\psi \cdot \sigma_{16} \cdot \psi$, $\psi_1 \cdot \sigma \cdot \psi_1$ and $\psi_2 \cdot \sigma \cdot \psi_2$ are all identically 0, while $\psi_2 \cdot \sigma \cdot \psi_1$ is not.

The two mass branches [D] are, both taken on the branch $C_2 \rightarrow -C_1$, $C_4 \rightarrow -C_3$:

- **Psi16ab**: $M \rightarrow H M_{ab} \sqrt{C_1 C_3}$, which keeps a free dimensionless ratio M_{ab} ;
- **Psi16aa**: $M \rightarrow H \sqrt{C_1 C_3}$, the case $M_{ab} = 1$.

In the (x_0, x_4) chart the advanced and retarded coordinates are $\xi_{\text{adv}} = x_0 - x_4$ and $\xi_{\text{ret}} = x_0 + x_4$. Every component of **Psi16aa** is a constant times one of $\exp(\pm H(C_1 \xi_{\text{ret}} + C_3 \xi_{\text{adv}}))$ or $\exp(\pm H(C_3 \xi_{\text{ret}} + C_1 \xi_{\text{adv}}))$ [D, re-read for this document from the saved kernel state; the notebook prints **Short** forms]. The special case $C_3 \rightarrow C_1$, **Psi16x0ONLY**, depends on x_0 alone. It is exactly [D, re-read in full from the saved kernel state], with $e_{\pm} \equiv e^{\pm 2C_1 H x_0}$:

$$\begin{aligned} \text{Psi16x0ONLY} = & (-c_{13}c_{14}e_+, c_{23}c_{24}e_-, -c_{33}c_{34}e_+, c_{43}c_{44}e_-, \\ & c_{21}c_{22}e_-, -c_{11}c_{12}e_+, c_{41}c_{42}e_-, -c_{31}c_{32}e_+, \\ & c_{13}c_{14}e_+, c_{23}c_{24}e_-, c_{33}c_{34}e_+, c_{43}c_{44}e_-, \\ & c_{21}c_{22}e_-, c_{11}c_{12}e_+, c_{41}c_{42}e_-, c_{31}c_{32}e_+). \end{aligned}$$

Triality turns each spinor of the solution into a vector. The genuine type-1 and type-2 spinors are $\psi_1 = \text{Psi16ab}[[1;;8]]$ and $\psi_2 = \text{Psi16ab}[[9;;16]]$. From them:

$$\text{vectorFormOfType2} = \text{triVecToSpin} \cdot \psi_2, \quad (26)$$

$$\text{vectorFormOfType1} = \eta \cdot (\psi_1^T \cdot \sigma \cdot \text{triSpinToVec}). \quad (27)$$

Assertions [A]:

- both triality vectors have eight components
- `vectorFormOfType2 == psi2` contracted on the `SPINOR` index of `triVecToSpin`
- `triVecToSpin` and `triSpinToVec` really are mutually inverse

Each vector is normal to a 7-plane through the origin of the split-octonion algebra, the set of position vectors `Zoct` with $\langle \text{vector}, \text{Zoct} \rangle = 0$. [A] `sevenPlane2` is annihilated by `vectorFormOfType2`; `sevenPlane1` is annihilated by `vectorFormOfType1`.

Two non-parallel 7-planes in an 8-space meet in a 6-plane, **M6**. [A] **M6** is a non-empty solution set; **M6** has six free parameters. The intersection determines `zoct[7]` and `zoct[8]` as linear combinations of the free `zoct[1]..zoct[6]`, with coefficients that depend on x_0 , x_4 and the constants C_1 , C_3 , M_{ab} , c_{jk} [D].

What is proved is that **M6** exists and is a 6-parameter set. The reading of **M6** as the home of “the three generations of Einstein–Rosen 2-plane bridges”, which gives the original notebook its title, is the author’s interpretation and is stated in prose only [P].

Finally [A]: `cfResolution[unit]`: the closed-form bridge equals the matrix inverse; `cfResolution[unit]`: the two bridges are mutually inverse; `cfResolution[unit]` reproduces `triVecToSpin`.

2.11 The canonical frame field (Section 15), restated

At every point of the curved $4 + 4$ spacetime there is a local flat $4 + 4$ Minkowski coordinate system. The frame field installs it; in eight dimensions a vielbein is simply an 8-dimensional vierbein. The frame is stored with the curved index μ as the row and the flat index a as the column. The canonical frame is the one the original notebook records as the value of its symbol `gtrye`:

$$e = \text{frameCanonical} = \text{diag}(\tan 6Hx_0, q, q, q, 1, p, p, p), \quad (28)$$

$$q = \frac{e^{-a_4(Hx_4)}}{(\sin 6Hx_0)^{1/6}}, \quad p = \frac{e^{+a_4(Hx_4)}}{(\sin 6Hx_0)^{1/6}}, \quad g = e \eta e^T, \quad (29)$$

$$ds^2 = \tan^2(6Hx_0) dx_0^2 + q^2(dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 - p^2(dx_5^2 + dx_6^2 + dx_7^2). \quad (30)$$

Assertions [A]:

- CANONICAL [structural]: frame . coframe == ID8 (certifies only that the frame is invertible)
- CANONICAL [definition]: g == frame . eta4488 . Transpose[frame] (bridge [d]; this IS how g was defined, so it cannot fail)
- CANONICAL [has content]: g == diag(Tan[6 H x0]^2, q^2, q^2, q^2, -1, -p^2, -p^2, -p^2), the stated line element
- CANONICAL [has content]: Diagonal[g] == Diagonal[eta4488] * Diagonal[frame]^2 exactly
- CANONICAL [has content]: g has signature (4,4), GIVEN a4 real and $0 < 6 H x_0 < \pi/2$
- Inverse[g] == Transpose[coframe] . eta4488 . coframe
- g is symmetric
- the metric is non-degenerate
- FABLE-5.1: det g == Sec[6 H x0]^2 (positive, as signature 4+4 requires)

So $\sqrt{|\det g|} = \sec 6Hx_0$ on $0 < 6Hx_0 < \pi/2$. This is the notebook's volume factor `sqrtDetgTele` = `Sec[6 H x0]`, defined in Section 21 and used throughout Part VI (Section 15 itself displays `Abs[Sec[6 H x0]]`).

Two facts follow from the definitions of q and p . First, $qp = (\sin 6Hx_0)^{-1/3}$: the two exponentials cancel [P, Section 15]. Hence $q^3 p^3 = 1/\sin 6Hx_0$, and the 8-volume element factorizes. [A]:

- FABLESCALAR [has content]: `Sqrt[det g_8] == q^3 * (Tan[6 H x0] p^3)`: observed 3-volume density times hidden 4-volume density
- FABLESCALAR [has content]: the 8-volume element is independent of x_4 : $d(q^3 p^3)/dx_4 == 0$
- FABLESCALAR [has content]: the hidden 4-volume density scales as `Exp[+3 a4[H x4]]` and the observed 3-volume as `Exp[-3 a4[H x4]]`: `Vhid Exp[-3 a4]` and `q^3 Exp[+3 a4]` are x_4 -independent

One Text cell of Section 23 writes $q^3 p^3 = 1/\sqrt{\sin 6Hx_0}$. That is a typo in the prose: from $qp = (\sin 6Hx_0)^{-1/3}$ the cube is $1/\sin 6Hx_0$, which is what makes $\sqrt{\det g} = q^3 \tan(6Hx_0) p^3 = \sec 6Hx_0$. The assertions above are unaffected.

The curved Dirac matrices are $\gamma^\mu = \sum_a \text{coframe}[[a, \mu]] T_{16}[a]$, with `coframe` = e^{-1} . On the canonical frame they are as follows (read for this document from the saved kernel state; they follow directly from the diagonal coframe):

$$\begin{aligned} \gamma^0 &= \cot(6Hx_0) T_{16}[0], & \gamma^i &= \frac{1}{q} T_{16}[i] = e^{a_4} (\sin 6Hx_0)^{1/6} T_{16}[i], \\ \gamma^4 &= T_{16}[4], & \gamma^h &= \frac{1}{p} T_{16}[h] = e^{-a_4} (\sin 6Hx_0)^{1/6} T_{16}[h], \end{aligned} \quad (31)$$

for $i = 1, 2, 3$ and $h = 5, 6, 7$. [A] `{gammaCurved[mu], gammaCurved[nu]} == 2 gInv[mu, nu] ID16`; and, from Part VI, FABLE [has content]: `(gamma^4)^2 == -ID16`, so `gamma^4 d_4 Psi' == H V'(s) Psi'` is solved by `d_4 Psi' == -H V'(s) gamma^4 Psi'`.

The free function a_4 . a_4 is a free, differentiable scalar function of Hx_4 . The original notebook never defines it; it occurs only inside the recorded frame. It is carried symbolically everywhere and is never given a value, and every result holds for arbitrary a_4 . The notebook says so on screen: the log prints NOTE a_4 is an undefined scalar function inherited from the original notebook together with the explanation, which also names 1a. Near its end Part VI checks [A] PART VI [control]: a_4 is STILL UNDEFINED - nothing in Part VI gave it a value; and the FLRW scale factor a_F never entered the canonical frame.

The physical reading. The observer is the geodesic observer $u = \partial/\partial x_4$: $g_{44} = -1$, and x_0 is constant along the worldline [P, Section 23]. The cosmological time is $t = x_4$, the proper time of that observer [P, Section 25]. The observed sheet is x_1, x_2, x_3 ; the hidden spacelike coordinate is x_0 ; the hidden timelike sheet is x_5, x_6, x_7 . Part VI makes the kinematic identification $a(t) \sim q \sim e^{-a_4(Ht)}$, $H_{\text{obs}} = -H a'_4(Ht)$. Assertions [A]:

- KINEMATIC [definition]: $\ln q == -a_4[H x_4] + \text{const}(x_0)$: $D[\text{Log}[q], x_4] == -H a'_4[H x_4] == H_{\text{obs}}$
- KINEMATIC [has content]: the second sheet contracts at exactly the opposite rate: $D[\text{Log}[p], x_4] == +H a'_4[H x_4] == -H_{\text{obs}}$
- KINEMATIC [has content]: $H_{\text{obs}} > 0$ exactly when $a'_4 < 0$ (and $H_{\text{obs}} < 0$ when $a'_4 > 0$) - the sign HYPOTHESIS of this Part, stated, not derived

So which sheet grows is set by the sign of a'_4 , which is never fixed. Section 15's prose describes the 3-space contracting while the deflating directions expand. Part VI reads the same frame the other way round (the observed sheet expands when $a'_4 < 0$) and states that sign as a hypothesis [P].

What the canonical spin connection contributes. The spin connection of this frame follows from the zero-torsion vielbein postulate. Only the facts that the classical fable uses are restated here:

$$\Gamma_{\mu}^{\text{spin}} = \frac{1}{8} \omega_{\mu ab} [T_{16}[a], T_{16}[b]], \quad D_{\mu} \Psi = \partial_{\mu} \Psi + \Gamma_{\mu}^{\text{spin}} \Psi, \quad (32)$$

where each $\Gamma_{\mu}^{\text{spin}}$ is block diagonal: it never mixes type-1 with type-2. Its 24 non-zero components [D, Section 16] are listed below, with $c = \cos 6Hx_0$, $s = \sin 6Hx_0$, $A = a_4(Hx_4)$, $A' = a'_4(Hx_4)$, $j = 1, 2, 3$ and $k = 5, 6, 7$:

component	value
$\omega_{j;0j} = -\omega_{j;j0}$	$Hc^2/(e^A s^{13/6})$
$\omega_{j;4j} = -\omega_{j;j4}$	$HA'/(e^A s^{1/6})$
$\omega_{k;0k} = -\omega_{k;k0}$	$-e^A Hc^2/s^{13/6}$
$\omega_{k;4k} = -\omega_{k;k4}$	$e^A HA'/s^{1/6}$

Assertions [A]:

- CANONICAL [has content]: $\omega_{\mu, a, b} == -\omega_{\mu, b, a}$ (metric compatibility)
- INDEPENDENT CHECK: the frame-only derivation reproduces omegaCanonical exactly
- [has content] the spinor connection is compatible with omega:
 $[\text{Gamma}^{\text{spin}}_{\mu}, \text{gamma}^a] + \omega_{\mu}^a{}_b \text{gamma}^b == 0$
- $\text{Gamma}^{\text{spin}}[\mu]$ is block diagonal: it never mixes type-1 with type-2

- [has content] and therefore the curved Dirac matrices are covariantly constant: $D_\mu \gamma^\mu_{\text{Curved}} = 0$

From Part V (the fable-5.1 bridge), three results are used below [A]:

- FABLE-5.1 [THE RESULT]: $\gamma^\mu \Gamma^\mu_{\text{spin}} = -(1/2) T^\mu \gamma_\mu$, one vector term, with the Weitzenböck torsion vector $T_\mu = \partial_\mu \ln \sin 6Hx_0$, i.e. FABLE-5.1 torsion vector has exactly one non-zero component, $6H \cot[6Hx_0]$ along x_0 . Hence $\gamma^\mu \Gamma^\mu_{\text{spin}} = -3H \cot^2(6Hx_0) T_{16}[0]$.
- FABLE-5.1 [THE RESULT]: $\text{DiracCanonical}[\text{Sqrt}[\text{Sin}[6Hx_0]] \Psi'] = \text{Sqrt}[\text{Sin}[6Hx_0]] \text{DiracFable51}[\Psi']$, for a generic Ψ'
- FABLE-5.1 [THE RESULT]: $\Psi^T \sigma_{16} \gamma^\mu \Gamma^\mu_{\text{spin}} \Psi = 0$ for EVERY Ψ - the spin connection drops out of the Lagrangian, because FABLE-5.1: the reason is Section 11's identity, $\Psi^T \sigma_{16} T_{16}[a] \Psi = 0$ ($\sigma_{16} T_{16}[a]$ antisymmetric). Note that this reason is a statement about **commuting** components.

The Ricci scalar of the canonical metric is [D, Section 16]

$$R = -3H^2 \left((55 + 7 \cos 12Hx_0) \cot^2(6Hx_0) \csc^2(6Hx_0) - 2a'_4(Hx_4)^2 \right). \quad (33)$$

2.12 The classical field fable of 2026-09-16 (Part VI, Section 24)

Part VI (Sections 23–25) places two matter fields on the pre-universe. It asks of each the cosmologist's question: what are the energy density ρ , the pressure P , and $w = P/\rho$? The first field, **fableScalar**, is a real scalar with potential $V(\phi)$. The second, **fable**, is the subject of this document.

fableScalar, in brief. It has $\hat{L}_\phi = -\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi)$ and the Hilbert tensor $T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi + g_{\mu\nu} \hat{L}_\phi$. Its central theorem is [A] FABLESCALAR [THE RESULT]: the null energy condition $\rho + P_1 = 2KE$ holds IDENTICALLY, for every $\phi(x_0, x_4, x_5)$ and every V , so it cannot cross $w = -1$ with positive energy density.

2.12.1 The field and its Lagrangian

fable is a **real, commuting** 16-component spinor $\Psi(x)$ that transforms under $\text{Spin}(4,4)$ exactly as Ψ_{16} . It uses $T_{16}[a]$, σ_{16} , the curved γ^μ , the scalar bilinear $s = \Psi^T \sigma_{16} \Psi$, and a general potential $V(s)$:

$$L_\Psi = \sqrt{\det g} \hat{L}_\Psi, \quad \hat{L}_\Psi = \frac{1}{H} \Psi^T \sigma_{16} \gamma^\mu D_\mu \Psi - V(s), \quad (34)$$

$$D_\mu = \partial_\mu + \Gamma^\mu_{\text{spin}}, \quad s = \Psi^T \sigma_{16} \Psi. \quad (35)$$

In Part VI's generic computations $\Psi = \Psi(x_0, x_4)$ has sixteen arbitrary component functions. Assertions [A]:

- FABLE [definition]: $s = \Psi^T \sigma_{16} \Psi$ is a genuine scalar with content: on the constant spinor $e_1 + e_5$, $s = -2$
- FABLE [has content]: L_{hat} with the covariant derivative $= L_{\text{hat}}$ with the partial derivative, for the generic $\Psi(x_0, x_4)$ (Part V's theorem, re-asserted here)

The fidelity anchor. The author's Lagrangian is the special case $V(s) = -(2M/H)s$, taken on the flat frame ($e \rightarrow I_8$) and with no volume factor ($\sqrt{g} \rightarrow 1$). With that potential $-V = +(2M/H)s$, which is exactly L_a 's mass term. [A]:

- FABLE [fidelity]: with $V = -(2M/H)s$, frame $\rightarrow$ ID8 and $\text{Sqrt}[g] \rightarrow 1$, on the model's Psi16, the Lagrangian helper IS the author's La[] of Section 11
- FABLE [fidelity]: and its Euler-Lagrange equations, formed with the author's own eL operator, ARE the sixteen equations eLa of Section 12
- FABLE [solver regression]: the new helper cfEulerLagrange reproduces eLa too
- FABLE-5.1: the covariant Lagrangian with frame $\rightarrow$ ID8 and volume factor $\rightarrow 1$ IS the author's La of Section 11, term for term

2.12.2 The field equation

The equation is obtained by explicit variation of $\sqrt{g} \hat{L}_\Psi$. The kinetic matrix is $A^\mu = \sigma_{16} \gamma^\mu$, which is antisymmetric; the mass matrix σ_{16} is symmetric. [A]:

- FABLE [THE RESULT]: the Euler-Lagrange equations by explicit variation == $\text{Sqrt}[g] \left[(2/H)(A^\mu d_\mu \Psi + (1/2) D \Psi) - 2 V'(s) \sigma_{16} \Psi \right]$, where $D = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} A^\mu)$
- FABLE [has content]: $A^\mu = \sigma_{16} \gamma^\mu$ is antisymmetric and σ_{16} is symmetric - the structure that makes the variation close
- FABLE [has content]: in general $(1/\text{Sqrt}[g]) d_\mu (\text{Sqrt}[g] \gamma^\mu) == [\gamma^\mu, \Gamma^{\text{spin}}_\mu]$ (covariant constancy of γ^μ , Section 17)
- FABLE [has content]: the HYPOTHESIS that makes it a product - the anticommutator $\{\gamma^\mu, \Gamma^{\text{spin}}_\mu\} == 0$ on the canonical frame (no totally antisymmetric part of the connection)
- FABLE [has content]: hence $(1/2)(1/\text{Sqrt}[g]) d_\mu (\text{Sqrt}[g] \gamma^\mu) == \gamma^\mu \Gamma^{\text{spin}}_\mu$, and on this geometry == $-3 H \cot[6 H x_0]^2 T_{16}[0]$
- FABLE [THE RESULT]: $EL == \text{Sqrt}[g] (2/H) \sigma_{16} \cdot (\gamma^\mu D_\mu \Psi - H V'(s) \Psi)$: the field equation of fable is EXACTLY the Levi-Civita covariant Dirac equation $\gamma^\mu D_\mu \Psi == H V'(s) \Psi$
- FABLE [has content]: the connection term does not vanish on an x_0 -independent spinor - witnessed on the constant spinor e1: a spinor homogeneous in everything but x_4 is NOT a solution

So the field equation of the classical fable is

$$\gamma^\mu D_\mu \Psi = H V'(s) \Psi, \quad \text{i.e. on the canonical frame} \quad (36)$$

$$\begin{aligned} \cot(6Hx_0) T_{16}[0] \partial_0 \Psi + \frac{1}{q} T_{16}[i] \partial_i \Psi + T_{16}[4] \partial_4 \Psi + \frac{1}{p} T_{16}[h] \partial_h \Psi \\ - 3H \cot^2(6Hx_0) T_{16}[0] \Psi = H V'(s) \Psi \end{aligned} \quad (37)$$

(summed over $i = 1, 2, 3$ and $h = 5, 6, 7$; for $\Psi(x_0, x_4)$ only the ∂_0 and ∂_4 terms survive).

The rescaling. [A]:

- FABLE [definition]: under $\Psi = \text{Sqrt}[\text{Sin}[6 H x_0]] \Psi'$, $s == \text{Sin}[6 H x_0] \Psi'^T \sigma_{16} \Psi'$

- FABLE [THE RESULT]: $(\gamma^\mu D_\mu - H V'(s)) [\text{Sqrt}[\text{Sin}[6 H x_0]] \Psi'] == \text{Sqrt}[\text{Sin}[6 H x_0]] (\gamma^\mu d_\mu \Psi' - H V'(s) \Psi')$ for a GENERIC $\Psi'(x_0, x_4)$: the rescaling removes the connection term exactly
- FABLE [has content]: for $\Psi' = \Psi'(x_4)$ the field equation reduces to $\text{Sqrt}[\text{Sin}[6 H x_0]] (\gamma^4 d_4 \Psi' - H V'(s) \Psi')$, $s = \text{Sin}[6 H x_0]$ $\Psi'^T \sigma_{16} \Psi'$
- FABLE [has content]: the reduced equation still depends on x_0 through $V'(\text{Sin}[6 H x_0] s')$: for a NONLINEAR V a $\Psi'(x_4)$ alone is not a solution - witnessed with $V = s^2$ on $\Psi' = e_1 + e_5$
- FABLE [has content]: for the mass term $V = -(2M/H) s$, V' is constant and $\Psi'(x_4)$ IS an exact solution

2.12.3 The energy–momentum tensor

The tensor is built with the covariant derivative and symmetrized:

$$T_{\mu\nu}^{\text{cov}} = -\frac{1}{2H} (\Psi^T \sigma_{16} \gamma_\mu D_\nu \Psi + \Psi^T \sigma_{16} \gamma_\nu D_\mu \Psi) + g_{\mu\nu} \hat{L}_\Psi, \quad (38)$$

with $\gamma_\mu = g_{\mu\nu} \gamma^\nu$. Assertions [A]:

- FABLE [definition]: T_{cov} is symmetric by construction
- FABLE [has content]: the diagonal of T_{cov} equals the diagonal of the partial-derivative tensor, for the generic $\Psi(x_0, x_4)$: ρ and every pressure are the same in both
- FABLE [has content]: but the two tensors DIFFER off the diagonal - witnessed on the (x_3, x_4) component $T[[4,5]]$, a momentum density along the observed sheet, for the constant spinor $e_1 + e_{13}$ and $V = s^2$
- FABLE [THE RESULT]: $\nabla^\mu T_{\mu\nu} == 0$ ON SHELL in all eight components, for a generic $\Psi(x_0, x_4)$ - the covariant tensor is the conserved one

“On shell” means the following. The field equation is solved for $\partial_4 \Psi = F = -\gamma^4 (H V'(s) \Psi - \gamma^0 \partial_0 \Psi - \frac{1}{2} \text{div} G \Psi)$, where $\text{div} G = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \gamma^\mu)$ (on this geometry $\frac{1}{2} \text{div} G = -3H \cot^2(6Hx_0) T_{16}[0]$, by the assertion quoted above), and `cfOnShell` replaces every x_4 -derivative of the field by the corresponding derivative of F , repeatedly, until none is left. [A] FABLE [definition]: F solves the field equation for $d_4 \Psi$: $\gamma^4 F + \gamma^0 d_0 \Psi + (1/2) \text{div} G \Psi - H V'(s) \Psi == 0$; FABLE [definition]: `cfOnShell` leaves no x_4 -derivative of the field in ρ . The partial-derivative tensor was checked symbolically while Part VI was designed: its x_0 and x_4 components are conserved and the other six are not. That check takes minutes and is not repeated in the notebook [P].

2.12.4 ρ , P , and w on shell

The observer is $u = \partial/\partial x_4$, so $\rho = T_{44}$ ($T[[5,5]]$) and $P_i = T^i_i$ (no sum, $i = 1, 2, 3$). Two kinetic bilinears organize the answer:

$$K_\Psi = \frac{1}{H} \Psi^T \sigma_{16} \gamma^4 \partial_4 \Psi \quad (\text{along the observer's time}), \quad (39)$$

$$K_h = -\frac{1}{H} \Psi^T \sigma_{16} \gamma^0 \partial_0 \Psi \quad (\text{along the hidden coordinate } x_0). \quad (40)$$

Assertions [A]:

- FABLE [THE RESULT]: on shell, $L_{\text{hat}} == s V'(s) - V(s)$
- FABLE [THE RESULT]: on shell, $\rho == V(s) + K_h$, $K_h = -(1/H) \Psi^T \sigma_16 \gamma^0 d_0 \Psi$
- FABLE [THE RESULT]: on shell, $P_1 == s V'(s) - V(s)$, and $P_2 == P_3 == P_1$
- FABLE [has content]: on shell, $K_{\Psi} == s V'(s) + K_h$ - the x_4 kinetic bilinear is fixed by the potential and the hidden one
- FABLE [has content]: $K_h == 0$ for $\Psi = \text{Sqrt}[\text{Sin}[6 H x_0]] \Psi'(x_4)$ ($d_0 \Psi$ is proportional to Ψ , and $\Psi^T \sigma_16 \gamma^0 \Psi == 0$)
- FABLE [THE RESULT]: so for those solutions $\rho == V(s)$, hence $w == s V'(s)/V(s) - 1$, and $\rho > 0$ requires $V(s) > 0$

In summary:

$$\rho = V(s) + K_h, \quad P = sV'(s) - V(s), \quad (41)$$

$$\text{and for } \Psi = \sqrt{\sin 6 H x_0} \Psi'(x_4): \quad \rho = V(s), \quad w = \frac{sV'(s)}{V(s)} - 1. \quad (42)$$

2.12.5 No dilution on the pre-universe

Assertions [A]:

- FABLE [THE RESULT]: on shell, $ds/dx_4 == 2 \Psi^T \sigma_16 \gamma^4 \gamma^0 d_0 \Psi$ for a generic $\Psi(x_0, x_4)$
- FABLE [has content]: the two other terms of $d_4 \Psi$ drop out because $\sigma_16 \gamma^4$ and $\sigma_16 \gamma^4 \gamma^0$ are ANTISYMMETRIC
- FABLE [control]: that bilinear is NOT identically zero for an x_0 -dependent Ψ - witnessed on $\Psi = e_1 + x_0 e_2$: ' $ds/dx_4 == 0$ for generic $\Psi(x_0, x_4)$ ' would be FALSE
- FABLE [THE RESULT]: for $\Psi = \text{Sqrt}[\text{Sin}[6 H x_0]] \Psi'(x_4)$ it vanishes: s is CONSTANT IN x_4 - no dilution on the pre-universe
- FABLE [THE RESULT]: the same from the reduced equation: $ds'/dx_4 == 0$ on shell for $\Psi'(x_4)$, because $\Psi'^T \sigma_16 \gamma^4 \Psi' == 0$

So on those solutions s , V , ρ and w are all constant along x_4 . This is the spinor analogue of the scalar's "no Hubble friction" (for the scalar, [A] FABLESCALAR [THE RESULT]: for $\phi = \phi(x_4)$ the field equation is $\phi'' == -V'(\phi)$: NO HUBBLE FRICTION on the pre-universe). For the scalar, Section 23's prose gives the reason as geometric: the factor $\sqrt{g} g^{44} = -\sec 6 H x_0$ does not depend on x_4 , because the 8-volume element is x_4 -independent [P].

2.12.6 Dust for the author's mass term

[A]:

- FABLE [THE RESULT]: the author's mass term $V = -(2M/H) s$ is DUST: on shell $P_1 == 0$ identically and $\rho == -(2M/H) s$ ($w == 0$, exactly)
- FABLE [has content]: that density is not zero - witnessed on $\Psi' = e_1 + e_5$ at $M = 1$ ($\rho = (2M/H) * 2 \text{Sin}[6 H x_0]$ there)

The density $\rho = -(2M/H)s$ is positive only on the branch $M s < 0$ [P, Section 24].

2.12.7 $w = sV'/V - 1$ for the six potentials, and the classical phantom crossing

With $u \equiv (s/s_1)^2$:

mass:	$V = ms,$	$w = 0,$
lambda-mass:	$V = V_0 + ms,$	$w = -\frac{V_0}{V_0 + ms},$
power:	$V = ms + \lambda s^n,$	$w = \frac{ms + n\lambda s^n}{ms + \lambda s^n} - 1,$
hilltop:	$V = V_0 - \mu(s - s_*)^2,$	$w = \frac{-2\mu s(s - s_*)}{V_0 - \mu(s - s_*)^2} - 1,$
lorentz:	$V = V_0 + \frac{ms}{1 + u},$	$w = \frac{ms(1 - u)}{(1 + u)^2(V_0 + ms/(1 + u))} - 1,$
expdamp:	$V = V_0 + ms e^{-s/s_1},$	$w = \frac{ms e^{-s/s_1}(1 - s/s_1)}{V_0 + ms e^{-s/s_1}} - 1.$

Assertions [A]:

- FABLE [has content]: the closed forms of $w_{\text{Psi}}(s) = s V'(s)/V(s) - 1$ for the six potentials of the numerical work (mass, lambda-mass, power, hilltop, lorentz, expdamp)
- FABLE [THE RESULT]: mass $\rightarrow w == 0$ (dust); $V = \text{lam } s^n$ alone $\rightarrow w == n - 1$ (accelerating for $n < 2/3$, phantom for $n < 0$); $n = 0.236 \rightarrow w == -0.764$ exactly
- FABLE [THE RESULT]: $w + 1 == s V'/V$ vanishes where V' does, with $V > 0$ there: lorentz and expdamp cross $w = -1$ at $s = s_1$, hilltop at $s = s_*$ - the phantom divide is crossed with no wrong-sign kinetic term
- FABLE [has content]: lorentz and expdamp are bounded below by $V_0 > 0$ for $s \geq 0$ ($\rho > 0$ on the whole history), and $w < -1$ for $s > s_1$ (the past), $w > -1$ for $0 < s < s_1$ (the present)
- FABLE [has content]: the limits - lorentz and expdamp: $w \rightarrow -1$ both as $s \rightarrow \text{infinity}$ and as $s \rightarrow 0$; lambda-mass: $w \rightarrow 0$ (dust) as $s \rightarrow \text{infinity}$ and $\rightarrow -1$ (Lambda) as $s \rightarrow 0$
- FABLE [has content]: expdamp with $(V_0, m, s_1) = (1.566, 0.839, 2.21)$ at $s = 1$ (today) gives $w == -0.8608$, Unite's $w_0 = -0.861$ to three decimals

The assumptions for the sign statements are $V_0 > 0$, $m > 0$, $s_1 > 0$. This is the **classical phantom crossing**: $w + 1 = sV'/V$ changes sign where V' does, with $V > 0$ there, and no kinetic term has the wrong sign. It is the spinor-quantum mechanism of Cai and Wang, Class. Quantum Grav. 25 (2008) 165014. The stability of perturbations is not examined in Part VI [P]. The V_0 terms of lambda-mass, lorentz and expdamp are a bare cosmological constant and are reported as such [D, the summary table printed in Section 25].

2.12.8 The FLRW reference model (Model B)

Because $ds/dx_4 = 0$ on the pre-universe, a history $w(a)$ needs a different background. Part VI builds it as a **separate** frame, $\text{frameFLRW} = \text{diag}(1, a_F(x_4), a_F(x_4), a_F(x_4), 1, 1, 1, 1)$, with its own scale factor a_F , and feeds it to the same spin-connection solver. The canonical frame and a_4 are not touched. Assertions [A]:

- FLRW [definition]: the reference frame does not mention a_4 , and the canonical frame is untouched

- FLRW [solver regression]: vielbein postulate residual is zero
- FLRW [has content]: omega is antisymmetric (metric compatible)
- FLRW [THE RESULT]: $\gamma^\mu \Gamma_{\text{spin}_\mu} == (3/2) (a'/a) \gamma^4$ - the Hubble term of the FLRW Dirac equation
- FLRW [definition]: the rule solves $\gamma^\mu D_\mu \Psi == H V'(s) \Psi$ for $d_4 \Psi$
- FLRW [THE RESULT]: $d(a^3 s)/dx4 == 0$ on shell, for EVERY V : $s = s_0 a^{-3}$, the dust dilution law, and $w_\Psi(a) = s V'(s)/V(s) - 1$ on it
- FLRW [control]: without the Hubble term, $d(a^3 s)/dx4 == 3 (a'/a) a^3 s$ instead (i.e. $ds/dx4 == 0$, the pre-universe law): the dilution IS the Hubble term

Model B, solved independently with NDSolve. The units are $8\pi G = 1$, $H_0 = 1$ and $\rho_{\text{crit}} = 3$. The independent variable is $N = \ln a$, and $s = e^{-3N}$ is exact, so only the cosmic time needs integrating:

$$\frac{dt}{dN} = \frac{1}{H}, \quad H^2 = \Omega_m e^{-3N} + \Omega_r e^{-4N} + \frac{V(s)}{3}, \quad \Omega_m = \frac{3}{10}, \quad \Omega_r = \frac{84}{10^6}, \quad (43)$$

$$V = \text{expdamp with } (V_0, m, s_1) = \left(\frac{1566}{1000}, \frac{839}{1000}, \frac{221}{100}\right), \quad (44)$$

$$N \in [-7, 0], \quad 701 \text{ rows}, \quad t(N_0) = \frac{1}{2H(N_0)} \text{ added.} \quad (45)$$

All inputs are exact rationals, and WorkingPrecision -> 24. Assertions [A]:

- MODEL B [definition]: 701 rows from $N = -7$ to 0 , $s(0) == 1$ and $\rho_\Psi(0) == V(1) == 1.566 + 0.839 \text{ Exp}[-1/2.21]$
- MODEL B [THE RESULT]: $w \leq -1$ at every grid point with $s > s_1$ and $w > -1$ at every grid point with $s < s_1$: the crossing is at $s = s_1$, $a = s_1^{(-1/3)} = 0.7677$
- MODEL B [has content]: a genuine phantom phase: $\min w = -1.302$ (to 0.001) at $a = 0.543$, and $w(a = 1) = -0.8608$
- MODEL B [has content]: $V(s) > 0$ at every row - $\rho_\Psi > 0$ on the whole history
- MODEL B [fidelity]: the reference CSV has the expected header and the same 701 values of N
- MODEL B [fidelity]: w agrees with fable-cosmology/reference/mathematica_spinor_exp damp.csv at EVERY row to $1e-9$
- MODEL B [fidelity]: so do t , H and ρ_Ψ , to $1e-9$ (s differs only by the file's 16-digit rounding of numbers of order $1e9$)

The run prints [D] (one line in the log, broken here):

Model B (expdamp):

```
w(a=1) = -0.86084564037764806560795392619028548152`20.
min w = -1.30209881490918676547015655743122703056`20. at N = -0.61
(a = 0.54335086907449978711266408781657974806`20.)
t(a=1) = 0.96433642136311524576147498023448633382`20.
H(a=1) = 0.99998205003431888956547950340514648567`20.
```

Model B applies no normalization: $V(1) = 1.566 + 0.839 e^{-1/2.21} \approx 2.0996 = 3 \Omega_{\Psi_0}$ with $\Omega_{\Psi_0} = 0.6999$ [P, Section 24]. That is why the printed $H(a=1)$ is 0.99998 and not exactly 1: $H(a=1)^2 = 3/10 + 84/10^6 + V(1)/3 \approx 0.99996$ (this arithmetic was added for this document; it is not a notebook statement).

2.12.9 What is proved on the pre-universe, and what belongs to the reference model

Proved on the 8-manifold, for arbitrary a_4 , with every identity closed symbolically:

- the Lagrangian and its fidelity to L_a and $\mathbf{eLa}$;
- the covariant Dirac equation and its rescaling;
- the covariant energy-momentum tensor and its conservation;
- $\rho = V + K_h$ and $P = sV' - V$;
- dust for the author's mass term;
- the constancy of s in x_4 ;
- the volume factorization.

The history $s = s_0 a^{-3}$, the phantom crossing at $a = 0.768$, and all contact with the Unite fits $(w_0, w_a) = (-0.861, -0.60)$ belong to 4-dimensional physics on the separate FLRW frame [P, Section 25].

The volume bookkeeping is proved [A]:

- VOLUME [has content]: the reduced density $\rho_4 = V^{-1} \rho_8$ is NOT separately conserved: $d\rho_4/dx_4 + 3 H_{\text{obs}} (\rho_4 + P_4) = 3 H_{\text{obs}} P_4$, for every w
- VOLUME [has content]: equivalently $d\rho_4/dx_4 = -3 H_{\text{obs}} \rho_4$: ρ_4 falls as a^{-3} for EVERY w - a volume effect, the same for a cosmological constant as for dust

Which of ρ_4 and ρ_8 gravitates is left undetermined in Part VI, because Part VI has no gravitational sector: no Einstein equations, and a_4 free [P, Section 25].

2.13 Why the classical fable must be refined

The fable of Part VI is a field of **real, commuting** numbers. A fermion needs **anticommuting** (Grassmann) components, and the change is not cosmetic. It rests on two algebraic facts about the author's Clifford algebra.

F1, the symmetry of CT . Write $\Gamma_{(r)}$ for an antisymmetrized product of r of the $T_{16}[a]$. With $C_- = \sigma_{16}$:

- σ_{16} is symmetric;
- $\sigma_{16} T_{16}[a]$ is **antisymmetric** (this much is already Part I's **sigma16 . T16[A]** is **antisymmetric** for **A = 0..7**);
- $\sigma_{16} \Gamma_{ab}$ is antisymmetric;
- $\sigma_{16} \Gamma_{abc}$ and $\sigma_{16} \Gamma_{abcd}$ are symmetric;
- $\sigma_{16} T_{16}[8]$ is symmetric;
- $\sigma_{16} T_{16}[a] \sigma_{16}^{-1} = -T_{16}[a]^T$.

The second conjugation is $C_+ = \sigma_{16} T_{16}[8]$:

- it is symmetric, and $C_+ T_{16}[a]$ is **symmetric**;
- $C_+ \Gamma_{ab}$ and $C_+ \Gamma_{abc}$ are antisymmetric;
- $C_+ T_{16}[a] C_+^{-1} = +T_{16}[a]^T$.

The Spin(4,4)-invariant bilinear forms on the 16-spinor form a two-dimensional space, spanned by σ_{16} and $\sigma_{16}T_{16}[8]$ ($C_+T_{16}[8] = \sigma_{16}$). Both are symmetric, and both are chirality-diagonal. [VII] REFINEMENT [definition]: $T_{16}[8]^2 == \text{sigma16}^2 == C_+^2 == \text{ID16}$, C_+ is symmetric, and $C_+ \cdot T_{16}[8] == \text{sigma16}$ - the four candidate mass matrices are only two, REFINEMENT [has content]: $C_- T_{16}[a] C_-^{-1} == -T_{16}[a]^T$ and $C_+ T_{16}[a] C_+^{-1} == +T_{16}[a]^T$ for $a = 0..7$, REFINEMENT [THE RESULT]: the symmetry of C Gamma depends only on the rank r of Gamma: C_- : S A A S S A A S S, C_+ : S S A A S S A A S ($r = 0..8$), REFINEMENT [THE RESULT]: the Spin(4,4)-invariant bilinear forms on the 16-spinor form a 2-dimensional space, spanned by $C_- = \text{sigma16}$ and $C_+ = \text{sigma16} T_{16}[8]$, REFINEMENT [has content]: every invariant form is SYMMETRIC and chirality-diagonal, and the two chiral pieces PL sigma16 PL and PR sigma16 PR are invariant separately.

F2, what anticommuting components do to a bilinear. For Grassmann θ :

- $\theta^T N \theta$ sees only the **antisymmetric** part of N ;
- $\theta^T N d\theta = \frac{1}{2} d(\theta^T N \theta) + \theta^T N_{\text{sym}} d\theta$.

For commuting components the roles are exactly reversed: $\theta^T N \theta$ sees only the symmetric part, and the Euler-Lagrange expression of $\theta^T N \phi$ is $2N_{\text{antisym}}\phi$. That reversal is why Part VI's real commuting fable has both a mass term (σ_{16} symmetric) and a kinetic term ($\sigma_{16}T_{16}[a]$ antisymmetric). [VII] GRASSMANN [THE RESULT]: $\text{theta}^T N \text{theta} == \text{theta}^T N_A \text{theta}$ and $\text{theta}^T N_S \text{theta} == 0$ for a general 16×16 N - an anticommuting bilinear sees only the ANTISYMMETRIC part, GRASSMANN [THE RESULT]: $\text{theta}^T N \phi - (1/2) d(\text{theta}^T N \text{theta}) == \text{theta}^T N_S \phi$ for a general N , GRASSMANN [THE RESULT]: the Euler-Lagrange expression of $L = \text{theta}^T N \phi$ (left derivatives) is $(N + N^T) \phi$: EMPTY field equations exactly when N is antisymmetric, GRASSMANN [control]: for COMMUTING components the roles are reversed - $\text{theta}^T N \text{theta}$ sees only N_S , and the Euler-Lagrange expression of $\text{theta}^T N \phi$ is $2 N_A \phi$.

The consequences:

- A real Grassmann 16-spinor with the author's σ_{16} has no dynamics.** Its mass term $\theta^T \sigma_{16} \theta$ vanishes identically, because σ_{16} is symmetric. Its kinetic term $\theta^T \sigma_{16} T_{16}[a] d\theta$ is a total derivative, because $\sigma_{16} T_{16}[a]$ is antisymmetric. So its Euler-Lagrange expression is identically empty, even though the kinetic term itself is not zero. The statement holds on **every** frame. $\sigma_{16} \Gamma_{abc}$ is symmetric, so the connection term $\theta^T \sigma_{16} \{\gamma^\mu, \Gamma_\mu^{\text{spin}}\} \theta$ vanishes too. [VII] REFINEMENT (a) [THE RESULT]: real Grassmann + sigma16 : both invariant mass terms vanish, and $\text{theta}^T \text{sigma16} T_{16}[a] \phi == (1/2) d(\text{theta}^T \text{sigma16} T_{16}[a] \text{theta})$ for every a (a total derivative), REFINEMENT (a) [has content]: its Euler-Lagrange expression is identically empty in every direction - while the kinetic term itself is NOT zero, REFINEMENT (a) [THE RESULT]: FRAME-INDEPENDENT - $\text{sigma16} \{T_{16}[a], S^{bc}\}$ is SYMMETRIC for all a, b, c (so $\text{theta}^T \text{sigma16} \{\gamma^\mu, \Gamma_\mu\} \text{theta} == 0$ on every frame), and $d_\mu(\text{Sqrt}[g] \text{sigma16} \gamma^\mu) == \text{Sqrt}[g] \text{sigma16} [\gamma^\mu, \Gamma_\mu]$ on the canonical frame.
- A Majorana 16-spinor built with C_+ propagates but is massless.** Its kinetic term is genuine, but every invariant bilinear vanishes on it, so there is no s and $V(s)$ is undefined. It has exactly one quartic Lorentz scalar: the quartic scalars $v \cdot v$, $B \cdot B$ and $C \cdot C$ built from it are all proportional to one quartic. Its commuting shadow has no kinetic term at all. [VII] REFINEMENT (b) [THE RESULT]: Majorana with C_+ : the kinetic term is genuine - Euler-Lagrange expression $2 C_+ T_{16}[a] \phi$ with $C_+ T_{16}[a]$ invertible - but every invariant bilinear vanishes (massless, no $V(s)$), REFINEMENT (b) [has content]: the scope - the quartic Lorentz scalars

v.v, B.B, C.C of the Majorana spinor are NOT zero and are all proportional to ONE quartic (same monomials, one constant ratio each), REFINEMENT (b) [has content]: and the commuting shadow of the C+ Majorana spinor has NO kinetic term: for commuting components $\theta^T C+ T_{16}[a] \phi$ is the total derivative $(1/2) d(\theta^T C+ T_{16}[a] \theta)$.

- (c) **Weyl and Majorana–Weyl 8-spinors (one chirality) cannot propagate.** Both C s are chirality-diagonal and γ^μ is chirality-odd, so $P_L C T_{16}[a] P_L = P_R C T_{16}[a] P_R = 0$. [VII] REFINEMENT (c) [THE RESULT]: Weyl and Majorana–Weyl: $PL C T_{16}[a] PL == PR C T_{16}[a] PR == 0$ for $C = C-, C+$ and every a (C chirality-diagonal, $T_{16}[a]$ chirality-odd) – no kinetic term.
- (d) **Fermion fable is the complex 16-component spinor.** It is $\Psi = (\theta_1 + i\theta_2)/\sqrt{2}$, built from two real Grassmann 16-spinors, with the conjugate $\bar{\Psi} = \Psi^\dagger \sigma_{16}$. Here † is the conjugation of the complex Grassmann algebra; the Hilbert-space adjoint is fixed in the quantization, section 6.
- It has two scalar bilinears. $s = \bar{\Psi}\Psi = i\theta_1^T \sigma_{16} \theta_2$ is non-zero and Hermitian under † , and so is $p = \bar{\Psi} T_{16}[8] \Psi$. (With the Hilbert-space adjoint that the quantization fixes, p is odd under the fundamental symmetry J and becomes an anti-Hermitian operator, while s stays Hermitian.)
 - The symmetrized kinetic term is Hermitian and genuine.
 - It reduces **exactly** to Part VI’s real commuting fable: its Lagrangian, its field equation, and its covariant energy–momentum tensor. Through that reduction it reduces to the author’s L_a and eLa.

[VII] REFINEMENT (d) [THE RESULT]: $s = \text{Psibar Psi} == i \theta_1^T \sigma_{16} \theta_2$: NON-ZERO (16 monomials) and HERMITIAN; $p = \text{Psibar } T_{16}[8] \text{ Psi}$ is non-zero and Hermitian too, REFINEMENT (d) [THE RESULT]: its symmetric part is $A_S == (i/2) \Omega$, $\Omega = [[0, K], [-K, 0]]$ real SYMMETRIC and invertible: a genuine kinetic term, REFINEMENT (d) [has content]: the symmetrized kinetic term is HERMITIAN and equals $\text{Psi}^\dagger K \phi - (1/2) d(\text{Psi}^\dagger K \text{ Psi})$; the unsymmetrized one is not Hermitian, FIDELITY [1] [fidelity]: $\text{Psi} \rightarrow \text{psi}, \text{Psibar} \rightarrow \text{psi}^T \sigma_{16}$ turns the symmetrized covariant L_{hat} into Part VI’s cfLhatFable EXACTLY (canonical frame, generic V), FIDELITY [2] [fidelity]: on the flat frame with $V = -(2M/H) s$ and no volume factor, the same limit on Psi_{16} IS the author’s L_a of Section 11, FIDELITY [3] [fidelity]: $\sigma_{16} \cdot EL_{\text{Psibar}} + EL_{\text{Psi}}$ in the real limit == Part VI’s explicit variation cfELFable , component for component, FIDELITY [3] [fidelity]: and on the flat frame with $V = -(2M/H) s$ the same combination IS the author’s eLa of Section 12, FIDELITY [4] [fidelity]: the energy-momentum tensor That in the real limit IS Part VI’s covariant tensor cfTCov , all 64 components.

The claim must be stated with its scope. The complex 16-spinor is the **minimal** field that:

1. uses the author’s σ_{16} conjugation;
2. propagates;
3. admits a potential $V(s)$;
4. reduces to Part VI’s classical fable as its real commuting shadow.

It is **not** the unique field with dynamics: a Majorana spinor with C_+ also propagates. The author chose the complex 16-spinor on 2026-09-24.

One more consequence matters for everything that follows. Part VI’s mechanism for the spin connection dropping out of the Lagrangian was $\Psi^T \sigma_{16} T_{16}[a] \Psi = 0$, which is a property of commuting components. It does not carry over to complex Grassmann fields. There, the drop of the connection is a property of the matrix $\{\gamma^\mu, \Gamma_\mu^{\text{spin}}\}$ itself, and it has to be proved again. Whether the classical results of this section, among them w and the phantom crossing, survive quantization is decided in sections 7 and 8. The complete refinement, with every assertion of Section 26, is section 3.

3 The refinement: why fable must be a complex 16-spinor

This is Section 26 of the notebook (manifest `claude-fable/cells_part7.wl`, Input cells 173–179, 38 assertions). It turns the classical, real, commuting fable of section 2.12 into a fermion.

Two conjugations, two symbols. $\Psi^\dagger$ is the **classical** conjugate of the field, the conjugation of the Grassmann algebra, the one that appears in the Lagrangian; the Dirac conjugate is $\bar{\Psi} = \Psi^\dagger \sigma_{16}$. After quantization (section 6) the Hilbert-space adjoint is a **different** operation, $\Psi^\dagger = \Psi^\dagger J$. The two are never written with the same symbol [**prose VII, introduction of Part VII**].

Two kinds of algebra, and when each is used [**prose VII**]. Statements whose content *is* the anticommutation of the field (that a mass term or a kinetic term vanishes, is Hermitian, or is a total derivative) are proved in a genuine Grassmann (exterior) algebra, built and tested in Section 26 (subsection 3.3). Statements about **bilinear** expressions (Lagrangians, Euler–Lagrange equations, currents, energy–momentum tensors, conservation laws) are proved with Ψ and $\bar{\Psi}$ represented by two **independent** vectors of ordinary functions of the coordinates, with $\bar{\Psi}$ always written to the **left** of Ψ . That commuting proxy is exact for such identities: in every term $\bar{\Psi}$ is the leftmost and Ψ the rightmost odd factor, so the left derivative with respect to $\bar{\Psi}$ and the right derivative with respect to Ψ are the ordinary derivatives, and no reordering of odd factors ever occurs. Where reality matters (Hermiticity under †), $\Psi = u + iv$ with real u, v and $\bar{\Psi} = (u - iv)^T \sigma_{16}$.

3.1 The Clifford symmetry table

A spinor bilinear $\psi^T C \Gamma \psi$ is built from a “charge conjugation” matrix C and one of the 256 ordered products Γ of the Dirac matrices (Section 6’s basis `base16`, grouped by the rank r , the number of factors). There are two candidates for C :

$$\begin{aligned} C_- &= \sigma_{16}, & C_- T_{16}[a] C_-^{-1} &= -T_{16}[a]^T & (\text{the author’s spinor metric}), \\ C_+ &= \sigma_{16} T_{16}[8], & C_+ T_{16}[a] C_+^{-1} &= +T_{16}[a]^T. \end{aligned}$$

Every product $C\Gamma$ is either symmetric (S) or antisymmetric (A), and the answer depends only on the rank:

rank r	0	1	2	3	4	5	6	7	8
number of products, $\binom{8}{r}$	1	8	28	56	70	56	28	8	1
$C_- \Gamma = \sigma_{16} \Gamma$	S	A	A	S	S	A	A	S	S
$C_+ \Gamma = \sigma_{16} T_{16}[8] \Gamma$	S	S	A	A	S	S	A	A	S

For each C , 136 products are symmetric and 120 antisymmetric, as it must be for 16×16 matrices ($16 \cdot 17/2 = 136$, $16 \cdot 15/2 = 120$). The “four candidate” mass matrices σ_{16} , C_+ , $\sigma_{16} T_{16}[8]$, $C_+ T_{16}[8]$ are only two, because $T_{16}[8]^2 = \text{ID}_{16}$: $\sigma_{16} T_{16}[8] = C_+$ and $C_+ T_{16}[8] = \sigma_{16}$. Why this

table decides everything: for **commuting** components $\psi^T N \psi$ sees only the symmetric part of N ; for **anticommuting** (Grassmann) components only the antisymmetric part.

[proved VII §26]

- REFINEMENT [definition]: $T16[8]^2 == \text{sigma16}^2 == C+^2 == \text{ID16}$, $C+$ is symmetric, and $C+ \cdot T16[8] == \text{sigma16}$ - the four candidate mass matrices are only two
- REFINEMENT [has content]: $C- T16[a] C-^{-1} == -T16[a]^T$ and $C+ T16[a] C+^{-1} == +T16[a]^T$ for $a = 0..7$
- REFINEMENT [THE RESULT]: the symmetry of C Gamma depends only on the rank r of Gamma: $C-: S A A S S A A S S$, $C+: S S A A S S A A S$ ($r = 0..8$) (the assertion also checks that the counts per rank are $\binom{8}{r}$)
- REFINEMENT [has content]: for each C , 136 of the 256 products are symmetric and 120 antisymmetric

3.2 The Lorentz-invariant bilinears: exactly two, both symmetric

A bilinear $\psi^T N \psi$ (real field) or $\Psi^\dagger N \Psi$ (complex field) is a $\text{Spin}(4,4)$ scalar exactly when $S^T N + N S = 0$ for all 28 generators $S = \text{SAB}$ of Section 5 (the generators are real, so the condition is the same in both cases). Solving these 7168 linear equations (28×256) for the 256 entries of N gives a **two**-dimensional space, spanned by C_- and C_+ , equivalently by the two chiral pieces $P_L \sigma_{16} P_L$ and $P_R \sigma_{16} P_R$, one for each split-octonion spinor type. Every invariant N is **symmetric** and **chirality-diagonal** (it commutes with $T_{16}[8]$). That is the complete list of candidates for a mass term; nothing is left out [prose VII §26].

[proved VII §26]

- REFINEMENT [THE RESULT]: the $\text{Spin}(4,4)$ -invariant bilinear forms on the 16-spinor form a 2-dimensional space, spanned by $C- = \text{sigma16}$ and $C+ = \text{sigma16 } T16[8]$
- REFINEMENT [has content]: every invariant form is **SYMMETRIC** and **chirality-diagonal**, and the two chiral pieces $PL \text{ sigma16 } PL$ and $PR \text{ sigma16 } PR$ are invariant separately

3.3 A Grassmann algebra, built and tested

To prove statements about anticommuting fields the notebook uses an honest exterior algebra, not a sign convention applied by hand. An element is an **Association** from **monomials** to coefficients; a monomial is the sorted list of the generator indices it contains, $\{\}$ being the unit. The product of two monomials is zero if they share a generator; otherwise it is the sorted union with the sign of the permutation that sorts the concatenation (**Signature**). The operations are [prose VII §26]:

operation	what it computes
<code>grAdd, grScale, grMul</code>	sums, scalar multiples and products
<code>grSame[x, y]</code>	equality of two elements ($x - y$ has no monomial left)
<code>grBil[xs, N, ys]</code>	the bilinear $\sum_{ab} x_a N_{ab} y_b$ of two lists of elements
<code>grD[x, dmap]</code>	the even derivation that maps generator i to generator <code>dmap[i]</code> (with $\theta_a \mapsto \phi_a$ it is d/dt , with $\phi_a = d\theta_a/dt$)
<code>grDL[x, i]</code>	the left derivative with respect to generator i
<code>grConj[x]</code>	the classical conjugation † : $(xy)^\dagger = y^\dagger x^\dagger$, generators self-conjugate (real), coefficients complex-conjugated

operation	what it computes
<code>grEL[L, thIdx, phIdx, dmap]</code>	the Euler–Lagrange expression $\partial L/\partial\theta_c - d(\partial L/\partial\phi_c)$, with left derivatives

Before any physics the algebra is tested on facts known independently (random elements drawn with `SeedRandom[20260924]`). **[proved VII §26]**

- GRASSMANN [solver regression]: generators anticommute and square to zero (all pairs of 1..6)
- GRASSMANN [solver regression]: the product is associative, $(x\ y)\ z == x\ (y\ z)$, on elements of degree 1, 2, 3 (and the products tested are not zero)
- GRASSMANN [solver regression]: graded commutativity $x\ y == (-1)^{(|x|\ |y|)}\ y\ x$ (degrees 1, 2, 3), and $x^2 == 0$ for x odd
- GRASSMANN [solver regression]: the conjugation is an involutive anti-automorphism, $(x\ y)^{\text{ddag}} == y^{\text{ddag}}\ x^{\text{ddag}}$, $(x^{\text{ddag}})^{\text{ddag}} == x$
- GRASSMANN [solver regression]: `grD` is an even derivation, $d(x\ y) == d(x)\ y + x\ d(y)$; `grDL` obeys $d_i(x\ y) == d_i(x)\ y + (-1)^{|x|}\ x\ d_i(y)$

Then the two lemmas that drive the section are proved for a completely general, symbolic 16×16 matrix N (N_A, N_S its antisymmetric and symmetric parts, θ sixteen real Grassmann generators, $\phi = d\theta/dt$ sixteen more):

$$\begin{aligned}\theta^T N \theta &= \theta^T N_A \theta \quad (\text{and } \theta^T N_S \theta = 0), \\ \theta^T N \phi - \frac{1}{2} d(\theta^T N \theta) &= \theta^T N_S \phi, \\ \text{Euler–Lagrange expression of } L = \theta^T N \phi &= (N + N^T) \phi = 2N_S \phi.\end{aligned}$$

So a first-order Grassmann kinetic term is a **total derivative**, with empty field equations, precisely when its matrix is antisymmetric. For **commuting** variables the roles are exactly reversed (the control). **[proved VII §26]**

- GRASSMANN [THE RESULT]: $\theta^T N \theta == \theta^T N_A \theta$ and $\theta^T N_S \theta == 0$ for a general 16×16 N – an anticommuting bilinear sees only the ANTISYMMETRIC part
- GRASSMANN [THE RESULT]: $\theta^T N \phi - (1/2) d(\theta^T N \theta) == \theta^T N_S \phi$ for a general N
- GRASSMANN [THE RESULT]: the Euler–Lagrange expression of $L = \theta^T N \phi$ (left derivatives) is $(N + N^T) \phi$: EMPTY field equations exactly when N is antisymmetric
- GRASSMANN [control]: for COMMUTING components the roles are reversed – $\theta^T N \theta$ sees only N_S , and the Euler–Lagrange expression of $\theta^T N \phi$ is $2 N_A \phi$

3.4 The real options, and why each fails

With the table and the lemmas the verdicts are immediate, and each is proved in the Grassmann algebra.

(a) **A real Grassmann 16-spinor with the author’s σ_{16} has no dynamics, on any frame.** Its only invariant mass terms vanish, $\theta^T \sigma_{16} \theta = \theta^T C_+ \theta = 0$ (both matrices are symmetric). In every direction its flat kinetic term $\theta^T \sigma_{16} T_{16}[a] d\theta$ is the total derivative $\frac{1}{2} d(\theta^T \sigma_{16} T_{16}[a] \theta)$

($\sigma_{16}T_{16}[a]$ is antisymmetric), with an identically empty Euler–Lagrange expression; the total derivative itself is not zero. On a curved frame the covariant kinetic term is

$$\sqrt{g}\theta^T\sigma_{16}\gamma^\mu D_\mu\theta = (\text{total derivative}) + \frac{1}{2}\sqrt{g}\theta^T\sigma_{16}\{\gamma^\mu, \Gamma_\mu\}\theta,$$

because $\partial_\mu(\sqrt{g}\sigma_{16}\gamma^\mu) = \sqrt{g}\sigma_{16}[\gamma^\mu, \Gamma_\mu]$; and the second term vanishes on **every** frame, because $\{T_{16}[a], S^{bc}\}$ is a rank-3 product (or zero) and σ_{16} times a rank-3 product is symmetric. [proved VII §26]

- REFINEMENT (a) [THE RESULT]: real Grassmann + sigma16: both invariant mass terms vanish, and $\text{theta}^T \text{sigma16 } T_{16}[a] \text{ phi} == (1/2) \text{d}(\text{theta}^T \text{sigma16 } T_{16}[a] \text{ theta})$ for every a (a total derivative)
- REFINEMENT (a) [has content]: its Euler-Lagrange expression is identically empty in every direction - while the kinetic term itself is NOT zero
- REFINEMENT (a) [THE RESULT]: FRAME-INDEPENDENT - sigma16 $\{T_{16}[a], S^{bc}\}$ is SYMMETRIC for all a, b, c (so $\text{theta}^T \text{sigma16 } \{\gamma^\mu, \Gamma_\mu\} \text{ theta} == 0$ on every frame), and $\text{d}_\mu(\text{Sqrt}[g] \text{sigma16 } \gamma^\mu) == \text{Sqrt}[g] \text{sigma16 } [\gamma^\mu, \Gamma_\mu]$ on the canonical frame

(b) **A Majorana 16-spinor built with C_+ propagates, but is massless and has no $V(s)$.** Now $C_+T_{16}[a]$ is symmetric, so the kinetic term is genuine: its Euler–Lagrange expression is $2C_+T_{16}[a]\phi$, and $C_+T_{16}[a]$ is invertible. But both invariant forms are symmetric, so there is **no** Lorentz-scalar bilinear: no mass, no s , no $V(s)$. Its quartic Lorentz scalars are not zero, and they are all **one** scalar: with the vector $v_a = \theta^T\sigma_{16}T_{16}[a]\theta$, the 2-form $B_{ab} = \theta^TC_+T_{16}[a]T_{16}[b]\theta$ and the 3-form $C_{abc} = \theta^TC_+T_{16}[a]T_{16}[b]T_{16}[c]\theta$, the contractions $v\cdot v$, $B\cdot B$ and $C\cdot C$ are proportional to each other (same monomials, one constant ratio each). Its commuting shadow has no kinetic term ($C_+T_{16}[a]$ symmetric is a total derivative for commuting components). [proved VII §26]

- REFINEMENT (b) [THE RESULT]: Majorana with C_+ : the kinetic term is genuine - Euler-Lagrange expression $2 C_+ T_{16}[a] \text{ phi}$ with $C_+ T_{16}[a]$ invertible - but every invariant bilinear vanishes (massless, no $V(s)$)
- REFINEMENT (b) [has content]: the scope - the quartic Lorentz scalars $v.v$, $B.B$, $C.C$ of the Majorana spinor are NOT zero and are all proportional to ONE quartic (same monomials, one constant ratio each)
- REFINEMENT (b) [has content]: and the commuting shadow of the C_+ Majorana spinor has NO kinetic term: for commuting components $\text{theta}^T C_+ T_{16}[a] \text{ phi}$ is the total derivative $(1/2) \text{d}(\text{theta}^T C_+ T_{16}[a] \text{ theta})$

(c) **A Weyl or Majorana–Weyl 8-spinor (one chirality, P_L or P_R) cannot propagate.** Both invariant forms are chirality-**diagonal** and every $T_{16}[a]$ is chirality-**off**-diagonal, so $P_L C T_{16}[a] P_L = P_R C T_{16}[a] P_R = 0$: no kinetic term at all. [proved VII §26]

- REFINEMENT (c) [THE RESULT]: Weyl and Majorana–Weyl: $P_L C T_{16}[a] P_L == P_R C T_{16}[a] P_R == 0$ for $C = C-, C_+$ and every a (C chirality-diagonal, $T_{16}[a]$ chirality-odd) - no kinetic term

(d) **Why Part VI’s real commuting fable escapes (a), and why it cannot be quantized as a fermion.** For commuting components the symmetric σ_{16} gives a non-zero mass term and the antisymmetric $\sigma_{16}T_{16}[a]$ gives a genuine kinetic term (the control of subsection 3.3). That is why Part VI’s classical theory is consistent, and why it cannot be quantized as a fermion: a commuting spinor violates the spin–statistics connection [prose VII §26].

3.5 The complex 16-spinor

Take two real Grassmann 16-spinors θ_1, θ_2 (32 real generators) and

$$\Psi = \frac{\theta_1 + i\theta_2}{\sqrt{2}}, \quad \Psi^\dagger = \frac{(\theta_1 - i\theta_2)^T}{\sqrt{2}}, \quad \bar{\Psi} = \Psi^\dagger \sigma_{16}.$$

Now everything the real options lacked is present:

- $s = \bar{\Psi}\Psi = i\theta_1^T \sigma_{16} \theta_2$ is **non-zero** (16 monomials) and **Hermitian** (real under †). The second invariant $p = \bar{\Psi} T_{16}[8] \Psi = \Psi^\dagger C_+ \Psi$ is non-zero and Hermitian as well: the complex 16-spinor has **two** scalar bilinears.
- s is a sum of 16 **commuting nilpotent** even elements x_a ($x_a^2 = 0$), so every polynomial $V(s)$ is a finite polynomial: $s^{17} = 0$.
- The kinetic term is genuine: $\Psi^\dagger K \phi = \Theta^T A \Phi$ with $\Theta = (\theta_1, \theta_2)$, $\Phi = d\Theta$, $K = \sigma_{16} T_{16}[a]$, and the symmetric part of the 32×32 matrix A is $A_S = \frac{i}{2} \begin{pmatrix} 0 & K \\ -K & 0 \end{pmatrix} =: \frac{i}{2} \Omega$, which is **not** zero. Ω is real symmetric and invertible (it is the symplectic matrix of section 6.3).
- The symmetrized kinetic term $\frac{1}{2}(\Psi^\dagger K \phi - \phi^\dagger K \Psi)$ is **Hermitian**, and it differs from $\Psi^\dagger K \phi$ by the total derivative $-\frac{1}{2} d(\Psi^\dagger K \Psi)$; the unsymmetrized one is **not** Hermitian.

Invariance of s and p under $\text{Spin}(4,4)$ is the statement $S^T N + N S = 0$ of subsection 3.2 (the Lorentz generators are real). The self-interaction is taken to depend on s only; section 6.7 shows why p cannot enter. **[proved VII §26]**

- REFINEMENT (d) [definition]: `Psi^ddag == (theta1 - i theta2)^T/Sqrt[2]` in the algebra
- REFINEMENT (d) [THE RESULT]: `s = Psibar Psi == i theta1^T sigma16 theta2`: NON-ZERO (16 monomials) and HERMITIAN; `p = Psibar T16[8] Psi` is non-zero and Hermitian too
- REFINEMENT (d) [has content]: `s` is a sum of 16 commuting nilpotent even elements `x_a` (`x_a x_b == x_b x_a`, `x_a^2 == 0`), so `s^17 == 0` and `V(s)` is a finite polynomial
- REFINEMENT (d) [definition]: the 32×32 matrix `A` read off the algebra reproduces `Psi^ddag K phi` exactly
- REFINEMENT (d) [THE RESULT]: its symmetric part is `A_S == (i/2) Omega`, `Omega = [[0, K], [-K, 0]]` real SYMMETRIC and invertible: a genuine kinetic term
- REFINEMENT (d) [has content]: the symmetrized kinetic term is HERMITIAN and equals `Psi^ddag K phi - (1/2) d(Psi^ddag K Psi)`; the unsymmetrized one is not Hermitian
- REFINEMENT (d) [has content]: the same holds in every direction `a = 0..7`: the symmetrized `Psi^ddag sigma16 T16[a] phi` is Hermitian, with symmetric part `(i/2)[[0, sigma16 T16[a]], [-sigma16 T16[a], 0]] != 0`

3.6 Four four-dimensional Dirac fermions

The observer's Clifford algebra is generated by $T_{16}[1], T_{16}[2], T_{16}[3]$ (the observed space) and $T_{16}[4]$ (the observer's time). Its 16 ordered products are linearly independent, so over the complex numbers it is the full matrix algebra $M_4(\mathbb{C})$, the algebra of 4×4 Dirac matrices. Its **commutant**

in $M_{16}(\mathbb{C})$, computed by solving $XT_{16}[i] = T_{16}[i]X$ ($i = 1..4$), is again 16-dimensional, generated by the four “hidden” (flavour) gammas

$$h_A = \Omega_4 T_{16}[A], \quad A = 0, 5, 6, 7, \quad \Omega_4 = T_{16}[1]T_{16}[2]T_{16}[3]T_{16}[4], \quad \Omega_4^2 = -\text{ID}16, \\ \{h_A, h_B\} = -2\eta_{AB} \text{ID}16,$$

which commute with $T_{16}[1..4]$: a second Clifford algebra, of the hidden directions. The two algebras meet only in the multiples of the identity (the rank of their union is $16 + 16 - 1 = 31$). By the double-commutant theorem

$$\mathbb{C}^{16} = \mathbb{C}^4 (\text{the observer's Dirac index}) \otimes \mathbb{C}^4 (\text{the hidden flavour index}).$$

Under the observer's Lorentz group $\text{Spin}(3,1)$, generated by $T_{16}[i]T_{16}[j]$ with i, j in $\{1, 2, 3, 4\}$, fermion fable is **four 4-dimensional Dirac fermions**. Per momentum that is $4 \times 2 = 8$ particle states (and 8 antiparticle states): the degeneracy $g = 8$ of sections 6 and 8. [**proved VII §26**]

- 4D [THE RESULT]: the 16 ordered products of $T_{16}[1..4]$ are linearly independent (the observer's algebra is $M_4(\mathbb{C})$), and its commutant in $M_{16}(\mathbb{C})$ is 16-dimensional
- 4D [has content]: $\Omega_4^2 == -\text{ID}16$, the hidden gammas $h_A = \Omega_4 T_{16}[A]$ ($A = 0, 5, 6, 7$) commute with $T_{16}[1..4]$ and obey $\{h_A, h_B\} == -2 \eta_{AB}$
- 4D [THE RESULT]: the 16 products of the hidden gammas span the commutant, and the two algebras meet only in the multiples of $\text{ID}16$ (rank of the union $16 + 16 - 1 = 31$): $\mathbb{C}^{16} = \mathbb{C}^4(x) \mathbb{C}^4$, four 4D Dirac fermions
- 4D [has content]: the observer's Lorentz generators $T_{16}[i] T_{16}[j]$ ($i < j$ in $1..4$) commute with every hidden gamma

3.7 The refined Lagrangian, and the fidelity chain to Part VI, L_a and eLa

The Lagrangian of fermion fable is the **symmetrized covariant** one (section 4 shows why no other form is admissible for a complex field):

$$L = \sqrt{\det g} \hat{L}, \quad \hat{L} = \frac{1}{2H} \left[\bar{\Psi} \gamma^\mu D_\mu \Psi - (D_\mu \bar{\Psi}) \gamma^\mu \Psi \right] - V(s), \quad s = \bar{\Psi} \Psi, \\ D_\mu \Psi = \partial_\mu \Psi + \Gamma_\mu^{\text{spin}} \Psi, \quad D_\mu \bar{\Psi} = \partial_\mu \bar{\Psi} - \bar{\Psi} \Gamma_\mu^{\text{spin}},$$

and its energy–momentum tensor (section 7) is

$$\hat{T}_{\mu\nu} = -\frac{1}{4H} \left[\bar{\Psi} \gamma_\mu D_\nu \Psi - (D_\nu \bar{\Psi}) \gamma_\mu \Psi + (\mu \leftrightarrow \nu) \right] + g_{\mu\nu} \hat{L}.$$

Four helpers implement them for an arbitrary frame (curved gammas and spin connection passed in): `cfLhatFermion`, `cfEquationFermion` ($\gamma^\mu D_\mu \Psi - HV'(s)\Psi$), `cfAdjointEquationFermion` ($(D_\mu \bar{\Psi}) \gamma^\mu + HV'(s)\bar{\Psi}$) and `cfTFermion`. Before they are used for anything new they are anchored to what the notebook already knows. Set $\Psi \rightarrow \psi$ (a real commuting spinor) and $\bar{\Psi} \rightarrow \psi^T \sigma_{16}$ (the “bosonic shadow”). Then:

1. $\hat{L}$ becomes Part VI's `cfLhatFable` **exactly**, on the canonical frame, with generic V ;
2. on the flat frame with $V = -(2M/H)s$ and no volume factor, it becomes the author's `La[]` of Section 11 (section 2.7);
3. the Euler–Lagrange expressions of the complex field, combined as $\sigma_{16} \cdot \text{EL}_{\bar{\Psi}} + \text{EL}_{\Psi}$ (the chain rule for $\bar{\Psi} = \psi^T \sigma_{16}$), become Part VI's explicit variation `cfELFable`, and on the flat frame the author's sixteen equations `eLa` of Section 12 (section 2.8);

4. $\hat{T}$ becomes Part VI's covariant tensor `cfTCov` (section 2.12.3), all 64 components.

[proved VII §26]

- FIDELITY [1] [fidelity]: `Psi -> psi, Psibar -> psi^T sigma16` turns the symmetrized covariant `Lhat` into Part VI's `cfLhatFable` EXACTLY (canonical frame, generic V)
- FIDELITY [2] [fidelity]: on the flat frame with $V = -(2M/H)$ s and no volume factor, the same limit on `Psi16` IS the author's `La[]` of Section 11
- FIDELITY [3] [fidelity]: `sigma16 . EL_Psibar + EL_Psi` in the real limit == Part VI's explicit variation `cfELFable`, component for component
- FIDELITY [3] [fidelity]: and on the flat frame with $V = -(2M/H)$ s the same combination IS the author's `eLa` of Section 12
- FIDELITY [4] [fidelity]: the energy-momentum tensor `That` in the real limit IS Part VI's covariant tensor `cfTCov`, all 64 components

Every new result of this effort therefore contains Part VI, and through it the author's L_a and `eLa`, as its real commuting limit.

3.8 The conclusion, with its scope

The complex 16-spinor is the **minimal** field that

1. uses the author's σ_{16} conjugation,
2. propagates,
3. admits a potential $V(s)$ with $s = \bar{\Psi}\Psi$, and
4. reduces to Part VI's classical fable as its real commuting shadow.

It is **not** the unique field with dynamics: the C_+ Majorana spinor propagates too, but it is massless, has no bilinear $V(s)$, and has no commuting shadow with a kinetic term. The statement of minimality is the conclusion of Section 26 as a whole; no single assertion states it. It rests on the assertions of subsections 3.1–3.7 [prose VII §26, assembled from the assertions quoted].

4 The Lagrangian and its Hermiticity

This is the first cell of Section 27 (Input cell 180, 10 assertions). From here on Ψ and $\bar{\Psi}$ are two independent 16-component fields of **all eight** coordinates (the commuting proxy of section 3, $\bar{\Psi}$ always on the left).

4.1 The four candidates

Four candidate Lagrangians differ by where the derivative acts and whether it is covariant:

$$L_{\text{sym}} = \frac{1}{2H} \left[\bar{\Psi} \gamma^\mu D_\mu \Psi - (D_\mu \bar{\Psi}) \gamma^\mu \Psi \right] - V(s) \quad (\text{the Lagrangian of fermion fable}),$$

$L_{\text{sym},\partial}$ = the same with ∂_μ in place of D_μ ,

$$L_{\text{un}} = \frac{1}{H} \bar{\Psi} \gamma^\mu D_\mu \Psi - V(s),$$

$$L_{\text{un},\partial} = \frac{1}{H} \bar{\Psi} \gamma^\mu \partial_\mu \Psi - V(s),$$

with $D_\mu \Psi = \partial_\mu \Psi + \Gamma_\mu \Psi$, $D_\mu \bar{\Psi} = \partial_\mu \bar{\Psi} - \bar{\Psi} \Gamma_\mu$, and $\Gamma_\mu = \text{GammaSpinCanonical}[[\mu+1]]$, the canonical spin connection of section 9. (The notebook's labels call them `Lsym`, `Lsym_d`, `Lun` and `Lun_d`.) The action is $S = \int d^8x \sqrt{\det g} L_{\text{sym}}$, with $\sqrt{\det g} = \sec 6Hx_0$ on the canonical frame. There is **no factor i** in front of the kinetic term, as in the author's L_a : section 4.2(b) shows that none is needed.

4.2 The facts, each asserted on the canonical frame for generic $\Psi(x_0..x_7)$, $\bar{\Psi}(x_0..x_7)$

(a) **The connection is real and $\sigma_{16}\Gamma_\mu$ is antisymmetric**, i.e. $\Gamma_\mu^T \sigma_{16} = -\sigma_{16} \Gamma_\mu$. So $D_\mu \bar{\Psi} = \partial_\mu \bar{\Psi} - \bar{\Psi} \Gamma_\mu$ is exactly the Dirac conjugate of $D_\mu \Psi$. [proved VII §27] LAGRANGIAN (a) [definition]: every `Gamma^spin_mu` is real and `sigma16 Gamma_mu` is antisymmetric (`Gamma_mu^T sigma16 == -sigma16 Gamma_mu`): `D_mu Psibar` is the Dirac conjugate of `D_mu Psi`

(b) **L_{sym} is Hermitian; L_{un} is not.** With $\Psi = u + iv$ and $\bar{\Psi} = (u - iv)^T \sigma_{16}$, L_{sym} is real. No factor i is needed, because $\sigma_{16}\gamma^\mu$ is real and antisymmetric: $(\Psi^\dagger K d\Psi)^\dagger = -(d\Psi^\dagger) K \Psi$ for real antisymmetric K , so the symmetrized combination is real. L_{un} is not Hermitian: its imaginary part is not zero (a witness). [proved VII §27]

- LAGRANGIAN (b) [THE RESULT]: `Lsym` is HERMITIAN - real for `Psi = u + i v`, `Psibar = (u - i v)^T sigma16`, generic `u`, `v` of all eight coordinates
- LAGRANGIAN (b) [control]: `Lun` is NOT Hermitian - its imaginary part is not zero, witnessed on the constant spinors `u = e1`, `v = e13` (with `V = s^2`)

(c) **L_{sym} contains the connection only through $\{\gamma^\mu, \Gamma_\mu\}$, and that matrix vanishes for each μ .** Expanding D ,

$$L_{\text{sym}} - L_{\text{sym},\partial} = \frac{1}{2H} \sum_\mu \bar{\Psi} \{\gamma^\mu, \Gamma_\mu\} \Psi \quad (\text{identically, off shell}).$$

On the canonical frame $\{\gamma^\mu, \Gamma_\mu\} = 0$ for **each** μ separately: every Γ_μ is a combination of $T_{16}[\mu]T_{16}[0]$ and $T_{16}[\mu]T_{16}[4]$, with no totally antisymmetric part (section 9.8). So $L_{\text{sym}} = L_{\text{sym},\partial}$ there: the connection drops out of the symmetrized Lagrangian on every frame on which the **matrix** $\{\gamma^\mu, \Gamma_\mu\}$ vanishes (diagonal frames), and **not** in general. [proved VII §27] LAGRANGIAN (c) [THE RESULT]: `Lsym - Lsym_d == (1/(2H)) Sum_mu Psibar {gamma^mu, Gamma_mu} Psi` identically, and `{gamma^mu, Gamma_mu} == 0` for EACH `mu` on the canonical frame: `Lsym == Lsym_d`

(d) **L_{un} and L_{sym} differ by a total divergence.** [proved VII §27] LAGRANGIAN (d) [has content]: `Lun - Lsym == (1/(2H)) (1/Sqrt[g]) d_mu( Sqrt[g] Psibar gamma^mu Psi )`, a total divergence

(e) **But L_{un} and $L_{\text{un},\partial}$ differ by a term that is not zero for a complex field.**

$$L_{\text{un}} - L_{\text{un},\partial} = \frac{1}{H} \bar{\Psi} \gamma^\mu \Gamma_\mu \Psi = -3 \cot^2(6Hx_0) \bar{\Psi} T_{16}[0] \Psi.$$

This vanishes only in the real commuting limit, which is the scope of the statement “ $\hat{L}$ with $D = \hat{L}$ with ∂ ” of Parts V and VI (section 2.12.1). [proved VII §27]

- LAGRANGIAN (e) [has content]: `Lun - Lun_d == (1/H) Psibar gamma^mu Gamma_mu Psi == -3 Cot[6 H x0]^2 Psibar T16[0] Psi`

- LAGRANGIAN (e) [control]: that term is NOT zero for a complex field (witness $u = e1$, $v = e13$), and IS zero in the real commuting limit (the scope of Parts V-VI)

(f) So $L_{un,\partial}$ is inadmissible. Varying $\bar{\Psi}$ gives $\gamma^\mu \partial_\mu \Psi = HV' \Psi$ with the connection term missing. Varying Ψ gives an adjoint equation that **contains** it, $(\partial_\mu \bar{\Psi}) \gamma^\mu + \bar{\Psi} [\gamma^\mu, \Gamma_\mu] = -HV' \bar{\Psi}$. The two are **not** Dirac conjugates of each other: they differ by

$$\bar{\Psi} [\gamma^\mu, \Gamma_\mu] = 2\bar{\Psi} \gamma^\mu \Gamma_\mu = -6H \cot^2(6Hx_0) \bar{\Psi} T_{16}[0]$$

(a witness). Only a Lagrangian whose two variations are conjugate defines a consistent theory. [proved VII §27]

- LAGRANGIAN (f) [THE RESULT]: $L_{un,d}$ varied with respect to Ψ gives $(\text{Sqrt}[g]/H)(\gamma^\mu d_\mu \Psi - H V' \Psi)$: the connection term is MISSING
- LAGRANGIAN (f) [THE RESULT]: varied with respect to Ψ it gives $-(\text{Sqrt}[g]/H)((d_\mu \bar{\Psi}) \gamma^\mu + \bar{\Psi} [\gamma^\mu, \Gamma_\mu] + H V' \bar{\Psi})$, with $\bar{\Psi} [\gamma^\mu, \Gamma_\mu] == 2 \bar{\Psi} \gamma^\mu \Gamma_\mu$: the connection term is PRESENT
- LAGRANGIAN (f) [control]: so $L_{un,d}$ is INADMISSIBLE - its two field equations are not Dirac conjugates: they differ by $2 \bar{\Psi} \gamma^\mu \Gamma_\mu = -6 H \cot^2 \bar{\Psi} T_{16}[0]$, non-zero (witness $u = e1$, $v = 0$, $V = s^2$)

4.3 Why the connection drops out, and why this is not Part VI's mechanism

For the real commuting field of Part VI the connection dropped out because $\Psi^T \sigma_{16} T_{16}[a] \Psi = 0$ for commuting components ($\sigma_{16} T_{16}[a]$ antisymmetric; section 2.11). For a complex field that bilinear identity is false: $\bar{\Psi} T_{16}[0] \Psi = \Psi^\dagger (\sigma_{16} T_{16}[0]) \Psi$ with $\sigma_{16} T_{16}[0]$ real antisymmetric is i times the Hermitian form of $-i\sigma_{16} T_{16}[0]$, which is not zero in general [derived here]. The drop-out of the connection from L_{sym} is therefore a property of the **matrix** $\{\gamma^\mu, \Gamma_\mu\}$ (fact (c)), not of a bilinear identity. The general identity behind it is $\sum_\mu \{\gamma^\mu, \Gamma_\mu\} = \frac{1}{2} \omega_{[cab]} \gamma^{cab}$: only the totally antisymmetric part of the connection can enter the symmetrized Lagrangian (section 9.8).

Hermiticity rests on two facts: $\sigma_{16} \gamma^a$ and $\sigma_{16} \Gamma_\mu$ are **real antisymmetric**. The first is Part I's $\text{sigma16} . T_{16}[A]$ is antisymmetric for $A = 0..7$; the second is fact (a).

5 The field equations in the primordial gravitational field, written out

This is Section 27 of the notebook (Input cells 181, 182 and 184) and item (2) of the pre-universe cell of Section 29 (Input cell 197). The **primordial gravitational field** is the author's canonical frame of section 2.11:

$$\begin{aligned} \text{frameCanonical} &= \text{diag}(\tan 6Hx_0, q, q, q, 1, p, p, p), \\ q &= \frac{e^{-a_4(Hx_4)}}{\sin^{1/6}(6Hx_0)}, \quad p = \frac{e^{+a_4(Hx_4)}}{\sin^{1/6}(6Hx_0)} \quad (a_4 \text{ free, never given a value}), \\ ds^2 &= \tan^2(6Hx_0) dx_0^2 + q^2(dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 - p^2(dx_5^2 + dx_6^2 + dx_7^2), \\ \sqrt{|\det g|} &= \sec 6Hx_0, \\ \gamma^0 &= \cot(6Hx_0) T_{16}[0], \quad \gamma^i = \frac{1}{q} T_{16}[i] \quad (i = 1, 2, 3), \\ \gamma^4 &= T_{16}[4], \quad \gamma^h = \frac{1}{p} T_{16}[h] \quad (h = 5, 6, 7), \end{aligned}$$

on $0 < 6Hx_0 < \pi/2$. The spin connection enters through $\gamma^\mu \Gamma_\mu = -3H \cot^2(6Hx_0) T_{16}[0]$ (section 9.8).

5.1 The derivation

Vary $\bar{\Psi}$ and Ψ independently in $L = \sqrt{g} L_{\text{sym}}$ (the left derivative with respect to $\bar{\Psi}$ and the right derivative with respect to Ψ are the ordinary derivatives of the commuting proxy). For generic fields of all eight coordinates:

$$\begin{aligned} \text{EL}_{\bar{\Psi}} &= \frac{\sqrt{g}}{H} (\gamma^\mu D_\mu \Psi - H V'(s) \Psi), \\ \text{EL}_\Psi &= -\frac{\sqrt{g}}{H} ((D_\mu \bar{\Psi}) \gamma^\mu + H V'(s) \bar{\Psi}), \end{aligned}$$

so the field equation and the adjoint field equation of fermion fable are

$$\boxed{\gamma^\mu D_\mu \Psi = H V'(s) \Psi, \quad (D_\mu \bar{\Psi}) \gamma^\mu = -H V'(s) \bar{\Psi}}$$

The **sign** of the adjoint equation is derived, not assumed, and it is exactly the one consistency requires: with $\Psi = u + iv$ and $\bar{\Psi} = (u - iv)^T \sigma_{16}$, the adjoint equation is the Dirac conjugate of the field equation, (adjoint residual) = $-(\text{field residual})^* \cdot \sigma_{16}$, because $T_{16}[a]^T \sigma_{16} = -\sigma_{16} T_{16}[a]$. The opposite sign fails that test (control). The derivation uses only the covariant constancy of γ^μ , in the form $\frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \gamma^\mu) = [\gamma^\mu, \Gamma_\mu]$ (Section 17); the anticommutator is not needed for the symmetrized Lagrangian. **[proved VII §27]**

- FIELD EQUATION [has content]: the identity the derivation uses - $(1/\text{Sqrt}[g]) \text{d_mu}(\text{Sqrt}[g] \text{gamma}^\mu) == [\text{gamma}^\mu, \text{Gamma}_\mu]$ (covariant constancy, Section 17)
- FIELD EQUATION [THE RESULT]: $\text{EL_Psibar} == (\text{Sqrt}[g]/H) (\text{gamma}^\mu \text{D_mu Psi} - H V'(s) \text{Psi})$ for generic Psi, Psibar of all eight coordinates: the field equation is $\text{gamma}^\mu \text{D_mu Psi} == H V'(s) \text{Psi}$
- FIELD EQUATION [THE RESULT]: $\text{EL_Psi} == -(\text{Sqrt}[g]/H) ((\text{D_mu Psibar}) \text{gamma}^\mu + H V'(s) \text{Psibar})$: the adjoint equation is $(\text{D_mu Psibar}) \text{gamma}^\mu == -H V'(s) \text{Psibar}$
- FIELD EQUATION [has content]: the adjoint equation IS the Dirac conjugate of the field equation: with $\text{Psi} = u + i v$, $\text{Psibar} = (u - i v)^T \text{sigma16}$, adjoint residual == $-(\text{field residual})^* \cdot \text{sigma16}$
- FIELD EQUATION [control]: the opposite sign, $(\text{D Psibar}) \text{gamma} == +H V' \text{Psibar}$, is NOT the conjugate - witnessed on $u = e1 + e5$, $v = 0$ with $V = s^2$ ($s = -2$ there)

5.2 The 16×16 matrix form on the canonical frame

For fields of all eight coordinates, with $\partial_\mu = \partial/\partial x_\mu$, sums over $i = 1, 2, 3$ and $h = 5, 6, 7$:

$$\begin{aligned} &\cot(6Hx_0) T_{16}[0] \partial_0 \Psi + \frac{1}{q} \sum_i T_{16}[i] \partial_i \Psi + T_{16}[4] \partial_4 \Psi + \frac{1}{p} \sum_h T_{16}[h] \partial_h \Psi \\ &\quad - 3H \cot^2(6Hx_0) T_{16}[0] \Psi = H V'(s) \Psi, \\ &\cot(6Hx_0) \partial_0 \bar{\Psi} T_{16}[0] + \frac{1}{q} \sum_i \partial_i \bar{\Psi} T_{16}[i] + \partial_4 \bar{\Psi} T_{16}[4] + \frac{1}{p} \sum_h \partial_h \bar{\Psi} T_{16}[h] \\ &\quad - 3H \cot^2(6Hx_0) \bar{\Psi} T_{16}[0] = -H V'(s) \bar{\Psi}. \end{aligned}$$

In the adjoint equation the connection enters as $-\bar{\Psi}\Gamma_\mu\gamma^\mu = +\bar{\Psi}\gamma^\mu\Gamma_\mu$ (per- μ anticommutator), so it carries the **same** coefficient $-3H \cot^2$ as in the field equation. [proved VII §27]

- WRITTEN OUT (i) [THE RESULT]: the field equation on the canonical frame is $\text{Cot } T_{16}[0] \text{ d_0 Psi} + (1/q) T_{16}[i] \text{ d_i Psi} + T_{16}[4] \text{ d_4 Psi} + (1/p) T_{16}[h] \text{ d_h Psi} - 3 H \text{Cot}^2 T_{16}[0] \text{ Psi} == H V'(s) \text{ Psi}$
- WRITTEN OUT (i) [THE RESULT]: the adjoint equation is $\text{Cot d_0 Psibar } T_{16}[0] + \dots - 3 H \text{Cot}^2 \text{ Psibar } T_{16}[0] == -H V'(s) \text{ Psibar}$ (the connection enters with the sign fixed by $\{\gamma^\mu, \Gamma_\mu\} = 0$)

5.3 The split-octonion 8+8 form

Write $\Psi = (\psi_1, \psi_2)$: ψ_1 = the upper eight components (type-1 split-octonion spinor, chirality -1), ψ_2 = the lower eight (type-2, chirality $+1$). With $T_{16}[a] = \begin{pmatrix} 0 & \bar{\tau}_a \\ \tau_a & 0 \end{pmatrix}$ and $\Gamma_\mu^{\text{spin}} = \text{diag}(\Gamma_\mu^{(1)}, \Gamma_\mu^{(2)})$ (Section 17; the spin connection never mixes the two types), **covariantly, on any frame**:

$$e_a^\mu \bar{\tau}_a (\partial_\mu + \Gamma_\mu^{(2)}) \psi_2 = H V'(s) \psi_1 \quad (\text{rows 1..8}),$$

$$e_a^\mu \tau_a (\partial_\mu + \Gamma_\mu^{(1)}) \psi_1 = H V'(s) \psi_2 \quad (\text{rows 9..16}),$$

$$s = -\psi_1^\dagger \sigma \psi_1 + \psi_2^\dagger \sigma \psi_2, \quad \bar{\psi}_1 = -\psi_1^\dagger \sigma, \quad \bar{\psi}_2 = \psi_2^\dagger \sigma,$$

$$(\partial_\mu \bar{\psi}_2 - \bar{\psi}_2 \Gamma_\mu^{(2)}) \tau_a e_a^\mu = -H V'(s) \bar{\psi}_1,$$

$$(\partial_\mu \bar{\psi}_1 - \bar{\psi}_1 \Gamma_\mu^{(1)}) \bar{\tau}_a e_a^\mu = -H V'(s) \bar{\psi}_2$$

($\bar{\Psi} = (\bar{\psi}_1, \bar{\psi}_2) = \Psi^\dagger \sigma_{16}$, $\sigma_{16} = \text{diag}(-\sigma, \sigma)$; $\tau_a = \text{tau}[a]$, $\bar{\tau}_a = \text{taubar}[a]$.) On the canonical frame ($\tau_0 = \bar{\tau}_0 = \text{ID8}$), the connection term is $-3H \cot^2 \psi_2$ in the upper rows and $-3H \cot^2 \psi_1$ in the lower rows:

$$\begin{aligned} \cot \partial_0 \psi_2 + \frac{1}{q} \sum_i \bar{\tau}_i \partial_i \psi_2 + \bar{\tau}_4 \partial_4 \psi_2 + \frac{1}{p} \sum_h \bar{\tau}_h \partial_h \psi_2 - 3H \cot^2 \psi_2 &= H V'(s) \psi_1, \\ \cot \partial_0 \psi_1 + \frac{1}{q} \sum_i \tau_i \partial_i \psi_1 + \tau_4 \partial_4 \psi_1 + \frac{1}{p} \sum_h \tau_h \partial_h \psi_1 - 3H \cot^2 \psi_1 &= H V'(s) \psi_2, \end{aligned}$$

with $\cot = \cot(6Hx_0)$. [proved VII §27]

- WRITTEN OUT (ii) [THE RESULT]: the split-octonion 8+8 form - the type-1 rows are the taubar equations for psi2, the type-2 rows the tau equations for psi1 ($\text{tau}[0] == \text{taubar}[0] == \text{ID8}$)
- WRITTEN OUT (ii) [THE RESULT]: the covariant 8+8 form - rows 1..8 are $e_a^\mu \text{taubar}[a] (\text{d_mu} + \Gamma_\mu^{(2)}) \text{psi2} - H V' \text{psi1}$, rows 9..16 are $e_a^\mu \text{tau}[a] (\text{d_mu} + \Gamma_\mu^{(1)}) \text{psi1} - H V' \text{psi2}$; and the conjugate rows
- WRITTEN OUT (ii) [has content]: with $\text{Psibar} = \text{Psi}^\dagger \text{sigma16}$ and $\text{sigma16} = \text{diag}(-\text{sigma}, \text{sigma})$: $s == -\text{psi1}^\dagger \text{sigma} \text{psi1} + \text{psi2}^\dagger \text{sigma} \text{psi2}$ ($\text{psibar1} = -\text{psi1}^\dagger \text{sigma}$, $\text{psibar2} = \text{psi2}^\dagger \text{sigma}$)

5.4 All sixteen component equations

Each $T_{16}[a]$ is a signed permutation matrix, so component equation r contains exactly **one** derivative per coordinate, of the component that $T_{16}[a]$ maps to r ; the connection term multiplies the same component as the x_0 -derivative; and $-HV'$ multiplies ψ_r itself. For fields of all eight coordinates the sixteen equations form **one** coupled block. (The four blocks of four of Section 13, section 2.9, belong to the reduction to fields of (x_0, x_4) only.) The term lists `cfFieldEqTerms` and `cfAdjEqTerms` from which the equations below are rendered reproduce all sixteen field equations and all sixteen adjoint equations exactly. [proved VII §27]

- WRITTEN OUT (iii) [definition]: the term lists reproduce all sixteen field equations and all sixteen adjoint equations exactly
- WRITTEN OUT (iii) [has content]: every component equation has exactly one derivative per coordinate $x_0..x_7$, the connection term multiplies the component carrying the x_0 -derivative, and $-H V'$ multiplies ψ_r itself
- WRITTEN OUT (iii) [has content]: for fields of all eight coordinates the sixteen equations form ONE coupled block (the four blocks of four of Section 13 belong to the (x_0, x_4) reduction)

How to read the equations below. They are the TeX export written by `claude-fable/export_fermion_fable_tex.wls` from the notebook's own asserted objects (`provenance-latex/generated/fermion_fable_field_equations.tex`, included here unchanged; the same equations in plain text are the `.txt` file of the same name).

- ψ_k ($k = 0..15$) is component k of Ψ ; $\bar{\psi}_k$ is component k of $\bar{\Psi}$. The index is the notebook's 0-based component index (array position $k + 1$).
- $\partial_\mu \psi_k$ is $\partial \psi_k / \partial x_\mu$, $\mu = 0..7$.
- a_4 is the free function. Hence $e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} = 1/q$ multiplies the observed derivatives $\partial_1, \partial_2, \partial_3$, and $e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} = 1/p$ multiplies the hidden derivatives $\partial_5, \partial_6, \partial_7$.
- $V'(s)$ is evaluated at $s = \bar{\Psi}\Psi$; H is the author's Protected constant.
- Equation E_r is row r of $\gamma^\mu D_\mu \Psi - HV'(s)\Psi = 0$, with the V' term moved to the right.

$$\begin{aligned}
(E_0) \quad & \cot(6Hx_0) \partial_0 \psi_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_8 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{13} \\
& - \partial_4 \psi_{13} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{14} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{14} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{15} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{15} - 3H \cot^2(6Hx_0) \psi_8 \\
& = HV'(s) \psi_0
\end{aligned}$$

$$\begin{aligned}
(E_1) \quad & \cot(6Hx_0) \partial_0 \psi_9 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{12} \\
& + \partial_4 \psi_{12} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{14} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{14} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{15} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{15} - 3H \cot^2(6Hx_0) \psi_9 \\
& = HV'(s) \psi_1
\end{aligned}$$

$$\begin{aligned}
(E_2) \quad & \cot(6Hx_0) \partial_0 \psi_{10} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{12} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{12} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{13} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{13} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{15} + \partial_4 \psi_{15} - 3H \cot^2(6Hx_0) \psi_{10} \\
& = HV'(s) \psi_2
\end{aligned}$$

$$\begin{aligned}
(E_3) \quad & \cot(6Hx_0) \partial_0 \psi_{11} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{11} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{12} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{12} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{13} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{13} \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{14} - \partial_4 \psi_{14} - 3H \cot^2(6Hx_0) \psi_{11} \\
& = HV'(s) \psi_3
\end{aligned}$$

$$\begin{aligned}
(E_4) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_9 + \partial_4 \psi_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{10} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{11} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{11}
\end{aligned}$$

$$\begin{aligned}
& + \cot(6Hx_0) \partial_0 \psi_{12} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{12} - 3H \cot^2(6Hx_0) \psi_{12} \\
& = HV'(s) \psi_4
\end{aligned}$$

$$\begin{aligned}
(E_5) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_8 - \partial_4 \psi_8 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{10} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{11} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{11} \\
& + \cot(6Hx_0) \partial_0 \psi_{13} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{13} - 3H \cot^2(6Hx_0) \psi_{13} \\
& = HV'(s) \psi_5
\end{aligned}$$

$$\begin{aligned}
(E_6) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_8 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_9 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_9 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{11} - \partial_4 \psi_{11} \\
& + \cot(6Hx_0) \partial_0 \psi_{14} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{14} - 3H \cot^2(6Hx_0) \psi_{14} \\
& = HV'(s) \psi_6
\end{aligned}$$

$$\begin{aligned}
(E_7) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_8 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_9 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{10} + \partial_4 \psi_{10} \\
& + \cot(6Hx_0) \partial_0 \psi_{15} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{15} - 3H \cot^2(6Hx_0) \psi_{15} \\
& = HV'(s) \psi_7
\end{aligned}$$

$$\begin{aligned}
(E_8) \quad & \cot(6Hx_0) \partial_0 \psi_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_0 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_5 \\
& + \partial_4 \psi_5 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_6 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_6 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_7 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_7 - 3H \cot^2(6Hx_0) \psi_0 \\
& = HV'(s) \psi_8
\end{aligned}$$

$$\begin{aligned}
(E_9) \quad & \cot(6Hx_0) \partial_0 \psi_1 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_4 \\
& - \partial_4 \psi_4 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_6 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_6 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_7 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_7 - 3H \cot^2(6Hx_0) \psi_1 \\
& = HV'(s) \psi_9
\end{aligned}$$

$$\begin{aligned}
(E_{10}) \quad & \cot(6Hx_0) \partial_0 \psi_2 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_4 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_4 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_5 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_5 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_7 - \partial_4 \psi_7 - 3H \cot^2(6Hx_0) \psi_2 \\
& = HV'(s) \psi_{10}
\end{aligned}$$

$$\begin{aligned}
(E_{11}) \quad & \cot(6Hx_0) \partial_0 \psi_3 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_3 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_4 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_4 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_5 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_5 \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_6 + \partial_4 \psi_6 - 3H \cot^2(6Hx_0) \psi_3 \\
& = HV'(s) \psi_{11}
\end{aligned}$$

$$\begin{aligned}
(E_{12}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_1 - \partial_4 \psi_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_2 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_3 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_3 \\
& + \cot(6Hx_0) \partial_0 \psi_4 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_4 - 3H \cot^2(6Hx_0) \psi_4 \\
& = HV'(s) \psi_{12}
\end{aligned}$$

$$\begin{aligned}
(E_{13}) \quad & -e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_0 + \partial_4 \psi_0 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_2 \\
& -e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_3 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_3 \\
& + \cot(6Hx_0) \partial_0 \psi_5 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_5 - 3H \cot^2(6Hx_0) \psi_5 \\
& = HV'(s) \psi_{13}
\end{aligned}$$

$$\begin{aligned}
(E_{14}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_0 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_1 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_1 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_3 + \partial_4 \psi_3 \\
& + \cot(6Hx_0) \partial_0 \psi_6 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_6 - 3H \cot^2(6Hx_0) \psi_6 \\
& = HV'(s) \psi_{14}
\end{aligned}$$

$$\begin{aligned}
(E_{15}) \quad & -e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_0 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_1 \\
& -e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_2 - \partial_4 \psi_2 \\
& + \cot(6Hx_0) \partial_0 \psi_7 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_7 - 3H \cot^2(6Hx_0) \psi_7 \\
& = HV'(s) \psi_{15}
\end{aligned}$$

5.5 All sixteen adjoint equations

Equation $\bar{E}_r$ is column r of $(D_\mu \bar{\Psi}) \gamma^\mu + HV'(s) \bar{\Psi} = 0$, with the V' term moved to the right (provenance-latex/generated/fermion_fable_adjoint_equations.tex, unchanged):

$$\begin{aligned}
(\bar{E}_0) \quad & \cot(6Hx_0) \partial_0 \bar{\psi}_8 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_8 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_{13} \\
& + \partial_4 \bar{\psi}_{13} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_{14} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_{14} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_{15} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_{15} - 3H \cot^2(6Hx_0) \bar{\psi}_8 \\
& = -HV'(s) \bar{\psi}_0
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_1) \quad & \cot(6Hx_0) \partial_0 \bar{\psi}_9 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_{12} \\
& - \partial_4 \bar{\psi}_{12} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_{14} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_{14} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_{15} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_{15} - 3H \cot^2(6Hx_0) \bar{\psi}_9 \\
& = -HV'(s) \bar{\psi}_1
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_2) \quad & \cot(6Hx_0) \partial_0 \bar{\psi}_{10} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_{12} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_{12} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_{13} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_{13} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_{15} - \partial_4 \bar{\psi}_{15} - 3H \cot^2(6Hx_0) \bar{\psi}_{10} \\
& = -HV'(s) \bar{\psi}_2
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_3) \quad & \cot(6Hx_0) \partial_0 \bar{\psi}_{11} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_{11} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_{12} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_{12} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_{13} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_{13} \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_{14} + \partial_4 \bar{\psi}_{14} - 3H \cot^2(6Hx_0) \bar{\psi}_{11} \\
& = -HV'(s) \bar{\psi}_3
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_4) \quad & -e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_9 - \partial_4 \bar{\psi}_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_{10} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_{11} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_{11}
\end{aligned}$$

$$\begin{aligned}
& + \cot(6Hx_0) \partial_0 \bar{\psi}_{12} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_{12} - 3H \cot^2(6Hx_0) \bar{\psi}_{12} \\
& = -HV'(s) \bar{\psi}_4
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_5) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_8 + \partial_4 \bar{\psi}_8 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_{10} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_{11} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_{11} \\
& + \cot(6Hx_0) \partial_0 \bar{\psi}_{13} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_{13} - 3H \cot^2(6Hx_0) \bar{\psi}_{13} \\
& = -HV'(s) \bar{\psi}_5
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_6) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_8 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_8 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_9 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_9 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_{11} + \partial_4 \bar{\psi}_{11} \\
& + \cot(6Hx_0) \partial_0 \bar{\psi}_{14} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_{14} - 3H \cot^2(6Hx_0) \bar{\psi}_{14} \\
& = -HV'(s) \bar{\psi}_6
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_7) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_8 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_8 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_9 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_{10} - \partial_4 \bar{\psi}_{10} \\
& + \cot(6Hx_0) \partial_0 \bar{\psi}_{15} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_{15} - 3H \cot^2(6Hx_0) \bar{\psi}_{15} \\
& = -HV'(s) \bar{\psi}_7
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_8) \quad & \cot(6Hx_0) \partial_0 \bar{\psi}_0 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_0 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_5 \\
& - \partial_4 \bar{\psi}_5 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_6 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_6 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_7 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_7 - 3H \cot^2(6Hx_0) \bar{\psi}_0 \\
& = -HV'(s) \bar{\psi}_8
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_9) \quad & \cot(6Hx_0) \partial_0 \bar{\psi}_1 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_4 \\
& + \partial_4 \bar{\psi}_4 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_6 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_6 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_7 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_7 - 3H \cot^2(6Hx_0) \bar{\psi}_1 \\
& = -HV'(s) \bar{\psi}_9
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_{10}) \quad & \cot(6Hx_0) \partial_0 \bar{\psi}_2 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_4 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_4 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_5 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_5 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_7 + \partial_4 \bar{\psi}_7 - 3H \cot^2(6Hx_0) \bar{\psi}_2 \\
& = -HV'(s) \bar{\psi}_{10}
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_{11}) \quad & \cot(6Hx_0) \partial_0 \bar{\psi}_3 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_3 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_4 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_4 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_5 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_5 \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_6 - \partial_4 \bar{\psi}_6 - 3H \cot^2(6Hx_0) \bar{\psi}_3 \\
& = -HV'(s) \bar{\psi}_{11}
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_{12}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_1 + \partial_4 \bar{\psi}_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_2 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_3 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_3 \\
& + \cot(6Hx_0) \partial_0 \bar{\psi}_4 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_4 - 3H \cot^2(6Hx_0) \bar{\psi}_4 \\
& = -HV'(s) \bar{\psi}_{12}
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_{13}) \quad & -e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_0 - \partial_4 \bar{\psi}_0 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_2 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_3 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_3 \\
& + \cot(6Hx_0) \partial_0 \bar{\psi}_5 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_5 - 3H \cot^2(6Hx_0) \bar{\psi}_5 \\
& = -HV'(s) \bar{\psi}_{13}
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_{14}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_0 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_0 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_1 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_1 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_3 - \partial_4 \bar{\psi}_3 \\
& + \cot(6Hx_0) \partial_0 \bar{\psi}_6 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_6 - 3H \cot^2(6Hx_0) \bar{\psi}_6 \\
& = -HV'(s) \bar{\psi}_{14}
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_{15}) \quad & -e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_0 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_0 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_1 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_2 + \partial_4 \bar{\psi}_2 \\
& + \cot(6Hx_0) \partial_0 \bar{\psi}_7 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_7 - 3H \cot^2(6Hx_0) \bar{\psi}_7 \\
& = -HV'(s) \bar{\psi}_{15}
\end{aligned}$$

For every r , E_r and $\bar{E}_r$ contain the same eight (derivative direction, component index) pairs, $\partial_\mu \psi_k$ against $\partial_\mu \bar{\psi}_k$, with the same coefficients. The terms with derivatives along the spacelike directions $x_0 \dots x_3$ have the **same** sign in both; those along the timelike directions $x_4 \dots x_7$ have the **opposite** sign; the connection term $-3H \cot^2$ multiplies the same component with the same sign; and the V' term changes sign. This is what the transposition requires: $T_{16}[a]$ is a real signed permutation matrix with $T_{16}[a]^2 = \eta_{aa}$, so $T_{16}[a]^T = \eta_{aa} T_{16}[a]$ [derived here; the rule was checked for this page on all 128 derivative terms and 16 connection terms of the two exported files by a short script; it is not a notebook assertion]. The adjoint equations are the Dirac conjugates of the field equations (FIELD EQUATION [has content]): the adjoint equation IS the Dirac conjugate of the field equation: with $\text{Psi} = u + i v$, $\text{Psibar} = (u - i v)^T \text{sigma16}$, $\text{adjoint residual} == -(\text{field residual})^* \cdot \text{sigma16}$, section 5.1).

5.6 The 8+8 form, component by component

Rows 0–7 are the $\bar{\tau}$ equations for ψ_2 , rows 8–15 the τ equations for ψ_1 ; $\psi_k^{(1)} = \psi_k$ and $\psi_k^{(2)} = \psi_{k+8}$ ($k = 0..7$) (provenance-latex/generated/fermion_fable_field_equations_8plus8.tex, unchanged):

$$\begin{aligned}
& \cot(6Hx_0) \partial_0 \psi^{(2)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \sum_{i=1}^3 \bar{\tau}_i \partial_i \psi^{(2)} + \bar{\tau}_4 \partial_4 \psi^{(2)} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \sum_{h=5}^7 \bar{\tau}_h \partial_h \psi^{(2)} - 3H \cot^2(6Hx_0) \psi^{(2)} = HV'(s) \psi^{(1)}, \\
& \cot(6Hx_0) \partial_0 \psi^{(1)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \sum_{i=1}^3 \tau_i \partial_i \psi^{(1)} + \tau_4 \partial_4 \psi^{(1)} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \sum_{h=5}^7 \tau_h \partial_h \psi^{(1)} - 3H \cot^2(6Hx_0) \psi^{(1)} = HV'(s) \psi^{(2)}, \\
& s = -\psi^{(1)\dagger} \sigma \psi^{(1)} + \psi^{(2)\dagger} \sigma \psi^{(2)}.
\end{aligned}$$

Component form (rows 0–7: the $\bar{\tau}$ equations for $\psi^{(2)}$; rows 8–15: the τ equations for $\psi^{(1)}$):

$$(E_0) \quad \cot(6Hx_0) \partial_0 \psi_0^{(2)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_0^{(2)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_5^{(2)}$$

$$\begin{aligned}
& -\partial_4\psi_5^{(2)} + e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_2\psi_6^{(2)} + e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_5\psi_6^{(2)} \\
& - e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_1\psi_7^{(2)} + e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_6\psi_7^{(2)} - 3H\cot^2(6Hx_0)\psi_0^{(2)} \\
& = HV'(s)\psi_0^{(1)}
\end{aligned}$$

$$\begin{aligned}
(E_1) \quad & \cot(6Hx_0)\partial_0\psi_1^{(2)} + e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_7\psi_1^{(2)} + e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_3\psi_4^{(2)} \\
& + \partial_4\psi_4^{(2)} - e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_1\psi_6^{(2)} - e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_6\psi_6^{(2)} \\
& - e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_2\psi_7^{(2)} + e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_5\psi_7^{(2)} - 3H\cot^2(6Hx_0)\psi_1^{(2)} \\
& = HV'(s)\psi_1^{(1)}
\end{aligned}$$

$$\begin{aligned}
(E_2) \quad & \cot(6Hx_0)\partial_0\psi_2^{(2)} + e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_7\psi_2^{(2)} - e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_2\psi_4^{(2)} \\
& - e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_5\psi_4^{(2)} + e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_1\psi_5^{(2)} + e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_6\psi_5^{(2)} \\
& - e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_3\psi_7^{(2)} + \partial_4\psi_7^{(2)} - 3H\cot^2(6Hx_0)\psi_2^{(2)} \\
& = HV'(s)\psi_2^{(1)}
\end{aligned}$$

$$\begin{aligned}
(E_3) \quad & \cot(6Hx_0)\partial_0\psi_3^{(2)} + e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_7\psi_3^{(2)} + e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_1\psi_4^{(2)} \\
& - e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_6\psi_4^{(2)} + e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_2\psi_5^{(2)} - e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_5\psi_5^{(2)} \\
& + e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_3\psi_6^{(2)} - \partial_4\psi_6^{(2)} - 3H\cot^2(6Hx_0)\psi_3^{(2)} \\
& = HV'(s)\psi_3^{(1)}
\end{aligned}$$

$$\begin{aligned}
(E_4) \quad & - e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_3\psi_1^{(2)} + \partial_4\psi_1^{(2)} + e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_2\psi_2^{(2)} \\
& - e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_5\psi_2^{(2)} - e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_1\psi_3^{(2)} - e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_6\psi_3^{(2)} \\
& + \cot(6Hx_0)\partial_0\psi_4^{(2)} - e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_7\psi_4^{(2)} - 3H\cot^2(6Hx_0)\psi_4^{(2)} \\
& = HV'(s)\psi_4^{(1)}
\end{aligned}$$

$$\begin{aligned}
(E_5) \quad & e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_3\psi_0^{(2)} - \partial_4\psi_0^{(2)} - e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_1\psi_2^{(2)} \\
& + e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_6\psi_2^{(2)} - e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_2\psi_3^{(2)} - e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_5\psi_3^{(2)} \\
& + \cot(6Hx_0)\partial_0\psi_5^{(2)} - e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_7\psi_5^{(2)} - 3H\cot^2(6Hx_0)\psi_5^{(2)} \\
& = HV'(s)\psi_5^{(1)}
\end{aligned}$$

$$\begin{aligned}
(E_6) \quad & - e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_2\psi_0^{(2)} + e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_5\psi_0^{(2)} + e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_1\psi_1^{(2)} \\
& - e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_6\psi_1^{(2)} - e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_3\psi_3^{(2)} - \partial_4\psi_3^{(2)} \\
& + \cot(6Hx_0)\partial_0\psi_6^{(2)} - e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_7\psi_6^{(2)} - 3H\cot^2(6Hx_0)\psi_6^{(2)} \\
& = HV'(s)\psi_6^{(1)}
\end{aligned}$$

$$\begin{aligned}
(E_7) \quad & e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_1\psi_0^{(2)} + e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_6\psi_0^{(2)} + e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_2\psi_1^{(2)} \\
& + e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_5\psi_1^{(2)} + e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_3\psi_2^{(2)} + \partial_4\psi_2^{(2)} \\
& + \cot(6Hx_0)\partial_0\psi_7^{(2)} - e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_7\psi_7^{(2)} - 3H\cot^2(6Hx_0)\psi_7^{(2)} \\
& = HV'(s)\psi_7^{(1)}
\end{aligned}$$

$$(E_8) \quad \cot(6Hx_0)\partial_0\psi_0^{(1)} - e^{-a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_7\psi_0^{(1)} + e^{a_4(Hx_4)}\sqrt[6]{\sin(6Hx_0)}\partial_3\psi_5^{(1)}$$

$$\begin{aligned}
& + \partial_4 \psi_5^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_6^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_6^{(1)} \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_7^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_7^{(1)} - 3H \cot^2(6Hx_0) \psi_0^{(1)} \\
& = HV'(s) \psi_0^{(2)}
\end{aligned}$$

$$\begin{aligned}
(E_9) \quad & \cot(6Hx_0) \partial_0 \psi_1^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_1^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_4^{(1)} \\
& - \partial_4 \psi_4^{(1)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_6^{(1)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_6^{(1)} \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_7^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_7^{(1)} - 3H \cot^2(6Hx_0) \psi_1^{(1)} \\
& = HV'(s) \psi_1^{(2)}
\end{aligned}$$

$$\begin{aligned}
(E_{10}) \quad & \cot(6Hx_0) \partial_0 \psi_2^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_2^{(1)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_4^{(1)} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_4^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_5^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_5^{(1)} \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_7^{(1)} - \partial_4 \psi_7^{(1)} - 3H \cot^2(6Hx_0) \psi_2^{(1)} \\
& = HV'(s) \psi_2^{(2)}
\end{aligned}$$

$$\begin{aligned}
(E_{11}) \quad & \cot(6Hx_0) \partial_0 \psi_3^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_3^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_4^{(1)} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_4^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_5^{(1)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_5^{(1)} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_6^{(1)} + \partial_4 \psi_6^{(1)} - 3H \cot^2(6Hx_0) \psi_3^{(1)} \\
& = HV'(s) \psi_3^{(2)}
\end{aligned}$$

$$\begin{aligned}
(E_{12}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_1^{(1)} - \partial_4 \psi_1^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_2^{(1)} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_2^{(1)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_3^{(1)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_3^{(1)} \\
& + \cot(6Hx_0) \partial_0 \psi_4^{(1)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_4^{(1)} - 3H \cot^2(6Hx_0) \psi_4^{(1)} \\
& = HV'(s) \psi_4^{(2)}
\end{aligned}$$

$$\begin{aligned}
(E_{13}) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_0^{(1)} + \partial_4 \psi_0^{(1)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_2^{(1)} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_2^{(1)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_3^{(1)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_3^{(1)} \\
& + \cot(6Hx_0) \partial_0 \psi_5^{(1)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_5^{(1)} - 3H \cot^2(6Hx_0) \psi_5^{(1)} \\
& = HV'(s) \psi_5^{(2)}
\end{aligned}$$

$$\begin{aligned}
(E_{14}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_0^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_0^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_1^{(1)} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_1^{(1)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_3^{(1)} + \partial_4 \psi_3^{(1)} \\
& + \cot(6Hx_0) \partial_0 \psi_6^{(1)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_6^{(1)} - 3H \cot^2(6Hx_0) \psi_6^{(1)} \\
& = HV'(s) \psi_6^{(2)}
\end{aligned}$$

$$\begin{aligned}
(E_{15}) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_0^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_0^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_1^{(1)} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_1^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_2^{(1)} - \partial_4 \psi_2^{(1)} \\
& + \cot(6Hx_0) \partial_0 \psi_7^{(1)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_7^{(1)} - 3H \cot^2(6Hx_0) \psi_7^{(1)} \\
& = HV'(s) \psi_7^{(2)}
\end{aligned}$$

5.7 The rescaled form: the spin connection disappears

Put $\Psi = \sqrt{\sin 6Hx_0} \Psi'$ and $\bar{\Psi} = \sqrt{\sin 6Hx_0} \bar{\Psi}'$. Then $s = \sin(6Hx_0) s'$ with $s' = \bar{\Psi}'\Psi'$, and

$$\begin{aligned} \gamma^\mu D_\mu \Psi - H V'(s) \Psi &= \sqrt{\sin 6Hx_0} \left(\cot(6Hx_0) T_{16}[0] \partial_0 \Psi' + \frac{1}{q} \sum_i T_{16}[i] \partial_i \Psi' + T_{16}[4] \partial_4 \Psi' \right. \\ &\quad \left. + \frac{1}{p} \sum_h T_{16}[h] \partial_h \Psi' - H V'(\sin(6Hx_0) s') \Psi' \right) \end{aligned}$$

with **no connection term at all**, and the same for the adjoint equation. This works because the connection term is a gradient: $\gamma^\mu \Gamma_\mu = -\frac{1}{2} T_\mu \gamma^\mu$ with the Weitzenböck torsion vector $T_\mu = \partial_\mu \ln \sin(6Hx_0)$ (section 9.10), and multiplying by $\exp(\frac{1}{2} \ln \sin) = \sqrt{\sin}$ removes it. [**proved VII §27**]

- WRITTEN OUT (iv) [THE RESULT]: Psi = Sqrt[Sin[6Hx0]] Psi' turns the field equation into Sqrt[Sin] (Cot T16[0] d_0 Psi' + (1/q) T16[i] d_i Psi' + T16[4] d_4 Psi' + (1/p) T16[h] d_h Psi' - H V'(Sin s') Psi'): NO connection term
- WRITTEN OUT (iv) [THE RESULT]: and the adjoint equation into Sqrt[Sin] (Cot d_0 Psi'bar T16[0] + ... + H V'(Sin s') Psi'bar), with no connection term either
- WRITTEN OUT (iv) [definition]: under the rescaling s == Sin[6 H x0] s', s' = Psi'bar Psi'

The sixteen rescaled equations (ψ'_k = component k of Ψ' ; V' evaluated at $s = \sin(6Hx_0) \bar{\Psi}'\Psi'$; provenance-latex/generated/fermion_fable_field_equations_rescaled.tex, unchanged):

$$\begin{aligned} (E'_0) \quad & \cot(6Hx_0) \partial_0 \psi'_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_8 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_{13} \\ & - \partial_4 \psi'_{13} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_{14} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_{14} \\ & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_{15} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_{15} \\ & = HV'(s) \psi'_0 \\ (E'_1) \quad & \cot(6Hx_0) \partial_0 \psi'_9 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_{12} \\ & + \partial_4 \psi'_{12} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_{14} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_{14} \\ & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_{15} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_{15} \\ & = HV'(s) \psi'_1 \\ (E'_2) \quad & \cot(6Hx_0) \partial_0 \psi'_{10} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_{12} \\ & - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_{12} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_{13} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_{13} \\ & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_{15} + \partial_4 \psi'_{15} \\ & = HV'(s) \psi'_2 \\ (E'_3) \quad & \cot(6Hx_0) \partial_0 \psi'_{11} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_{11} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_{12} \\ & - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_{12} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_{13} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_{13} \\ & + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_{14} - \partial_4 \psi'_{14} \\ & = HV'(s) \psi'_3 \\ (E'_4) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_9 + \partial_4 \psi'_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_{10} \end{aligned}$$

$$\begin{aligned}
& -e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_{11} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_{11} \\
& + \cot(6Hx_0) \partial_0 \psi'_{12} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_{12} \\
& = HV'(s) \psi'_4
\end{aligned}$$

$$\begin{aligned}
(E'_5) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_8 - \partial_4 \psi'_8 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_{10} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_{11} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_{11} \\
& + \cot(6Hx_0) \partial_0 \psi'_{13} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_{13} \\
& = HV'(s) \psi'_5
\end{aligned}$$

$$\begin{aligned}
(E'_6) \quad & -e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_8 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_9 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_9 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_{11} - \partial_4 \psi'_{11} \\
& + \cot(6Hx_0) \partial_0 \psi'_{14} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_{14} \\
& = HV'(s) \psi'_6
\end{aligned}$$

$$\begin{aligned}
(E'_7) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_8 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_9 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_{10} + \partial_4 \psi'_{10} \\
& + \cot(6Hx_0) \partial_0 \psi'_{15} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_{15} \\
& = HV'(s) \psi'_7
\end{aligned}$$

$$\begin{aligned}
(E'_8) \quad & \cot(6Hx_0) \partial_0 \psi'_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_0 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_5 \\
& + \partial_4 \psi'_5 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_6 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_6 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_7 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_7 \\
& = HV'(s) \psi'_8
\end{aligned}$$

$$\begin{aligned}
(E'_9) \quad & \cot(6Hx_0) \partial_0 \psi'_1 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_4 \\
& - \partial_4 \psi'_4 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_6 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_6 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_7 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_7 \\
& = HV'(s) \psi'_9
\end{aligned}$$

$$\begin{aligned}
(E'_{10}) \quad & \cot(6Hx_0) \partial_0 \psi'_2 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_4 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_4 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_5 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_5 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_7 - \partial_4 \psi'_7 \\
& = HV'(s) \psi'_{10}
\end{aligned}$$

$$\begin{aligned}
(E'_{11}) \quad & \cot(6Hx_0) \partial_0 \psi'_3 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_3 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_4 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_4 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_5 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_5 \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_6 + \partial_4 \psi'_6 \\
& = HV'(s) \psi'_{11}
\end{aligned}$$

$$\begin{aligned}
(E'_{12}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_1 - \partial_4 \psi'_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_2 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_3 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_3 \\
& + \cot(6Hx_0) \partial_0 \psi'_4 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_4
\end{aligned}$$

$$\begin{aligned}
&= HV'(s) \psi'_{12} \\
(E'_{13}) \quad &- e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_0 + \partial_4 \psi'_0 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_2 \\
&- e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_3 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_3 \\
&+ \cot(6Hx_0) \partial_0 \psi'_5 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_5 \\
&= HV'(s) \psi'_{13} \\
(E'_{14}) \quad &e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_0 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_1 \\
&+ e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_1 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_3 + \partial_4 \psi'_3 \\
&+ \cot(6Hx_0) \partial_0 \psi'_6 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_6 \\
&= HV'(s) \psi'_{14} \\
(E'_{15}) \quad &- e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_0 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_1 \\
&- e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_2 - \partial_4 \psi'_2 \\
&+ \cot(6Hx_0) \partial_0 \psi'_7 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_7 \\
&= HV'(s) \psi'_{15}
\end{aligned}$$

5.8 The x_4 -only family and the condition $V''s' = 0$

The rescaled field with no x_0 -dependence, $\Psi = \sqrt{\sin 6Hx_0} \Psi'(x_4)$, satisfies the field equation at every x_0 **if and only if** $\partial_{x_0} V'(\sin(6Hx_0) s') = 0$, and $\partial_{x_0} V'(\sin s') = 6H \cos(6Hx_0) s' V''(\sin s')$. So it is a solution exactly when $V''s' = 0$:

- either V is **linear** (the author's mass term $V = -(2M/H)s$, $V' = -2M/H$ constant), or
- the data are **null**, $s' = \bar{\Psi}'\Psi' = 0$.

For the complex field s' is **conserved** by the x_4 -only equations $\partial_4 \Psi' = -HV' T_{16}[4] \Psi'$, $\partial_4 \bar{\Psi}' = HV' \bar{\Psi}' T_{16}[4]$, for **any** V . So null data give x_4 -only solutions for any V , with $\rho = V(0)$. For a nonlinear V and $s' \neq 0$ the family fails (witness: $V = s^2$). **[proved VII §29]**

- PRE-UNIVERSE (2) [THE RESULT]: `Psi = Sqrt[Sin] Psi'(x4)` satisfies the field equation iff `d_x0 V'(Sin[6Hx0] s') == 0`, and `d_x0 V'(Sin s') == 6 H Cos[6 H x0] s' V''(Sin s')`: `V linear OR s' == 0`
- PRE-UNIVERSE (2) [has content]: `s' = Psi'bar Psi'` is conserved by the x_4 -only equations (`d_4 Psi' = -H V' T16[4] Psi'`, `d_4 Psi'bar = H V' Psi'bar T16[4]`) for ANY V ; so null data $s' = 0$ are solutions for any V , with $\rho == V(0)$; and for $V = s^2$ with $s' \neq 0$ the x_0 -dependence is real (witness)

5.9 The operator form of the field equation, and its ordering

These are definitions, not identities **[prose VII §27]**. The Hamiltonian (section 6.11) contains $V(:s:)$, defined by the spectral calculus of the Hermitian operator $s(x) = \Psi^\dagger \beta \Psi$ of section 6.5 (regularized and normal-ordered). No ordering prescription is needed there, and a non-polynomial V is covered. The operator field equation is

$$\gamma^\mu D_\mu \Psi = H :V'(s) \Psi:$$

where $V'(s)\Psi$ is the **Weyl-symmetrized** product, the average over the positions at which Ψ can be inserted in $V'(s)$.

- It is **exact** for linear V (the author's mass term).
- For nonlinear V it differs from the naive product by Hartree and Fock contractions with the coincident J -vacuum propagator. That propagator is a c-number **matrix** with two parts: a **scalar** part, which is divergent and renormalizes V' ; and a γ^4 part, the charge density of the Dirac sea, which is a constant chemical-potential shift. The γ^4 part is removed by charge-symmetric ordering or by a phase $\Psi \rightarrow e^{i\mu x_4} \Psi$, **not** by renormalizing V .
- In the Kohn–Sham mean field of section 8, $V'(s)$ is replaced by $V'(\langle s \rangle)$ and the equation is linear.

5.10 The U(1) current and the particle-number current

L_{sym} is invariant under $\Psi \rightarrow e^{i\alpha} \Psi$, $\bar{\Psi} \rightarrow e^{-i\alpha} \bar{\Psi}$. The Noether current is

$$J^\mu = \frac{\partial L}{\partial(\partial_\mu \Psi)} (i\Psi) + (-i\bar{\Psi}) \frac{\partial L}{\partial(\partial_\mu \bar{\Psi})} = \frac{\sqrt{g}}{H} i \bar{\Psi} \gamma^\mu \Psi,$$

so $j^\mu = i \bar{\Psi} \gamma^\mu \Psi$ is the U(1) current. It is real, because $i\sigma_{16}\gamma^\mu$ is Hermitian. **Off shell**, for generic fields of all eight coordinates,

$$\frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} j^\mu) = i (\bar{E} \Psi + \bar{\Psi} E), \quad E, \bar{E} = \text{the two field-equation residuals},$$

so the current is conserved on shell. [**proved VII §27**]

- CURRENT [THE RESULT]: the Noether current of the U(1) phase is (Sqrt[g]/H) i Psibar gamma^mu Psi
- CURRENT [has content]: i sigma16 gamma^mu is Hermitian for every mu, so j^mu is real
- CURRENT [THE RESULT]: OFF SHELL (1/Sqrt[g]) d_mu(Sqrt[g] j^mu) == i (Ebar Psi + Psibar E) for generic fields of all eight coordinates - conserved on shell

The **particle-number current** is $n^\mu := -\frac{i}{H} \bar{\Psi} \gamma^\mu \Psi = -j^\mu/H$. As section 6.15 shows, $N = \int \sqrt{g} n^4 d^7x$ is the normal-ordered number of particles minus antiparticles, so the charge of j carries a conventional minus sign: $\int \sqrt{g} j^4 d^7x = -HN$ [**prose VII §27**]. So one particle (one occupied positive-frequency J -mode, $N = +1$) carries j -charge $-H$, and one antiparticle $+H$ [derived here from the two definitions; the design review, QK-10, states the same sign as “a positive-frequency J -mode carries j -charge -1 ”, that is, in units of H].

5.11 The only internal symmetry is the vector phase

The commutant of all eight $T_{16}[a]$ in $M_{16}(\mathbb{C})$ is one-dimensional (multiples of ID16), so the only internal U(1) is the vector phase. In particular the **axial** phase $\Psi \rightarrow \exp(i\alpha T_{16}[8]) \Psi$ leaves $s = \bar{\Psi} \Psi$ invariant but **not** the kinetic term ($T_{16}[8]$ anticommutes with every gamma): $\exp(-i\alpha T_8) T_{16}[a] \exp(i\alpha T_8) = T_{16}[a] \exp(2i\alpha T_8)$, so $\bar{\Psi} \gamma^\mu \partial_\mu \Psi$ changes at first order by $2i\alpha \bar{\Psi} \gamma^\mu T_{16}[8] \partial_\mu \Psi$, which is not zero. This is the reverse of the four-dimensional intuition, where the axial phase preserves the kinetic term and breaks the mass term. [**proved VII §27**]

- SYMMETRY [THE RESULT]: the only internal U(1) is the vector phase - the commutant of all eight T16[a] in M16(C) is one-dimensional (multiples of ID16)
- SYMMETRY [has content]: the axial phase exp(i alpha T16[8]) leaves s invariant but not the kinetic term: exp(-i alpha T8) T16[a] exp(i alpha

T8) == T16[a] exp(2 i alpha T8), so Psibar gamma^mu d_mu Psi changes at first order by 2 i alpha Psibar gamma^mu T16[8] d_mu Psi, which is not zero (witness)

6 Canonical quantization in 4+4 dimensions

This is Section 28 of the notebook (Input cells 185–191, 56 assertions). “Extended to 4+4 dimensions” means: the time is one of the four timelike directions, x_4 , and the slice of constant time is seven-dimensional and contains the other three timelike directions. Everything below follows from the symmetrized Lagrangian of section 4; the result is summarized first.

The result in one paragraph. The Dirac bracket gives the anticommutator $\{\Psi_a(x), \Psi_b^\dagger(y)\} = \frac{H}{\sqrt{|g|}} (-i\sigma_{16}\gamma^4)_{ab} \delta^7(x-y)$. Its matrix is $J = -iT_{16}[0]T_{16}[1]T_{16}[2]T_{16}[3]T_{16}[4]$, which is Hermitian with square 1 and has signature (8, 8): the naive Fock space would have negative-norm states. J acts only on a hidden flavour index, $\sigma_{16} = J\beta$ with $\beta = -iT_{16}[4]$, and the Hilbert adjoint $\Psi^\dagger := \Psi^\dagger J$ makes the anticommutator positive, $\{\Psi, \Psi^\dagger\} = \frac{H}{\sqrt{|g|}} \delta^7$. This works exactly when the field carries no momentum along the three hidden timelike directions x_5, x_6, x_7 (the admissibility theorem), and there J is unique. The admissible theory is a dimensional reduction with a finite hidden coordinate volume V_{hid} . Its Fock space is built on J -eigenmodes: per momentum, 8 positive-frequency and 8 negative-frequency modes; the ground state is the filled Dirac sea; the first excitations are 8 particles and 8 antiparticles. A finite Jordan–Wigner Fock space checks all of this exactly.

6.1 Time, slices, and why the slice is not a Cauchy surface

Time is x_4 , the observer’s proper time. On the canonical frame $g_{44} = -1$ (lapse 1) and $g_{4\mu} = 0$ for $\mu \neq 4$ (shift 0). A slice $x_4 = \text{const}$ is **seven**-dimensional with signature (4, 3): x_0, x_1, x_2, x_3 spacelike and x_5, x_6, x_7 **timelike**. It is therefore not a Cauchy surface, and the quantization is carried out where it is well posed (the admissible sector, section 6.10). **[proved VII §28]** QUANTIZATION [definition]: lapse 1 and shift 0 ($g_{44} == -1$, $g_{4\mu} == 0$), and the slice $x_4 = \text{const}$ has signature (4, 3) - $x_0..x_3$ spacelike, $x_5..x_7$ timelike (given a_4 real and $0 < 6 H x_0 < \text{Pi}/2$)

6.2 The momenta, the kinetic form G , and the signature (8, 8)

From L_{sym} the canonical momenta are

$$\Pi_\Psi = \frac{\partial L}{\partial(\partial_4 \Psi)} = \frac{\sqrt{g}}{2H} \bar{\Psi} \gamma^4, \quad \Pi_{\bar{\Psi}} = \frac{\partial L}{\partial(\partial_4 \bar{\Psi})} = -\frac{\sqrt{g}}{2H} \gamma^4 \Psi.$$

The symplectic potential $\Pi_\Psi d\Psi + d\bar{\Psi} \Pi_{\bar{\Psi}}$ differs from that of the unsymmetrized Lagrangian, $\frac{\sqrt{g}}{H} \bar{\Psi} \gamma^4 d\Psi$, by the exact variation $\delta[\frac{\sqrt{g}}{2H} \bar{\Psi} \gamma^4 \Psi]$: the two define the **same** symplectic form, hence the same brackets. With $\bar{\Psi} = \Psi^\dagger \sigma_{16}$ the time-derivative part of the Lagrangian is

$$\Psi^\dagger K \partial_4 \Psi, \quad K = \frac{\sqrt{g}}{H} \sigma_{16} \gamma^4 = i \tilde{G}, \quad \tilde{G} = \frac{\sqrt{g}}{H} G, \quad G = -i \sigma_{16} \gamma^4.$$

On the canonical frame $\gamma^4 = T_{16}[4]$, so

$$G = -i \sigma_{16} T_{16}[4] = -i T_{16}[0]T_{16}[1]T_{16}[2]T_{16}[3]T_{16}[4] =: J.$$

G is Hermitian, $G^2 = \text{ID}_{16}$, with eigenvalues $+1$ (8 times) and -1 (8 times): the kinetic form is **indefinite**, signature (8, 8). The canonical frame has $\Gamma_4 = 0$, so the time-derivative part, which alone determines the anticommutator, contains no spin connection. **[proved VII §28]**

- QUANTIZATION [THE RESULT]: the momenta of Lsym are $\text{Pi_Psi} == (\text{Sqrt}[g]/(2H)) \text{Psibar } \gamma^4$ and $\text{Pi_Psibar} == -(\text{Sqrt}[g]/(2H)) \gamma^4 \text{Psi}$
- QUANTIZATION [has content]: $\text{theta_un} - \text{theta_sym} == \text{delta}[(\text{Sqrt}[g]/(2H)) \text{Psibar } \gamma^4 \text{Psi}]$ exactly: the same symplectic form, the same brackets
- QUANTIZATION [THE RESULT]: $G = -i \text{sigma16 } \gamma^4 == J = -i \text{T16}[0].\text{T16}[1].\text{T16}[2].\text{T16}[3].\text{T16}[4]$ on the canonical frame; J is Hermitian, $J^2 == \text{ID16}$, eigenvalues +1 (8) and -1 (8): signature (8,8)
- QUANTIZATION [has content]: $\text{Gamma}_4 == 0$ - the time-derivative part of the Lagrangian, which fixes the anticommutator, contains no spin connection

6.3 The Dirac bracket and the canonical anticommutator

Write $\Psi = (\theta_1 + i\theta_2)/\sqrt{2}$ with real Grassmann θ_1, θ_2 and $\Theta = (\theta_1, \theta_2)$. In the Grassmann algebra of subsection 3.3, for $K = c\sigma_{16}T_{16}[4]$ with any scalar c ,

$$\Psi^\dagger K d\Psi = \frac{i}{2} \Theta^T \Omega_c d\Theta + \frac{1}{4} d(\theta_1^T K \theta_1 + \theta_2^T K \theta_2),$$

$$\Omega_c = \begin{pmatrix} 0 & K \\ -K & 0 \end{pmatrix} \quad (\text{real, symmetric, invertible}; \quad \Omega_c^{-1} = \begin{pmatrix} 0 & -K^{-1} \\ K^{-1} & 0 \end{pmatrix}).$$

So up to a total time derivative the Lagrangian is the first-order system $\frac{i}{2} \Theta^T \Omega \Theta' - h$. Its momenta are proportional to Θ ; the constraints π - (momentum as a function of Θ) are second class with constraint matrix proportional to Ω ; and the Dirac bracket of the Θ 's is proportional to Ω^{-1} . The constant is fixed by the elementary case $\Omega = 1$, $\frac{i}{2} \xi \xi'$, whose quantization is $\{\xi, \xi\} = 1$. Hence

$$\begin{aligned} \{\Theta_i, \Theta_j\} &= (\Omega^{-1})_{ij}, \\ \{\Psi_a, \Psi_b^\dagger\} &= \frac{1}{2} [(\Omega^{-1})_{11} + (\Omega^{-1})_{22} - i(\Omega^{-1})_{12} + i(\Omega^{-1})_{21}]_{ab} = i(K^{-1})_{ab} = (\tilde{G}^{-1})_{ab}, \\ \{\Psi_a, \Psi_b\} &= 0. \end{aligned}$$

The same sign and normalization are confirmed independently, with no convention for graded Poisson brackets, by the Fock space of section 6.14: with this anticommutator the Heisenberg equation generated by the canonical Hamiltonian reproduces the field equation, and with the opposite sign it gives the time-reversed one. On the canonical frame $J^{-1} = J$ and $\sqrt{g} = \sec 6Hx_0$, so

$$\boxed{\{\Psi_a(x), \Psi_b^\dagger(y)\}_{x_4=y_4} = \frac{H}{\sqrt{g}} J_{ab} \delta^7(x-y) = H \cos(6Hx_0) J_{ab} \delta^7(x-y), \quad \{\Psi_a, \Psi_b\} = 0}$$

which is $\frac{H}{\sqrt{|g|}} (-i\sigma_{16}\gamma^4)_{ab}$, because $(-i\sigma_{16}T_{16}[4])^{-1} = -i\sigma_{16}T_{16}[4]$. The spin connection does not appear: the anticommutator comes from the time-derivative term alone, and $\Gamma_4 = 0$. **[proved VII §28]**

- DIRAC BRACKET [THE RESULT]: in the Grassmann algebra $\text{Psi}^\dagger \text{ddag } K \text{ phi} == (i/2) \text{Theta}^T \text{Omega_c } \text{Phi} + (1/4) d(\text{theta1}^T K \text{ theta1} + \text{theta2}^T K \text{ theta2})$, for $K = c \text{sigma16 } \text{T16}[4]$ with a free scalar c
- DIRAC BRACKET [has content]: Omega_c is real symmetric and invertible, $\text{Omega_c}^{-1} == [[0, -K^{-1}], [K^{-1}, 0]]$
- DIRAC BRACKET [THE RESULT]: $\{\text{Theta_i}, \text{Theta_j}\} = (\text{Omega}^{-1})_{ij}$ gives $\{\text{Psi_a}, \text{Psi}^\dagger \text{ddag_b}\} == i (K^{-1})_{ab} == (\text{Gtilde}^{-1})_{ab}$ and $\{\text{Psi_a}, \text{Psi_b}\} == 0$

- ANTICOMMUTATOR [THE RESULT]: on the canonical frame ($c = \text{Sqrt}[g]/H$) $\{\Psi_i, \Psi_i^{\dagger}\} == (H/\text{Sqrt}[g]) J == H \text{Cos}[6 H x_0] J (J^{-1} == J)$, which IS the design's $(H/\text{Sqrt}[g])(-i \text{sigma16 gamma}^4)$

The exported anticommutator, with its matrix (provenance-latex/generated/fermion_fable_anticommutator.tex, unchanged):

$$\{\Psi_a(x), \Psi_b^{\dagger}(y)\}_{x_4=y_4} = \frac{H}{\sqrt{|g|}} J_{ab} \delta^7(x-y) = H \cos(6Hx_0) J_{ab} \delta^7(x-y), \quad \{\Psi_a, \Psi_b\} = 0,$$

$$\{\Psi_a(x), \Psi_b^{\dagger}(y)\}_{x_4=y_4} = \frac{H}{\sqrt{|g|}} \delta_{ab} \delta^7(x-y)$$

with $\Psi^{\dagger} = \Psi^{\dagger} J$; the anticommutator matrix is $\frac{H}{\sqrt{|g|}} J = H \cos(6Hx_0) J$ with

$$J = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

6.4 The admissible truncation, V_{hid} , and the rescaled field

The admissible theory (section 6.10) is a **truncation**: a dimensional reduction to fields $\Psi = \Psi(x_0, x_1, x_2, x_3, x_4)$, independent of x_5, x_6, x_7 . It needs a **finite** hidden coordinate volume $V_{\text{hid}} = \int dx_5 dx_6 dx_7$. That is an assumption, and it has a consequence that must be stated: compact timelike directions contain closed timelike curves. In the truncation the delta function becomes $\delta^4(x_{0..3} - y_{0..3})$ and the factor $H/\sqrt{g}$ becomes $H/(V_{\text{hid}}\sqrt{g})$:

$$\{\Psi_a(x), \Psi_b^{\dagger}(y)\}_{x_4=y_4} = \frac{H}{V_{\text{hid}}\sqrt{|g|}} G_{ab} \delta^4(x_{0..3} - y_{0..3}), \quad G = -i \sigma_{16} \gamma^4.$$

For the rescaled field $\Psi' = \Psi/\sqrt{\sin 6Hx_0}$ of section 5.7 the factor is $H \cos(6Hx_0)/\sin(6Hx_0) = H \cot(6Hx_0)$. On a frame whose $\sqrt{g}$ depends on x_4 one quantizes $\chi = (\sqrt{g}/H)^{1/2} \Psi$, with $\{\chi, \chi^{\dagger}\} = G \delta^7$; on the canonical frame $\sqrt{g}$ does not depend on x_4 and the two coincide [**prose VII §28**]. [**proved VII §28**] ANTICOMMUTATOR [has content]: the rescaled field $\Psi' = \Psi/\text{Sqrt}[\text{Sin}[6 H x_0]]$ has $\{\Psi', \Psi'^{\dagger}\} == H \text{Cot}[6 H x_0] J \delta^7$, and $\text{Sqrt}[g]$ is x_4 -independent (so quantizing $\chi = (\text{Sqrt}[g]/H)^{(1/2)} \Psi$ changes nothing on the canonical frame)

(For the time-dependent warped frames of the interacting system, Part VIII of the notebook, Section 32, proves that the rescaled field has an x_4 -independent anticommutator: RESCALING [THE RESULT]: $\{\Psi_i, \Psi_i^{\dagger}\} == (H/\text{Sqrt}[|\det g|]) G$ on the warped frame (Section 28's cfAntiPsiPsiDd at $c = \text{Sqrt}[|\det g|]/H$), and for $\chi = \Psi/f$, $f = \text{Sqrt}[\text{Sin} (A^3 B^3 C)^{(-1/2)}]$: $\{\chi_i, \chi_i^{\dagger}\} == H \text{Cot}[6 H x_0] G$ - the $\text{Sqrt}[\det g]$ cancels, independent of x_4 and of A, B, C and RESCALING [has content]: on the warped frame $G = -i \text{sigma16 gamma}^4$ is Section 28's cfGK (= J) - the same matrix on every member of the family ($\text{gamma}^4 == \text{T16}[4]$, lapse 1).)

6.5 What J is, and the Hilbert adjoint

$J = -i T_{16}[0] \cdots T_{16}[4] = -i \Omega_4 T_{16}[0] = -i h_0$, with $\Omega_4 = T_{16}[1] T_{16}[2] T_{16}[3] T_{16}[4]$ of section 3.6. So J is one of the hidden **flavour** gammas: it lies in the commutant of the observer's Clifford

algebra, and on $\mathbb{C}^{16} = \mathbb{C}^4(\text{observer}) \otimes \mathbb{C}^4(\text{flavour})$ it acts on the **flavour factor only**, where it has signature $(2, 2)$ (verified on the image of a rank-4 primitive idempotent of the observer's algebra). With $\beta := -iT_{16}[4]$ (Hermitian, $\beta^2 = 1$):

$$\sigma_{16} = J\beta, \quad \beta T_{16}[j] \beta = -T_{16}[j] \quad \text{and} \quad T_{16}[j] \text{ Hermitian} \quad (j = 0..3),$$

so β is the observer's Dirac-adjoint matrix. Define the **Hilbert adjoint** by the J -involution, $\Psi^\dagger := \Psi^\dagger J$. Then

$$\bar{\Psi} = \Psi^\dagger \sigma_{16} = \Psi^\dagger \beta \quad (\text{the standard 4D Dirac adjoint}),$$

$$s = \Psi^\dagger \beta \Psi,$$

$$\Psi^\dagger G \partial_4 \Psi = \Psi^\dagger \partial_4 \Psi \quad (\text{the time-kinetic term becomes } \frac{\sqrt{g}}{H} i \Psi^\dagger \partial_4 \Psi),$$

$$\{\Psi_a, \Psi_b^\dagger\} = \frac{H}{\sqrt{g}} \delta_{ab} \delta^7 \quad (\text{POSITIVE}).$$

So fermion fable is four flavours times a 4D Dirac field, and J -quantization replaces the classical flavour metric of signature $(2, 2)$ by the identity. The adjoint is **mode-independent**, and it must be defined this way: a mode-by-mode prescription is basis-dependent unless every mode is a J -eigenvector (section 6.13). [**proved VII §28**]

- J [THE RESULT]: J == -i Omega4 T16[0] == -i h_0 is a hidden flavour gamma: it lies in the commutant of T16[1..4] (in the span of the hidden-gamma products of Section 26)
- J [has content]: on the flavour factor J has signature (2,2): cfIdemObs is a rank-4 projector commuting with J, and J restricted to its image has eigenvalues +1, +1, -1, -1
- J [THE RESULT]: sigma16 == J beta with beta = -i T16[4] Hermitian, beta^2 == 1; T16[0..3] Hermitian with beta T16[j] beta == -T16[j]: beta is the observer's Dirac adjoint
- J [THE RESULT]: with Psi^dagger := Psi^ddag J: Psibar = Psi^dagger beta (J sigma16 == beta), Psi^ddag G = Psi^dagger (J G == ID16), {Psi, Psi^dagger} == (H/Sqrt[g]) ID16 (G^-1 J == ID16): POSITIVE

J and β , as exported (provenance-latex/generated/fermion_fable_krein_J.tex, unchanged):

$$J = -iT_{16}[0]T_{16}[1]T_{16}[2]T_{16}[3]T_{16}[4] = -i\sigma_{16}T_{16}[4] = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\beta = -iT_{16}[4] = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \sigma_{16} = J\beta, \quad J^2 = \beta^2 = 1, \quad \bar{\Psi} = \Psi^\dagger \beta.$$

6.6 Why this is legitimate for fermions, and why it would fail for bosons

For Grassmann fields the $*$ -structure is a free choice: the Berezin integral treats Ψ and $\bar{\Psi}$ as independent, and the J -involution is that choice. The filled Dirac sea makes the Hamiltonian bounded below. For a **commuting** 16-component field the same involution gives $[b, b^\dagger] = 1$, but the Hamiltonian $\sum_u \omega_u b_u^\dagger b_u$ then has 8 negative-frequency modes per momentum and no exclusion principle to fill them: it is unbounded below. The only local involutions compatible with a Hermitian s and a Hermitian Hamiltonian are $+J$ and $-J$ (the uniqueness theorem, section 6.10), and $-J$ gives every mode negative norm; making the bosonic Hamiltonian bounded below would turn 4 modes into negative-norm modes. For fermions the negative-frequency modes are filled and the obstruction disappears [**prose VII §28; design review QK-5**].

The J -odd bilinears $(\hat{T}_{\mu h}, j^h, \bar{\Psi} T_{16}[8] \Psi)$ become **anti-Hermitian** operators (section 6.8). So the classical real theory of Part VI is the classical limit of the J -**even** sector only. (A reflection-positivity claim that appeared in the design was dropped by the review: it was not verified.)

6.7 Chirality, and why V depends on s only

J anticommutes with $T_{16}[8]$: J exchanges the two split-octonion types. Each chirality subspace is G -**null** ($P_L G P_L = P_R G P_R = 0$: the Krein form pairs type-1 with type-2 only), so the J -vacuum is not chirality-graded; but the two chirality subspaces **are** orthogonal in the Hilbert product ($P_L P_R = 0$, with P_L, P_R Hermitian). Neither $p = \bar{\Psi} T_{16}[8] \Psi$ (anti-Hermitian under J) nor ip (anti-Hermitian under the classical conjugation $\ddagger$, so it would make the Lagrangian complex) can enter V consistently: $V = V(s)$. [**proved VII §28**] J [has content]: chirality - $\{J, T_{16}[8]\} = 0$, each chirality subspace is G -NULL ($PL G PL = PR G PR = 0$), and the two chiralities are orthogonal in the Hilbert product (PL, PR Hermitian, $PL PR = 0$)

6.8 The Hermiticity rule and the J table

Under the J -adjoint, a bilinear $\Psi^\dagger M \Psi$ with M classically Hermitian has $(\Psi^\dagger M \Psi)^\dagger = \Psi^\dagger (J M J) \Psi$: it is **Hermitian** iff $[J, M] = 0$ and **anti-Hermitian** iff $\{J, M\} = 0$. The table records J against every matrix that occurs [**proved VII §28**]:

matrix	relation to J	consequence
σ_{16}	commutes	s is Hermitian
$T_{16}[0], \dots, T_{16}[4]$	commute	
$T_{16}[5], T_{16}[6], T_{16}[7]$	anticommute	
$\sigma_{16} \gamma^\mu, \mu = 0..4$	commute	$j^\mu, \mu = 0..4$, Hermitian
$\sigma_{16} \gamma^h, h = 5, 6, 7$	anticommute	j^h anti-Hermitian
each $\gamma^\mu \Gamma_\mu$ (no sum), $\mu = 0..7$	commute	the connection term of the Dirac operator is J -even
Γ_0, Γ_4 (both zero), $\Gamma_1, \Gamma_2, \Gamma_3$	commute	
$\Gamma_5, \Gamma_6, \Gamma_7$	anticommute	boosts mixing x_0 or x_4 with the hidden timelike sheet
$T_{16}[8]$ and $C_+ = \sigma_{16} T_{16}[8]$	anticommute	$p = \bar{\Psi} T_{16}[8] \Psi$ is anti-Hermitian

- **J TABLE [THE RESULT]**: J commutes with σ_{16} , $T_{16}[0..4]$, $\sigma_{16} \gamma^\mu$ ($\mu = 0..4$), every $\gamma^\mu \Gamma_\mu$ and $\Gamma_1.. \Gamma_3$; it **ANTI**commutes with $T_{16}[5..7]$, $\sigma_{16} \gamma^h$ ($h = 5..7$), $\Gamma_5.. \Gamma_7$, $T_{16}[8]$ and C_+
- **J [has content]**: the Hermiticity rule - for classically Hermitian M , $(J M)^\dagger = J M$ iff $[J, M] = 0$ and $= -J M$ iff $\{J, M\} = 0$: so $s = \Psi^\dagger \sigma_{16} \Psi$ is Hermitian and $p = \Psi^\dagger C_+ \Psi$ is **ANTI**-Hermitian

Consequences: s is Hermitian and p anti-Hermitian (so V may depend on s but not on p); the hidden current components j^h and the energy-momentum components $\hat{T}_{\mu h}$ ($\mu = 0..4$) are built from anticommuting matrices and are anti-Hermitian: their expectation values are purely imaginary in every state and vanish in every state invariant under the hidden rotations (the J -vacuum and the Kohn–Sham states of section 8), consistent with $G_{\mu h} = 0$ on the canonical frame. $\hat{T}_{hh'}$ ($h \neq h'$) vanishes identically for fields independent of $x_5..x_7$, and $\hat{T}_{hh} = g_{hh}\hat{L}$ is Hermitian (section 7.5).

(The same J table holds on the dynamical frames of the interacting system; Part VIII, Section 32, proves it: J [THE RESULT]: [J, γ^μ Gamma_mu] == 0 for EACH mu separately (h = 5,6,7 included), on the warped and on the Bianchi-I frame and J [THE RESULT]: [J, Gamma_mu] == 0 for mu = 0..4 and {J, Gamma_h} == 0 for h = 5, 6, 7, on the warped and on the Bianchi-I frame.)

6.9 The symmetry that survives, and what is lost

Of the 28 Lorentz generators S_{ab} , **13 commute** with J : Spin(4,1) of $x_0..x_4$ (10 generators) and Spin(3) of $x_5..x_7$ (3). The other **15 anticommute**: the 12 boosts between $x_0..x_3$ and $x_5..x_7$, and the 3 **rotations** between x_4 and $x_5..x_7$ ($x_4..x_7$ are all timelike, so those planes are rotations). The Krein form G itself is invariant ($S^T G + G S = 0$) under exactly the 21 generators with $a, b \neq 4$: Spin(4,3), the group of the slice. [proved VII §28] (The J structure is defined in the diagonal gauge of the canonical frame and transforms as $J \rightarrow S^{-1} J S$ under a change of frame gauge S ; design review QK-6.)

- J [THE RESULT]: 13 of the 28 Lorentz generators commute with J (a, b both in {0..4}: Spin(4,1), or both in {5,6,7}: Spin(3)); the other 15 anticommute - 12 boosts $x_0..x_3 \leftrightarrow x_5..x_7$ and 3 rotations $x_4 \leftrightarrow x_5..x_7$
- J [has content]: the Krein form G is invariant ($S^T G + G S = 0$) under exactly the 21 generators with $a, b \neq 4$ - Spin(4,3), the group of the slice

6.10 The admissible sector: a theorem

Take a plane wave at momentum k (flat local analysis, the frame factors frozen). The one-particle Hamiltonian is $h_k = \sigma_{16} (m - i \gamma \cdot k)$, with $\gamma \cdot k = \sum_{a \neq 4} k_a T_{16}[a]$, and the evolution generator is $E_k = G^{-1} h_k = G h_k$, $i \partial_4 \Psi = E_k \Psi$. Write $k_s^2 = k_0^2 + k_1^2 + k_2^2 + k_3^2$ and $k_h^2 = k_5^2 + k_6^2 + k_7^2$.

1. For **every** k , $E_k^2 = (k_s^2 - k_h^2 + m^2) \text{ID16}$: the dispersion relation is $\omega^2 = k_s^2 - k_h^2 + m^2$.
2. **Admissible** k ($k_h = 0$): $[J, E_k] = 0$ and J itself gives the positive quantization ($GJ = \text{ID16}$).
3. Hidden momentum **below** threshold ($k_h^2 < k_s^2 + m^2$): a momentum-dependent, boosted $J' = S^{-1} J S$ (S the spin boost in the plane of k_s and k_h that removes k_h) commutes with E_k and with β , $J'^2 = 1$, and $GJ' = S^2$ is positive definite: each such momentum can be quantized positively on its own (checked at $k = 5e_1 + 3e_5$, $m = 3$, $\omega^2 = 25 - 9 + 9 = 25$, boost with $\tanh \theta = 3/5$).
4. **At** threshold E_k is not zero but $E_k^2 = 0$: E_k is nilpotent and cannot be diagonalized (Jordan blocks of size 2, rank 8); there is no mode expansion (checked at $k = 4e_1 + 5e_5$, $m = 3$).
5. **Above** threshold $\omega^2 < 0$: the frequencies are imaginary (growth along x_4 , the ill-posed ultrahyperbolic Cauchy problem) and the modes are **G -neutral** ($u^\dagger G u = 0$), so they carry no norm at all. Example: $k = (0, \frac{3}{4}, 1, 0 \mid k_5 = 2)$, $m = 1$: $\omega^2 = -23/16$, $\omega = \pm i\sqrt{23}/4 = \pm 1.19896 i$.

6. **No** single momentum-independent positive J' exists once a hidden momentum is present. If $V(s)$ and the local energy-momentum tensor are to be Hermitian, J' must commute with β (because $s = \Psi^\dagger G \beta \Psi$) and with E_k for all k . Certificate: for the four admissible unit momenta and one hidden momentum, **every** X commuting with β and all five E_k has $\text{Tr}(GX) = 0$, so GX is never positive definite. Analytic proof for a purely hidden momentum: $[E_k, \beta] = 2i k_5 T_{16}[5]$, so J' commutes with $T_{16}[5]$; $T_{16}[5]$ anticommutes with G , hence $T_{16}[5]^\dagger (GJ') T_{16}[5] = -(GJ')$: a positive matrix would be congruent to a negative one.
7. J is **unique**: the joint commutant of β and all admissible E_k is 8-dimensional and every one of its elements commutes with J ; if X is in it with $X^2 = 1$ and $GX = JX$ definite, then $(JX)^2 = 1$, and a definite Hermitian matrix with square 1 is $\pm \text{ID}16$: $X = J$ (positive metric) or $X = -J$ (every mode of negative norm).

The general identity behind (6) is $[E_k, \beta] = 2i T_k$, $T_k = \sum_a k_a T_{16}[a]$, for every k .

THEOREM (admissibility). Let W be the linear span of the momenta carried by the modes, a subspace of the slice $\mathbb{R}^{(4,3)}$. A positive (Hilbert-space) Fock quantization in which the local $s(x)$, the local energy-momentum tensor and the free Hamiltonian are all Hermitian exists **if and only if** W is **spacelike-definite**. Then $J' = S^{-1} J S$ with S a slice boost carrying W into $\text{span}(x_0..x_3)$, and J' is unique (with $J'^2 = 1$) when $W = \text{span}(x_0..x_3)$. On the canonical frame the fields depend on all of $x_0..x_3$, so the condition is: **no momentum along** x_5, x_6, x_7 , and $J = -i T_{16}[0]T_{16}[1]T_{16}[2]T_{16}[3]T_{16}[4]$ is unique.

The admissible theory is a **truncation** (a dimensional reduction $\Psi = \Psi(x_0, \dots, x_4)$), not a superselection sector: $V(s)$ contains terms like $b_{k_h}^\dagger b_{-k_h}^\dagger b_0 b_0$ that would connect it to hidden momenta. What survives of the multi-time (Tomonaga-Schwinger) picture is the isometry fact: the metric, the frame and the connection do not depend on x_5, x_6, x_7 , so the hidden translations commute with the x_4 -evolution, $[P_4, P_h] = 0$ [**prose VII §28**]. [**proved VII §28**]

- **ADMISSIBILITY (1)** [THE RESULT]: $E_k^2 == (k_s^2 - k_h^2 + m^2) \text{ID}16$ for every k : $\omega^2 = k_s^2 - k_h^2 + m^2$
- **ADMISSIBILITY (2)** [THE RESULT]: for admissible k ($k_h = 0$), $[J, E_k] == 0$, E_k is J -Hermitian, and $G J == \text{ID}16$ is positive: J quantizes every admissible momentum positively
- **ADMISSIBILITY (3)** [THE RESULT]: below threshold ($k = 5 e1 + 3 e5$, $m = 3$) the boosted $J' = S^{-1} J S$ commutes with E_k and β , $J'^2 == 1$, and $G J' == S^2$ is Hermitian positive definite
- **ADMISSIBILITY (4)** [THE RESULT]: AT threshold ($k = 4 e1 + 5 e5$, $m = 3$) $E_k \neq 0$ but $E_k^2 == 0$: nilpotent, not diagonalizable (rank 8: Jordan blocks of size 2)
- **ADMISSIBILITY (5)** [THE RESULT]: ABOVE threshold ($k = (0, 3/4, 1, 0 \mid k_5 = 2)$, $m = 1$) $\omega^2 == -23/16$: every eigenvalue of E_k is $\pm i \sqrt{23}/4 = \pm 1.19896 i$ and every eigenvector is G -NEUTRAL ($u^\dagger G u == 0$)
- **ADMISSIBILITY** [has content]: the general identity $[E_k, \beta] == 2 i T_k$, $T_k = \sum_a k_a T_{16}[a]$, for every k (admissible and hidden components alike)
- **ADMISSIBILITY (6)** [THE RESULT]: traceless certificate - every X commuting with β , the four admissible E_k and one hidden-momentum E_k has $\text{Tr}(G X) == 0$, so no positive $G J'$ exists; J itself is not in that commutant

- ADMISSIBILITY (6) [THE RESULT]: the analytic proof for a purely hidden momentum - $[E_k, \beta] = 2 i k^5 T_{16}[5]$; $T_{16}[5]$ anticommutes with G and $T_{16}[5]^\dagger G T_{16}[5] = -G$ (so a J' commuting with $T_{16}[5]$ has $T_{16}[5]^\dagger (G J') T_{16}[5] = -(G J')$)
- ADMISSIBILITY (7) [THE RESULT]: J is UNIQUE - the joint commutant of β and the admissible E_k is 8-dimensional, contains J , and every element of it commutes with J (so $X^2 = 1$ with $J X > 0$ forces $X = J$)
- ADMISSIBILITY [has content]: $[P_4, P_h] = 0$ - the metric, the frame and the spin connection do not depend on x_5, x_6, x_7

The theorem statement combines items (1)–(7); the items are asserted individually, the “if and only if” is their combination [prose VII §28].

6.11 The Hamiltonian on the canonical frame, and the spin connection

The Hamiltonian **density** of the symmetrized Lagrangian, $\Pi_\Psi \partial_4 \Psi + \partial_4 \bar{\Psi} \Pi_\Psi - L$, is

$$\sqrt{g} \left[-\frac{1}{2H} \sum_{k \neq 4} (\bar{\Psi} \gamma^k \partial_k \Psi - \partial_k \bar{\Psi} \gamma^k \Psi) + V(s) \right]$$

with **no spin connection** in it, for fields of all eight coordinates. (This corrects design revision 1, which said that the connection “enters the Hamiltonian”.) Varying it gives the one-particle operator. In the admissible sector ($\Psi = \Psi(x_0, \dots, x_4)$) the field equation is $i \tilde{G} \partial_4 \Psi = h \Psi$ with

$$h \Psi = \sqrt{g} \sigma_{16} \left[V'(s) \Psi - \frac{1}{H} \left(\sum_{j=0}^3 \gamma^j \partial_j \Psi + \sum_{\mu} \gamma^\mu \Gamma_\mu \Psi \right) \right]$$

($V'(s)$ is the mean-field $V'(\langle s \rangle)$ here). The connection appears here, through the equation of motion, and it has a job: since $\frac{1}{2} \sum_{\mu \neq 4} \partial_\mu (\sqrt{g} \sigma_{16} \gamma^\mu) = \sqrt{g} \sigma_{16} \gamma^\mu \Gamma_\mu$ and $\partial_4 (\sqrt{g} \sigma_{16} \gamma^4) = 0$, the derivative part of h is $-\frac{1}{H} \sum_j (A_j \partial_j + \frac{1}{2} \partial_j A_j)$ with $A_j = \sqrt{g} \sigma_{16} \gamma^j$ real **antisymmetric**, and for such an operator $\langle f, h g \rangle - \langle h f, g \rangle$ is a total derivative. The connection term is exactly the term that makes the one-particle Hamiltonian Hermitian. [proved VII §28]

- HAMILTONIAN [THE RESULT]: the Hamiltonian density $\Pi_\Psi \partial_4 \Psi + \partial_4 \bar{\Psi} \Pi_\Psi - L$ of the symmetrized Lagrangian is $\text{Sqrt}[g] [-(1/(2H)) \sum_{k \neq 4} (\bar{\Psi} \gamma^k \partial_k \Psi - \partial_k \bar{\Psi} \gamma^k \Psi) + V]$: NO spin connection (fields of all eight coordinates)
- HAMILTONIAN [THE RESULT]: $(\text{Sqrt}[g]/H) \sigma_{16} (\gamma^\mu D_\mu \Psi - H V' \Psi) = i \tilde{G} \partial_4 \Psi - h \Psi$ for admissible $\Psi(x_0..x_4)$: the field equation is $i \tilde{G} \partial_4 \Psi = h \Psi$
- HAMILTONIAN [THE RESULT]: $(1/2) \sum_{\mu \neq 4} d_\mu (\text{Sqrt}[g] \sigma_{16} \gamma^\mu) = \text{Sqrt}[g] \sigma_{16} \sum \gamma^\mu \Gamma_\mu$, and $d_4 (\text{Sqrt}[g] \sigma_{16} \gamma^4) = 0$: the connection term is the one that makes h Hermitian
- HAMILTONIAN [has content]: $A_0 = \text{Sqrt}[g] \sigma_{16} \gamma^0$ is real antisymmetric, and for $h_0 = -(A_0 d_0 + (1/2) A_0')$: $\langle f, h_0 g \rangle - \langle h_0 f, g \rangle = -d_0 (f^\dagger A_0 g)$, a pure boundary term

J commutes with h : every matrix in h ($\sigma_{16}, \sigma_{16} \gamma^j$ for $j = 0..3, \sigma_{16} \gamma^\mu \Gamma_\mu$) commutes with J , for every a_4 , every x_0 -profile and every V . A hidden derivative term would anticommute (control). The x_4 -dependence of h enters only through the observed frame factor $1/q$; it mixes positive and negative frequencies (a **Bogoliubov** transformation, particle production by the x_4 -dependence of a_4) **within** the admissible sector, and since it commutes with J that mixing is unitary on the Fock space. [proved VII §28]

- HAMILTONIAN [THE RESULT]: $[J, h] == 0$ on the admissible sector, for every a_4 , every x_0 -profile and every V' - including the spin-connection term
- HAMILTONIAN [control]: a hidden derivative term breaks it - $[J, h] != 0$ once Ψ depends on x_5 (witnessed on $\Psi = x_5 e_1$)
- HAMILTONIAN [has content]: the x_4 -dependence of h enters only through $1/q$: $d_4 h \Psi == (H a_4' [H x_4]) \times$ (the observed-direction part of h), which commutes with J

6.12 The x_0 direction and the boundary condition at the wall

For the rescaled field $\Psi' = \Psi/\sqrt{\sin 6Hx_0}$ the measure becomes flat in the proper distance $z = -\ln(\cos 6Hx_0)/(6H)$ along x_0 :

$$\sqrt{g} dx_0 |\Psi|^2 = |\Psi'|^2 dz \quad (\sec \cdot \sin = \tan = dz/dx_0).$$

z runs over $[0, \infty)$ as $6Hx_0$ runs over $[0, \pi/2)$; the x_0 part of the rescaled Dirac operator is exactly $T_{16}[0] \partial_z$, and the frame factors are bounded, $\sin(6Hx_0) = \sqrt{1 - e^{-12Hz}}$. So the one-particle space is $L^2(dz)$. The operator is **regular** at the wall $z = 0$, so a boundary condition is needed there, and **limit-point** at $z \rightarrow \infty$ (by the standard theorem that a one-dimensional Dirac system with locally integrable coefficients is always limit-point at an infinite endpoint), so no condition is needed or allowed there. The condition at the wall, in this real-Clifford convention, is the bag-type

$$T_{16}[0] \Psi' = +\Psi' \quad \text{or} \quad T_{16}[0] \Psi' = -\Psi' \quad \text{at } z = 0 \quad (\text{no factor } i, \text{ because } T_{16}[0]^2 = +1).$$

It kills the normal flux $\Psi'^\dagger T_{16}[0] \Psi'$ (because $\{\beta, T_{16}[0]\} = 0$) and commutes with J . It is part of the canonical quantization. [**proved VII §28**] It also gives $\bar{\Psi}\Psi = 0$ at the wall [derived here: $\sigma_{16} = J\beta$ anticommutes with $T_{16}[0]$, because J commutes and β anticommutes with it, so $P\sigma_{16}P = 0$ on either eigenspace P of $T_{16}[0]$; design review QK-9].

- BOUNDARY [has content]: the x_0 direction in proper distance $z = -\text{Log}[\text{Cos}[6Hx_0]]/(6H)$: $dz/dx_0 == \text{Tan}[6Hx_0] == \text{Sec Sin}$ (so $\text{Sqrt}[g] dx_0 |\Psi|^2 = |\Psi'|^2 dz$), $z(0) == 0$ and $z \rightarrow \text{infinity}$ at $6Hx_0 \rightarrow \pi/2$, $\gamma^0 d_0 == T_{16}[0] d_z$, and $\text{Sin}[6Hx_0] == \text{Sqrt}[1 - \text{Exp}[-12Hz]]$ (bounded frame factors)
- BOUNDARY [THE RESULT]: the bag condition $T_{16}[0] \Psi' = \pm \Psi'$ at the wall - $T_{16}[0]^2 == 1$, $\{\beta, T_{16}[0]\} == 0$, the normal-flux form vanishes on both eigenspaces ($P\beta T_{16}[0]P == 0$), and $[J, T_{16}[0]] == 0$

(The full family of self-adjoint, Lorentz-covariant, J -sector-preserving wall conditions is $P = T_{16}[0]Q$, with Q any Hermitian involution in the 8-dimensional commutant of $T_{16}[0..4]$, which contains J ; the uniform choices $Q = \pm 1$ are the two above. That refinement belongs to the treatment of the states along the hidden coordinate in the coupled system; it was adopted by the design review, DFT-8, and is not asserted in Part VII.)

6.13 Krein modes, and why the mode basis must consist of J -eigenvectors

For an admissible momentum $k = (k_0, k_1, k_2, k_3)$ and mass m (flat local analysis; on the canonical frame k is the physical momentum, the coordinate momentum along $x_1..x_3$ divided by q), put $A = Gh = E_k$ and $w = \sqrt{k \cdot k + m^2}$. Then $A^2 = w^2$, $[A, J] = 0$, A is Hermitian, and the four Hermitian projectors

$$P_{s,\eta} = \frac{1}{4} (\text{ID}_{16} + sA/w)(\text{ID}_{16} + \eta J), \quad s, \eta = +1, -1,$$

have rank 4 each (trace 4), are mutually orthogonal, sum to ID_{16} , and $\sum \eta P_{s,\eta} = J = G^{-1}$: the **completeness relation** $\sum_u \eta_u u u^\dagger = G^{-1}$ of any J -orthonormal mode basis. So each frequency

$\pm w$ is 8-fold degenerate, and the G -form restricted to each frequency eigenspace has signature $(4, 4)$. For **every** mode u in the image of $P_{s,\eta}$, normalized by $u^\dagger G u = \eta$, the projector identities $PXP = cP$ give, exactly and for arbitrary k and m ,

$$\begin{aligned}\eta u^\dagger h u &= s w = \omega_u & (\text{energy} = \text{frequency}), \\ \eta u^\dagger \sigma_{16} u &= m/\omega_u & (\text{scalar density}), \\ \eta u^\dagger (\partial h / \partial k_j) u &= k_j / \omega_u = \partial \omega_u / \partial k_j & (\text{velocity: Hellmann–Feynman}),\end{aligned}$$

with $\partial h / \partial k_j = -i \sigma_{16} T_{16}[j]$. The identities are proved with w a symbol and w^2 reduced to $k \cdot k + m^2$ (exact polynomial remainder); nothing is numerical. **[proved VII §28]**

- KREIN MODES [has content]: $A = G h$ satisfies $A^2 == w^2 \text{ ID16}$ and $[A, J] == 0$, and A is Hermitian (G and h commute) for every admissible k and m
- KREIN MODES [THE RESULT]: the four $P_{\{s,\eta\}}$ are projectors of trace (rank) 4, mutually orthogonal, summing to ID16 , and $\text{Sum } \eta P_{\{s,\eta\}} == J == G^{-1}$ (completeness)
- KREIN MODES [THE RESULT]: for every mode, $\eta u^\dagger h u == s w$ (energy = frequency), $\eta u^\dagger \sigma_{16} u == m/\omega_u$, $\eta u^\dagger (\partial h / \partial k_j) u == k_j / \omega_u$: $PXP == cP$ with $c = \eta s w$, $\eta s m/w$, $\eta s k_j/w$
- KREIN MODES [has content]: $\partial h / \partial k_j == -i \sigma_{16} T_{16}[j]$, and $d \omega_u / d k_j == k_j / \omega_u$ for $\omega_u = \pm w$
- KREIN MODES [THE RESULT]: every mode of $P_{\{s,\eta\}}$ is a J -eigenvector: $J P_{\{s,\eta\}} == \eta P_{\{s,\eta\}}$

The mode basis must consist of J -eigenvectors. The Krein prescription $b_u^\dagger := \eta_u b_u^\dagger$ depends on the basis. It reproduces the J -involution $\Psi^\dagger = \Psi^\dagger J$ only if every mode is a J -eigenvector, $Ju = \eta_u u$, which is what the $P_{s,\eta}$ deliver ($JP_{s,\eta} = \eta P_{s,\eta}$), and which is possible because $[J, Gh_k] = 0$ for momenta along $x_0..x_3$. A G -orthonormal basis mixed by a Krein boost inside one frequency eigenspace still diagonalizes the evolution, but it induces the involution $J_U = G U U^\dagger \neq J$, and under it s is **not** Hermitian (control, exact, at $k = 3e_1$, $m = 4$, with the boost $\cosh = 5/4$, $\sinh = 3/4$). A generic generalized-eigenvector routine does not choose J -eigenvectors; the Fock construction of section 6.14 does. **[proved VII §28]** KREIN MODES [control]: a Krein-boosted pair $u_1 = (5/4)u_+ + (3/4)u_-$, $u_2 = (3/4)u_+ + (5/4)u_-$ in the $\omega_u = +5$ eigenspace is still G -orthonormal and still consists of modes, but not of J -eigenvectors; the induced $J_U = G U U^\dagger$ differs from J , and under it s is NOT Hermitian

6.14 A finite Fock space at one momentum (Jordan–Wigner): the Dirac sea, 8 particles and 8 antiparticles

At the momentum $k = 3e_1$ with $m = 4$ (so $\omega = 5$ exactly) the sixteen modes of one momentum are represented on $\mathbb{C}^{2^{16}} = \mathbb{C}^{65536}$ by Jordan–Wigner operators $f_1..f_{16}$ (**SparseArray**, exact integers), which obey the positive CAR $\{f_i, f_j^\dagger\} = \delta_{ij}$, $\{f_i, f_j\} = 0$. Two realizations are checked.

Field level. $\Psi_a := f_a$ (the factor $H/\sqrt{g}$ set to 1), $\Psi^\dagger = f^\dagger$, $\Psi^\dagger := \Psi^\dagger J$. Then:

- $\{\Psi_a, \Psi_b^\dagger\} = J_{ab} = (G^{-1})_{ab}$: exactly the Dirac-bracket anticommutator, realized on a positive Fock space;
- the free Hamiltonian $H_{\text{free}} = \Psi^\dagger h \Psi = \Psi^\dagger (Jh) \Psi$ and $s = \Psi^\dagger \sigma_{16} \Psi = \Psi^\dagger \beta \Psi$ are Hermitian operators, and so is the interacting $H_{\text{free}} + \frac{3}{7} s^2$; $p = \Psi^\dagger C_+ \Psi$ is anti-Hermitian (control); the one-body matrix Jh has eigenvalues $+5$ and -5 , eight times each;

- the **Heisenberg equation** holds: $[H_{\text{free}}, \Psi_a] = -(G^{-1}h\Psi)_a$ for all 16 components, i.e. $i\partial_4\Psi = i[H, \Psi]$ reproduces $iG\partial_4\Psi = h\Psi$;
- with the **opposite** sign of the anticommutator ($\Psi^\dagger := -\Psi^\dagger J$) the same construction gives $H' = -H_{\text{free}}$ and the time-**reversed** equation (control). This is the convention-free confirmation of the sign derived with the Dirac bracket.

Mode level. $b_u := f_u$ for the sixteen J -eigenmodes of section 6.13 ($u = 1..8$: $\omega = +5$; $u = 9..16$: $\omega = -5$; $Ju = \eta_u u$ with $\eta = +1$ for four of each and -1 for the other four):

- the Krein conjugate $b_u^\dagger := \eta_u b_u^\dagger$ obeys $\{b_u, b_v^\dagger\} = \eta_u \delta_{uv}$, the indefinite CAR that the field anticommutator demands through $\sum_u \eta_u u u^\dagger = G^{-1}$, and the J -involution $b^\dagger = \eta b^\dagger$ restores $\{b, b^\dagger\} = 1$;
- the naive Krein “norm” $\langle 0|b_u b_u^\dagger|0\rangle = \eta_u$ is **negative** for $\eta = -1$: that is the indefinite metric the J -involution removes (the Fock norm $\langle 0|b_u b_u^\dagger|0\rangle = 1$);
- $\text{Ham} = \sum_u \omega_u \eta_u b_u^\dagger b_u = \sum_u \omega_u b_u^\dagger b_u$, and $[\text{Ham}, b_u] = -\omega_u b_u$ ($b_u(x_4) = b_u e^{-i\omega_u x_4}$, the time dependence of the classical mode);
- the spectrum on the 65536 states: a **unique ground state** at $-8\omega = -40$, the **Dirac sea** (all eight negative-frequency modes filled); the normal-ordered $:\text{Ham}: = \text{Ham} + 40 \geq 0$;
- the first excited level, $\omega = 5$ above the sea (energy -35), is **16-fold: 8 particle states** (one positive-frequency mode added, $N = +1$) and **8 antiparticle states** (one negative-frequency mode emptied, a hole, $N = -1$), with $N = \sum_u b_u^\dagger b_u - 8$;
- in every Fock basis state tested (the sea and the 16 first excited states) the expectation of $b_u^\dagger b_v$ is $\eta_u n_u \delta_{uv}$: the rule $\langle \Psi^\dagger M \Psi \rangle = \sum_{\text{occupied}} \eta_u u^\dagger M u$ used in section 8.

[proved VII §28]

- FOCK [solver regression]: the Jordan-Wigner operators obey the positive CAR $\{f_i, f_j^\dagger\} == \delta_{ij}$, $\{f_i, f_j\} == 0$ on C^{65536}
- FOCK (field level) [THE RESULT]: $\{\Psi_a, \Psi^\dagger b\} == J_{ab} == (G^{-1})_{ab}$ with $\Psi = f$, $\Psi^\dagger = f^\dagger J$ - the Dirac-bracket anticommutator, realized on a positive Fock space
- FOCK (field level) [THE RESULT]: $J h$ is Hermitian with eigenvalues $+5$ (8) and -5 (8); H_{free} and s are Hermitian operators, so is $H_{\text{free}} + (3/7) s^2$; $p = \Psi^\dagger C \Psi$ is ANTI-Hermitian
- FOCK (field level) [THE RESULT]: Heisenberg - $[H_{\text{free}}, \Psi_a] == -(G^{-1} h \Psi)_a$ for all 16 components, so $i d_4 \Psi = i [H, \Psi]$ IS the field equation $i G d_4 \Psi = h \Psi$
- FOCK (field level) [control]: with the opposite sign of the anticommutator the Hamiltonian is $-H_{\text{free}}$ and $[H', \Psi_a] == +(G^{-1} h \Psi)_a$: the time-REVERSED equation
- FOCK (mode level) [THE RESULT]: $\{b_u, b_v^\dagger\} == \eta_u \delta_{uv}$ (the indefinite CAR demanded by $\sum \eta_u u u^\dagger = G^{-1}$), and the J -involution $b^\dagger = \eta b^\dagger$ restores $\{b, b^\dagger\} == 1$
- FOCK (mode level) [has content]: the naive Krein 'norm' $\langle 0|b_u b_u^\dagger|0\rangle$ equals η_u - NEGATIVE for the four $\eta = -1$ modes of each frequency - while the Fock norm $\langle 0|b_u b_u^\dagger|0\rangle == 1$

- FOCK (mode level) [THE RESULT]: $\text{Sum } \omega_u \eta_u b_u^{\dagger} b_u == \text{Sum } \omega_u b_u^{\dagger} b_u$, and $[\text{Ham}, b_u] == -\omega_u b_u$ for every u ($b_u(x_4) = b_u e^{-i \omega_u x_4}$)
- FOCK (mode level) [THE RESULT]: the spectrum - a UNIQUE ground state at -8 $\omega = -40$, the Dirac sea (modes 9..16 filled, 1..8 empty); $:\text{Ham}: = \text{Ham} + 40 \geq 0$
- FOCK (mode level) [THE RESULT]: the first excited level, $\omega = 5$ above the sea, is 16-fold: 8 particle states ($N = +1$) and 8 antiparticle states ($N = -1$), with $N = \text{Sum } b^{\dagger} b - 8$
- FOCK (mode level) [THE RESULT]: in every Fock basis state tested (the sea and the 16 first excited states) $\langle n | b_u^{\dagger} b_v | n \rangle == \eta_u n_u \delta_{uv}$ - the expectation rule of Section 29

The Jordan–Wigner CAR check on the 65536-dimensional space took 0.749 s and the Heisenberg check 1.868 s; the whole Fock cell (Input cell 191) took 9.858 s (claude-fable/run_fermion_fable_part7.log, lines [0.749 s], [1.868 s], CELL 191 t=9.858 s).

6.15 The particle number

The particle-number density is

$$n^4 = -\frac{i}{H} \bar{\Psi} \gamma^4 \Psi = \frac{1}{H} \Psi^{\dagger} G \Psi = \frac{1}{H} \Psi^{\dagger} \Psi \geq 0$$

and its mode form $\sum_u \eta_u b_u^{\dagger} b_u = \sum_u b_u^{\dagger} b_u$ is the particle number (normal-ordered: particles minus antiparticles). [proved VII §28] CHARGE [THE RESULT]: $n^4 = -(i/H) \bar{\Psi} \gamma^4 \Psi == (1/H) \Psi^{\dagger} G \Psi == (1/H) \Psi^{\dagger} \Psi$, whose mode form $\text{Sum}_u \eta_u b_u^{\dagger} b_u == \text{Sum}_u b_u^{\dagger} b_u$ is the particle number (normal-ordered: particles - antiparticles)

6.16 The dynamical-frame field χ

On a frame whose volume factor depends on x_4 the field to quantize is $\chi = (\sqrt{g}/H)^{1/2} \Psi$, with $\{\chi, \chi^{\dagger}\} = G \delta^7$ (section 6.4). On the canonical frame $\sqrt{g} = \sec 6Hx_0$ does not depend on x_4 , and quantizing χ changes nothing (ANTICOMMUTATOR [has content]: the rescaled field $\Psi' = \Psi/\text{Sqrt}[\text{Sin}[6Hx_0]]$ has $\{\Psi', \Psi'^{\dagger}\} == H \text{Cot}[6Hx_0] J \delta^7$, and $\text{Sqrt}[g]$ is x_4 -independent (so quantizing $\chi = (\text{Sqrt}[g]/H)^{(1/2)} \Psi$ changes nothing on the canonical frame)). On the time-dependent warped frames of the interacting system (Part VIII), the rescaled $\chi = \Psi/(\sqrt{\sin 6Hx_0} (A^3 B^3 C)^{-1/2})$ has $\{\chi, \chi^{\dagger}\} = H \cot(6Hx_0) G$, independent of x_4 and of the scale factors (RESCALING [THE RESULT]: $\{\Psi, \Psi^{\dagger}\} == (H/\text{Sqrt}[|\det g|]) G$ on the warped frame (Section 28's cfAntiPsiPsiDd at $c = \text{Sqrt}[|\det g|]/H$), and for $\chi = \Psi/f$, $f = \text{Sqrt}[\text{Sin}] (A^3 B^3 C)^{-1/2}$: $\{\chi, \chi^{\dagger}\} == H \text{Cot}[6Hx_0] G$ - the $\text{Sqrt}[\det g]$ cancels, independent of x_4 and of A, B, C).

7 The energy–momentum tensor operator

This is the first half of Section 29 of the notebook (Input cells 192–194, 16 assertions).

7.1 The definition, and where it comes from

The energy-momentum tensor of fermion field is

$$\hat{T}_{\mu\nu} = -\frac{1}{4H} \left[\bar{\Psi} \gamma_\mu D_\nu \Psi - (D_\nu \bar{\Psi}) \gamma_\mu \Psi + (\mu \leftrightarrow \nu) \right] + g_{\mu\nu} \hat{L}, \quad \gamma_\mu = g_{\mu\nu} \gamma^\nu$$

normal-ordered with respect to the J -vacuum of section 6 when it is an operator. It is the **Hilbert** (symmetric-tetrad) tensor of the symmetrized action, $T^{\mu\nu} = \frac{2}{\sqrt{g}} \frac{\delta S}{\delta g_{\mu\nu}}$, and it equals the covariant expression above **off shell**. The notebook checks this the direct way: it perturbs the canonical frame by a symmetric first-order tetrad perturbation $h(x_0, x_4)$ in the (μ, ν) slot, recomputes to first order everything that depends on the frame (the metric, the Christoffel symbols, the spin connection from the vielbein postulate, the curved gammas, $\sqrt{g}$), and takes the Euler-Lagrange derivative with respect to h . Three facts come out:

1. The result is the tensor above, checked on the diagonal (x_1, x_1) , (x_4, x_4) and off the diagonal (x_1, x_4) , (x_0, x_5) , (x_5, x_4) .
2. The variation of the spin connection contributes **exactly nothing**. Its totally antisymmetric part does not change under a symmetric tetrad variation, and the symmetrized Lagrangian sees the connection only through that part, $\frac{1}{2} \omega_{[cab]} \gamma^{cab}$. The covariant-derivative terms of the tensor come instead from the frame dependence of $\{\gamma^\mu, \Gamma_\mu\}$, which vanishes on the background while its first variation does not.
3. The Hilbert tensor is **not** the partial-derivative tensor T_{par} (control, at (x_1, x_4)): one never varies a partial-derivative Lagrangian with respect to the frame.

[proved VII §29]

- EMT [THE RESULT]: the Hilbert (symmetric-tetrad) tensor of the symmetrized action IS That, OFF shell - components (x1,x1), (x4,x4), (x1,x4), (x0,x5), (x5,x4) (the five components took 0.693s)
- EMT [has content]: the spin-connection variation contributes NOTHING - dropping it leaves the Hilbert tensor unchanged at (x1,x4), (x0,x5), (x5,x4)
- EMT [control]: the Hilbert tensor is NOT the partial-derivative tensor: at (x1,x4) they differ - witnessed on constant spinors (Psibar = e1, Psi = e11: (T16[4] T16[1] T16[0])_{1,11} != 0)

7.2 Symmetric and Hermitian

$\hat{T}$ is symmetric, and it is Hermitian under the classical conjugation: real for $\Psi = u + iv$, $\bar{\Psi} = (u - iv)^T \sigma_{16}$, in all 64 components. [proved VII §29]

- EMT [definition]: That is symmetric
- EMT [THE RESULT]: That is Hermitian under the classical conjugation: real for Psi = u + i v, Psibar = (u - i v)^T sigma16, all 64 components

7.3 Conservation on shell, and why the commuting proxy proves it for the fermion

Solve the two field equations for the x_4 -derivatives:

$$\begin{aligned} \partial_4 \Psi &= F = -\gamma^4 (H V'(s) \Psi - \gamma^0 \partial_0 \Psi - (\gamma^\mu \Gamma_\mu) \Psi), \\ \partial_4 \bar{\Psi} &= \bar{F} = (H V'(s) \bar{\Psi} + \partial_0 \bar{\Psi} \gamma^0 - \bar{\Psi} (\Gamma_\mu \gamma^\mu)) \gamma^4, \end{aligned}$$

and let `cfOnShellC` replace every x_4 -derivative of either field (of any order) by the corresponding derivative of F or $\bar{F}$, to a fixed point. Then $\nabla^\mu \hat{T}_{\mu\nu} = 0$ in all eight components, for generic $\Psi(x_0, x_4)$, $\bar{\Psi}(x_0, x_4)$ and generic V .

Why a commuting computation proves the Grassmann statement. After the substitution, every term of the divergence is a bilinear with $\bar{\Psi}$ (or a derivative of it) on the left and Ψ (or a derivative) on the right, multiplied by functions of the **even** composites s and ∂s (through V' , V'' and their derivatives). No two odd factors are ever exchanged, so the same algebra holds verbatim for anticommuting components. (The argument is not valid for arbitrary quartic identities; it is valid here because of that structure.) A configuration that is not a solution has a non-zero divergence (control).

The connection does not drop out of the tensor. $\hat{T}$ differs from the partial-derivative tensor by

$$\hat{T}_{\mu\nu} - T_{\mu\nu}^{\text{par}} = -\frac{1}{4H} \bar{\Psi}(\{\gamma_\mu, \Gamma_\nu\} + \{\gamma_\nu, \Gamma_\mu\})\Psi,$$

which vanishes on the diagonal (so ρ and every pressure are the same in both) but not off it, and only $\hat{T}$ is conserved. (The symbolic divergence of T_{par} is not attempted: it takes minutes, and the off-diagonal difference already shows the two tensors are different.) [**proved VII §29**]

- EMT CONSERVATION [definition]: `F` and `Fbar` solve the field equation and the adjoint equation for the `x4`-derivatives
- EMT CONSERVATION [THE RESULT]: `nabla^mu That_{mu nu} == 0 ON SHELL` in all eight components, for generic `Psi(x0, x4)`, `Psibar(x0, x4)` and generic `V` (the divergence took 0.767s and its on-shell reduction 2.599s)
- EMT CONSERVATION [control]: OFF shell the divergence is not zero - witnessed on `Psi = (x4 + x0^2) e1 + e13`, `Psibar = e1 + x4 e5` with `V = s^2`, which is not a solution
- EMT [THE RESULT]: `That - T_par == -(1/(4H)) Psibar ({gamma_mu, Gamma_nu} + {gamma_nu, Gamma_mu}) Psi`: equal on the diagonal (every `{gamma_mu, Gamma_mu}` vanishes), different off it (the `(x1, x4)` witness above)

The statement is proved for fields of (x_0, x_4) . That is the family in which ρ , P and the pre-universe results of section 8 are evaluated.

7.4 On shell: the Lagrangian, the density, the pressures and the trace

With `cfOnShellC`, for generic $\Psi(x_0, x_4)$, $\bar{\Psi}(x_0, x_4)$ and generic V :

$$\begin{aligned} \frac{1}{2H} \left[\bar{\Psi} \gamma^\mu D_\mu \Psi - (D_\mu \bar{\Psi}) \gamma^\mu \Psi \right] &= s V'(s) \quad (\text{the kinetic bilinear; NOT zero}), \\ \hat{L} &= s V'(s) - V(s), \\ \rho = \hat{T}_{44} &= V(s) + K_h, \quad K_h = -\frac{1}{2H} (\bar{\Psi} \gamma^0 \partial_0 \Psi - \partial_0 \bar{\Psi} \gamma^0 \Psi), \\ T^i_i &= T^h_h = s V'(s) - V(s) \\ &\quad (i = 1, 2, 3 \text{ observed; } h = 5, 6, 7 \text{ hidden; no sum}), \\ T^0_0 &= s V'(s) - V(s) + K_h, \\ T^\mu_\mu &= 7 s V'(s) - 8 V(s) \quad (\text{off shell: } 7L_{\text{kin}} - 8V). \end{aligned}$$

The observer is $u = \partial/\partial x_4$, so $\rho = T_{44}$ (the array element `[[5,5]]`) and the pressure along a direction j is $P_j = T^j_j$ (no sum). $P_i = P_h$ holds because $\{\gamma^\mu, \Gamma_\mu\} = 0$ for **each** direction

separately: the kinetic parts of T^i_i and T^h_h vanish for fields of (x_0, x_4) , and both reduce to $\hat{L}$. The trace is consistent with the list: $T^4_4 = -\rho$, so $-(V+K_h)+(sV'-V+K_h)+6(sV'-V) = 7sV'-8V$ [derived here]. In the real commuting limit K_h is Part VI's cfKh (section 2.12.4), and for the rescaled solutions $\Psi = \sqrt{\sin 6Hx_0} \Psi'(x_4)$ (and the same for $\bar{\Psi}$) $K_h = 0$ and $\rho = V(s)$: Part VI's formulas, now for the complex field. [proved VII §29]

- EMT ON SHELL [THE RESULT]: the kinetic bilinear == s V'(s) (not zero) and Lhat == s V'(s) - V(s)
- EMT ON SHELL [THE RESULT]: rho = That_44 == V(s) + K_h
- EMT ON SHELL [THE RESULT]: T^i_i == T^h_h == s V'(s) - V(s) for the three observed and the three hidden directions, and T^0_0 == s V'(s) - V(s) + K_h
- EMT [has content]: the trace T^mu_mu == 7 Lkin - 8 V off shell, == 7 s V'(s) - 8 V(s) on shell
- EMT [fidelity]: K_h in the real commuting limit IS Part VI's cfKh, and K_h == 0 on the rescaled solutions Sqrt[Sin] Psi'(x4), Sqrt[Sin] Psi'bar(x4), where rho == V(s)

7.5 Which components are Hermitian operators: the J -parity table

For fields independent of x_5, x_6, x_7 (the admissible sector, here $\Psi(x_0, \dots, x_4)$, $\bar{\Psi}(x_0, \dots, x_4)$), replace $\Psi \rightarrow J\Psi$ and $\bar{\Psi} \rightarrow \bar{\Psi}J$ ($J^2 = 1$, s is unchanged). A component that goes into itself is built from J -even matrices and is a **Hermitian** operator; one that goes into minus itself is **anti-Hermitian** (section 6.8). The notebook computes the parity of all 64 components (3.561 s) and asserts the table:

component	J -parity	operator
$\hat{T}_{\mu\nu}$, μ, ν in $\{0..4\}$	even	Hermitian
$\hat{T}_{\mu h}$, μ in $\{0..4\}$, h in $\{5, 6, 7\}$	odd	anti-Hermitian: purely imaginary expectation values, zero in every state invariant under the hidden rotations ($\hat{T}_{\mu h}$ is a vector under them)
$\hat{T}_{hh'}$, $h \neq h'$	zero identically	—
$\hat{T}_{hh}$	even	Hermitian; $\hat{T}_{hh} = g_{hh}\hat{L}$ exactly, because $\{\gamma_h, \Gamma_h\} = 0$

The hidden momenta $P_h = \int \sqrt{g} \hat{T}^4_h$ are J -odd operators with zero expectation in the J -eigenmode Fock states of section 6 [prose VII §29]. The anti-Hermitian components are consistent with $G_{\mu h} = 0$ on the canonical frame. [proved VII §29]

- EMT HERMITICITY [THE RESULT]: J-parity table - (mu, nu) in {0..4}: EVEN (Hermitian); (mu in 0..4, h in 5..7): ODD (anti-Hermitian); (h, h'), h != h': ZERO identically; (h, h): even
- EMT HERMITICITY [has content]: for x5..x7-independent fields That_{hh} == g_hh Lhat exactly, because {gamma_h, Gamma_h} == 0

7.6 The real commuting limit

Set $\Psi \rightarrow \psi$ (real, commuting) and $\bar{\Psi} \rightarrow \psi^T \sigma_{16}$. Then $\hat{T}$ becomes Part VI's covariant tensor T_{cov} of section 2.12.3 exactly, all 64 components: FIDELITY [4] [fidelity]: the energy-momentum tensor That in the real limit IS Part VI's covariant tensor cftCov, all 64 components. The classical tensor of 2026-09-16 is therefore the commuting shadow of the operator derived here.

8 Density, pressure and equations of state

This is the second half of Section 29 of the notebook (Input cells 195–197, 23 assertions), with the numerically stable forms used by the solver and the vacuum-energy caveat added from the design review.

8.1 The operators

The observer is $u = \partial/\partial x_4$. The energy-density and pressure **operators** are the normal-ordered components of section 7:

$$\begin{aligned}\rho_{\text{op}} &= :\hat{T}_{44}: && \text{(the energy density measured by the observer),} \\ P_{j,\text{op}} &= :\hat{T}^j_j: \quad (\text{no sum}) \quad j = 1, 2, 3: \text{ the observed sheet; } j = 0: \text{ the hidden spacelike direction;} \\ & && h = 5, 6, 7: \text{ the hidden timelike sheet,} \\ w &= \langle P_{\text{obs}} \rangle / \langle \rho \rangle && P_{\text{obs}} = \text{the pressure along the observed sheet.}\end{aligned}$$

On shell, for fields of (x_0, x_4) : $\rho = V(s) + K_h$, $P_i = P_h = s V'(s) - V(s)$, $P_0 = s V'(s) - V(s) + K_h$ (section 7.4). These are operator identities in the sense of section 5.9 (ordering by the Weyl rule; exact for linear V).

8.2 Expectation values: the mode sums, and one particle

Use the **canonical normalization** $\chi = \Psi/\sqrt{H}$. Then $V(s) = V(H\sigma) =: W(\sigma)$ with $\sigma = \langle \bar{\chi}\chi \rangle$, and $m = W'(\sigma)$ is the mass in the mode equation. On a flat-frame plane wave $\chi = u e^{i(k \cdot x - \omega x_4)}$, $\bar{\chi} = \bar{u} e^{-i(k \cdot x - \omega x_4)}$ ($\bar{u} = u^\dagger \sigma_{16}$), the kinetic parts of the tensor are

$$\begin{aligned}T_{44} + \hat{L} &= \omega \bar{u} (-i T_{16}[4]) u = \omega u^\dagger G u, \\ T^j_j - \hat{L} &= k_j \bar{u} (-i T_{16}[j]) u = k_j u^\dagger (\partial h / \partial k_j) u \quad (j = 0..3, \text{ flat frame}).\end{aligned}$$

By the expectation rule $\langle \Psi^\dagger N \Psi \rangle = \sum_{\text{occupied}} \eta_u u^\dagger N u$ (section 6.14) and the exact mode identities of section 6.13, each occupied positive-frequency mode contributes ω to the energy density, k_j^2/ω to the pressure along j , and m/ω to σ ; each hole in the sea contributes the same with $\omega \rightarrow |\omega|$. So for a state of N modes in a coordinate volume Vol:

$$\rho = \sum \frac{\omega}{\text{Vol}} + (W - \sigma W'), \quad P_j = \sum \frac{k_j^2}{\omega \text{Vol}} + (\sigma W' - W), \quad \sigma = \sum \frac{m}{\omega \text{Vol}}.$$

For **one particle**, $w = (k^2/3)/\omega^2$ (isotropic average) on top of the background $W - \sigma W'$. [proved VII §29]

- **MODE SUMS [THE RESULT]:** on a flat-frame plane wave (canonical normalization), $T_{44} + \text{Lhat} == \omega \text{ubar} (-i T_{16}[4]) u$ and $T^j_j - \text{Lhat} == k_j \text{ubar} (-i T_{16}[j]) u$ ($j = 0..3$): energy and pressure are the frequency and k_j dh/dk_j bilinears
- **MODE SUMS [has content]:** with $\text{ubar} = u^\dagger \text{sigma16}$ these are $\omega u^\dagger G u$ and $k_j u^\dagger (dh/dk_j) u$, so by Section 28 each occupied mode contributes ω , k_j^2/ω and (to σ) m/ω

8.3 The vacuum: normal ordering, the sea energy, and the cosmological-constant problem

$\langle 0 | \hat{T}_{\mu\nu} | 0 \rangle = 0$ by normal ordering with respect to the J -vacuum. That statement is clean only for a **static** background ($a_4 = \text{const}$) or in the adiabatic limit. On an x_4 -dependent

background normal ordering is ambiguous: with respect to the instantaneous vacuum it breaks covariant conservation, and with respect to a fixed in-vacuum it leaves a divergence. A conserved renormalized $\langle T \rangle$ needs adiabatic subtraction or point-splitting [prose VII §29].

The **unrenormalized sea energy** per unit volume, $-g \int^\Lambda \frac{d^3k}{(2\pi)^3} \omega$, is minus the Fermi-sea energy at $k_F = \Lambda$:

$$-\varepsilon_{\text{KS}}(\Lambda) = -\frac{g}{16\pi^2} \left[2\Lambda^4 + 2m^2\Lambda^2 + \frac{m^4}{4} - m^4 \ln \frac{2\Lambda}{m} \right] + o(1) \quad (\Lambda \rightarrow \infty).$$

Its finite, mass-dependent part contains $-\frac{g}{32\pi^2} m^4 \ln m^2$. It is absorbed into V (the renormalized, “no-sea” functional). **The cosmological-constant problem is not solved here.** [proved VII §29] VACUUM [THE RESULT]: the unrenormalized sea energy `-eps_KS(Lambda) == -(g/(16 pi^2))(2 Lambda^4 + 2 m^2 Lambda^2 + m^4/4 - m^4 Log[2 Lambda/m]) + o(1)` as `Lambda -> infinity`; its m -dependent finite part contains `-(g/(32 pi^2)) m^4 Log[m^2]`

For a mass that varies (the coupled system), the sea energy cannot be absorbed silently (design review DFT-4, adopted). The renormalized one-loop Dirac-sea energy that the no-sea functional drops (relativistic Hartree approximation, Chin 1977), per 3-volume, with reference mass M and counterterms through m^4 , is

$$\Delta E_{\text{vac}}(m) = -\frac{g}{16\pi^2} \left[m^4 \ln \frac{m}{M} + M^3(M-m) - \frac{7}{2} M^2(M-m)^2 + \frac{13}{3} M(M-m)^3 - \frac{25}{12} (M-m)^4 \right]$$

[file: `fable-cosmology/rust/fable_fermion/src/kohn_sham.rs`, function `vacuum_energy`; its unit test `vacuum_energy_vanishes_to_fifth_order_at_the_reference_mass` checks that it vanishes like $(m-M)^5$ at $m=M$ and is even in m]. Since 2bc936d the solver evaluates it as $\Delta E_{\text{vac}} = -\frac{g}{16\pi^2} M^4 F(u)$ with $u = (m-M)/M$, summing the power series $F(u) = u^5/5 - u^6/30 + u^7/105 - \dots$ for $|u| \leq 0.75$ and the closed form (with `log1p`) otherwise, because the closed form above cancels to rounding noise near $m=M$; the unit test `vacuum_energy_matches_high_precision_references_near_and_far_from_the_reference_mass` compares it with 120-digit references, and the commit records agreement to 1.8×10^{-15} [file: the same function, `vac_bracket` and their tests; commit 2bc936d]. It is neither zero nor absorbable into W without changing the gap equation; the solver reports it (column `vac_over_rho_c`) and does not include it in the dynamics. Reading W as the fully renormalized effective potential absorbs it formally, but that is a fine-tuning. Its size [derived here]: at $m=0$ the bracket is $M^4(1 - \frac{7}{2} + \frac{13}{3} - \frac{25}{12}) = -M^4/4$, so $\Delta E_{\text{vac}}(0) - \Delta E_{\text{vac}}(M) = gM^4/(64\pi^2)$. With $g=8$ and the unit $E_c = \rho_{c0}^{1/4} = 2.46261 \times 10^{-3}$ eV (asserted in Part VIII: `fable4d RUNS [definition]: the units computed from CODATA 2018 - E_c = rho_c0^(1/4) == 2.46261e-3 eV, Omega_r0 == 9.2096e-5, Omega_b0 = 0.02237/h^2 == 0.049243 (to the quoted digits)`), a mass variation of order m changes the vacuum energy by $\frac{gm^4}{64\pi^2}/\rho_{c0} = 3.44, 3.44 \times 10^4, 3.44 \times 10^8, 3.44 \times 10^{12}$ for $m = 0.01, 0.1, 1, 10$ eV (it scales as m^4). So for $m \gtrsim 10$ meV it exceeds the critical density. (The design review quoted 3.44×10^{-4} and 3.44 at 0.01 and 0.1 eV; those two figures are off by 10^4 and are corrected here.)

8.4 The homogeneous Fermi sea (the Kohn–Sham ground state) in closed form

Fill the positive-frequency modes with $|k| < k_F$ along the observed sheet (zero modes along x_0 — the $\sqrt{\sin}$ profile — and along x_5, x_6, x_7). The degeneracy is $\mathbf{g} = \mathbf{8} = 4$ flavours $\times 2$ spins (sections 3.6 and 6: eight positive-frequency modes per momentum). With $\omega_F = \sqrt{k_F^2 + m^2}$ and $\mathcal{L} = \ln((k_F + \omega_F)/m)$ ($m > 0$; everything depends on m^2 except σ , which has the sign of m):

$$n = \frac{g k_F^3}{6\pi^2} = g \int \frac{d^3k}{(2\pi)^3} 1,$$

$$\begin{aligned}
\sigma_{KS} &= \frac{g m}{4\pi^2} [k_F \omega_F - m^2 \mathcal{L}] &= g \int \frac{d^3 k}{(2\pi)^3} \frac{m}{\omega}, \\
\varepsilon_{KS} &= \frac{g}{16\pi^2} [k_F \omega_F (2k_F^2 + m^2) - m^4 \mathcal{L}] &= g \int \frac{d^3 k}{(2\pi)^3} \omega, \\
P_{KS} &= \frac{g}{48\pi^2} [k_F \omega_F (2k_F^2 - 3m^2) + 3m^4 \mathcal{L}] &= g \int \frac{d^3 k}{(2\pi)^3} \frac{k^2}{3\omega}.
\end{aligned}$$

Each closed form is proved equal to its integral: its k_F -derivative is the integrand times $gk_F^2/(2\pi^2)$ and it vanishes at $k_F = 0$; independently, Mathematica's **Integrate** returns the same function (with $\text{artanh}(k_F/\omega_F)$, which equals $\mathcal{L}$); and a numerical quadrature at $k_F = 3/2$, $m = 1$, $g = 8$ agrees to 10^{-12} (a cross-check; the proof is the symbolic one). **[proved VII §29]**

- KOHN-SHAM [definition]: the degeneracy $g = 8$ - per momentum the positive-frequency eigenspace has rank 8 (Section 28's $P_{\{+,+\}} + P_{\{+,-\}}$), four flavours times two spins
- KOHN-SHAM [THE RESULT]: each closed form has k_F -derivative $(g/(2\pi^2)) k_F^2 f(k_F)$ with $f = 1, m/\omega, \omega, k^2/(3\omega)$, and vanishes at $k_F = 0$ - so it IS the integral $g \int d^3 k / (2\pi)^3 f$
- KOHN-SHAM [has content]: **Integrate** returns the same functions, with $\text{ArcTanh}[k_F/w_F]$ in place of $\mathcal{L}$ - and $\text{ArcTanh}[k_F/w_F] == \mathcal{L}$ (equal derivatives, both 0 at $k_F = 0$)
- KOHN-SHAM [control]: numerical quadrature agrees at $k_F = 3/2$, $m = 1$, $g = 8$ to $1e-12$ (a cross-check; the proof is the symbolic one above)

The exported closed forms at $g = 8$ ($k_F = k_F$; provenance-latex/generated/fermion_fable_eos.tex, unchanged):

$$\begin{aligned}
n &= \frac{4k_F^3}{3\pi^2} \\
\sigma_{KS} &= \frac{2m \left(k_F \sqrt{k_F^2 + m^2} - m^2 \log \left(\frac{\sqrt{k_F^2 + m^2} + k_F}{m} \right) \right)}{\pi^2} \\
\varepsilon_{KS} &= \frac{k_F \sqrt{k_F^2 + m^2} (2k_F^2 + m^2) - m^4 \log \left(\frac{\sqrt{k_F^2 + m^2} + k_F}{m} \right)}{2\pi^2} \\
P_{KS} &= \frac{k_F \sqrt{k_F^2 + m^2} (2k_F^2 - 3m^2) + 3m^4 \log \left(\frac{\sqrt{k_F^2 + m^2} + k_F}{m} \right)}{6\pi^2} \\
\rho &= \varepsilon_{KS} + W - \sigma W', \quad P_{obs} = P_{KS} + \sigma W' - W, \quad P_{hid} = \sigma W' - W, \quad \rho + P_{obs} = w_F n, \quad w_F = \sqrt{k_F^2 + m^2} \\
w &= P_{KS}/\varepsilon_{KS} \in [0, 1/3] \text{ for the mass term } W = m\sigma \quad (w \rightarrow k_F^2/(5m^2) \text{ as } k_F/m \rightarrow 0, \quad w \rightarrow 1/3 \text{ as } k_F/m \rightarrow \infty)
\end{aligned}$$

The identities that organize them:

$$\begin{aligned}
\varepsilon_{KS} + P_{KS} &= \omega_F n && \text{(the zero-temperature Gibbs relation,} \\
&&& \text{chemical potential } \mu = \omega_F), \\
\varepsilon_{KS} - 3P_{KS} &= m \sigma_{KS} && \text{(the trace identity),} \\
\partial \varepsilon_{KS} / \partial m &= \sigma_{KS}, \quad \partial \varepsilon_{KS} / \partial k_F = \omega_F dn / dk_F.
\end{aligned}$$

[proved VII §29] KOHN-SHAM [THE RESULT]: $\text{eps} + P == w_F n$ (Gibbs), $\text{eps} - 3 P == m \sigma$ (trace), $d \text{eps} / dm == \sigma$, $d \text{eps} / dk_F == w_F dn / dk_F$

8.5 The numerically stable forms

The closed forms cancel catastrophically for $x = k_F/|m| \ll 1$: $\varepsilon \sim mn$, $\sigma \sim n$ and $P \sim nk_F^2/(5m)$ are small differences of large terms. The solver therefore uses three regimes, all written with the integral representations

$$S(x) = 2 \int_0^x \frac{t^2}{\sqrt{1+t^2}} dt, \quad E(x) = 8 \int_0^x t^2 \sqrt{1+t^2} dt, \quad Q(x) = 8 \int_0^x \frac{t^4}{\sqrt{1+t^2}} dt,$$

with $\sigma = g m^3 S/(4\pi^2)$, $\varepsilon = g m^4 E/(16\pi^2)$, $P = g m^4 Q/(48\pi^2)$ [file: `fable-cosmology/rust/fable_fermion/src/kohn_sham.rs`, module documentation and constants `X_LO = 0.6`, `X_HI = 100.0`]:

regime	form used
$x < X_{LO} = 0.6$	the convergent binomial series (radius 1) in x^2 , e.g. $\varepsilon = m n [1 + \frac{3}{10}x^2 - \frac{3}{56}x^4 + \dots]$, $\sigma = \text{sgn}(m) n [1 - \frac{3}{10}x^2 + \dots]$
$0.6 \leq x \leq X_{HI} = 100$	the closed forms (with $\text{asinh } x = \mathcal{L}$), losing at most about 1.3 digits at $x = 0.6$
$x > 100$ $m = 0$ exactly	the ultra-relativistic series in $y = m /k_F$, with the logarithm kept exact $\sigma = 0$, $\varepsilon = 3P = g k_F^4/(8\pi^2)$

The series coefficients are $\binom{-1/2}{j}$ for S and Q and $\binom{1/2}{j}$ for E :

$$S = \sum_j 2 \binom{-1/2}{j} \frac{x^{2j+3}}{2j+3}, \quad E = \sum_j 8 \binom{1/2}{j} \frac{x^{2j+3}}{2j+3}, \quad Q = \sum_j 8 \binom{-1/2}{j} \frac{x^{2j+5}}{2j+5}$$

(the `series_nr` function of the same file). The commit that finished the solver (c5892bd) records that these agree with quadrature to 10^{-12} from $x = 10^{-8}$ to 10^6 , and that `cargo test-release` passes 21 + 8 tests, among them `kohn_sham::tests::fermi_integrals_match_quadrature` and `kohn_sham::tests::series_and_closed_forms_are_continuous_at_the_switch_points`. After the defect fixes of 2bc936d the final solver passes 34 of 34 tests (23 unit, 11 integration), these two among them [commit 2bc936d; the `#[test]` functions of `src/*.rs` and `tests/cosmology.rs`]. (The design review had proposed a switch at $x = 0.25$; the implemented switch is 0.6.)

8.6 The mean field

With $W(\sigma) := V(H\sigma)$, the gap equation $m = W'(\sigma)$ and self-consistency $\sigma = \sigma_{KS}(k_F, m)$:

$$\begin{aligned} \rho &= \varepsilon_{KS} + W - \sigma W', \\ P_{\text{obs}} &= P_{KS} + \sigma W' - W, \\ P_{\text{hid}} = P_{x_0} &= \sigma W' - W && \text{(along } x_0 \text{ and along } x_5, x_6, x_7: \text{ zero modes} \\ &&& \text{carry no kinetic pressure),} \\ \rho + P_{\text{obs}} &= \omega_F n && \text{(exact),} \\ d\rho/dn &= \omega_F + (\sigma_{KS} - \sigma) W'' \sigma'(n), && \text{i.e. } = \omega_F \text{ at the self-consistent point.} \end{aligned}$$

[proved VII §29] MEAN FIELD [THE RESULT]: `rho = eps + W - sigma W'`, `P_obs = P_KS + sigma W' - W` give `rho + P_obs == w_F n` exactly; and `d rho/dn == w_F + (sigma_KS - sigma) W'' sigma'(n)`, i.e. `== w_F` at the self-consistent point `sigma = sigma_KS` (`m = W'(sigma)` throughout)

The split $\rho = \varepsilon_{KS} + U$ with $U := W - \sigma W'$ separates a **quasiparticle** part (ε_{KS} , P_{KS} , with $0 \leq P_{KS}/\varepsilon_{KS} \leq 1/3$) from a **mean-field condensate** U , which has $P = -U$ in the observed and in the hidden directions alike: an 8-dimensional vacuum-like energy that varies with σ [derived here from the formulas above].

8.7 The classical limit is the non-relativistic limit, and the sign rule

At the self-consistent point the trace identity gives, **exactly**,

$$\rho = W(\sigma) + 3P_{\text{KS}}, \quad P_{\text{obs}} = (\sigma W'(\sigma) - W(\sigma)) + P_{\text{KS}},$$

that is, Part VI's $\rho = V(s)$, $P = sV' - V$ (with $s = H\sigma$) plus the degeneracy pressure and three times it. Part VI is recovered exactly in the limit $P_{\text{KS}}/\varepsilon_{\text{KS}} \rightarrow 0$, which is $k_{\text{F}}/|m| \rightarrow 0$: the cold, **non-relativistic** limit of the Fermi sea. (It is not “ $k_{\text{F}} \rightarrow 0$ at fixed n ”, which is impossible, and not a coherent zero mode, which the Pauli principle forbids.) **[proved VII §29] CLASSICAL LIMIT [THE RESULT]:** at the self-consistent point $\rho == W + 3 P_{\text{KS}}$ and $P_{\text{obs}} == (\sigma W' - W) + P_{\text{KS}}$ exactly ($\sigma = \sigma_{\text{KS}}$, $m = W'$); Part VI's $\rho = V(s)$, $P = s V' - V$ is the limit $P_{\text{KS}}/\varepsilon_{\text{KS}} \rightarrow 0$, i.e. $k_{\text{F}}/m \rightarrow 0$

Expanded in $x = k_{\text{F}}/|m|$ with the series of section 8.5 and $W' = m$ **[derived here]:**

$$\begin{aligned} \sigma &= \text{sgn}(m) n \left[1 - \frac{3}{10} x^2 + \dots \right], \\ \rho &= W(\text{sgn}(m) n) + \frac{3}{10} n k_{\text{F}}^2 / |m| + \dots, \\ P_{\text{obs}} &= [\sigma W' - W] + n k_{\text{F}}^2 / (5|m|) + \dots, \\ P_{\text{hid}} &= \sigma W' - W \end{aligned}$$

($\rho = |m| n (1 + \frac{3}{10} x^2) + W(\sigma) - |m| n (1 - \frac{3}{10} x^2)$ with $W(\sigma) = W(\text{sgn}(m) n) - \frac{3}{10} |m| n x^2 + \dots$). **The sign rule:** self-consistency forces $\text{sign}(\sigma) = \text{sign}(m) = \text{sign}(W')$. This removes the Part VI branch $Ms > 0$. For the author's mass term $W = -2M\sigma$ ($V = -(2M/H)s$, $m = -2M$), $\rho \rightarrow 2|M|n > 0$ in the classical limit.

8.8 The author's mass term: w falls from 1/3 to 0

For $W = m\sigma$ ($V = -(2M/H)s$ gives $m = -2M$), $W - \sigma W' = 0$, so

$$\rho = \varepsilon_{\text{KS}} > 0, \quad P_{\text{obs}} = P_{\text{KS}} \geq 0, \quad 0 \leq w = P_{\text{KS}}/\varepsilon_{\text{KS}} \leq \frac{1}{3},$$

and w depends only on $x = k_{\text{F}}/|m|$: $w = I_2/(3I_1)$ with the $m = 1$ integrals

$$\begin{aligned} I_1 &= \frac{1}{8} (x w_x (2x^2 + 1) - \ln(x + w_x)), & I_2 &= \frac{1}{8} (x w_x (2x^2 - 3) + 3 \ln(x + w_x)), \\ I_3 &= \frac{1}{2} (x w_x - \ln(x + w_x)), & w_x &= \sqrt{x^2 + 1}. \end{aligned}$$

w rises **monotonically**, since $dw/dx = x^2 \Delta / (3 w_x I_1^2)$ with $\Delta = x^2 I_1 - w_x^2 I_2$, $\Delta(0) = 0$ and $\Delta' = 2x I_3 > 0$, from $w \rightarrow 0$ ($x \rightarrow 0$, $w = x^2/5 + O(x^4)$: dust) to $w \rightarrow 1/3$ ($x \rightarrow \infty$: radiation). As a gas of fixed particle number cools, w falls from 1/3 to 0.

The classical sign problem disappears. Part VI needed $Ms < 0$ for $\rho = -(2M/H)s > 0$. Quantum mechanically $\sigma_{\text{KS}} = g \int m/\omega$ is **odd** in m , and $m \sigma_{\text{KS}} = \frac{g m^4}{2\pi^2} I_3(k_{\text{F}}/|m|) \geq 0$ for either sign of m . With $m = -2M$, $M \sigma_{\text{KS}} \leq 0$: exactly the branch $Ms < 0$ that Part VI had to assume. **[proved VII §29]**

- **AUTHOR'S MASS TERM [THE RESULT]:** $W = m \sigma$ gives $W - \sigma W' == 0$, so $\rho == \varepsilon_{\text{KS}}$, $P == P_{\text{KS}}$, and $w = P_{\text{KS}}/\varepsilon_{\text{KS}}$ depends only on $x = k_{\text{F}}/m$: $w == I_2/(3 I_1)$, I_n the $m = 1$ integrals
- **AUTHOR'S MASS TERM [THE RESULT]:** $0 \leq w \leq 1/3 - P_{\text{KS}} \geq 0$ ($I_2' = x^4/w_x > 0$, $I_2(0) = 0$) and $\varepsilon - 3P = m \sigma = (g m^4/(2 \pi^2)) I_3 \geq 0$ ($I_3' = x^2/w_x > 0$, $I_3(0) = 0$)

- AUTHOR'S MASS TERM [THE RESULT]: w rises MONOTONICALLY - $dw/dx == x^2 \Delta / (3 w_x I_1^2)$ with $\Delta = x^2 I_1 - w_x^2 I_2$, $\Delta(0) = 0$ and $\Delta' = 2 x I_3 > 0$ - from $w \rightarrow 0$ ($x \rightarrow 0$, $w = x^2/5 + O(x^4)$: dust) to $w \rightarrow 1/3$ ($x \rightarrow \infty$: radiation)
- AUTHOR'S MASS TERM [has content]: the classical sign problem disappears - $\sigma_{KS} = g \int m/\omega$ is ODD in m and $m \sigma_{KS} = (g m^4 / (2 \pi^2)) I_3(k_F/|m|) \geq 0$ for either sign, so with $m = -2M$: $M \sigma_{KS} \leq 0$, the branch $M s < 0$ that Part VI had to assume

8.9 $w \geq -1$ always: no phantom crossing for the quantized fable

From $\rho + P_{\text{obs}} = \omega_F n$ (section 8.6), which is exact, and $\omega_F n \geq 0$:

$$w_f + 1 = \frac{\rho + P_{\text{obs}}}{\rho} = \frac{\omega_F n}{\rho} \geq 0 \quad \text{wherever } \rho > 0.$$

(w_f is the fable's equation of state $w = \langle P_{\text{obs}} \rangle / \langle \rho \rangle$ of section 8.1; $\omega_F = \sqrt{k_F^2 + m^2}$ is the Fermi energy.) The quantized fermion fable **cannot cross the phantom divide**, for any potential W [derived here from MEAN FIELD [THE RESULT]: $\rho = \epsilon + W - \sigma W'$, $P_{\text{obs}} = P_{KS} + \sigma W' - W$ give $\rho + P_{\text{obs}} == w_F n$ exactly; and $d\rho/dn == w_F + (\sigma_{KS} - \sigma) W'' \sigma'(n)$, i.e. $== w_F$ at the self-consistent point $\sigma = \sigma_{KS}$ ($m = W'(\sigma)$ throughout)]. The classical Part VI crossing at $V'(s_1) = 0$ (section 2.12.7) is unreachable. In the quantum theory $\rho + P_{\text{obs}} = \omega_F n > 0$ at every non-zero density, whatever the value of m . And for the `expdamp` potential the gap equation keeps $0 < \sigma < \min(n, s_1)$ strictly, so $m = W'(\sigma)$ never reaches the classical crossing point at finite density: $\sigma \rightarrow s_1$ and $m \rightarrow 0$ only as $n \rightarrow \infty$ (design review DFT-6; solver tests `kohn_sham::tests::expdamp_pins_sigma_below_s1_and_mass_goes_to_zero_at_high_density` and `kohn_sham::tests::no_phantom_crossing_rho_plus_p_obs_is_nonnegative`). The classical phantom crossing of 2026-09-16 therefore does **not** survive quantization. (What an observer who fits a cold-dark-matter-plus-dark-energy model to the coupled system can infer — an apparent crossing, when the dust is subtracted at today's dark-matter density and the effective mass grows — is a property of that fit, not of the fluid; it is part of the coupled system, Effort B, and is stated in section 1.)

8.10 On the pre-universe: the quantum gas cools, the classical field does not

Four facts, each proved on the canonical frame with a_4 free.

(1) Charge. The $U(1)$ current is conserved on shell for generic $\Psi(x_0, x_4)$, $\bar{\Psi}(x_0, x_4)$. For the x_0 -profile $\Psi = \sqrt{\sin 6Hx_0} \Psi'$ with Ψ' independent of x_0 , $\sqrt{g} j^0$ is independent of x_0 ($\sec \cdot \sin \cdot \cot = 1$), so $\partial_4(\sqrt{g} j^4) = 0$: the charge per coordinate volume does not depend on x_4 , and since $\sqrt{g}$ does not either, neither does the proper 8-density. The 7-volume is constant because the observed sheet grows exactly as the hidden sheet shrinks ($q^3 p^3$ is x_4 -independent). [**proved VII §29**]

- PRE-UNIVERSE (1) [THE RESULT]: the $U(1)$ current is conserved on shell, $(1/\text{Sqrt}[g]) d_\mu(\text{Sqrt}[g] j^\mu) == 0$, for generic $\Psi(x_0, x_4)$, $\bar{\Psi}(x_0, x_4)$
- PRE-UNIVERSE (1) [THE RESULT]: for $\Psi = \text{Sqrt}[\text{Sin}] \Psi'(x_4)$, $\text{Sqrt}[g] j^0$ is x_0 -independent ($\sec \sin \cot == 1$), so $d_4(\text{Sqrt}[g] j^4) == 0$ on shell; and $\text{Sqrt}[g]$ and $q^3 p^3$ are x_4 -independent

(2) Which x_4 -only fields solve the equations: $V''s' = 0$ (section 5.8).

(3) Redshift. A mode $\Psi = \sqrt{\sin} u(x_4) e^{ikx_1}$ (coordinate momentum k along the observed sheet) obeys $T_{16}[4] u' + i(k/q) T_{16}[1] u = HV' u$ exactly (mass term, V' constant): its **physical** momentum is k/q , and $d \ln(k/q)/dx_4 = H a'_4(Hx_4) = -H_{\text{obs}}$. The observed-sheet momenta redshift as $e^{a_4} \sim 1/a$, and the instantaneous frequency is $\sqrt{k^2/q^2 + m^2}$ ($E^2 = \omega^2$ at $k_{\text{phys}} = k/q$, section 6.10). **[proved VII §29]**

- PRE-UNIVERSE (3) [THE RESULT]: for $\text{Psi} = \text{Sqrt}[\text{Sin}] u(x_4) \text{Exp}[i k x_1]$ the field equation (mass term, $V' = \text{constant}$) is $\text{Sqrt}[\text{Sin}] \text{Exp}[i k x_1] (T_{16}[4] u' + i (k/q) T_{16}[1] u - H V' u)$: the physical momentum is k/q
- PRE-UNIVERSE (3) [THE RESULT]: $d \text{Log}[k/q]/dx_4 == H a_4' [H x_4] == -H_{\text{obs}}$ (observed momenta redshift as $\text{Exp}[a_4]$), and $E^2 == (k^2/q^2 + m^2)$ ID16 at the physical momentum

(4) Cooling. The occupied coordinate momenta are fixed (the mode labels are conserved; the x_4 -dependence of q only mixes $\pm\omega$, a Bogoliubov effect that vanishes adiabatically). So $x = k_{F,\text{phys}}/m$ falls as e^{a_4} : $d \ln x/dx_4 = H a'_4$, and since $w(x)$ is increasing, $\text{sign}(dw/dx_4) = \text{sign}(a'_4)$. In the regime $a'_4 < 0$ (the observed sheet expanding, the sign hypothesis of Part VI, section 2.11) the quantum fable gas **cools**, $w : 1/3 \rightarrow 0$, even though its 8-density does not dilute. The classical Part VI fable, by contrast, has $ds/dx_4 = 0$ on the rescaled solutions and $w = sV'/V - 1$ **frozen**. **[proved VII §29]**

- PRE-UNIVERSE (4) [THE RESULT]: $x = k_{F,\text{phys}}/m = x_c/(q m)$ obeys $d \text{Log}[x]/dx_4 == H a_4' [H x_4]$; since $dw/dx > 0$, $dw/dx_4 = w'(x) x H a_4' < 0$ exactly when $a_4' < 0$ (the observed sheet expanding): the quantum gas COOLS
- PRE-UNIVERSE (4) [control]: the classical Part VI fable does NOT cool - $ds/dx_4 == 0$ on its rescaled solutions (Part VI's result, re-asserted), and its $w = s V'/V - 1$ is a function of s alone: frozen

quantity	classical fable (Part VI)	fermion fable (Part VII, Kohn–Sham, mass term)
ρ	$V(s)$	$\varepsilon_{\text{KS}}(k_F, m) > 0$
P (observed)	$sV' - V$ ($= 0$: dust)	P_{KS} in $[0, \rho/3]$
w	$sV'/V - 1$, frozen ($ds/dx_4 = 0$)	$P_{\text{KS}}/\varepsilon_{\text{KS}}$: $1/3 \rightarrow 0$ as the observed sheet expands
8-density	constant	constant (charge per coordinate volume conserved)

(This table is the notebook's displayed summary at the end of Section 29 **[displayed VII §29]**.)

Finally, a_4 **is still undefined** after Part VII: PART VII [control]: a_4 is STILL UNDEFINED - nothing in Part VII gave it a value

9 The canonical spin connection: a complete discussion

This section covers the canonical spin connection of the 4+4 pre-universe in full: the frame it comes from, how it is computed, every non-zero component, its 16×16 spinor form, the gauge-covariant derivative of the 16-component spinor, how it relates to the three bridges of Part IV and to the fable-5.1 (Weitzenböck) bridge of Part V, and what it does and does not do for the complex fermion fable. It stands on its own. Every object is defined here and every result is restated here, with the place where it is proved.

Throughout, “Section n ” (capital S) means a section of the notebook. The numbered parts of this discussion are called “subsections” 9.1–9.13.

9.1 Sources, and how the status of each statement is marked

Where the proofs live. The proofs are in the Mathematica notebook `claude-fable/claude-fable_Einstein-Rosen-2-Planes.nb`. That notebook is generated from cell manifests and is never edited by hand:

manifest	Part	Sections	subject
<code>claude-fable/cells_part1.wl</code>	I	0–4	session, coordinates, <code>eta4488</code> , <code>tau</code> , <code>sigma</code>
<code>claude-fable/cells_part2.wl</code>	I	5–9	<code>T16</code> , <code>sigma16</code> , <code>SAB</code> , matrix bases, Cartan triality
<code>claude-fable/cells_part3.wl</code>	II	10–14	the author’s physics reproduced (<code>Psi16</code> , <code>La</code> , <code>eLa</code>)
<code>claude-fable/cells_part4.wl</code>	III, IV	15–17, 18–20	frame, Christoffel symbols, canonical spin connection, spinor covariant derivative; the three bridges
<code>claude-fable/cells_part5.wl</code>	V	21–22	the fable-5.1 (Weitzenböck) bridge; the master comparison
<code>claude-fable/cells_part6.wl</code>	VI	23–25	<code>fableScalar</code> and the real (classical) fable spinor
<code>claude-fable/cells_part7.wl</code>	VII	26–29	the complex fermion fable (new): refinement, Lagrangian and field equations, canonical quantization, energy–momentum tensor
<code>claude-fable/cells_part8.wl</code>	VIII	30–33	fable as a source of the Einstein equations of the primordial field (the coupled system; its spin connection is Section 32)

The run of Parts I–VI, `claude-fable/run_fable51_part6.log`, evaluates 172 Input cells straight out of the `.nb`. It reports `Assertions run: 358 passed: 358 failed: 0` and `cells w/ msgs : 0`, with `identities accepted on numerical evidence alone : 0` and `non-vanishing witnesses : 27`. The Part VII run is recorded in `claude-fable/run_fermion_fable_part7.log`. It evaluates 198 Input cells and reports `Assertions run: 528 passed: 528 failed: 0` and `cells w/ msgs : 0`, with `identities accepted on numerical evidence alone : 0` and `non-vanishing witnesses : 38` (Part VII adds 170 assertions and 11 witnesses to Parts I–VI). The acceptance run of Part VIII, `claude-fable/run_fermion_fable_part8.log`, reported `Assertions run: 642 passed: 642 failed: 0`; its two comparisons of the Rust solver with the Mathematica reference runs were skipped, because the solver’s CSVs did not exist yet. The final run of the whole notebook, `claude-fable/run_fermion_fable_final.log`, evaluates 217 Input cells and reports `Assertions run: 644 passed: 644 failed: 0` and `cells w/ msgs : 0`, with `identities accepted on numerical evidence alone : 0` and `non-vanishing witnesses : 46`.

How an identity is certified. Each identity is certified in up to three stages:

1. exact structural equality (the entry is literally 0);
2. Simplify under the geometric assumptions, with a 5-second budget;
3. a numerical check at three rational probe points to 30 significant digits.

Stage 3 has two uses, and the notebook counts them separately. The first is to accept an identity. The notebook reports *zero* of those, and its last assertion fails if that count is ever non-zero, so

every identity quoted below is closed symbolically. The second is to witness that something is *not* zero, through `cfNonZeroWitnessQ`. A witness succeeds only if every entry evaluates to an actual number at some probe point and at least one of them exceeds 10^{-10} in magnitude. That is a complete proof of non-vanishing.

The probe functions used in stage 3 are test inputs only, never definitions: `a4 -> Function[u, 3/7 + u/5 + Sin[u]/11]`, a probe for the boost rapidity of Bridge 1, and probe values of the Bridge 3 parameter `lambdaOct`.

Status marks used below.

mark	meaning
[proved P §n]	“label” or label a <code>cfAssert</code> of Part P, Section n, that prints PASS in <code>run_fable51_part6.log</code> (Parts I–VI), <code>run_fermion_fable_part7.log</code> (Part VII) or <code>run_fermion_fable_final.log</code> (Part VIII, the final run); the label is quoted verbatim
[displayed P §n]	the printed output of a cell of Part P, Section n (a table, a count or a closed form); an output, not an assertion
[prose P §n]	stated only in prose in the notebook (a Text cell, or a comment inside an Input cell), not asserted
[derived here]	a short derivation carried out in this section from results marked above, with every step shown

The notebook grades its own assertion labels. [definition] means true by construction. [structural] means true for any input of the right shape. [solver regression] is a check of the code, not of the geometry. [has content] could have failed. [THE RESULT] marks a headline result. [control] shows that the wrong alternative fails.

9.2 Conventions

- Coordinates $X = \{x_0, x_1, \dots, x_7\}$. The Wolfram array position is the coordinate index plus one, so `x4` is entry 5. Greek indices are curved and Latin indices are flat; both run 0..7. x_4 is the observer’s time ($g_{44} = -1$), x_1, x_2, x_3 are the observed 3-space, x_0 is the hidden spacelike direction, and x_5, x_6, x_7 are the hidden timelike sheet. The notebook’s own words for $x_5..x_7$ are “the superluminal deflating directions”.
- Flat metric: `eta4488 = DiagonalMatrix[{1,1,1,1,-1,-1,-1,-1}]`. [proved I §2] “signature of eta4488 is (4,4)”, “eta4488 . eta4488 == ID8”.
- The 8×8 generators have `tau[0] = ID8`. The conjugates are `taubar[A] = sigma . Transpose[sigma . tau[A]]`, with `sigma = ArrayFlatten[{{0,ID4},{ID4,0}}]` ([proved I §4] “sigma . sigma == ID8”, “sigma is symmetric”). It follows that `taubar[0] = ID8`.
- The 16×16 Dirac matrices are E. A. Lord’s reduced Brauer–Weyl doubling, `T16[A] = ArrayFlatten[{{0, taubar[A]}, {tau[A], 0}}]` for $A = 0..7$. They are real, and `T16[0] = ArrayFlatten[{{0,ID8},{ID8,0}}]`. [proved I §5] “{T16[A], T16[B]}/2 == eta4488[[A+1,B+1]] ID16”. Throughout, $T_{16}[a]$ is γ^a , with the flat index *up*. Since eta4488 is its own inverse, $\gamma_a = \eta_{ab} T_{16}[b]$ differs from $T_{16}[a]$ by the sign η_{aa} .
- Spinor metric: `sigma16 = T16[0].T16[1].T16[2].T16[3]`. [proved I §5] “sigma16 == ArrayFlatten[{{-sigma,0},{0,sigma}}]”, “sigma16 . T16[A] is antisymmetric for A = 0..7”, “sigma16 is symmetric and sigma16 . sigma16 == ID16”.
- Chirality: `T16[8] = T16[0].T16[1]...T16[7] = diag(-ID8, +ID8)`. [proved I §5] “PL projects onto the upper (type-1) components, PR onto the lower (type-2)

components". A 16-spinor is $\Psi = (\psi_1, \psi_2)$. The upper 8 components ψ_1 are a split-octonion type-1 spinor, and the lower 8 components ψ_2 are a type-2 spinor.

- $\text{so}(4,4)$ generators: $\text{SAB}[[A+1, B+1]] = (1/4) (\text{T16}[A].\text{T16}[B] - \text{T16}[B].\text{T16}[A])$. [proved I §5] "SAB is antisymmetric in its two flat indices", "sigma16 . SAB is antisymmetric", "T16[8] commutes with every SAB: the chiral halves are each $\text{Spin}(4,4)$ -invariant", "[SAB, T16] reproduces the vector representation".
- H , K and M are Protected constants of the original notebook.

9.3 The canonical frame (vielbein), the metric, and the volume factor

A vielbein here is simply an 8-dimensional vierbein, that is, an 8-dimensional frame field. The canonical frame is the diagonal matrix that the original notebook records as the value of the symbol it writes "gtrye with indices alpha and (A)" [prose III §15]:

$$e = \text{diag}(\tan 6Hx_0, q, q, q, 1, p, p, p), \quad (46)$$

$$q = \frac{e^{-a_4(Hx_4)}}{\sin^{1/6} 6Hx_0}, \quad p = \frac{e^{+a_4(Hx_4)}}{\sin^{1/6} 6Hx_0}. \quad (47)$$

In the notebook: `frameCanonical = DiagonalMatrix[{Tan[6 H x0], q, q, q, 1, p, p, p}], q = cfQminus (observed directions  $x_1, x_2, x_3$ ), p = cfQplus (hidden directions  $x_5, x_6, x_7$ ).`

Rows curved, columns flat. `frameCanonical[[mu+1, a+1]] =  $e_\mu^a$` . The coframe is the matrix inverse, stored with the flat index on the row: `coframeCanonical = Inverse[frameCanonical]`, `coframeCanonical[[a+1, mu+1]] =  $e_a^\mu$` . [proved III §15] "CANONICAL [structural]: frame . coframe == ID8 (certifies only that the frame is invertible)".

The free function a_4 . The original notebook never defines `a4`. It appears only inside the recorded frame. It is carried symbolically everywhere and is *never given a value*. The notebook prints a notice to this effect when Section 15 runs [prose III §15], and Part VI ends with [proved VI §25] "PART VI [control]: `a4` is STILL UNDEFINED - nothing in Part VI gave it a value; and the FLRW scale factor `aF` never entered the canonical frame". Every result in this section holds for an arbitrary differentiable a_4 . Below, a'_4 means `Derivative[1][a4][H x4]`.

Reciprocity. $qp = \sin^{-1/3} 6Hx_0$ does not depend on x_4 : the observed directions contract exactly as fast as the hidden directions expand [prose III §15].

The metric. The bridge between the curved and flat metrics is

$$[d] \quad g_{\mu\nu} = e_\mu^a \eta_{ab} e_\nu^b, \quad (48)$$

that is, `gCanonical = frameCanonical . eta4488 . Transpose[frameCanonical]`, and the line element is

$$ds^2 = \tan^2(6Hx_0) dx_0^2 + q^2 (dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 - p^2 (dx_5^2 + dx_6^2 + dx_7^2). \quad (49)$$

- [proved III §15] "CANONICAL [definition]: `g == frame . eta4488 . Transpose[frame] (bridge [d]; this IS how g was defined, so it cannot fail)`".
- [proved III §15] "CANONICAL [has content]: `g == diag(Tan[6 H x0]^2, q^2, q^2, q^2, -1, -p^2, -p^2, -p^2)`, the stated line element".

- [proved III §15] “CANONICAL [has content]: $\text{Diagonal}[g] == \text{Diagonal}[\text{eta4488}] * \text{Diagonal}[\text{frame}]^2$ exactly” (Sylvester’s law made explicit).
- [proved III §15] “CANONICAL [has content]: g has signature (4,4), GIVEN a_4 real and $0 < 6Hx_0 < \pi/2$ ”.
- [proved III §15] “Inverse[g] == Transpose[coframe] . eta4488 . coframe”, “g is symmetric”, “the metric is non-degenerate”.

The volume factor. Section 15 displays $\text{Det}[g] = \text{Sec}[6Hx_0]^2$ and $\text{Sqrt}[\text{Abs}[\text{Det}[g]]] = \text{Abs}[\text{Sec}[6Hx_0]]$ [displayed III §15]. [proved V §21] “FABLE-5.1: $\det g == \text{Sec}[6Hx_0]^2$ (positive, as signature 4+4 requires)”. On the domain $0 < 6Hx_0 < \pi/2$ the root is $\sec 6Hx_0$. From Part V on, the notebook uses $\text{sqrtDetgTele} = \text{Sec}[6Hx_0]$ rather than $\text{Abs}[\text{Sec}[\dots]]$, so that Abs is never differentiated [prose V §21]. The same result follows from the frame directly [derived here]:

$$\det e = \tan(6Hx_0) q^3 p^3 = \frac{\tan 6Hx_0}{\sin 6Hx_0} = \sec 6Hx_0 = \sqrt{|\det g|}. \quad (50)$$

The volume factor does not depend on x_4 .

The curved Dirac matrices. $\text{gammaCurvedCanonical}[[\mu+1]] = \text{Sum_a coframe}[[a,\mu]] T_{16}[a]$. The frame is diagonal, so [derived here]

$$\begin{aligned} \gamma^0 &= \cot(6Hx_0) T_{16}[0], \\ \gamma^j &= \frac{1}{q} T_{16}[j] = e^{a_4} \sin^{1/6}(6Hx_0) T_{16}[j] \quad (j = 1, 2, 3), \\ \gamma^4 &= T_{16}[4], \\ \gamma^k &= \frac{1}{p} T_{16}[k] = e^{-a_4} \sin^{1/6}(6Hx_0) T_{16}[k] \quad (k = 5, 6, 7). \end{aligned} \quad (51)$$

- [proved III §17] “{gammaCurved[mu], gammaCurved[nu]} == 2 gInv[mu,nu] ID16”.
- [proved VI §24] “FABLE [has content]: $(\text{gamma}^4)^2 == -\text{ID16}$, so $\text{gamma}^4 d_4 \text{Psi}' == H V'(s) \text{Psi}'$ is solved by $d_4 \text{Psi}' == -H V'(s) \text{gamma}^4 \text{Psi}'$ ”.

9.4 The generalized Christoffel symbols

These are the Levi-Civita symbols of g :

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2} g^{\rho s} (\partial_\mu g_{s\nu} + \partial_\nu g_{s\mu} - \partial_s g_{\mu\nu}). \quad (52)$$

$\text{cfChristoffel}[g, \text{ginv}, \text{coords}]$ computes the first derivatives $\text{dg}[[k,m,p]] = D[g[[m,p]], \text{coords}[[k]]]$ once, reuses them in the triple loop, and simplifies each entry under the notebook’s assumptions. The result is $\text{GammaCanonical}[[\text{rho}+1, \text{mu}+1, \text{nu}+1]]$. Section 16 lists its 37 non-zero entries (out of 512) [displayed III §16]; here $\text{Gamma}[r;m,n]$ stands for Γ^r_{mn} :

```
Gamma[0;0,0] = 12*H*Csc[12*H*x0]
Gamma[0;j,j] = (H*Cos[6*H*x0]^3)/(E^(2*a4[H*x4])*Sin[6*H*x0]^(10/3))   j = 1,2,3
Gamma[0;k,k] = -(E^(2*a4[H*x4])*H*Cos[6*H*x0]^3)/Sin[6*H*x0]^(10/3)     k = 5,6,7
Gamma[j;0,j] = Gamma[j;j,0] = -H*Cot[6*H*x0]                             j = 1,2,3
Gamma[j;j,4] = Gamma[j;4,j] = -H*a4'[H*x4]                               j = 1,2,3
Gamma[4;j,j] = -(H*a4'[H*x4])/(E^(2*a4[H*x4])*Sin[6*H*x0]^(1/3))         j = 1,2,3
Gamma[4;k,k] = -(E^(2*a4[H*x4])*H*a4'[H*x4])/Sin[6*H*x0]^(1/3)           k = 5,6,7
Gamma[k;0,k] = Gamma[k;k,0] = -H*Cot[6*H*x0]                             k = 5,6,7
Gamma[k;4,k] = Gamma[k;k,4] = H*a4'[H*x4]                                 k = 5,6,7
```

The count is $1 + 3 + 3 + 6 + 6 + 3 + 3 + 6 + 6 = 37$.

- [proved III §16] “Gamma is symmetric in its two lower indices (torsion-free affine connection)”. This check is structural: swapping the two lower indices in the formula reproduces the formula whenever g is symmetric.
- [proved VI §23] “HELPER [solver regression]: cfCovariantDivergence of the metric itself vanishes (metric compatibility of Section 16’s Gamma; a wrong index slot in the helper fails this)”.

9.5 The vielbein postulate, and how cfSpinConnection solves it

The spin connection is fixed by the zero-torsion (Levi-Civita) condition, written as the *vielbein postulate*. The total covariant derivative of the frame vanishes when Γ acts on the curved index and ω acts on the flat index [prose III §16]:

$$\partial_\mu e_\nu^a - \Gamma_{\mu\nu}^\rho e_\rho^a + \omega_\mu^a{}_b e_\nu^b = 0. \quad (53)$$

In the notebook: $D[e[\text{nu}, a], x[\text{mu}]] - \text{Gamma}[\text{rho}, \text{mu}, \text{nu}] e[\text{rho}, a] + \text{omegaMixed}[\text{mu}, a, b] e[\text{nu}, b] == 0$, with $\text{omegaMixed}[\text{mu}, a, b] = \omega_\mu^a{}_b$. Contracting with the coframe frees ν :

$$\omega_\mu^a{}_c = \sum_\nu \left(\sum_\rho \Gamma_{\mu\nu}^\rho e_\rho^a - \partial_\mu e_\nu^a \right) e_c^\nu. \quad (54)$$

The notebook’s solver is exactly this formula (Section 16):

```
cfSpinConnection[frame_, coords_, etaFlat_, gamma_] :=
Module[{n = Length[coords], cofr, dE},
  cofr = cfSimp[Inverse[frame]];
  dE = Table[D[frame[[nu, a]], coords[[mu]]], {mu, n}, {nu, n}, {a, n}];
  Table[
    cfSimp[Sum[(Sum[gamma[[r, mu, nu]] frame[[r, a]], {r, n}]
      - dE[[mu, nu, a]] cofr[[c, nu]], {nu, n}]],
    {mu, n}, {a, n}, {c, n}];

cfLowerFirstFlat[om_, etaFlat_] := Module[{n = Length[etaFlat]},
  Table[Sum[etaFlat[[a, c]] om[[mu, c, b]], {c, n}, {mu, n}, {a, n}, {b, n}];

(* omega_mu^a_b *)
omegaCanonicalMixed =
  cfSpinConnection[frameCanonical, X, eta4488, GammaCanonical];
(* omega_{mu a b} *)
omegaCanonical = cfLowerFirstFlat[omegaCanonicalMixed, eta4488];
```

The same routine produces every connection of the notebook that comes from the vielbein postulate: the canonical one, those of Bridges 1 and 2, the Levi-Civita part `omegaTriLC` of Bridge 3 (to which Section 20 then adds the contortion), and the fable-5.1 connection. The Christoffel symbols are computed once and passed in. So the comparisons below compare results, not different pieces of code [prose III §16, IV, V §21].

What each certificate is worth. The notebook grades its certificates as follows [prose III §16]:

- [proved III §16] “CANONICAL [solver regression]: vielbein postulate residual is zero”. This is an algebraic identity of the solver. `cfSpinConnection` contracts the postulate with `Inverse[frame]` and `cfVielbeinResidual` contracts it back with `frame`, and the two cancel for any affine connection and any invertible frame.
- [proved III §16] “CANONICAL [structural]: torsion vanishes”. Once the residual is zero, this reduces to the symmetry of Γ in its lower indices, which is itself structural. The torsion 2-form used is $T^a{}_{\mu\nu} = \partial_\mu e_\nu^a - \partial_\nu e_\mu^a + \omega_\mu^a{}_b e_\nu^b - \omega_\nu^a{}_b e_\mu^b$ (`cfTorsion`).

- **[proved III §16]** “CANONICAL [has content]: `omega[mu,a,b] == -omega[mu,b,a]` (metric compatibility)”. This is not imposed anywhere. It holds only because the affine connection fed in is the Levi-Civita connection of the metric the frame builds, so a wrong Christoffel symbol would break it. The connection is therefore $\mathfrak{so}(4,4)$ -valued.
- **[proved III §16]** “INDEPENDENT CHECK: the frame-only derivation reproduces `omegaCanonical` exactly” and “INDEPENDENT CHECK: and it is antisymmetric in its two raised flat indices”. The connection is re-derived from the frame alone by the anholonomy formula, which never mentions Γ or g (`cfSpinConnectionFromFrame`, with $e^{a\nu} = (\eta^{-1} \cdot \text{coframe})^{a\nu}$):

$$\begin{aligned} \omega_\mu^{ab} = & \frac{1}{2} e^{a\nu} (\partial_\mu e_\nu^b - \partial_\nu e_\mu^b) - \frac{1}{2} e^{b\nu} (\partial_\mu e_\nu^a - \partial_\nu e_\mu^a) \\ & - \frac{1}{2} e^{a\rho} e^{b\sigma} (\partial_\rho e_\sigma^c - \partial_\sigma e_\rho^c) \eta_{cd} e_\mu^d. \end{aligned} \quad (55)$$

- The canonical frame is diagonal and so equals its own transpose. Section 16 therefore cannot detect a transposition of the frame’s two slots. Section 18 closes that gap on a frame that is not symmetric (the canonical frame times a constant boost of rapidity $1/3$): **[proved IV §18]** “the slot-test frame is genuinely not symmetric”, “INDEPENDENT CHECK on a non-symmetric frame: the two solvers agree, so the curved/flat slots are placed correctly in both”, “and the result is the canonical connection conjugated by the constant boost, as the gauge law demands”.

9.6 The canonical spin connection: all its non-zero components

`omegaCanonical` has **24 non-zero components out of 512** **[displayed III §16]**. They are listed by `cfShowConnection[omegaCanonical, "omega"]`. In this subsection and the next only, write $c = \cos 6Hx_0$, $s = \sin 6Hx_0$ and $A = a_4(Hx_4)$. For $j = 1, 2, 3$ and $k = 5, 6, 7$, with all indices down, $\omega_{\mu ab} = \text{omega}[\mu; a, b] = \text{omegaCanonical}[[\mu+1, a+1, b+1]]$:

component (both flat indices down)	value
$\omega_{j0j} = -\omega_{jj0}$	$Hc^2/(e^A s^{13/6})$
$\omega_{j4j} = -\omega_{jj4}$	$Ha'_4/(e^A s^{1/6})$
$\omega_{k0k} = -\omega_{kk0}$	$-e^A Hc^2/s^{13/6}$
$\omega_{k4k} = -\omega_{kk4}$	$e^A Ha'_4/s^{1/6}$

That gives $6 \text{ directions} \times 2 \text{ planes} \times 2 \text{ orderings} = 24$ components. Every other component is zero. Part VII re-asserts the count and the pattern: **[proved VII §27]** `SPIN CONNECTION [has content]: metric compatible (omega_mu^{ab} antisymmetric) and every non-zero component is omega_mu^{\mu 0} or omega_mu^{\mu 4} (up to antisymmetry): omega_0 == omega_4 == 0`. That label writes the connection with both flat indices up, ω_μ^{ab} . Neither the antisymmetry nor the pattern of non-zero components depends on that position: raising both flat indices multiplies each component by $\eta_{aa}\eta_{bb} = \pm 1$ **[derived here]**. The **sign** of a listed component does depend on it: raising both indices flips the components in the planes with exactly one timelike index — $(4, j)$ and $(0, k)$ here — and leaves the $(0, j)$ and $(4, k)$ components unchanged. That is why the table above is written with both flat indices down, as `omegaCanonical` is, and why Part VIII, Section 32, lists the components of the connections of its dynamical frames the same way (section 10.3).

Structure [derived here], from the table.

- $\omega_0 = \omega_4 = 0$: the connection vanishes along the hidden direction x_0 and along the observer’s time x_4 .

- In each remaining direction μ (1, 2, 3, 5, 6, 7) only the two flat planes $(0, \mu)$ and $(4, \mu)$ are excited.
- Whether the generator of a plane is a rotation or a boost depends on the signs of `eta4488` in that plane. For $j = 1, 2, 3$, $(0, j)$ is a rotation (both directions spacelike) and $(4, j)$ is a boost. For $k = 5, 6, 7$, $(0, k)$ is a boost and $(4, k)$ is a rotation (both directions timelike).
- The x_0 -dependent components (planes $(0, \mu)$) do not involve a'_4 . The a_4 -dependent components (planes $(4, \mu)$) are proportional to a'_4 , so they all vanish when a_4 is constant.
- No component depends on $x_1, x_2, x_3, x_5, x_6, x_7$. The metric, the frame and the connection are invariant under translations along the observed and hidden sheets.
- The *mixed* components $\omega_\mu{}^a{}_b = \text{omegaCanonicalMixed}$ follow by raising the first flat index with `eta4488`. For example, $\omega_j{}^4{}_j = -\omega_{j4j}$ because $\eta_{44} = -1$.

Curvature. The curvature 2-form is $R_{\mu\nu}{}^a{}_b = \partial_\mu \omega_\nu{}^a{}_b - \partial_\nu \omega_\mu{}^a{}_b + [\omega_\mu, \omega_\nu]{}^a{}_b$ (`cfCurvature`). It has **156 non-zero components** [`displayed III §16`]. [`proved III §16`] “CANONICAL: curvature is antisymmetric in the two spacetime indices”. The Ricci scalar is [`displayed III §16`]

$$R = -3H^2 \left((55 + 7 \cos 12Hx_0) \cot^2(6Hx_0) \csc^2(6Hx_0) - 2a_4'^2 \right) \quad (56)$$

In the notebook:

```
RicciScalarCanonical = -3*H^2*((55 + 7*Cos[12*H*x0])*Cot[6*H*x0]^2*Csc[6*H*x0]^2
- 2*a4'[H*x4]^2)
```

This is equivalent to $-6H^2(24 \cot^2 6Hx_0 + 31 \cot^4 6Hx_0 - a_4'^2)$ [`derived here`], using $\cos 12Hx_0 = 1 - 2 \sin^2 6Hx_0$ and $\csc^2 = 1 + \cot^2$. It depends on a_4 only through $a_4'^2$. It is the frame-independent invariant that every bridge below is checked against.

Summary of the certified properties of the canonical connection. It is metric compatible (antisymmetric in its flat indices, a statement with content), it is torsion-free (structural), it is reproduced exactly by an independent frame-only derivation, and it is not flat.

9.7 The spinor connection Γ_μ and its exact normalization

The 16×16 spin-connection matrix is built by `cfSpinMatrix` in Section 17:

```
cfSpinMatrix[om_, mu_Integer] :=
  (1/8) Sum[om[[mu, a, b]] (T16[a - 1] . T16[b - 1] - T16[b - 1] . T16[a - 1]),
    {a, 8}, {b, 8}];
GammaSpinCanonical =
  Table[cfSimpArray[cfSpinMatrix[omegaCanonical, mu]], {mu, 8}];
```

It is fed `omegaCanonical`, the connection with *both flat indices down*. Since $T_{16}[a] = \gamma^a$ has its flat index *up*, the exact convention is

$$\begin{aligned} \Gamma_\mu &= \frac{1}{8} \omega_{\mu ab} [\gamma^a, \gamma^b] = \frac{1}{4} \sum_{a,b=0}^7 \omega_{\mu ab} T_{16}[a] T_{16}[b] \\ &= \frac{1}{2} \sum_{a < b} \omega_{\mu ab} T_{16}[a] T_{16}[b] = \frac{1}{2} \omega_{\mu ab} \text{SAB}[[a+1, b+1]]. \end{aligned} \quad (57)$$

The second line uses the antisymmetry of $\omega_{\mu ab}$ and $T_{16}[a] T_{16}[b] = -T_{16}[b] T_{16}[a]$ for $a \neq b$ [`derived here`]. The equivalent form with raised connection indices is $\frac{1}{4} \omega_\mu{}^{ab} \gamma_a \gamma_b$, with $\gamma_a = \eta_{ab} T_{16}[b]$.

Index warning. The expression $\frac{1}{4}\omega_\mu^{ab}T_{16}[a]T_{16}[b]$, with the raised connection contracted against the unlowered **T16**, is *not* the notebook's Γ_μ . It would multiply each plane by $\eta_{aa}\eta_{bb}$, and so flip the sign of the $(4,j)$ and $(0,k)$ terms [**derived here**].

The coefficient and the sign are not conventions left open. They are fixed against the vielbein postulate:

- [**proved III §17**] “[**definition**] $(1/8)\ \text{omega}\ [\text{gamma}^a, \text{gamma}^b] == (1/2)\ \text{omega}\ \text{SAB}$ (SAB is defined as $(1/4)[\text{gamma}, \text{gamma}]$, so this cannot fail)”.
- [**proved III §17**] “[**has content**] the spinor connection is compatible with omega: $[\text{Gamma}^{\text{spin_mu}}, \text{gamma}^a] + \text{omega_mu}^a{}_b \text{gamma}^b == 0$ ”. The spinor connection rotates the Dirac matrices exactly as ω rotates a vector. With $1/4$ in place of $1/8$, or with the opposite sign, this check fails.
- [**proved III §17**] “[**control**] with the opposite sign the compatibility check FAILS (witnessed): the sign is fixed by the vielbein postulate, not free”.
- [**proved VII §27**] the normalization used throughout Part VII, **SPIN CONNECTION** [**definition**]: $\text{Gamma}^{\text{spin_mu}} == \text{Sum}_{\{a<b\}} \text{omega_mu,ab}\ (1/2)\ \text{T16}[a]\ \text{T16}[b]$.

The explicit matrices [**derived here**], from the table of subsection 9.6 and the formula above. Only the planes $(0,\mu)$ and $(4,\mu)$ contribute. For $j = 1, 2, 3$ the ordered pair $(j, 4)$ gives $\omega_{jj4}T_{16}[j]T_{16}[4] = \omega_{j4j}T_{16}[4]T_{16}[j]$, so

$$\begin{aligned}\Gamma_0 &= \Gamma_4 = 0, \\ \Gamma_j &= \frac{1}{2} \left[\frac{Hc^2}{e^A s^{13/6}} T_{16}[0]T_{16}[j] + \frac{Ha'_4}{e^A s^{1/6}} T_{16}[4]T_{16}[j] \right] \quad (j = 1, 2, 3), \\ \Gamma_k &= \frac{1}{2} \left[-\frac{e^A Hc^2}{s^{13/6}} T_{16}[0]T_{16}[k] + \frac{e^A Ha'_4}{s^{1/6}} T_{16}[4]T_{16}[k] \right] \quad (k = 5, 6, 7).\end{aligned}\tag{58}$$

The provenance script `claude-fable/prov03_covariant_derivative.wls` (log `claude-fable/prov03.log`) prints the non-zero directions of $\text{Gamma}^{\text{spin}}$ as $\{1, 2, 3, 5, 6, 7\}$, each with 32 non-zero entries. That matches the two terms above: each $T_{16}[a]T_{16}[b]$ with $a \neq b$ is a signed permutation matrix with 16 entries, and the two terms occupy different positions.

The direct-sum structure. Every $T_{16}[a]$ is off-diagonal in the type-1/type-2 split, so every product of two of them is block-diagonal: `T16[A].T16[B] = ArrayFlatten[{{taubar[A].tau[B], 0}, {0, tau[A].taubar[B]}}`. The spin connection therefore never mixes the two split-octonion spinors.

- [**proved III §17**] “ $\text{Gamma}^{\text{spin}}[\mu]$ is block diagonal: it never mixes type-1 with type-2”.
- [**proved III §17**] “the type-1 block is $(1/8)\ \text{omega}\ (\text{taubar}^a \tau^b - \text{taubar}^b \tau^a)$ ”.
- [**proved III §17**] “the type-2 block is $(1/8)\ \text{omega}\ (\tau^a \text{taubar}^b - \tau^b \text{taubar}^a)$ ”.

Below, the upper (type-1) block is written $\Gamma_\mu^{(1)}$ and the lower (type-2) block $\Gamma_\mu^{(2)}$. In the notebook these are `GammaSpinType1` and `GammaSpinType2`.

9.8 The gauge-covariant derivative of the 16-component spinor

$$\begin{aligned}D_\mu \Psi &= \partial_\mu \Psi + \Gamma_\mu \Psi, & D_\mu \psi_1 &= \partial_\mu \psi_1 + \Gamma_\mu^{(1)} \psi_1, \\ D_\mu \psi_2 &= \partial_\mu \psi_2 + \Gamma_\mu^{(2)} \psi_2, & \text{Dirac}[\Psi] &= \gamma^\mu D_\mu \Psi.\end{aligned}\tag{59}$$

In the notebook these are `cfDcov16[psi, mu] = D[psi, X[[mu]]] + GammaSpinCanonical[[mu]] . psi, cfDcovType1, cfDcovType2` and `cfDiracOperator[psi] = Sum[gammaCurvedCanonical[[mu]] . cfDcov16[psi, mu], {mu, 8}]`.

- **[proved III §17]** “Dcov16 on the direct sum is the direct sum of Dcov1 and Dcov2” (on a generic spinor of all eight coordinates).

Covariant constancy of γ^μ . The Christoffel symbols act on the curved index, the spinor connection acts by commutator, and the frame connects the two. If any of the three had the wrong sign or index order, the curved Dirac matrices would not be covariantly constant:

$$D_\mu \gamma^\nu = \partial_\mu \gamma^\nu + \Gamma^\nu_{\mu\rho} \gamma^\rho + [\Gamma_\mu, \gamma^\nu] = 0. \quad (60)$$

- **[proved III §17]** “[has content] and therefore the curved Dirac matrices are covariantly constant: `D_mu gammaCurved^nu == 0`”.

The commutator identity. Contract $\mu = \nu$ and use $\Gamma^\mu_{\mu\rho} = \partial_\rho \ln \sqrt{g}$ **[derived here]**:

$$\frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \gamma^\mu) = [\gamma^\mu, \Gamma_\mu]. \quad (61)$$

- **[proved VI §24]** “FABLE [has content]: in general $(1/\text{Sqrt}[g]) \text{d_mu}(\text{Sqrt}[g] \text{gamma}^\mu) == [\text{gamma}^\mu, \text{Gamma}^\text{spin_mu}]$ (covariant constancy of gamma^μ , Section 17)”.

The connection term of the Dirac operator, direction by direction [derived here]. From the explicit Γ_μ of subsection 9.7 and the curved gammas of subsection 9.3, using $T_{16}[j]^2 = +\text{ID16}$, $T_{16}[k]^2 = -\text{ID16}$ and the anticommutation of distinct T_{16} :

$$\begin{aligned} \gamma^j \Gamma_j &= \frac{1}{2q} T_{16}[j] (w_0 T_{16}[0] T_{16}[j] + w_4 T_{16}[4] T_{16}[j]) \\ &= -\frac{1}{2q} (w_0 T_{16}[0] + w_4 T_{16}[4]), \end{aligned} \quad (62)$$

$$\begin{aligned} \gamma^k \Gamma_k &= \frac{1}{2p} T_{16}[k] (v_0 T_{16}[0] T_{16}[k] + v_4 T_{16}[4] T_{16}[k]) \\ &= +\frac{1}{2p} (v_0 T_{16}[0] + v_4 T_{16}[4]), \end{aligned} \quad (63)$$

where w_0, w_4 and v_0, v_4 are the coefficients in Γ_j and Γ_k . Since $w_0/q = Hc^2/s^2 = H \cot^2 6Hx_0$, $w_4/q = Ha'_4$, $v_0/p = -H \cot^2 6Hx_0$ and $v_4/p = Ha'_4$:

μ	$\gamma^\mu \Gamma_\mu$ (no sum)	$\{\gamma^\mu, \Gamma_\mu\}$ (no sum)
0	0	0
$j = 1, 2, 3$	$-\frac{1}{2} H \cot^2(6Hx_0) T_{16}[0] - \frac{1}{2} H a'_4 T_{16}[4]$	0
4	0	0
$k = 5, 6, 7$	$-\frac{1}{2} H \cot^2(6Hx_0) T_{16}[0] + \frac{1}{2} H a'_4 T_{16}[4]$	0

The anticommutator column follows because $\Gamma_j \gamma^j$ equals $+\frac{1}{2q}(w_0 T_{16}[0] + w_4 T_{16}[4])$, the negative of $\gamma^j \Gamma_j$, and similarly for k .

The sum. Summing the table **[derived here]** gives $3(-\frac{1}{2}) + 3(-\frac{1}{2}) = -3$ for the $T_{16}[0]$ terms and $3(-\frac{1}{2}) + 3(+\frac{1}{2}) = 0$ for the $T_{16}[4]$ terms. The observed and hidden a'_4 terms cancel:

$$\gamma^\mu \Gamma_\mu = -3H \cot^2(6Hx_0) T_{16}[0], \quad \sum_\mu \{\gamma^\mu, \Gamma_\mu\} = 0. \quad (64)$$

- [proved VI §24] “FABLE [has content]: the HYPOTHESIS that makes it a product - the anticommutator $\{\gamma^\mu, \Gamma^{\text{spin}_\mu}\} == 0$ on the canonical frame (no totally antisymmetric part of the connection)” (the sum over μ).
- [proved VI §24] “FABLE [has content]: hence $(1/2)(1/\text{Sqrt}[g]) d_\mu(\text{Sqrt}[g] \gamma^\mu) == \gamma^\mu \Gamma^{\text{spin}_\mu}$, and on this geometry $== -3 H \text{Cot}[6 H x_0]^2 T_{16}[0]$ ”.
- [proved VII §27] the per-direction form: every $\gamma^\mu \Gamma_\mu$ is $\alpha_\mu T_{16}[0] + \beta_\mu T_{16}[4]$, with $\sum \alpha_\mu = -3H \cot^2$ and $\sum \beta_\mu = 0$, SPIN CONNECTION [THE RESULT]: direction by direction $\gamma^\mu \Gamma_\mu == \alpha_\mu T_{16}[0] + \beta_\mu T_{16}[4]$; Sum $\alpha_\mu == -3 H \text{Cot}[6 H x_0]^2$ and Sum $\beta_\mu == 0$ (the observed and hidden a4 terms cancel); and the anticommutator vanishes for *each* μ separately, LAGRANGIAN (c) [THE RESULT]: $\text{Lsym} - \text{Lsym}_d == (1/(2H)) \sum_\mu \text{Psibar} \{\gamma^\mu, \Gamma_\mu\} \text{Psi}$ identically, and $\{\gamma^\mu, \Gamma_\mu\} == 0$ for EACH μ on the canonical frame: $\text{Lsym} == \text{Lsym}_d$.

Why the anticommutator vanishes [derived here]. For $a \neq b$, $\{\gamma^c, \gamma^a \gamma^b\} = 2\gamma^a \gamma^b \gamma^c + 2\eta^{ca} \gamma^b - 2\eta^{cb} \gamma^a$. This is $2\gamma^{cab}$ (the antisymmetrized product) when c differs from both a and b , and 0 when $c = a$ or $c = b$. Hence

$$\sum_\mu \{\gamma^\mu, \Gamma_\mu\} = \frac{1}{2} \omega_{[cab]} \gamma^{cab}, \quad \omega_{cab} = e_c^\mu \omega_{\mu ab}, \quad (65)$$

and only the totally antisymmetric part of ω_{cab} survives. For the diagonal canonical frame, c equals the direction μ , and every non-zero $\omega_{\mu ab}$ has μ equal to a or b (the planes $(0, \mu)$, $(4, \mu)$). So the totally antisymmetric part is zero, and it is zero direction by direction.

Contrast with an expanding frame. On the separate 4-dimensional FLRW reference frame of Part VI, the same computation leaves a Hubble term: [proved VI §24] “FLRW [THE RESULT]: $\gamma^\mu \Gamma_\mu == (3/2) (a'/a) \gamma^4$ - the Hubble term of the FLRW Dirac equation”. On the canonical frame the analogous $T_{16}[4]$ terms cancel between the observed sheet and the hidden sheet ($\sum \beta_\mu = 0$), because qp does not depend on x_4 [derived here]: the per-direction coefficient $-\frac{1}{2} H a'_4$ of the table equals $\frac{1}{2} \partial_4 \ln q$, and $+\frac{1}{2} H a'_4$ equals $\frac{1}{2} \partial_4 \ln p$, so the sum is $\frac{3}{2} \partial_4 \ln(qp) = 0$.

On the model’s own spinor Psi16 of Section 11.

- [proved III §17] “[definition] $\text{Dcov}[\mu] \text{Psi16} - \text{D}[\text{Psi16}, x_\mu] == \Gamma^{\text{spin}[\mu]}$. Psi16”.
- [proved III §17] “in the six directions on which Psi16 does not depend, $\text{Dcov} \text{Psi16}$ is purely the connection term”.
- [proved III §17] “the connection term does not vanish on Psi16: the connection matrices themselves are non-zero (witnessed)”.
- [proved VI §24] “FABLE [has content]: the connection term does not vanish on an x_0 -independent spinor - witnessed on the constant spinor e_1 : a spinor homogeneous in everything but x_4 is NOT a solution”.

Section 17 constructs the operator and applies it. It does not re-derive the flat field equations of Part II from a covariant Lagrangian [prose III §17].

9.9 The three bridges of Part IV

The gauge law [prose IV], verified per bridge. Suppose a new frame is `frameNew = frameOld . L` with $L(x)$ invertible on the flat indices. Substituting into the vielbein postulate, with $M = \text{Transpose}[L]$, gives

$$\omega_{\mu}^{\text{new}} = M \omega_{\mu}^{\text{old}} M^{-1} - (\partial_{\mu} M) M^{-1} \quad (\text{mixed components}). \quad (66)$$

The second term is the inhomogeneous gauge term, and it vanishes exactly when L is constant. The law is not assumed anywhere. Each new connection is derived independently by `cfSpinConnection`, and the law is then verified against the result. A frame `frameCanonical . L` with L preserving the flat metric describes the *same geometry*, and its Levi-Civita connection is the canonical one in a different gauge, *not* a new connection [prose IV].

9.9.1 Bridge 1: a locally boosted frame, the same connection in a local gauge (Section 18)

A new object, not in the original notebook, is introduced here: the rapidity $\theta = \text{cfBoostRapidity}[x_0, x_4]$, an arbitrary function. It generates a boost in the flat (0,4) plane:

$$\Lambda[\theta] = \text{ID8} + (\cosh \theta - 1)(E_{00} + E_{44}) + \sinh \theta (E_{04} + E_{40}) = e^{\theta K}, \quad (67)$$

where $K = E_{04} + E_{40}$ is the (0,4) boost generator on flat indices, and `frameBoost = frameCanonical . Lambda`.

- [proved IV §18] “Lambda is in $O(4,4)$: `Lambda . eta . Transpose[Lambda] == eta`”, “Lambda is NOT in $O(8)$: it is a genuine boost, not a rotation”, “`Lambda == MatrixExp[theta K]` with K the (0,4) boost generator”.
- [proved IV §18] “BRIDGE 1 reproduces the SAME curved metric g ”, “BRIDGE 1 frame really is different from the canonical frame”.
- [proved IV §18] “BRIDGE 1: vielbein postulate holds identically”, “BRIDGE 1: `omega[mu,a,b] == -omega[mu,b,a]`”, “BRIDGE 1: torsion still vanishes”.
- [proved IV §18] “COMPARISON 1: `omegaBoost == M omegaCanonical Inverse[M] - D[M,x_mu] Inverse[M]`”, “COMPARISON 1: the WHOLE difference is the inhomogeneous gauge term”, “COMPARISON 1: the difference equals `-D[theta, x_mu] * (boost generator)`”.
- [proved IV §18] “COMPARISON 1: curvature transforms covariantly, `R' == M R Inverse[M]`”, “COMPARISON 1: BRIDGE 1 has the SAME Ricci scalar as the canonical frame”.
- [displayed IV §18] `omegaBoost` has 28 non-zero components. The difference `deltaBoost` has 4: $-\partial_{\mu}\theta K$ for $\mu = 0$ and $\mu = 4$, in the two orderings of the plane (0,4).

At the spinor level, the spin lift is `S = MatrixExp[theta SAB[[1,5]]]`, with no extra factor 1/2, because `SAB` already carries 1/4 [prose IV §18]:

- [proved IV §18] “BRIDGE 1 spin lift: `Inverse[S] . S == ID16`”, “BRIDGE 1 spin lift: `S` preserves the spinor metric, `Transpose[S] . sigma16 . S == sigma16`”, “BRIDGE 1 spin lift [has content]: `Inverse[S] . gamma^a . S == Lambda^a_b gamma^b` (S is the spin lift of Λ)”.
- [proved IV §18] “COMPARISON 1 at the spinor level [THE GAUGE LAW]: `Gamma'_mu == S Gamma_mu Inverse[S] - D[S,x_mu] Inverse[S]`”, “COMPARISON 1 at the spinor level: the inhomogeneous term is `-D[theta,x_mu] SAB[[1,5]]`, the spin image of `-D[theta,x_mu] K`”, “COMPARISON 1 at the spinor level: and that is exactly `cfSpinMatrix` applied to the vector difference `deltaBoost`”.

- [proved IV §18] “BRIDGE 1 [has content]: $D'_{\mu}(S\psi) == S D_{\mu}\psi$ for a generic spinor - one covariant derivative, two gauges”, “BRIDGE 1: the curved Dirac matrices of the boosted frame are $S \gamma^{\mu} \text{Inverse}[S]$ ”, “BRIDGE 1: so the Dirac operator is one operator in two gauges, $\text{Dirac}'[S\psi] == S \text{Dirac}[\psi]$ ”.

A solution of the Dirac equation in one gauge is therefore a solution in the other, after $\psi \rightarrow S\psi$. Part V identifies the inhomogeneous term as the fable-5.1 connection of the boosted frame (subsection 9.10).

9.9.2 Bridge 2: the null (light-cone) frame, the same connection in a constant gauge (Section 19)

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} \text{ID4} & \text{ID4} \\ \text{ID4} & -\text{ID4} \end{pmatrix}, \quad U = U^T, \quad U \cdot U = \text{ID8}, \quad (68)$$

with $\text{etaNull} = \text{Transpose}[U] \cdot \text{eta4488} \cdot U$ and $\text{frameNull} = \text{frameCanonical} \cdot U$. The eight flat directions are paired as (0, 4), (1, 5), (2, 6) and (3, 7), one spacelike and one timelike in each pair, and each pair is replaced by its two null combinations [prose IV §19].

- [proved IV §19] “U is an involution: $U \cdot U == \text{ID8}$ ”, “U is symmetric”, “THE NULL TANGENT METRIC IS EXACTLY THE SPINOR METRIC σ ”, “etaNull is symmetric and squares to ID8”, “all eight frame directions are null: etaNull has zero diagonal”. The identity $\text{etaNull} == \sigma$ is a statement about the flat metric, not about the connection.
- [proved IV §19] “BRIDGE 2 reproduces the SAME curved metric g ”, “BRIDGE 2: vielbein postulate holds identically”, “BRIDGE 2: $\omega_{[\mu,a,b]} == -\omega_{[\mu,b,a]}$ with indices lowered by etaNull”, “BRIDGE 2: torsion vanishes”.
- [proved IV §19] “COMPARISON 2: $\omega_{\text{Null}} == U \cdot \omega_{\text{Canonical}} \cdot U$ exactly”, “COMPARISON 2: the inhomogeneous gauge term is ZERO because U is constant”, “COMPARISON 2: $R_{\text{Null}} == U \cdot R_{\text{Canonical}} \cdot U$ ”, “COMPARISON 2: BRIDGE 2 has the SAME Ricci scalar as the canonical frame”.
- [displayed IV §19] ω_{Null} has 48 non-zero components. A constant change of basis spreads the canonical 24 over more entries, and the difference from $U\omega U$ is zero.

9.9.3 Bridge 3: triality frame with totally antisymmetric split-octonion torsion, a different connection (Section 20)

The flat index is a *type-1 split-octonion spinor index*, reached through the constant triality bridge triVecToSpin of Section 9. [proved I §9] “THE TRIALITY METRIC IS EXACTLY THE SPINOR METRIC: $\text{etaTri} == \sigma$ ”. The frame is $\text{frameTri} = \text{frameCanonical} \cdot \text{triVecToSpin}$, with flat tangent metric σ , and the connection is

$$\omega_{\mu ab}^{\text{Oct}} = \omega_{\mu ab}^{\text{TriLC}} + \lambda m_{abc}^{\text{Skew}} (e_{\text{Tri}})_{\mu}^c, \quad (e_{\text{Tri}})_{\mu}^c = \text{frameTri}[[\mu, c]]. \quad (69)$$

Here ω_{TriLC} is the Levi-Civita connection in the triality frame. $m^{\text{Skew}} = m^{\text{SkewSpin}}$ is the totally antisymmetrized split-octonion structure constant m^{Lower} , transported to the spinor basis. $\lambda = \text{lambdaOct}$ is a free real torsion-strength parameter introduced in Section 20, not in the original notebook.

- [proved IV §20] “the triality bridge is Euclidean-orthogonal: $\text{Transpose}[P] == \text{Inverse}[P]$ ”, “but it is NOT in $O(4,4)$: $P \cdot \text{eta4488} \cdot \text{Transpose}[P] != \text{eta4488}$ (it carries eta4488 to σ)”, “BRIDGE 3 reproduces the SAME curved metric g ”, “the triality tangent metric is σ ”.

- [proved IV §20] “the antisymmetrized structure constants are totally antisymmetric”, “they agree with mLower on the seven imaginary directions”, “mSkewSpin is totally antisymmetric” (48 non-zero components [displayed IV §20]).
- [proved IV §20] “BRIDGE 3 (lambda=0): vielbein postulate holds identically”, “BRIDGE 3 (lambda=0): omega[mu,a,b] == -omega[mu,b,a]”, “BRIDGE 3 (lambda=0) is the constant triality conjugate of the canonical connection”.
- [proved IV §20] “BRIDGE 3: omegaOct[mu,a,b] == -omegaOct[mu,b,a] (still metric compatible)”, “BRIDGE 3: at lambda = 0 it reduces to the Levi-Civita connection”.
- [proved IV §20] “BRIDGE 3: torsion vanishes at lambda = 0”, “BRIDGE 3: torsion is NOT zero for lambda != 0”, “BRIDGE 3: the flat torsion T[a,b,c] is TOTALLY ANTISYMMETRIC”, “BRIDGE 3: T[a,b,c] == -2 lambda mSkewSpin[a,b,c]”.
- [proved IV §20] “COMPARISON 3: omegaOct - (triality conjugate of omegaCanonical) == contortion”, “COMPARISON 3: the difference is linear in lambda and vanishes at lambda = 0”. The difference is a *tensor* (a contortion), so this is a different connection.

Its curvature is quadratic in λ . With K the contortion, $R(\omega^{\text{Oct}}) = R(\omega^{\text{TriLC}}) + \lambda(dK + [\omega^{\text{TriLC}}, K]) + \lambda^2[K, K]$:

- [proved IV §20] “BRIDGE 3: at lambda = 0 the curvature is the triality conjugate of the canonical curvature”, “BRIDGE 3 [has content]: the curvature DOES depend on lambda (witnessed)”, “BRIDGE 3: the lambda-dependence is a polynomial of degree exactly two”.
- [proved IV §20] “BRIDGE 3: the lambda¹ term is the Levi-Civita covariant exterior derivative of the contortion, $dK + [\omega^{\text{TriLC}}, K]$ ”. Written out, it is $\partial_\mu K_\nu - \partial_\nu K_\mu + [\omega_\mu^{\text{TriLC}}, K_\nu] - [\omega_\nu^{\text{TriLC}}, K_\mu]$.
- [proved IV §20] “BRIDGE 3: the lambda² term is $[K_\mu, K_\nu]$, the contortion commuted with itself”.

The Ricci shift. $-42\lambda^2 = -\frac{1}{4}T_{abc}T^{abc}$:

- [proved IV §20] “BRIDGE 3: at lambda = 0 the Ricci scalar is the canonical one”.
- [proved IV §20] “BRIDGE 3 [THE RESULT]: $R(\omega^{\text{Oct}}) - R(\text{canonical}) == -(1/4) T_{\{abc\}} T^{\{abc\}}$, a constant shift set by the torsion alone”.
- [proved IV §20] “BRIDGE 3: equivalently $-\lambda^2 (m\text{Skew} \cdot m\text{Skew})$, with $m\text{Skew} \cdot m\text{Skew} == 42$, so the shift is exactly $-42 \lambda^2$ ”. Check [derived here]: $T = -2\lambda m^{\text{Skew}}$ gives $T \cdot T = 4\lambda^2 \cdot 42 = 168\lambda^2$, and $\frac{1}{4} \cdot 168 = 42$.

This connection resembles the Cartan–Schouten connections of a Lie group, but it differs in two ways that must not be assumed away [prose IV §20]. It is not flat. And the octonion structure constants violate the Jacobi identity, so they cannot be the anholonomy of any frame.

Spinor level. cfSpinMatrix is correct only for eta4488 vector indices. Bridge 3’s flat index is a triality-spinor index, so the matching Dirac matrices are $T16\text{Tri}[a] = \text{Sum_A triSpinToVec}[[a,A]] T16[A-1]$, and they are contracted by cfSpinMatrixTri [prose IV §20]:

- [proved IV §20] “THE TRIALITY DIRAC MATRICES OBEY THE sigma CLIFFORD RELATION: {T16Tri[a],T16Tri[b]} == 2 sigma[[a,b]] ID16”, “they are NOT the vector-frame Dirac matrices, so the distinction is real”.
- [proved IV §20] “at lambda = 0 the Bridge 3 spinor connection equals the canonical one”, “the extra spinor coupling vanishes at lambda = 0”, “the extra spinor coupling is non-zero for lambda != 0”, “the extra spinor coupling is still block diagonal (type-1 + type-2 preserved)”.
- The extra coupling (lambda/8) mSkew[a,b,c] frameTri[[mu,c]] [T16Tri[a], T16Tri[b]] is *quadratic* in the gammas. The third index is contracted with the frame, not with a third gamma. The cubic Einstein–Cartan axial term would appear only after contracting with γ^μ in the Dirac operator, and the notebook does *not* carry that contraction out [prose IV §20]. The extra coupling has 160 non-zero entries [displayed IV §20].

9.10 The fable-5.1 bridge of Part V: the Weitzenböck (teleparallel) connection

The fable-5.1 bridge keeps the metric, the frame, eta4488 and the bridge [d] unchanged, and changes only the rule for parallel transport. It declares the canonical frame itself to be parallel [prose V §21]:

$$(\Gamma^W)^\rho_{\mu\nu} = e_a^\rho \partial_\mu e_\nu^a, \quad (70)$$

in the notebook GammaWeitzenboeck[[rho, mu, nu]] = Sum_a coframe[[a, rho]] D[frame[[nu, a]], x[mu]].

- [proved V §21] “FABLE-5.1 [definition]: the frame is parallel - nabla^W e == 0 for all eight frame vectors”, “FABLE-5.1 [has content]: Gamma^W is metric compatible, nabla^W g == 0”, “FABLE-5.1 [has content]: Gamma^W is NOT symmetric in its lower indices - it has torsion”, “FABLE-5.1 [has content]: Gamma^W is NOT the Levi-Civita connection”.
- Γ^W has 13 non-zero components [displayed V §22].

The spin connection is zero, and so is the curvature. Γ^W is fed to the same solver. Because the first two terms of the vielbein postulate then cancel by construction, the solver returns $\omega = 0$:

- [proved V §21] “FABLE-5.1 [solver regression]: vielbein postulate residual is zero”, “FABLE-5.1 [THE RESULT]: the spin connection in the canonical frame is IDENTICALLY ZERO”, “FABLE-5.1: trivially antisymmetric, so trivially metric compatible”.
- [proved V §21] “FABLE-5.1 [has content]: the torsion is NOT zero”, “FABLE-5.1: $T^{\rho}_{\mu\nu} == \Gamma^W[\rho, \mu, \nu] - \Gamma^W[\rho, \nu, \mu]$ (torsion is the antisymmetric part of Gamma^W)”. With $\omega = 0$ the torsion is $T^a_{\mu\nu} = \partial_\mu e_\nu^a - \partial_\nu e_\mu^a$.
- [proved V §21] “FABLE-5.1 [THE RESULT]: the curvature is IDENTICALLY ZERO - a flat connection on a curved metric”.

The canonical connection is minus the contortion of this torsion. The contortion is $K = \Gamma^W - \Gamma^{LC}$, and in terms of the torsion $K_{\rho\mu\nu} = \frac{1}{2}(T_{\rho\mu\nu} + T_{\mu\rho\nu} + T_{\nu\rho\mu})$ (all indices down, first index then raised). Substituting $\Gamma^{LC} = \Gamma^W - K$ into the vielbein postulate leaves $\text{omegaCanonical}[\mu, a, b] = -K[\rho, \mu, \nu] e[\rho, a] \text{coframe}[[b, \nu]]$ [prose V §21].

- [proved V §21] “COMPARISON F [has content]: $\Gamma^W - \Gamma^{LC} == (1/2)(T_{\{rmn\}} + T_{\{mrn\}} + T_{\{nrm\}})$ with the index raised, the contortion formula”.
- [proved V §21] “COMPARISON F [THE RESULT]: $\omega\text{Canonical} == -\text{contortion}(\text{fable-5.1 torsion})$, pulled back to flat indices, exactly”.
- [proved V §21] “COMPARISON F: so $\omega\text{Canonical} - \omega\text{Fable51}$ is a TENSOR, not a gauge term ($\omega\text{Fable51} == 0$)”.
- contortionFable51 has 24 non-zero components, the same 24 as $\omega\text{Canonical}$. $\text{torsionFable51Flat}$ has 24, and RiemannFable51 has 0 [displayed V §22].

The kind of torsion. A torsion tensor splits into a vector part, an axial part and a tensor part:

- [proved V §21] “FABLE-5.1 torsion vector $T[\mu] = T[\text{nu}, \text{nu}, \mu]$ is a GRADIENT: $T[\mu] == D[\text{Log}[\text{Sin}[6 H x_0]], x[\mu]]$ ”.
- [proved V §21] “FABLE-5.1 torsion vector has exactly one non-zero component, $6 H \text{Cot}[6 H x_0]$ along x_0 ”.
- [proved V §21] “FABLE-5.1: the AXIAL (totally antisymmetric) part of the torsion vanishes - the complement of Bridge 3”, “FABLE-5.1: the torsion itself does not vanish, so the tensor part carries the rest”.

For any diagonal frame the torsion vector is, component by component (no sum) [derived here],

$$T_\mu = \sum_\nu e_a{}^\nu (\partial_\nu e_\mu{}^a - \partial_\mu e_\nu{}^a) = \partial_\mu \ln \frac{e_\mu{}^\mu}{\det e}. \quad (71)$$

On the canonical frame this is $\partial_0 \ln(\tan 6Hx_0 / \sec 6Hx_0) = \partial_0 \ln \sin 6Hx_0 = 6H \cot 6Hx_0$ for $\mu = 0$. It is zero for $\mu = 4$, because $e_4{}^4 = 1$ and $\det e = \sec 6Hx_0$ do not depend on x_4 . It is zero for $\mu = 1, 2, 3, 5, 6, 7$, because nothing depends on those coordinates. This agrees with the assertion.

$R = -T + B$ in closed form. With the torsion scalar $T = \frac{1}{4}T^{\rho\mu\nu}T_{\rho\mu\nu} + \frac{1}{2}T^{\rho\mu\nu}T_{\nu\mu\rho} - T^\mu T_\mu$ and the boundary term $B = \frac{2}{\sqrt{\det g}} \partial_\mu (\sqrt{\det g} T^\mu)$, where $\sqrt{\det g} = \sec 6Hx_0$:

$$R = -3H^2 \left((55 + 7 \cos 12Hx_0) \cot^2(6Hx_0) \csc^2(6Hx_0) - 2a_4'^2 \right), \quad (72)$$

$$T = -6H^2 (5 \cot^4(6Hx_0) + a_4'^2), \quad (73)$$

$$B = -36H^2 (5 + \cos 12Hx_0) \cot^2(6Hx_0) \csc^2(6Hx_0). \quad (74)$$

These closed forms are [displayed V §21] ($\text{RicciScalarCanonical}$, $\text{torsionScalarFable51}$, $\text{boundaryTermFable51}$). The following are [proved V §21]: “COMPARISON F [THE RESULT]: $R == -T + B$ - the teleparallel equivalent of GR, in closed form for this geometry”, “FABLE-5.1: T is not zero and B is not zero - neither side is trivial”. The Einstein–Hilbert and teleparallel actions therefore differ by a boundary term on this geometry.

The spinor: one vector term, removed by a rescaling. In the fable-5.1 gauge the spinor connection is zero, so the Dirac operator is $\gamma^\mu \partial_\mu$. The canonical Dirac operator carries in addition $\gamma^\mu \Gamma_\mu$, and this whole term is one vector term:

$$\gamma^\mu \Gamma_\mu = -\frac{1}{2} T_\mu \gamma^\mu = -\frac{1}{2} \partial_\mu (\ln \sin 6Hx_0) \gamma^\mu$$

$$= -3H \cot^2(6Hx_0) T_{16}[0], \quad (75)$$

$$\text{Dirac}_{LC}[\sqrt{\sin 6Hx_0} \Psi'] = \sqrt{\sin 6Hx_0} \gamma^\mu \partial_\mu \Psi' \quad \text{for every } \Psi'. \quad (76)$$

The first line checks against subsection 9.8 **[derived here]**: $-\frac{1}{2}(6H \cot)(\cot T_{16}[0]) = -3H \cot^2 T_{16}[0]$, because $\gamma^0 = \cot T_{16}[0]$. The matrix $\gamma^\mu \Gamma_\mu$ is $-3H \cot^2$ times the permutation matrix $T_{16}[0]$, so it has 16 non-zero entries.

- **[proved V §21]** “FABLE-5.1 [THE RESULT]: `gamma^mu Gamma^spin[mu] == -(1/2) T[mu] gamma^mu, one vector term`”, “FABLE-5.1: equivalently `-(1/2) D[Log[Sin[6 H x0]], x_mu] gamma^mu`”.
- **[proved V §21]** “FABLE-5.1 [THE RESULT]: `DiracCanonical[ Sqrt[Sin[6Hx0]] Psi' ] == Sqrt[Sin[6Hx0]] DiracFable51[Psi']`, for a generic `Psi'`”, “FABLE-5.1: the rescaling is NOT a no-op - the two Dirac operators differ on the constant spinor”.
- **[proved VI §24]** with a potential: “FABLE [THE RESULT]: `(gamma^mu D_mu - H V'(s)) [Sqrt[Sin[6 H x0]] Psi'] == Sqrt[Sin[6 H x0]] (gamma^mu d_mu Psi' - H V'(s) Psi')` for a `GENERIC Psi'(x0, x4)`: the rescaling removes the connection term exactly”.

The rescaling works because the torsion vector is a gradient. A vector-torsion term that is a gradient is removed by multiplying the spinor by $\exp(\frac{1}{2} \ln \sin 6Hx_0) = \sqrt{\sin 6Hx_0}$ **[derived here]**.

For the real spinor of Parts II–VI the connection drops out of the Lagrangian.

`sigma16` . `T16[a]` is antisymmetric, so $\Psi^T \sigma_{16} T_{16}[a] \Psi = 0$ for real commuting Ψ . Since $\gamma^\mu \Gamma_\mu$ is a multiple of γ^μ , the connection contributes nothing to that Lagrangian:

- **[proved V §21]** “FABLE-5.1 [THE RESULT]: `Psi^T sigma16 gamma^mu Gamma^spin[mu] Psi == 0` for EVERY `Psi` - the spin connection drops out of the Lagrangian”, “FABLE-5.1: the reason is Section 11’s identity, `Psi^T sigma16 T16[a] Psi == 0` (`sigma16 T16[a]` antisymmetric)”, “FABLE-5.1: but the connection term does NOT drop out of the Dirac OPERATOR (it is `-(1/2) T[mu] gamma^mu, non-zero`)”.
- **[proved V §21]** “FABLE-5.1: the covariant Lagrangian with frame `-> ID8` and volume factor `-> 1` IS the author’s La of Section 11, term for term”, “FABLE-5.1: and the covariant Lagrangian has NO connection term to omit - its kinetic term is already the fable-5.1 one”. The author’s flat La omits only the volume factor $\sec 6Hx_0$ and the coframe factors inside γ^μ **[prose V §21]**.

This mechanism is specific to a *real* field. subsection 9.12 shows that it does not carry over to the complex fermion fable.

Frame covariance: Bridge 1 explained. In Bridge 1’s boosted frame, the same Γ^W fed to the same solver gives a fable-5.1 spin connection that is pure gauge:

- **[proved V §21]** “FABLE-5.1 in the boosted frame: vielbein postulate residual is zero”, “FABLE-5.1 in the boosted frame [THE RESULT]: `omega == -D[M,x_mu] Inverse[M]`, pure gauge and nothing else”, “FABLE-5.1 in the boosted frame: the lemma’s prediction itself satisfies the vielbein postulate with `Gamma^W`”, “FABLE-5.1 in the boosted frame: the curvature is STILL identically zero”.
- **[proved V §21]** “and Bridge 1’s inhomogeneous term of Section 18 IS this object: `deltaBoost == omegaFable51Boost`”, “so `omegaBoost == (canonical,`

conjugated) + (fable-5.1 in the boosted frame)”.

- `omegaFable51BoostMixed` has 4 non-zero components, the same 4 as `deltaBoost` [displayed V §21].

9.11 Master comparison of the five connections (Section 22)

Section 22 prints two tables: the summary `cfConnectionSummary`, and a grid of the comparisons side by side [displayed V §22]. The table below is `cfConnectionSummary`, with three entries of the side-by-side grid added to the column “difference from canonical”: Bridge 1’s spinor-level law, and Bridge 3’s curvature and Ricci shift. Part VII restates the bridge results it relies on, so that it can be read alone: [proved VII §27] SPIN CONNECTION [has content]: Parts IV–V restated – Bridge 1’s difference is the pure gauge term, Bridge 2 is a constant conjugate, Bridge 3 has totally antisymmetric torsion that is not zero and SPIN CONNECTION [has content]: Part V restated – $\omega^W = 0$ in the canonical frame, $\omega^{LC} = -\text{contortion}$, $T_\mu = d_\mu \text{Log}[\text{Sin}[6 H x_0]]$, $R = -T + B$, and $\gamma^\mu \Gamma_\mu = -(1/2) T_\mu \gamma^\mu$. For all five connections the curved metric g is the same (“yes (reference)” for the canonical one), and the connection is metric compatible; those two columns of the notebook’s table read “yes” throughout and are not repeated below.

connection; frame field; flat metric	torsion zero? curvature zero?	difference from canonical	verdict
<code>omegaCanonical;</code> <code>frameCanonical</code> (diagonal); <code>eta4488</code>	torsion: yes; curvature: no	— (reference)	Levi-Civita
<code>omegaBoost</code> (Bridge 1); <code>frameCanonical</code> . <code>Lambda(x)</code> ; <code>eta4488</code>	torsion: yes; curvature: no	gauge term $-\partial_\mu \theta K$, which is the fable-5.1 connection of the boosted frame; at spinor level $\Gamma' = STS^{-1} - (\partial S)S^{-1}$ with the explicit spin lift S	SAME connection, local gauge
<code>omegaNull</code> (Bridge 2); <code>frameCanonical</code> . <code>U</code> ; <code>etaNull == sigma</code>	torsion: yes; curvature: no	constant similarity $U\omega U$	SAME connection, constant gauge
<code>omegaOct</code> (Bridge 3); <code>frameCanonical</code> . <code>triVecToSpin</code> ; <code>etaTri == sigma</code>	torsion: no , axial torsion $-2\lambda m^{\text{Skew}}$; curvature: no	contortion $\lambda m_{abc}^{\text{Skew}} (e_{\text{Tri}})_\mu{}^c$ (with $e_{\text{Tri}} = \text{frameTri}$), a tensor; curvature quadratic in λ ; Ricci shift $-42\lambda^2$	DIFFERENT connection
<code>omegaFable51</code> (fable-5.1); <code>frameCanonical</code> (parallel); <code>eta4488</code>	torsion: no , vector + tensor torsion, axial part zero; curvature: yes , flat	$\omega^{\text{canonical}} = -\text{contortion}(T_{\text{fable51}})$, a tensor; $\omega^{\text{fable51}} = 0$	DIFFERENT connection

Non-zero component counts (Section 22 fingerprint table and the counts printed in Sections 16–21, [displayed]):

object	non-zero components
GammaCanonical (Christoffel)	37
GammaWeitzenboeck	13
omegaCanonical	24
omegaBoost / its difference deltaBoost	28 / 4
omegaNull / its difference	48 / 0
omegaTriLC (Bridge 3, $\lambda = 0$)	84
contortionOct (Bridge 3)	78
torsionOctFlat (Bridge 3)	48
omegaFable51 (canonical frame)	0
contortionFable51 / torsionFable51Flat	24 / 24
RiemannCanonical	156
RiemannFable51	0

9.12 What the canonical connection does for the complex fermion fable

Everything in this subsection concerns the *complex* 16-component fermion fable of Part VII. Two kinds of statement appear. Some are consequences of results already proved in Parts I–VI, and the derivation is shown here. The others are assertions of the new Part VII, marked [**proved VII §n**], with the label quoted verbatim from `claude-fable/run_fermion_fable_part7.log`.

9.12.1 The field and the symmetrized Lagrangian

Fermion fable is a complex 16-component Grassmann spinor Ψ . It is the minimal field that uses the author’s σ_{16} conjugation, propagates, admits a potential $V(s)$, and reduces to Part VI’s real field as its real commuting shadow. It is not the unique field with dynamics. This is the conclusion of Section 26 as a whole (section 3.8); no single assertion states it. It rests on [**proved VII §26**] REFINEMENT (d) [THE RESULT]: `s = Psibar Psi == i theta1^T sigma16 theta2: NON-ZERO (16 monomials) and HERMITIAN; p = Psibar T16[8] Psi` is non-zero and Hermitian too, REFINEMENT (b) [THE RESULT]: Majorana with C+: the kinetic term is genuine - Euler-Lagrange expression 2 C+ T16[a] phi with C+ T16[a] invertible - but every invariant bilinear vanishes (massless, no $V(s)$), REFINEMENT (c) [THE RESULT]: Weyl and Majorana-Weyl: `PL C T16[a] PL == PR C T16[a] PR == 0` for $C = C-, C+$ and every a (C chirality-diagonal, $T16[a]$ chirality-odd) - no kinetic term and FIDELITY [1] [fidelity]: `Psi -> psi, Psibar -> psi^T sigma16` turns the symmetrized covariant $Lhat$ into Part VI’s `cflhatFable` EXACTLY (canonical frame, generic V), among the others quoted in section 3. Its Dirac conjugate and scalar bilinear are

$$\bar{\Psi} = \Psi^\dagger \sigma_{16}, \quad s = \bar{\Psi} \Psi = -\psi_1^\dagger \sigma \psi_1 + \psi_2^\dagger \sigma \psi_2, \quad (77)$$

where † is the classical conjugate transpose. The second form follows from $\sigma_{16} = \text{diag}(-\sigma, \sigma)$. In the quantized theory the Hilbert adjoint is $\Psi^\dagger := \Psi^\dagger J$ (subsection 9.12.6).

The only Lagrangian used is the *symmetrized covariant* one:

$$L = \sqrt{g} \hat{L}, \quad \hat{L} = \frac{1}{2H} \left[\bar{\Psi} \gamma^\mu D_\mu \Psi - (D_\mu \bar{\Psi}) \gamma^\mu \Psi \right] - V(s), \quad (78)$$

$$D_\mu \Psi = \partial_\mu \Psi + \Gamma_\mu \Psi, \quad D_\mu \bar{\Psi} = \partial_\mu \bar{\Psi} - \bar{\Psi} \Gamma_\mu.$$

$D_\mu \bar{\Psi}$ is the Dirac conjugate of $D_\mu \Psi$, and $\hat{L}$ is real, because $\sigma_{16} \gamma^a$ and $\sigma_{16} \Gamma_\mu$ are *real antisymmetric* matrices. That rests on [**proved I §5**] “`sigma16 . T16[A]` is antisymmetric for $A = 0..7$ ” and “`sigma16 . SAB` is antisymmetric”, together with $\Gamma_\mu = \frac{1}{2} \omega_{\mu ab} SAB$ with real ω (subsection 9.7). [**proved VII §27**] LAGRANGIAN (a) [definition]: `every Gamma^spin_mu` is real and `sigma16 Gamma_mu` is antisymmetric (`Gamma_mu^T sigma16 == -sigma16 Gamma_mu`): `D_mu Psibar` is the Dirac conjugate of `D_mu Psi`.

9.12.2 It drops out of the symmetrized Lagrangian because the matrix $\{\gamma^\mu, \Gamma_\mu\}$ vanishes

Expanding D in (78) gives [derived here]

$$\hat{L}(D) = \hat{L}(\partial) + \frac{1}{2H} \bar{\Psi} \left(\sum_\mu \{\gamma^\mu, \Gamma_\mu\} \right) \Psi. \quad (79)$$

The connection terms are $\bar{\Psi} \gamma^\mu \Gamma_\mu \Psi$ from the first product and $+\bar{\Psi} \Gamma_\mu \gamma^\mu \Psi$ from $-(D_\mu \bar{\Psi}) \gamma^\mu \Psi$. On the canonical frame $\{\gamma^\mu, \Gamma_\mu\} = 0$ for each μ (subsection 9.8), so $\hat{L}(D) = \hat{L}(\partial)$ identically. That is, “covariant = partial” holds for the symmetrized complex Lagrangian on this frame. It holds on any frame where the totally antisymmetric part $\omega_{[cab]}$ vanishes (65), and fails in general.

- [proved VII §27] LAGRANGIAN (c) [THE RESULT]: Lsym - Lsym_d == (1/(2H)) Sum_mu Psibar {gamma^mu, Gamma_mu} Psi identically, and {gamma^mu, Gamma_mu} == 0 for EACH mu on the canonical frame: Lsym == Lsym_d.

This is *not* Part VI’s mechanism. For the real commuting field, $\Psi^T \sigma_{16} \gamma^\mu \Gamma_\mu \Psi$ vanishes because $\sigma_{16} T_{16}[0]$ is antisymmetric (subsection 9.10). For a complex field, $\bar{\Psi} T_{16}[0] \Psi = \Psi^\dagger (\sigma_{16} T_{16}[0]) \Psi$ with $\sigma_{16} T_{16}[0]$ real antisymmetric is i times the Hermitian form of $-i \sigma_{16} T_{16}[0]$, which is not zero in general. So the drop-out for the complex field is a property of the *matrix* $\{\gamma^\mu, \Gamma_\mu\}$, not of a bilinear identity [derived here]. The other Lagrangians behave differently:

- The *unsymmetrized covariant* Lagrangian $\frac{1}{H} \bar{\Psi} \gamma^\mu D_\mu \Psi - V$ equals the symmetrized one plus $\frac{1}{2H} \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \bar{\Psi} \gamma^\mu \Psi)$, a total divergence. Its connection term is $\frac{1}{H} \bar{\Psi} \gamma^\mu \Gamma_\mu \Psi = -3 \cot^2(6Hx_0) \bar{\Psi} T_{16}[0] \Psi$, which is not zero. [proved VII §27] LAGRANGIAN (d) [has content]: Lun - Lsym == (1/(2H)) (1/Sqrt[g]) d_mu(Sqrt[g] Psibar gamma^mu Psi), a total divergence, LAGRANGIAN (e) [has content]: Lun - Lun_d == (1/H) Psibar gamma^mu Gamma_mu Psi == -3 Cot[6 H x0]^2 Psibar T16[0] Psi, LAGRANGIAN (e) [control]: that term is NOT zero for a complex field (witness u = e1, v = e13), and IS zero in the real commuting limit (the scope of Parts V-VI).
- The *unsymmetrized partial* Lagrangian $\frac{1}{H} \bar{\Psi} \gamma^\mu \partial_\mu \Psi - V$ is *inadmissible*. Its $\bar{\Psi}$ and Ψ equations are not Dirac conjugates of each other: they differ by $\bar{\Psi} \sum_\mu \{\gamma^\mu, \Gamma_\mu\} = 2 \bar{\Psi} \gamma^\mu \Gamma_\mu = -6H \cot^2(6Hx_0) \bar{\Psi} T_{16}[0]$. [proved VII §27] LAGRANGIAN (f) [THE RESULT]: Lun_d varied with respect to Psibar gives (Sqrt[g]/H)(gamma^mu d_mu Psi - H V' Psi): the connection term is MISSING, LAGRANGIAN (f) [THE RESULT]: varied with respect to Psi it gives -(Sqrt[g]/H)((d_mu Psibar) gamma^mu + Psibar [gamma^mu, Gamma_mu] + H V' Psibar), with Psibar [gamma^mu, Gamma_mu] == 2 Psibar gamma^mu Gamma_mu: the connection term is PRESENT, LAGRANGIAN (f) [control]: so Lun_d is INADMISSIBLE - its two field equations are not Dirac conjugates: they differ by 2 Psibar gamma^mu Gamma_mu = -6 H Cot^2 Psibar T16[0], non-zero (witness u = e1, v = 0, V = s^2).

9.12.3 It enters the field equation as $-3H \cot^2 T_{16}[0]$, and the rescaling removes it

Varying $\bar{\Psi}$ and Ψ independently in the symmetrized action gives the field equation and its conjugate:

$$\gamma^\mu D_\mu \Psi = H V'(s) \Psi, \quad (D_\mu \bar{\Psi}) \gamma^\mu = -H V'(s) \bar{\Psi}. \quad (80)$$

[proved VII §27] FIELD EQUATION [THE RESULT]: EL_Psibar == (Sqrt[g]/H)(gamma^mu D_mu Psi - H V'(s) Psi) for generic Psi, Psibar of all eight coordinates: the field equation is gamma^mu D_mu Psi == H V'(s) Psi, FIELD EQUATION [THE RESULT]: EL_Psi == -(Sqrt[g]/H)((D_mu Psibar) gamma^mu + H V'(s) Psibar): the adjoint equation is (D_mu Psibar) gamma^mu == -H V'(s) Psibar. For the real field of Part VI the

same equation is [proved VI §24] “FABLE [THE RESULT]: $EL == \text{Sqrt}[g] (2/H) \text{sigma16}$. ($\gamma^\mu D_\mu \Psi - H V'(s) \Psi$): the field equation of fable is EXACTLY the Levi-Civita covariant Dirac equation $\gamma^\mu D_\mu \Psi == H V'(s) \Psi$ ”.

On the canonical frame the connection enters as follows [derived here], from (64). In the equation it is $\gamma^\mu \Gamma_\mu \Psi = -3H \cot^2(6Hx_0) T_{16}[0] \Psi$. In the conjugate it is $-\bar{\Psi} \Gamma_\mu \gamma^\mu = +\bar{\Psi} \gamma^\mu \Gamma_\mu = -3H \cot^2(6Hx_0) \bar{\Psi} T_{16}[0]$, using the per- μ anticommutator. Because $T_{16}[0] = \begin{pmatrix} 0 & \text{ID}_8 \\ \text{ID}_8 & 0 \end{pmatrix}$, in the 8+8 split-octonion form $\Psi = (\psi_1, \psi_2)$:

$$e_a^\mu \bar{\tau}_a (\partial_\mu + \Gamma_\mu^{(2)}) \psi_2 = H V'(s) \psi_1 \quad (\text{upper 8 rows}), \quad (81)$$

$$e_a^\mu \tau_a (\partial_\mu + \Gamma_\mu^{(1)}) \psi_1 = H V'(s) \psi_2 \quad (\text{lower 8 rows}), \quad (82)$$

where the connection term is $-3H \cot^2(6Hx_0) \psi_2$ in the upper rows and $-3H \cot^2(6Hx_0) \psi_1$ in the lower rows, and $\tau_a = \text{tau}[a]$, $\bar{\tau}_a = \text{taubar}[a]$, $s = -\psi_1^\dagger \sigma \psi_1 + \psi_2^\dagger \sigma \psi_2$, $\bar{\psi}_1 = -\psi_1^\dagger \sigma$ and $\bar{\psi}_2 = \psi_2^\dagger \sigma$. The conjugate equations are $(\partial \bar{\psi}_2 - \bar{\psi}_2 \Gamma^{(2)}) \tau_a e_a^\mu = -H V' \bar{\psi}_1$ and $(\partial \bar{\psi}_1 - \bar{\psi}_1 \Gamma^{(1)}) \bar{\tau}_a e_a^\mu = -H V' \bar{\psi}_2$. [proved VII §27] WRITTEN OUT (ii) [THE RESULT]: the split-octonion 8+8 form - the type-1 rows are the taubar equations for ψ_2 , the type-2 rows the tau equations for ψ_1 ($\text{tau}[0] == \text{taubar}[0] == \text{ID}_8$), WRITTEN OUT (ii) [THE RESULT]: the covariant 8+8 form - rows 1..8 are $e_a^\mu \text{taubar}[a] (d_\mu + \Gamma_\mu^{(2)}) \psi_2 - H V' \psi_1$, rows 9..16 are $e_a^\mu \text{tau}[a] (d_\mu + \Gamma_\mu^{(1)}) \psi_1 - H V' \psi_2$; and the conjugate rows, WRITTEN OUT (ii) [has content]: with $\text{Psibar} = \Psi^\dagger \text{sigma16}$ and $\text{sigma16} = \text{diag}(-\text{sigma}, \text{sigma})$: $s == -\text{psi1}^\dagger \text{sigma} \text{psi1} + \text{psi2}^\dagger \text{sigma} \text{psi2}$ ($\text{psibar1} = -\text{psi1}^\dagger \text{sigma}$, $\text{psibar2} = \text{psi2}^\dagger \text{sigma}$).

The rescaling. The rescaling of Part V applies unchanged, because it is a statement about the operator. $\Psi = \sqrt{\sin 6Hx_0} \Psi'$ turns the equation into $\sqrt{\sin 6Hx_0} (\gamma^\mu \partial_\mu \Psi' - H V'(s) \Psi') = 0$ with $s = \sin(6Hx_0) \bar{\Psi}' \Psi'$. The connection term is gone. [proved VII §27] WRITTEN OUT (iv) [THE RESULT]: $\Psi = \text{Sqrt}[\text{Sin}[6Hx_0]] \Psi'$ turns the field equation into $\text{Sqrt}[\text{Sin}] (\text{Cot } T_{16}[0] d_0 \Psi' + (1/q) T_{16}[i] d_i \Psi' + T_{16}[4] d_4 \Psi' + (1/p) T_{16}[h] d_h \Psi' - H V'(\text{Sin } s') \Psi')$: NO connection term, WRITTEN OUT (iv) [THE RESULT]: and the adjoint equation into $\text{Sqrt}[\text{Sin}] (\text{Cot } d_0 \bar{\Psi}' T_{16}[0] + \dots + H V'(\text{Sin } s') \bar{\Psi}')$, with no connection term either. The x_4 -only family $\Psi = \sqrt{\sin 6Hx_0} \Psi'(x_4)$ solves the equation if and only if $V'' s' = 0$. That holds for linear V (the author's mass term), or for null data $s' = 0$. For the complex field s' is conserved on the x_4 -only equation, so null data give x_4 -only solutions for any V . [proved VII §29] PRE-UNIVERSE (2) [THE RESULT]: $\Psi = \text{Sqrt}[\text{Sin}] \Psi'(x_4)$ satisfies the field equation iff $d_{x0} V'(\text{Sin}[6Hx_0] s') == 0$, and $d_{x0} V'(\text{Sin } s') == 6 H \text{Cos}[6 H x_0] s' V''(\text{Sin } s')$: V linear OR $s' == 0$, PRE-UNIVERSE (2) [has content]: $s' = \bar{\Psi}' \Psi'$ is conserved by the x_4 -only equations ($d_4 \Psi' = -H V' T_{16}[4] \Psi'$, $d_4 \bar{\Psi}' = H V' \bar{\Psi}' T_{16}[4]$) for ANY V ; so null data $s' = 0$ are solutions for any V , with $\rho == V(0)$; and for $V = s'^2$ with $s' != 0$ the x_0 -dependence is real (witness). The real-field version is [proved VI §24] “FABLE [has content]: for the mass term $V = -(2M/H) s$, V' is constant and $\Psi'(x_4)$ IS an exact solution”.

The frame-independent form. Covariant constancy of γ^μ (61) together with a vanishing $\sum_\mu \{\gamma^\mu, \Gamma_\mu\}$ gives, for any frame on which that anticommutator vanishes,

$$\gamma^\mu \Gamma_\mu = \frac{1}{2} \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \gamma^\mu) = -\frac{1}{2} T_\mu \gamma^\mu, \quad T_\mu = \partial_\mu \ln \frac{e_\mu^\mu}{\det e} \quad (\text{no sum}), \quad (83)$$

where T_μ is the Weitzenböck torsion vector of a diagonal frame (71). The second equality is a one-line computation for a diagonal frame, since $\sqrt{g} = \det e$ and $\gamma^\mu = T_{16}[\mu]/e_\mu^\mu$ [derived here]. This links the canonical connection directly to the fable-5.1 bridge: the connection term of the

Dirac operator is the Weitzenböck torsion vector. On the canonical frame $T_\mu = \partial_\mu \ln \sin 6Hx_0$ (subsection 9.10), and the rescaling by $\exp(\frac{1}{2} \ln \sin) = \sqrt{\sin 6Hx_0}$ removes it. [proved VII §27] SPIN CONNECTION [has content]: Part V restated - $\omega^W = 0$ in the canonical frame, $\omega^{LC} = -\text{contortion}$, $T_\mu = d_\mu \text{Log}[\text{Sin}[6 H x_0]]$, $R = -T + B$, and $\gamma^\mu \Gamma_\mu = -(1/2) T_\mu \gamma^\mu$ (on the canonical frame); and on the time-dependent warped frames of the coupled system, Part VIII, Section 32: WEITZENBOECK [THE RESULT]: the torsion vector of the warped frame is a GRADIENT, $T_\mu = d_\mu \text{Log}[\text{Sin}[6 H x_0]/(A^3 B^3 C)]$; on the canonical member it is Part V's torsionVectorFable51, WEITZENBOECK [THE RESULT]: $\gamma^\mu \Gamma_\mu = -(1/2) T_\mu \gamma^\mu$ on the warped frame - one vector term, a gradient.

9.12.4 It enters neither the canonical anticommutator nor the symmetrized Hamiltonian density

The admissible theory is a dimensional reduction $\Psi = \Psi(x_0, x_1, x_2, x_3, x_4)$. It uses a finite hidden coordinate volume $V_{\text{hid}} = \int dx_5 dx_6 dx_7$, which is an assumption: compact timelike directions contain closed timelike curves. Time is x_4 . The canonical anticommutator on a slice $x_4 = y_4$ is

$$\{\Psi_a(x), \Psi_b^\dagger(y)\} = \frac{H}{V_{\text{hid}} \sqrt{|g|}} G_{ab} \delta^4(x_{0..3} - y_{0..3}), \quad G = -i \sigma_{16} \gamma^4. \quad (84)$$

[proved VII §28] DIRAC BRACKET [THE RESULT]: $\{\Theta_i, \Theta_j\} = (\Omega^{-1})_{ij}$ gives $\{\Psi_a, \Psi_b^\dagger\} = i (K^{-1})_{ab} = (G^{-1})_{ab}$ and $\{\Psi_a, \Psi_b\} = 0$, ANTICOMMUTATOR [THE RESULT]: on the canonical frame ($c = \sqrt{g}/H$) $\{\Psi, \Psi^\dagger\} = (H/\sqrt{g}) J = H \cos[6 H x_0] J$ ($J^{-1} = J$), which IS the design's $(H/\sqrt{g})(-i \sigma_{16} \gamma^4)$. No spin connection appears in it. It is fixed by the coefficient of $\partial_4 \Psi$ in the Lagrangian, and $\Gamma_4 = 0$ in any case. On the canonical frame $\gamma^4 = T_{16}[4]$, so $G = -i T_{16}[0] T_{16}[1] T_{16}[2] T_{16}[3] T_{16}[4] = J$ [derived here]. J is the fundamental symmetry of subsection 9.12.6.

The Hamiltonian density of the symmetrized Lagrangian is

$$\mathcal{H} = \sqrt{g} \left[-\frac{1}{2H} \sum_{k \neq 4} (\bar{\Psi} \gamma^k \partial_k \Psi - \partial_k \bar{\Psi} \gamma^k \Psi) + V(s) \right], \quad (85)$$

and it contains *no spin connection* [proved VII §28] HAMILTONIAN [THE RESULT]: the Hamiltonian density $\Pi_\Psi d_4 \Psi + d_4 \bar{\Pi}_\Psi \bar{\Psi} - L$ of the symmetrized Lagrangian is $\sqrt{g} [-(1/(2H)) \sum_{k \neq 4} (\bar{\Psi} \gamma^k d_k \Psi - d_k \bar{\Psi} \gamma^k \Psi) + V]$: NO spin connection (fields of all eight coordinates). This follows from subsection 9.12.2: the symmetrized $\hat{L}$ does not depend on the connection. The connection re-emerges in the equation of motion. Integrating the symmetrized spatial derivative terms by parts produces $\gamma^\mu \Gamma_\mu = \frac{1}{2} \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \gamma^\mu)$, because the volume factor and the curved gammas depend on x_0 . [proved VII §28] HAMILTONIAN [THE RESULT]: $(\sqrt{g}/H) \sigma_{16} (\gamma^\mu d_\mu \Psi - H V' \bar{\Psi}) = i G d_4 \Psi - h \bar{\Psi}$ for admissible $\Psi(x_{0..4})$: the field equation is $i G d_4 \Psi = h \bar{\Psi}$, HAMILTONIAN [THE RESULT]: $(1/2) \sum_{\mu \neq 4} d_\mu (\sqrt{g} \sigma_{16} \gamma^\mu) = \sqrt{g} \sigma_{16} \sum \gamma^\mu \Gamma_\mu$, and $d_4 (\sqrt{g} \sigma_{16} \gamma^4) = 0$: the connection term is the one that makes h Hermitian.

The same rescaling organizes the one-particle space. $\sqrt{g} dx_0 |\Psi|^2 = |\Psi'|^2 dz$ for $\Psi' = \Psi/\sqrt{\sin 6Hx_0}$, where $z = -\ln \cos(6Hx_0)/(6H)$ is proper distance along x_0 . So the one-particle space of Ψ' is $L^2(dz)$. The x_0 -operator is regular at the wall $z = 0$ and needs a boundary condition there. The Lorentz-covariant, J -compatible ones are $T_{16}[0] \Psi' = \pm \Psi'$ at $z = 0$. It is limit-point at $z \rightarrow \infty$, where no condition is needed. [proved VII §28] BOUNDARY [has content]: the x_0 direction in proper distance $z = -\text{Log}[\cos[6 H x_0]]/(6 H)$:

$dz/dx_0 == \text{Tan}[6Hx_0] == \text{Sec Sin}$ (so $\text{Sqrt}[g] \, dx_0 \, |\Psi|^2 = |\Psi'|^2 \, dz$), $z(0) == 0$ and $z \rightarrow \text{infinity}$ at $6Hx_0 \rightarrow \text{Pi}/2$, $\gamma^0 \, d_0 == T_{16}[0] \, d_z$, and $\text{Sin}[6Hx_0] == \text{Sqrt}[1 - \text{Exp}[-12Hx_0]]$ (bounded frame factors), **BOUNDARY [THE RESULT]**: the bag condition $T_{16}[0] \, \Psi' = \pm \Psi'$ at the wall - $T_{16}[0]^2 == 1$, $\{\beta, T_{16}[0]\} == 0$, the normal-flux form vanishes on both eigenspaces ($P \beta T_{16}[0] P == 0$), and $[J, T_{16}[0]] == 0$.

9.12.5 It does not drop out of the energy-momentum tensor

The energy-momentum tensor operator is

$$\hat{T}_{\mu\nu} = -\frac{1}{4H} \left[\bar{\Psi} \gamma_\mu D_\nu \Psi - (D_\nu \bar{\Psi}) \gamma_\mu \Psi + (\mu \leftrightarrow \nu) \right] + g_{\mu\nu} \hat{L}, \quad \gamma_\mu = g_{\mu\nu} \gamma^\nu. \quad (86)$$

This is the Hilbert (symmetric-tetrad) tensor of the symmetrized action, and it equals the covariant tensor T_{cov} off shell. It is *built with D , not ∂* . The difference from the tensor T_{par} built with partial derivatives is **[derived here]**

$$T_{\mu\nu}^{\text{cov}} - T_{\mu\nu}^{\text{par}} = -\frac{1}{4H} \bar{\Psi} (\{\gamma_\mu, \Gamma_\nu\} + \{\gamma_\nu, \Gamma_\mu\}) \Psi. \quad (87)$$

The $g_{\mu\nu} \hat{L}$ terms agree by subsection 9.12.2. For $\mu = \nu$ the difference is $-\frac{1}{2H} \bar{\Psi} \{\gamma_\mu, \Gamma_\mu\} \Psi = 0$ by the per- μ anticommutator, because $\gamma_\mu = g_{\mu\mu} \gamma^\mu$ on the diagonal frame. So the energy density and all the pressures are the same with D or ∂ . Not every off-diagonal component differs. The $(0, 4)$ component coincides, because $\Gamma_0 = \Gamma_4 = 0$. The $(5, 6)$ component coincides too, because there the two anticommutators cancel **[derived here]**: $\gamma_5 = -p T_{16}[5]$, $\gamma_6 = -p T_{16}[6]$, the coefficients of Γ_5 and Γ_6 are equal, and $T_{16}[5] T_{16}[a] T_{16}[6] = -T_{16}[6] T_{16}[a] T_{16}[5]$ for $a = 0, 4$. The pair at which the notebook shows the two tensors to differ is given in the third item below.

- **[proved VII §29]** Hilbert = T_{cov} , via a first-order symmetric perturbation of the frame, EMT **[THE RESULT]**: the Hilbert (symmetric-tetrad) tensor of the symmetrized action IS That, OFF shell - components (x1,x1), (x4,x4), (x1,x4), (x0,x5), (x5,x4).
- **[proved VII §29]** the piece coming from the variation of the spin connection is zero, EMT **[has content]**: the spin-connection variation contributes NOTHING - dropping it leaves the Hilbert tensor unchanged at (x1,x4), (x0,x5), (x5,x4). The Lagrangian depends on the connection only through $\sum_\mu \{\gamma^\mu, \Gamma_\mu\} = \frac{1}{2} \omega_{[cab]} \gamma^{cab}$ (65), and under a symmetric variation of the tetrad $\delta \omega_{[cab]} = 0$. So the spin-connection variation contributes *exactly nothing*. The covariant terms of $\hat{T}$ come from the dependence of $\{\gamma^\mu, \Gamma_\mu\}$ on the frame. A partial-derivative Lagrangian must never be varied with respect to the frame. (Part VI's Text cell described this contribution as "a term antisymmetric in μ , ν that drops from the symmetrised tensor". The statement above is the sharper one.)
- **[proved VII §29]** Hilbert $\neq T_{\text{par}}$, witnessed on the non-vacuous pair (x_1, x_4) (the pairs $(0, 4)$ and $(5, 6)$ coincide on the canonical frame, as shown above), EMT **[control]**: the Hilbert tensor is NOT the partial-derivative tensor: at (x1,x4) they differ - witnessed on constant spinors ($\text{Psibar} = e_1$, $\Psi = e_{11}$: $(T_{16}[4] T_{16}[1] T_{16}[0])_{1,11} \neq 0$).
- **[proved VII §29]** T_{cov} is conserved on shell, EMT **CONSERVATION [THE RESULT]**: $\text{nabla}^\mu \text{That}_{\{\mu \nu\}} == 0$ ON SHELL in all eight components, for generic $\Psi(x_0, x_4)$, $\text{Psibar}(x_0, x_4)$ and generic V .

The real-field analogues are already proved. **[proved VI §24]** "FABLE [definition]: T_{cov} is symmetric by construction". **[proved VI §24]** "FABLE [has content]: the diagonal of T_{cov} equals the diagonal of the partial-derivative tensor, for the generic

Psi(x0, x4): rho and every pressure are the same in both". [proved VI §24] "FABLE [has content]: but the two tensors DIFFER off the diagonal - witnessed on the (x3, x4) component T[[4,5]], a momentum density along the observed sheet, for the constant spinor e1 + e13 and V = s^2". [proved VI §24] "FABLE [THE RESULT]: nabla^mu T_{mu nu} == 0 ON SHELL in all eight components, for a generic Psi(x0, x4) - the covariant tensor is the conserved one". That only the covariant tensor is conserved, and T_{par} is not, is [prose VI §24]: the divergence of T_{par} is not computed symbolically.

9.12.6 The connection and the fundamental symmetry J

$$J = -i T_{16}[0]T_{16}[1]T_{16}[2]T_{16}[3]T_{16}[4], \quad J^2 = \text{ID16}. \quad (88)$$

J is the fundamental symmetry that gives the positive quantization of the admissible sector. With $\beta = -i T_{16}[4]$, $\sigma_{16} = J\beta$ holds, and the Hilbert adjoint is $\Psi^\dagger = \Psi^\ddagger J$. [proved VII §28] J [THE RESULT]: sigma16 == J beta with beta = -i T16[4] Hermitian, beta^2 == 1; T16[0..3] Hermitian with beta T16[j] beta == -T16[j]: beta is the observer's Dirac adjoint, J [THE RESULT]: with Psi^dagger := Psi^ddag J: Psibar = Psi^dagger beta (J sigma16 == beta), Psi^ddag G = Psi^dagger (J G == ID16), {Psi, Psi^dagger} == (H/Sqrt[g]) ID16 (G^-1 J == ID16): POSITIVE.

$J^2 = \text{ID16}$ follows directly [derived here]. The product $P = T_{16}[0] \cdots T_{16}[4]$ of five mutually anticommuting matrices satisfies $P^2 = (-1)^{5 \cdot 4/2} T_{16}[0]^2 \cdots T_{16}[4]^2 = (+1)(1)(1)(1)(1)(-1) = -\text{ID16}$, so $(-iP)^2 = +\text{ID16}$.

The commutation rules also follow by counting sign changes [derived here]. A single $T_{16}[a]$ anticommutes with the four factors of P other than itself and commutes with itself, if $a \in \{0..4\}$. It anticommutes with all five if $a \in \{5, 6, 7\}$. Hence J commutes with $T_{16}[0..4]$ and anticommutes with $T_{16}[5], T_{16}[6], T_{16}[7]$. Applying this to the explicit Γ_μ of subsection 9.7:

object	relation to J	reason
$\Gamma_0 = \Gamma_4 = 0$	commute	zero
Γ_j ($j = 1, 2, 3$)	commute	made of $T_{16}[0]T_{16}[j]$, $T_{16}[4]T_{16}[j]$, all factors in $\{0..4\}$
Γ_k ($k = 5, 6, 7$)	anticommute	made of $T_{16}[0]T_{16}[k]$, $T_{16}[4]T_{16}[k]$: one factor in $\{0..4\}$, one in $\{5, 6, 7\}$
each $\gamma^\mu \Gamma_\mu$ (no sum), and their sum	commute	each is $\alpha_\mu T_{16}[0] + \beta_\mu T_{16}[4]$ (subsection 9.8)

The hidden components Γ_5, Γ_6 and Γ_7 are J -odd. They enter the Dirac operator only through $\gamma^k \Gamma_k$, a product of two J -odd matrices, which is J -even. So every term of the admissible-sector Hamiltonian, the connection term included, is J -even. The J -odd objects, namely the $(0, h)$, (i, h) and $(4, h)$ components of T_{cov} and the hidden momenta P_h ($h = 5, 6, 7$), are nonzero operators with zero expectation in J -eigenmode Fock states. [proved VII §28] J TABLE [THE RESULT]: J commutes with sigma16, T16[0..4], sigma16 gamma^mu (mu = 0..4), every gamma^mu Gamma_mu and Gamma_1..Gamma_3; it ANTICommutes with T16[5..7], sigma16 gamma^h (h = 5..7), Gamma_5..Gamma_7, T16[8] and C+. The same J table holds on the dynamical (warped and Bianchi-I) frames of the coupled system. Part VIII, Section 32, proves it: J [THE RESULT]: [J, gamma^mu Gamma_mu] == 0 for EACH mu separately (h = 5, 6, 7 included), on the warped and on the Bianchi-I frame, J [THE RESULT]: [J, Gamma_mu] == 0 for mu = 0..4 and {J, Gamma_h} == 0 for h = 5, 6, 7, on the warped and on the Bianchi-I frame and J [has content]: the anticommuting Gamma_h are not zero (witnessed on A = 1 + x4/2, B = 1 + x4/3, C = 1 + x4^2/3), so {J, Gamma_h} == 0 is not vacuous.

9.13 Status ledger

statement	status	where (notebook Part and Section; assertion tag)
frame e , $g = e \eta e^T$, stated line element, signature (4,4)	proved	III §15
a_4 is free and never given a value	proved	VI §25, VII §29 and VIII (controls); notice printed in III §15
$\det g = \sec^2 6Hx_0$; $\sqrt{ \det g } = \sec 6Hx_0$ on $0 < 6Hx_0 < \pi/2$	displayed; proved	displayed III §15; proved V §21
37 Christoffel symbols; symmetric in the lower indices; metric compatible	displayed; proved	III §16; VI §23
the vielbein postulate is solved by <code>cfSpinConnection</code> (residual zero)	proved (solver regression)	III §16
the 24 non-zero components of <code>omegaCanonical</code>	displayed; independent frame-only derivation proved; count 24 proved	III §16; VII §27 (SPIN CONNECTION [has content])
antisymmetry in the flat indices (metric compatibility)	proved (has content)	III §16; VII §27
torsion-free	proved (structural)	III §16
curvature: 156 components; Ricci scalar $\Gamma_\mu = \frac{1}{8}\omega_{\mu ab}[T_{16}[a], T_{16}[b]]$; sign and coefficient fixed by compatibility	displayed	III §16
block-diagonal; type-1 and type-2 blocks; <code>Dcov16</code> = direct sum	proved (with control)	III §17; VII §27 (SPIN CONNECTION [definition])
covariant constancy of γ^μ	proved	III §17
$\frac{1}{\sqrt{g}}\partial_\mu(\sqrt{g}\gamma^\mu) = [\gamma^\mu, \Gamma_\mu]$	proved	VI §24; VII §27 (FIELD EQUATION [has content])
$\sum_\mu \{\gamma^\mu, \Gamma_\mu\} = 0$; $\gamma^\mu \Gamma_\mu = -3H \cot^2(6Hx_0) T_{16}[0]$	proved	VI §24; VII §27
per-direction $\gamma^\mu \Gamma_\mu = \alpha_\mu T_{16}[0] + \beta_\mu T_{16}[4]$, $\sum \alpha_\mu = -3H \cot^2(6Hx_0)$, $\sum \beta_\mu = 0$;	derived here; proved	VII §27 (SPIN CONNECTION [THE RESULT], LAGRANGIAN (c) [THE RESULT])
anticommutator zero for each μ		
Bridges 1 and 2: the same connection in a local / constant gauge, at vector and spinor level	proved	IV §18, IV §19; restated VII §27
Bridge 3: a different connection; torsion $-2\lambda m^{\text{Skew}}$; curvature of degree 2 in λ ; Ricci shift $-42\lambda^2 = -\frac{1}{4}T \cdot T$	proved	IV §20; restated VII §27
fable-5.1: $\omega = 0$, zero curvature, $\omega^{LC} = -\text{contortion}(T)$, $T_\mu = \partial_\mu \ln \sin 6Hx_0$, axial part zero, $R = -T + B$	proved	V §21; restated VII §27
$\gamma^\mu \Gamma_\mu = -\frac{1}{2}T_\mu \gamma^\mu$; the rescaling $\Psi = \sqrt{\sin 6Hx_0} \Psi'$	proved	V §21; VI §24; VII §27 (SPIN CONNECTION [has content], WRITTEN OUT (iv)); VIII §32 on the warped frames (WEITZENBOECK, RESCALING)
Bridge 1's gauge term is the fable-5.1 connection of the boosted frame	proved	V §21
master table of the five connections	displayed	V §22
complex fable: the connection drops out of the symmetrized L because the matrix $\{\gamma^\mu, \Gamma_\mu\}$ vanishes	proved	VII §27 (LAGRANGIAN (c) [THE RESULT])
unsymmetrized covariant L = symmetrized + divergence; unsymmetrized partial L inadmissible	proved	VII §27 (LAGRANGIAN (d), (e), (f))
field equations; connection term $-3H \cot^2(6Hx_0) T_{16}[0]$; 8+8 form; rescaling; frame-independent form	proved	VII §27 (FIELD EQUATION, WRITTEN OUT (i)-(iv), SPIN CONNECTION); VIII §32
no connection in the canonical anticommutator or in the symmetrized Hamiltonian density; the connection makes $\hbar$ Hermitian	proved	VII §28 (QUANTIZATION [has content]: <code>Gamma_4 == 0</code> , ANTICOMMUTATOR, HAMILTONIAN)

statement	status	where (notebook Part and Section; assertion tag)
EMT: Hilbert = T_{cov} ; spin-connection variation zero; Hilbert $\neq T_{\text{par}}$; T_{cov} conserved	proved	VII §29 (EMT, EMT CONSERVATION)
J commutes with each $\gamma^\mu \Gamma_\mu$ and with $\Gamma_0 \dots \Gamma_4$, and anticommutes with $\Gamma_5 \dots \Gamma_7$	derived here; proved	VII §28 (J TABLE [THE RESULT]); VIII §32 on the warped and Bianchi-I frames (J [THE RESULT])

10 How it was verified

10.1 Part VII of the notebook: sections, cells, assertions

Part VII is the manifest `claude-fable/cells_part7.wl` (1777 lines; 170 `cfAssert` calls). Built into the notebook by `claude-fable/build_tools.py` (section 11.1), it adds 26 Input cells, 173–198, to the 172 of Parts I–VI. The section-to-cell map is printed by `claude-fable/nb_section_map.wls` (`claude-fable/nb_section_map.log`, section 11.4):

notebook Section	subject	Input cells	assertions	seconds (sum of the cells' times)	this page
26	the refinement: why fermion fable is a complex 16-spinor	173–179	38	1.957	3
27	the Lagrangian, the field equations in the primordial gravitational field, and the canonical spin connection	180–184	36	2.291	4, 5, 9
28	canonical quantization in 4+4 dimensions: the Dirac bracket, the Krein structure J , and the admissible sector	185–191	56	12.311	6
29	the energy–momentum tensor operator: definition, conservation, and ρ , P and w	192–198	40	17.874	7, 8
Part VII		26 cells	170	34.433	

The counts and times per cell, read from `claude-fable/run_fermion_fable_part7.log` (each cell's PASS lines are printed before its `CELL n t=...` line):

cell	173	174	175	176	177	178	179	180	181	182	183	184	185
assertions	4	2	9	7	7	4	5	10	5	3	5	13	5
seconds	0.016	0.541	0.117	0.592	0.053	0.064	0.574	1.005	0.403	0.395	0.209	0.279	0.062
cell	186	187	188	189	190	191	192	193	194	195	196	197	198
assertions	5	9	10	9	6	12	5	4	7	3	11	9	1
seconds	0.017	0.344	0.129	0.221	1.680	9.858	2.248	4.431	4.080	2.098	4.945	0.072	0.000

The 170 assertions by tag (the text of a label before its first `[`), counted from the log:

tag	count	tag	count
REFINEMENT	6	ADMISSIBILITY (all items)	10
GRASSMANN	9	HAMILTONIAN	7
REFINEMENT (a), (b), (c), (d)	3, 3, 1, 7	BOUNDARY	2

tag	count	tag	count
4D	4	KREIN MODES	6
FIDELITY	5	FOCK, FOCK (field level), FOCK (mode level)	1, 4, 6
LAGRANGIAN (a)–(f)	1, 2, 1, 1, 2, 3	CHARGE	1
FIELD EQUATION	5	EMT	8
CURRENT	3	EMT CONSERVATION, EMT ON SHELL, EMT HERMITICITY	3, 3, 2
SPIN CONNECTION	5	MODE SUMS, VACUUM	2, 1
WRITTEN OUT (i)–(iv)	2, 3, 3, 3	KOHN-SHAM, MEAN FIELD	5, 1
SYMMETRY	2	AUTHOR’S MASS TERM, CLASSICAL LIMIT	4, 1
QUANTIZATION	5	PRE-UNIVERSE (1)–(4)	2, 2, 2, 2
DIRAC BRACKET, ANTICOMMUTATOR	3, 2	PART VII	1
J, J TABLE	8, 1	“no identity in this notebook was accepted on numerical evidence alone”	1

By grade: the labels carry [THE RESULT] (a headline result), [has content] (a statement that could have failed), [definition], [structural], [solver regression] (a check of the code), [control] (the wrong alternative is shown to fail) and [fidelity] (agreement with the author’s or an earlier Part’s objects).

The final tally and the run summary, verbatim from the end of `claude-fable/run_fermion_fable_part7.log`:

```

identities accepted on numerical evidence alone : 0    (stage 3 returned True)
non-vanishing witnesses                        : 38    (cfNonZeroWitnessQ found a probe point at which
↪ every entry is a                               number and one of them is non-zero: a complete
                                                    ↪ proof of non-vanishing)

PASS no identity in this notebook was accepted on numerical evidence alone
Assertions run: 528   passed: 528   failed: 0
CELL 198  t=0.000 s
===== RUN SUMMARY =====
cells evaluated : 198
total seconds   : 232.146
cells w/ msgs   : 0
---- slowest cells ----
  cell 80  89.913 s
  cell 88  27.790 s
  cell 86  11.167 s
  cell 87  10.303 s
  cell 191  9.858 s
  cell 137  8.740 s
  cell 160  6.445 s
  cell 121  5.401 s
  cell 196  4.945 s
  cell 193  4.431 s
  cell 107  4.158 s
  cell 194  4.080 s
===== ASSERTIONS =====
assertions run   : 528
passed          : 528
FAILED          : 0
=====
RUN-DONE

```

So through Part VII: 198 Input cells, **528 assertions, 528 passed, 0 failed, 0 cells with messages, 0 identities accepted on numerical evidence alone** (every identity is closed symbolically; the notebook’s last assertion is `no identity in this notebook was accepted on numerical evidence alone`), and **38 non-vanishing witnesses** (19 from Parts I–V, 8 more from Part VI, 11 more from Part VII: the Part V and Part VI tallies in the same log print

19 and 27). Parts I–VI account for 358 of the assertions; Part VII adds 170. The whole run took 232.146s; the slowest cells are Part II's (cell 80, the bilinears of the author's closed-form solution, 89.913s). The last line, RUN-DONE, is not printed by `run_from_nb.wls`: it was appended by the shell wrapper that ran it.

10.2 Every Part VII assertion, verbatim

This is the complete output of Input cells 173–198 in `claude-fable/run_fermion_fable_part7.log` (lines 634–844), from the line that closes cell 172 to the line that closes cell 198. Every label is printed after PASS; a bracketed `[t s]` line is a timing printed by `cfTimed`. Lines longer than the page are broken, and the break is marked with a hook arrow.

```
CELL 172 t=0.000 s
PASS REFINEMENT [definition]: T16[8]^2 == sigma16^2 == C+^2 == ID16, C+ is symmetric, and C+ . T16[8]
  ↪ == sigma16 -- the four candidate mass matrices are only two
PASS REFINEMENT [has content]: C- T16[a] C-^-1 == -T16[a]^T and C+ T16[a] C+^-1 == +T16[a]^T for a =
  ↪ 0..7
PASS REFINEMENT [THE RESULT]: the symmetry of C Gamma depends only on the rank r of Gamma: C-: S A A
  ↪ S S A A S S, C+: S S A A S S A A S (r = 0..8)
PASS REFINEMENT [has content]: for each C, 136 of the 256 products are symmetric and 120
  ↪ antisymmetric
CELL 173 t=0.016 s
PASS REFINEMENT [THE RESULT]: the Spin(4,4)-invariant bilinear forms on the 16-spinor form a
  ↪ 2-dimensional space, spanned by C- = sigma16 and C+ = sigma16 T16[8]
PASS REFINEMENT [has content]: every invariant form is SYMMETRIC and chirality-diagonal, and the two
  ↪ chiral pieces PL sigma16 PL and PR sigma16 PR are invariant separately
CELL 174 t=0.541 s
PASS GRASSMANN [solver regression]: generators anticommute and square to zero (all pairs of 1..6)
PASS GRASSMANN [solver regression]: the product is associative, (x y) z == x (y z), on elements of
  ↪ degree 1, 2, 3 (and the products tested are not zero)
PASS GRASSMANN [solver regression]: graded commutativity x y == (-1)^(|x| |y|) y x (degrees 1, 2,
  ↪ 3), and x^2 == 0 for x odd
PASS GRASSMANN [solver regression]: the conjugation is an involutive anti-automorphism, (x y)^ddag
  ↪ == y^ddag x^ddag, (x^ddag)^ddag == x
PASS GRASSMANN [solver regression]: grD is an even derivation, d(x y) == d(x) y + x d(y); grDL obeys
  ↪ d_i(x y) == d_i(x) y + (-1)^|x| x d_i(y)
PASS GRASSMANN [THE RESULT]: theta^T N theta == theta^T N_A theta and theta^T N_S theta == 0 for a
  ↪ general 16x16 N -- an anticommuting bilinear sees only the ANTISYMMETRIC part
PASS GRASSMANN [THE RESULT]: theta^T N phi - (1/2) d(theta^T N theta) == theta^T N_S phi for a
  ↪ general N
PASS GRASSMANN [THE RESULT]: the Euler-Lagrange expression of L = theta^T N phi (left derivatives) is
  ↪ (N + N^T) phi: EMPTY field equations exactly when N is antisymmetric
PASS GRASSMANN [control]: for COMMUTING components the roles are reversed -- theta^T N theta sees
  ↪ only N_S, and the Euler-Lagrange expression of theta^T N phi is 2 N_A phi
CELL 175 t=0.117 s
PASS REFINEMENT (a) [THE RESULT]: real Grassmann + sigma16: both invariant mass terms vanish, and
  ↪ theta^T sigma16 T16[a] phi == (1/2) d(theta^T sigma16 T16[a] theta) for every a (a total
  ↪ derivative)
PASS REFINEMENT (a) [has content]: its Euler-Lagrange expression is identically empty in every
  ↪ direction -- while the kinetic term itself is NOT zero
PASS REFINEMENT (a) [THE RESULT]: FRAME-INDEPENDENT -- sigma16 {T16[a], S^bc} is SYMMETRIC for all a,
  ↪ b, c (so theta^T sigma16 {gamma^mu, Gamma_mu} theta == 0 on every frame), and d_mu(Sqrt[g] sigma16
  ↪ gamma^mu) == Sqrt[g] sigma16 [gamma^mu, Gamma_mu] on the canonical frame
PASS REFINEMENT (b) [THE RESULT]: Majorana with C+: the kinetic term is genuine -- Euler-Lagrange
  ↪ expression 2 C+ T16[a] phi with C+ T16[a] invertible -- but every invariant bilinear vanishes
  ↪ (massless, no V(s))
PASS REFINEMENT (b) [has content]: the scope -- the quartic Lorentz scalars v.v, B.B, C.C of the
  ↪ Majorana spinor are NOT zero and are all proportional to ONE quartic (same monomials, one constant
  ↪ ratio each)
PASS REFINEMENT (b) [has content]: and the commuting shadow of the C+ Majorana spinor has NO kinetic
  ↪ term: for commuting components theta^T C+ T16[a] phi is the total derivative (1/2) d(theta^T C+
  ↪ T16[a] theta)
PASS REFINEMENT (c) [THE RESULT]: Weyl and Majorana-Weyl: PL C T16[a] PL == PR C T16[a] PR == 0 for C
  ↪ = C-, C+ and every a (C chirality-diagonal, T16[a] chirality-odd) -- no kinetic term
CELL 176 t=0.592 s
PASS REFINEMENT (d) [definition]: Psi^ddag == (theta1 - i theta2)^T/Sqrt[2] in the algebra
PASS REFINEMENT (d) [THE RESULT]: s = Psibar Psi == i theta1^T sigma16 theta2: NON-ZERO (16
  ↪ monomials) and HERMITIAN; p = Psibar T16[8] Psi is non-zero and Hermitian too
PASS REFINEMENT (d) [has content]: s is a sum of 16 commuting nilpotent even elements x_a (x_a x_b ==
  ↪ x_b x_a, x_a^2 == 0), so s^17 == 0 and V(s) is a finite polynomial
```

PASS REFINEMENT (d) [definition]: the 32x32 matrix A read off the algebra reproduces $\Psi^\dagger d d \Psi$ K phi
 $\hookrightarrow$ exactly
PASS REFINEMENT (d) [THE RESULT]: its symmetric part is $A_S = (i/2) \Omega$, $\Omega = [[0, K], [-K, 0]]$
 $\hookrightarrow$ real SYMMETRIC and invertible: a genuine kinetic term
PASS REFINEMENT (d) [has content]: the symmetrized kinetic term is HERMITIAN and equals $\Psi^\dagger d d \Psi$ K
 $\hookrightarrow$ $\Psi - (1/2) d(\Psi^\dagger d d \Psi)$; the unsymmetrized one is not Hermitian
PASS REFINEMENT (d) [has content]: the same holds in every direction $a = 0..7$: the symmetrized
 $\hookrightarrow$ $\Psi^\dagger d d \Psi$ sigma16 T16[a] phi is Hermitian, with symmetric part $(i/2)[[0, \text{sigma16 T16[a]}, [-\text{sigma16}$
 $\hookrightarrow$ T16[a], 0]] $\neq 0$
CELL 177 t=0.053 s
PASS 4D [THE RESULT]: the 16 ordered products of T16[1..4] are linearly independent (the observer's
 $\hookrightarrow$ algebra is M4(C)), and its commutant in M16(C) is 16-dimensional
PASS 4D [has content]: $\Omega^4 = -ID_{16}$, the hidden gammas $h_A = \Omega^4 T16[A]$ ($A = 0,5,6,7$) commute
 $\hookrightarrow$ with T16[1..4] and obey $\{h_A, h_B\} = -2 \eta_{AB}$
PASS 4D [THE RESULT]: the 16 products of the hidden gammas span the commutant, and the two algebras
 $\hookrightarrow$ meet only in the multiples of ID_{16} (rank of the union $16 + 16 - 1 = 31$): $C^{16} = C^4(x) C^4$, four
 $\hookrightarrow$ 4D Dirac fermions
PASS 4D [has content]: the observer's Lorentz generators T16[i] T16[j] ($i < j$ in 1..4) commute with
 $\hookrightarrow$ every hidden gamma
CELL 178 t=0.064 s
[0.024 s] Euler-Lagrange expressions of the complex fable with respect to Psibar, generic (x0, x4)
[0.023 s] Euler-Lagrange expressions of the complex fable with respect to Psi, generic (x0, x4)
PASS FIDELITY [1] [fidelity]: $\Psi \rightarrow \psi$, $\text{Psibar} \rightarrow \psi^T$ sigma16 turns the symmetrized covariant
 $\hookrightarrow$ Lhat into Part VI's cfLhatFable EXACTLY (canonical frame, generic V)
PASS FIDELITY [2] [fidelity]: on the flat frame with $V = -(2M/H)$ s and no volume factor, the same
 $\hookrightarrow$ limit on Psi16 IS the author's La[] of Section 11
PASS FIDELITY [3] [fidelity]: sigma16 . EL_Psibar + EL_Psi in the real limit == Part VI's explicit
 $\hookrightarrow$ variation cfELFable, component for component
PASS FIDELITY [3] [fidelity]: and on the flat frame with $V = -(2M/H)$ s the same combination IS the
 $\hookrightarrow$ author's eLa of Section 12
PASS FIDELITY [4] [fidelity]: the energy-momentum tensor That in the real limit IS Part VI's
 $\hookrightarrow$ covariant tensor cTCov, all 64 components
CELL 179 t=0.574 s
PASS LAGRANGIAN (a) [definition]: every Γ^{spin_μ} is real and sigma16 Γ_μ is antisymmetric
 $\hookrightarrow$ ($\Gamma_\mu^T \text{sigma16} = -\text{sigma16} \Gamma_\mu$): $D_\mu \text{Psibar}$ is the Dirac conjugate of $D_\mu \Psi$
PASS LAGRANGIAN (b) [THE RESULT]: Lsym is HERMITIAN -- real for $\Psi = u + i v$, $\text{Psibar} = (u - i v)^T$
 $\hookrightarrow$ sigma16, generic u, v of all eight coordinates
PASS LAGRANGIAN (b) [control]: Lun is NOT Hermitian -- its imaginary part is not zero, witnessed on
 $\hookrightarrow$ the constant spinors $u = e_1$, $v = e_{13}$ (with $V = s^2$)
PASS LAGRANGIAN (c) [THE RESULT]: Lsym - Lsym_d == $(1/(2H)) \sum_\mu \text{Psibar} \{\Gamma_\mu, \Gamma_\mu\} \Psi$
 $\hookrightarrow$ identically, and $\{\Gamma_\mu, \Gamma_\mu\} = 0$ for EACH μ on the canonical frame: Lsym == Lsym_d
PASS LAGRANGIAN (d) [has content]: Lun - Lsym == $(1/(2H)) (1/\sqrt{g}) d_\mu (\sqrt{g} \text{Psibar} \Gamma_\mu$
 $\hookrightarrow$ $\Psi)$, a total divergence
PASS LAGRANGIAN (e) [has content]: Lun - Lun_d == $(1/H) \text{Psibar} \Gamma_\mu \Gamma_\mu \Psi = -3 \text{Cot}[6 H$
 $\hookrightarrow$ $x]^2 \text{Psibar T16}[0] \Psi$
PASS LAGRANGIAN (e) [control]: that term is NOT zero for a complex field (witness $u = e_1$, $v = e_{13}$),
 $\hookrightarrow$ and IS zero in the real commuting limit (the scope of Parts V-VI)
PASS LAGRANGIAN (f) [THE RESULT]: Lun_d varied with respect to Psibar gives $(\sqrt{g}/H)(\Gamma_\mu d_\mu$
 $\hookrightarrow$ $\Psi - H V' \Psi)$: the connection term is MISSING
PASS LAGRANGIAN (f) [THE RESULT]: varied with respect to Psi it gives $-(\sqrt{g}/H)((d_\mu \text{Psibar})$
 $\hookrightarrow$ $\Gamma_\mu + \text{Psibar} [\Gamma_\mu, \Gamma_\mu] + H V' \text{Psibar})$, with $\text{Psibar} [\Gamma_\mu, \Gamma_\mu] = 2$
 $\hookrightarrow$ $\text{Psibar} \Gamma_\mu \Gamma_\mu$: the connection term is PRESENT
PASS LAGRANGIAN (f) [control]: so Lun_d is INADMISSIBLE -- its two field equations are not Dirac
 $\hookrightarrow$ conjugates: they differ by $2 \text{Psibar} \Gamma_\mu \Gamma_\mu = -6 H \text{Cot}^2 \text{Psibar T16}[0]$, non-zero
 $\hookrightarrow$ (witness $u = e_1$, $v = 0$, $V = s^2$)
CELL 180 t=1.005 s
[0.134 s] Euler-Lagrange expressions with respect to Psibar, generic fields of all eight coordinates
[0.118 s] Euler-Lagrange expressions with respect to Psi, generic fields of all eight coordinates
PASS FIELD EQUATION [has content]: the identity the derivation uses -- $(1/\sqrt{g}) d_\mu (\sqrt{g}$
 $\hookrightarrow$ $\Gamma_\mu) = [\Gamma_\mu, \Gamma_\mu]$ (covariant constancy, Section 17)
PASS FIELD EQUATION [THE RESULT]: $\text{EL_Psibar} = (\sqrt{g}/H)(\Gamma_\mu D_\mu \Psi - H V'(s) \Psi)$ for
 $\hookrightarrow$ generic Psi, Psibar of all eight coordinates: the field equation is $\Gamma_\mu D_\mu \Psi = H V'(s)$
 $\hookrightarrow$ Ψ
PASS FIELD EQUATION [THE RESULT]: $\text{EL_Psi} = -(\sqrt{g}/H)((D_\mu \text{Psibar}) \Gamma_\mu + H V'(s) \text{Psibar})$:
 $\hookrightarrow$ the adjoint equation is $(D_\mu \text{Psibar}) \Gamma_\mu = -H V'(s) \text{Psibar}$
PASS FIELD EQUATION [has content]: the adjoint equation IS the Dirac conjugate of the field equation:
 $\hookrightarrow$ with $\Psi = u + i v$, $\text{Psibar} = (u - i v)^T$ sigma16, adjoint residual == $-(\text{field residual})^*$.
 $\hookrightarrow$ sigma16
PASS FIELD EQUATION [control]: the opposite sign, $(D \text{Psibar}) \Gamma = +H V' \text{Psibar}$, is NOT the
 $\hookrightarrow$ conjugate -- witnessed on $u = e_1 + e_5$, $v = 0$ with $V = s^2$ ($s = -2$ there)
CELL 181 t=0.403 s
PASS CURRENT [THE RESULT]: the Noether current of the U(1) phase is $(\sqrt{g}/H) i \text{Psibar} \Gamma_\mu \Psi$
PASS CURRENT [has content]: i sigma16 Γ_μ is Hermitian for every μ , so j^μ is real
PASS CURRENT [THE RESULT]: OFF SHELL $(1/\sqrt{g}) d_\mu (\sqrt{g} j^\mu) = i (E \bar{\Psi} + \bar{\Psi} E)$ for
 $\hookrightarrow$ generic fields of all eight coordinates -- conserved on shell

CELL 182 t=0.395 s

PASS SPIN CONNECTION [has content]: metric compatible ($\omega_{\mu}^{\{ab\}}$ antisymmetric) and every
 $\hookrightarrow$ non-zero component is $\omega_{\mu}^{\{0\}}$ or $\omega_{\mu}^{\{4\}}$ (up to antisymmetry): $\omega_0 = \omega_4$
 $\hookrightarrow == 0$

PASS SPIN CONNECTION [definition]: $\Gamma^{\text{spin}}_{\mu} == \sum_{\{a<b\}} \omega_{\mu,ab} (1/2) T_{16}[a] T_{16}[b]$

PASS SPIN CONNECTION [THE RESULT]: direction by direction $\gamma_{\mu} \Gamma_{\mu} == \alpha_{\mu} T_{16}[0] +$
 $\hookrightarrow \beta_{\mu} T_{16}[4]$; $\sum \alpha_{\mu} == -3 H \cot[6 H x_0]^2$ and $\sum \beta_{\mu} == 0$ (the observed and hidden
 $\hookrightarrow$ a4 terms cancel)

PASS SPIN CONNECTION [has content]: Parts IV-V restated -- Bridge 1's difference is the pure gauge
 $\hookrightarrow$ term, Bridge 2 is a constant conjugate, Bridge 3 has totally antisymmetric torsion that is not
 $\hookrightarrow$ zero

PASS SPIN CONNECTION [has content]: Part V restated -- $\omega^W = 0$ in the canonical frame, ω^{LC}
 $\hookrightarrow == -\text{contortion}$, $T_{\mu} == d_{\mu} \log[\sin[6 H x_0]]$, $R == -T + B$, and $\gamma_{\mu} \Gamma_{\mu} == -(1/2) T_{\mu}$
 $\hookrightarrow \gamma_{\mu}$

CELL 183 t=0.209 s

PASS WRITTEN OUT (i) [THE RESULT]: the field equation on the canonical frame is $\cot T_{16}[0] d_0 \Psi +$
 $\hookrightarrow (1/q) T_{16}[i] d_i \Psi + T_{16}[4] d_4 \Psi + (1/p) T_{16}[h] d_h \Psi - 3 H \cot^2 T_{16}[0] \Psi == H V'(s) \Psi$

PASS WRITTEN OUT (i) [THE RESULT]: the adjoint equation is $\cot d_0 \bar{\Psi} T_{16}[0] + \dots - 3 H \cot^2$
 $\hookrightarrow \bar{\Psi} T_{16}[0] == -H V'(s) \bar{\Psi}$ (the connection enters with the sign fixed by $\{\gamma_{\mu}$,
 $\hookrightarrow \Gamma_{\mu}\} = 0$)

PASS WRITTEN OUT (ii) [THE RESULT]: the split-octonion 8+8 form -- the type-1 rows are the taubar
 $\hookrightarrow$ equations for ψ_2 , the type-2 rows the tau equations for ψ_1 ($\tau[0] == \text{taubar}[0] == \text{ID8}$)

PASS WRITTEN OUT (ii) [THE RESULT]: the covariant 8+8 form -- rows 1..8 are $e_a^{\mu} \text{taubar}[a] (d_{\mu} +$
 $\hookrightarrow \Gamma_{\mu}^{(2)}) \psi_2 - H V' \psi_1$, rows 9..16 are $e_a^{\mu} \tau[a] (d_{\mu} + \Gamma_{\mu}^{(1)}) \psi_1 - H V'$
 $\hookrightarrow \psi_2$; and the conjugate rows

PASS WRITTEN OUT (ii) [has content]: with $\bar{\Psi} = \Psi^{\dagger} \sigma_{16}$ and $\sigma_{16} = \text{diag}(-\sigma, \sigma)$,
 $\hookrightarrow \sigma = -\psi_1^{\dagger} \sigma_{16} \psi_1 + \psi_2^{\dagger} \sigma_{16} \psi_2$ ($\bar{\psi}_1 = -\psi_1^{\dagger} \sigma_{16}$, $\bar{\psi}_2 = \psi_2^{\dagger} \sigma_{16}$)

PASS WRITTEN OUT (iii) [definition]: the term lists reproduce all sixteen field equations and all
 $\hookrightarrow$ sixteen adjoint equations exactly

PASS WRITTEN OUT (iii) [has content]: every component equation has exactly one derivative per
 $\hookrightarrow$ coordinate $x_0 \dots x_7$, the connection term multiplies the component carrying the x_0 -derivative, and $-H$
 $\hookrightarrow V'$ multiplies ψ_r itself

PASS SYMMETRY [THE RESULT]: the only internal $U(1)$ is the vector phase -- the commutant of all eight
 $\hookrightarrow T_{16}[a]$ in $M_{16}(C)$ is one-dimensional (multiples of ID_{16})

PASS SYMMETRY [has content]: the axial phase $\exp(i \alpha T_{16}[8])$ leaves s invariant but not the
 $\hookrightarrow$ kinetic term: $\exp(-i \alpha T_8) T_{16}[a] \exp(i \alpha T_8) == T_{16}[a] \exp(2 i \alpha T_8)$, so $\bar{\Psi}$
 $\hookrightarrow \gamma_{\mu} d_{\mu} \Psi$ changes at first order by $2 i \alpha \bar{\Psi} \gamma_{\mu} T_{16}[8] d_{\mu} \Psi$, which is
 $\hookrightarrow$ not zero (witness)

PASS WRITTEN OUT (iii) [has content]: for fields of all eight coordinates the sixteen equations form
 $\hookrightarrow$ ONE coupled block (the four blocks of four of Section 13 belong to the (x_0, x_4) reduction)

PASS WRITTEN OUT (iv) [THE RESULT]: $\Psi = \sqrt{\sin[6 H x_0]} \Psi'$ turns the field equation into
 $\hookrightarrow \sqrt{\sin[6 H x_0]} (\cot T_{16}[0] d_0 \Psi' + (1/q) T_{16}[i] d_i \Psi' + T_{16}[4] d_4 \Psi' + (1/p) T_{16}[h] d_h \Psi' -$
 $\hookrightarrow H V'(\sin s') \Psi')$: NO connection term

PASS WRITTEN OUT (iv) [THE RESULT]: and the adjoint equation into $\sqrt{\sin[6 H x_0]} (\cot d_0 \bar{\Psi}' T_{16}[0] +$
 $\hookrightarrow \dots + H V'(\sin s') \bar{\Psi}')$, with no connection term either

PASS WRITTEN OUT (iv) [definition]: under the rescaling $s == \sin[6 H x_0] s'$, $s' = \bar{\Psi}' \Psi'$

CELL 184 t=0.279 s

PASS QUANTIZATION [definition]: lapse 1 and shift 0 ($g_{44} == -1$, $g_{4\mu} == 0$), and the slice $x_4 =$
 $\hookrightarrow$ const has signature (4,3) -- $x_0 \dots x_3$ spacelike, $x_5 \dots x_7$ timelike (given a4 real and $0 < 6 H x_0 <$
 $\hookrightarrow \pi/2$)

PASS QUANTIZATION [THE RESULT]: the momenta of L_{sym} are $\Pi_{\Psi} == (\sqrt{g}/(2H)) \bar{\Psi} \gamma^4$ and
 $\hookrightarrow \Pi_{\bar{\Psi}} == -(\sqrt{g}/(2H)) \gamma^4 \Psi$

PASS QUANTIZATION [has content]: $\theta_{\text{un}} - \theta_{\text{sym}} == \delta[(\sqrt{g}/(2H)) \bar{\Psi} \gamma^4 \Psi]$
 $\hookrightarrow$ exactly: the same symplectic form, the same brackets

PASS QUANTIZATION [THE RESULT]: $G = -i \sigma_{16} \gamma^4 == J = -i T_{16}[0] \cdot T_{16}[1] \cdot T_{16}[2] \cdot T_{16}[3] \cdot T_{16}[4]$
 $\hookrightarrow$ on the canonical frame; J is Hermitian, $J^2 == \text{ID}_{16}$, eigenvalues ± 1 (8) and -1 (8): signature
 $\hookrightarrow (8,8)$

PASS QUANTIZATION [has content]: $\Gamma_4 == 0$ -- the time-derivative part of the Lagrangian, which
 $\hookrightarrow$ fixes the anticommutator, contains no spin connection

CELL 185 t=0.062 s

PASS DIRAC BRACKET [THE RESULT]: in the Grassmann algebra $\Psi^{\dagger} K \Phi == (i/2) \theta^T \Omega_c \Phi$
 $\hookrightarrow + (1/4) d(\theta^T K \theta_1 + \theta^T K \theta_2)$, for $K = c \sigma_{16} T_{16}[4]$ with a free scalar c

PASS DIRAC BRACKET [has content]: Ω_c is real symmetric and invertible, $\Omega_c^{-1} == [[0,$
 $\hookrightarrow -K^{-1}], [K^{-1}, 0]]$

PASS DIRAC BRACKET [THE RESULT]: $\{\theta_i, \theta_j\} = (\Omega_c^{-1})_{ij}$ gives $\{\Psi_a, \Psi^{\dagger}_b\} == i$
 $\hookrightarrow (K^{-1})_{ab} == (G^{-1})_{ab}$ and $\{\Psi_a, \Psi_b\} == 0$

PASS ANTICOMMUTATOR [THE RESULT]: on the canonical frame ($c = \sqrt{g}/H$) $\{\Psi, \Psi^{\dagger}\} ==$
 $\hookrightarrow (H/\sqrt{g}) J == H \cos[6 H x_0] J$ ($J^{-1} == J$), which IS the design's $(H/\sqrt{g})(-i \sigma_{16}$
 $\hookrightarrow \gamma^4)$

PASS ANTICOMMUTATOR [has content]: the rescaled field $\Psi' = \Psi/\sqrt{\sin[6 H x_0]}$ has $\{\Psi',$
 $\hookrightarrow \Psi'^{\dagger}\} == H \cot[6 H x_0] J$ delta⁷, and $\sqrt{g}$ is x_4 -independent (so quantizing $\chi =$
 $\hookrightarrow (\sqrt{g}/H)^{(1/2)} \Psi$ changes nothing on the canonical frame)

CELL 186 t=0.017 s

PASS J [THE RESULT]: $J = -i \Omega_4 T_{16}[0] = -i h_0$ is a hidden flavour gamma: it lies in the
 $\hookrightarrow$ commutant of $T_{16}[1..4]$ (in the span of the hidden-gamma products of Section 26)

PASS J [has content]: on the flavour factor J has signature (2,2): cfIdemObs is a rank-4 projector
 $\hookrightarrow$ commuting with J, and J restricted to its image has eigenvalues +1, +1, -1, -1

PASS J [THE RESULT]: $\text{sigma16} = J \text{ beta}$ with $\text{beta} = -i T_{16}[4]$ Hermitian, $\text{beta}^2 = 1$; $T_{16}[0..3]$
 $\hookrightarrow$ Hermitian with $\text{beta } T_{16}[j] \text{ beta} = -T_{16}[j]$: beta is the observer's Dirac adjoint

PASS J [THE RESULT]: with $\text{Psi}^\dagger := \text{Psi}^\dagger \text{dag } J$: $\text{Psibar} = \text{Psi}^\dagger \text{dag } \text{beta}$ ($J \text{ sigma16} = \text{beta}$),
 $\hookrightarrow \text{Psi}^\dagger \text{dag } G = \text{Psi}^\dagger \text{dag } (J G = \text{ID16})$, $\{\text{Psi}, \text{Psi}^\dagger \text{dag}\} = (H/\text{Sqrt}[g]) \text{ID16}$ ($G^{-1} J = \text{ID16}$):
 $\hookrightarrow$ POSITIVE

PASS J TABLE [THE RESULT]: J commutes with sigma16 , $T_{16}[0..4]$, $\text{sigma16 } \gamma^\mu$ ($\mu = 0..4$), every
 $\hookrightarrow \gamma^\mu \text{Gamma}_\mu$ and $\text{Gamma}_1.. \text{Gamma}_3$; it ANTICOMMUTES with $T_{16}[5..7]$, $\text{sigma16 } \gamma^h$ ($h =$
 $\hookrightarrow 5..7$), $\text{Gamma}_5.. \text{Gamma}_7$, $T_{16}[8]$ and C^+

PASS J [has content]: the Hermiticity rule -- for classically Hermitian M, $(J M)^\dagger = J M$ iff
 $\hookrightarrow [J, M] = 0$ and $= -J M$ iff $\{J, M\} = 0$: so $s = \text{Psi}^\dagger \text{dag } \text{sigma16 } \text{Psi}$ is Hermitian and $p = \text{Psi}^\dagger \text{dag}$
 $\hookrightarrow C^+ \text{Psi}$ is ANTI-Hermitian

PASS J [THE RESULT]: 13 of the 28 Lorentz generators commute with J (a, b both in $\{0..4\}$: $\text{Spin}(4,1)$,
 $\hookrightarrow$ or both in $\{5,6,7\}$: $\text{Spin}(3)$); the other 15 anticommute -- 12 boosts $x_0..x_3 \leftrightarrow x_5..x_7$ and 3
 $\hookrightarrow$ rotations $x_4 \leftrightarrow x_5..x_7$

PASS J [has content]: the Krein form G is invariant ($S^T G + G S = 0$) under exactly the 21
 $\hookrightarrow$ generators with a, b != 4 -- $\text{Spin}(4,3)$, the group of the slice

PASS J [has content]: chirality -- $\{J, T_{16}[8]\} = 0$, each chirality subspace is G-NULL ($PL G PL = PR$
 $\hookrightarrow G PR = 0$), and the two chiralities are orthogonal in the Hilbert product (PL, PR Hermitian, $PL PR$
 $\hookrightarrow = 0$)

CELL 187 t=0.344 s

PASS ADMISSIBILITY (1) [THE RESULT]: $E_k^2 = (k_s^2 - k_h^2 + m^2) \text{ID16}$ for every k: $\omega^2 = k_s^2$
 $\hookrightarrow -k_h^2 + m^2$

PASS ADMISSIBILITY (2) [THE RESULT]: for admissible k ($k_h = 0$), $[J, E_k] = 0$, E_k is J-Hermitian,
 $\hookrightarrow$ and $G J = \text{ID16}$ is positive: J quantizes every admissible momentum positively

PASS ADMISSIBILITY (3) [THE RESULT]: below threshold ($k = 5 e_1 + 3 e_5$, $m = 3$) the boosted $J' = S^{-1} J$
 $\hookrightarrow S$ commutes with E_k and beta , $J'^2 = 1$, and $G J' = S^2$ is Hermitian positive definite

PASS ADMISSIBILITY (4) [THE RESULT]: AT threshold ($k = 4 e_1 + 5 e_5$, $m = 3$) $E_k \neq 0$ but $E_k^2 = 0$:
 $\hookrightarrow$ nilpotent, not diagonalizable (rank 8: Jordan blocks of size 2)

PASS ADMISSIBILITY (5) [THE RESULT]: ABOVE threshold ($k = (0, 3/4, 1, 0 \mid k_5 = 2)$, $m = 1$) $\omega^2 =$
 $\hookrightarrow -23/16$: every eigenvalue of E_k is $\pm i \text{Sqrt}[23]/4 = \pm 1.19896 i$ and every eigenvector is G-NEUTRAL
 $\hookrightarrow (u^\dagger G u = 0)$

PASS ADMISSIBILITY [has content]: the general identity $[E_k, \text{beta}] = 2 i T_k$, $T_k = \text{Sum}_a k_a$
 $\hookrightarrow T_{16}[a]$, for every k (admissible and hidden components alike)

PASS ADMISSIBILITY (6) [THE RESULT]: traceless certificate -- every X commuting with beta , the four
 $\hookrightarrow$ admissible E_k and one hidden-momentum E_k has $\text{Tr}(G X) = 0$, so no positive $G J'$ exists; J itself
 $\hookrightarrow$ is not in that commutant

PASS ADMISSIBILITY (6) [THE RESULT]: the analytic proof for a purely hidden momentum -- $[E_k, \text{beta}]$
 $\hookrightarrow = 2 i k_5 T_{16}[5]$; $T_{16}[5]$ anticommutes with G and $T_{16}[5]^\dagger G T_{16}[5] = -G$ (so a J' commuting
 $\hookrightarrow$ with $T_{16}[5]$ has $T_5^\dagger (G J') T_5 = -(G J')$)

PASS ADMISSIBILITY (7) [THE RESULT]: J is UNIQUE -- the joint commutant of beta and the admissible
 $\hookrightarrow E_k$ is 8-dimensional, contains J, and every element of it commutes with J (so $X^2 = 1$ with $J X >$
 $\hookrightarrow 0$ forces $X = J$)

PASS ADMISSIBILITY [has content]: $[P_4, P_h] = 0$ -- the metric, the frame and the spin connection do
 $\hookrightarrow$ not depend on x_5, x_6, x_7

CELL 188 t=0.129 s

PASS HAMILTONIAN [THE RESULT]: the Hamiltonian density $\text{Pi}_\text{Psi } d_4 \text{Psi} + d_4 \text{Psibar } \text{Pi}_\text{Psibar} - L$ of
 $\hookrightarrow$ the symmetrized Lagrangian is $\text{Sqrt}[g] [-(1/(2H)) \text{Sum}_{k \neq 4} (\text{Psibar } \gamma^k d_k \text{Psi} - d_k \text{Psibar}$
 $\hookrightarrow \gamma^k \text{Psi}) + V]$: NO spin connection (fields of all eight coordinates)

PASS HAMILTONIAN [THE RESULT]: $(\text{Sqrt}[g]/H) \text{sigma16 } (\gamma^\mu D_\mu \text{Psi} - H V' \text{Psi}) = i \text{Gtilde } d_4 \text{Psi}$
 $\hookrightarrow -h \text{Psi}$ for admissible $\text{Psi}(x_0..x_4)$: the field equation is $i \text{Gtilde } d_4 \text{Psi} = h \text{Psi}$

PASS HAMILTONIAN [THE RESULT]: $(1/2) \text{Sum}_{\mu \neq 4} d_\mu (\text{Sqrt}[g] \text{sigma16 } \gamma^\mu) = \text{Sqrt}[g] \text{sigma16}$
 $\hookrightarrow \text{Sum } \gamma^\mu \text{Gamma}_\mu$, and $d_4 (\text{Sqrt}[g] \text{sigma16 } \gamma^4) = 0$: the connection term is the one that
 $\hookrightarrow$ makes h Hermitian

PASS HAMILTONIAN [has content]: $A_0 = \text{Sqrt}[g] \text{sigma16 } \gamma^0$ is real antisymmetric, and for $h_0 =$
 $\hookrightarrow -(A_0 d_0 + (1/2) A_0')$: $\langle f, h_0 g \rangle - \langle h_0 f, g \rangle = -d_0(f^\dagger \text{dag } A_0 g)$, a pure boundary term

PASS BOUNDARY [has content]: the x_0 direction in proper distance $z = -\text{Log}[\text{Cos}[6 H x_0]]/(6 H)$: dz/dx_0
 $\hookrightarrow = \text{Tan}[6 H x_0] = \text{Sec Sin}$ (so $\text{Sqrt}[g] dx_0 |\text{Psi}|^2 = |\text{Psi}'|^2 dz$), $z(0) = 0$ and $z \rightarrow \text{infinity}$ at $6 H$
 $\hookrightarrow x_0 \rightarrow \text{Pi}/2$, $\gamma^0 d_0 = T_{16}[0] d_z$, and $\text{Sin}[6 H x_0] = \text{Sqrt}[1 - \text{Exp}[-12 H z]]$ (bounded frame
 $\hookrightarrow$ factors)

PASS BOUNDARY [THE RESULT]: the bag condition $T_{16}[0] \text{Psi}' = \pm \text{Psi}'$ at the wall -- $T_{16}[0]^2 = 1$,
 $\hookrightarrow \{\text{beta}, T_{16}[0]\} = 0$, the normal-flux form vanishes on both eigenspaces ($P \text{beta } T_{16}[0] P = 0$), and
 $\hookrightarrow [J, T_{16}[0]] = 0$

PASS HAMILTONIAN [THE RESULT]: $[J, h] = 0$ on the admissible sector, for every a_4 , every x_0 -profile
 $\hookrightarrow$ and every V' -- including the spin-connection term

PASS HAMILTONIAN [control]: a hidden derivative term breaks it -- $[J, h] \neq 0$ once Psi depends on x_5
 $\hookrightarrow$ (witnessed on $\text{Psi} = x_5 e_1$)

PASS HAMILTONIAN [has content]: the x_4 -dependence of h enters only through $1/q$: $d_4 h \text{Psi} = (H a_4' [H$
 $\hookrightarrow x_4]) x$ (the observed-direction part of h), which commutes with J

CELL 189 t=0.221 s

PASS KREIN MODES [has content]: $A = G h$ satisfies $A^2 = w^2 \text{ID}_{16}$ and $[A, J] = 0$, and A is Hermitian
 $\hookrightarrow$ (G and h commute) for every admissible k and m
PASS KREIN MODES [THE RESULT]: the four $P_{\{s, \eta\}}$ are projectors of trace (rank) 4, mutually
 $\hookrightarrow$ orthogonal, summing to ID_{16} , and $\sum \eta P_{\{s, \eta\}} = J = G^{-1}$ (completeness)
PASS KREIN MODES [THE RESULT]: for every mode, $\eta u^\dagger h u = s w$ (energy = frequency), η
 $\hookrightarrow u^\dagger \sigma_{16} u = m/\omega_u$, $\eta u^\dagger (dh/dk_j) u = k_j/\omega_u$: $P X P = c P$ with $c =$
 $\hookrightarrow \eta s w$, $\eta s m/w$, $\eta s k_j/w$
PASS KREIN MODES [has content]: $dh/dk_j = -i \sigma_{16} T_{16}[j]$, and $d\omega/dk_j = k_j/\omega$ for
 $\hookrightarrow \omega = +w$
PASS KREIN MODES [THE RESULT]: every mode of $P_{\{s, \eta\}}$ is a J -eigenvector: $J P_{\{s, \eta\}} = \eta$
 $\hookrightarrow P_{\{s, \eta\}}$
PASS KREIN MODES [control]: a Krein-boosted pair $u_1 = (5/4)u_+ + (3/4)u_-$, $u_2 = (3/4)u_+ + (5/4)u_-$ in
 $\hookrightarrow$ the $\omega = +5$ eigenspace is still G -orthonormal and still consists of modes, but not of
 $\hookrightarrow J$ -eigenvectors; the induced $J_U = G U U^\dagger$ differs from J , and under it s is NOT Hermitian

CELL 190 $t=1.680$ s
[0.749 s] the CAR of 16 Jordan-Wigner operators
PASS FOCK [solver regression]: the Jordan-Wigner operators obey the positive CAR $\{f_i, f_j^\dagger\} =$
 $\hookrightarrow \delta_{ij}$, $\{f_i, f_j\} = 0$ on C^{65536}
PASS FOCK (field level) [THE RESULT]: $\{\Psi_a, \Psi^\dagger_{\bar{a}}\} = J_{ab} = (G^{-1})_{ab}$ with $\Psi = f$, $\Psi^\dagger_{\bar{a}} =$
 $\hookrightarrow f^\dagger J$ -- the Dirac-bracket anticommutator, realized on a positive Fock space
PASS FOCK (field level) [THE RESULT]: $J h$ is Hermitian with eigenvalues $+5$ (8) and -5 (8); H_{free} and
 $\hookrightarrow s$ are Hermitian operators, so is $H_{\text{free}} + (3/7) s^2$; $p = \Psi^\dagger_{\bar{a}} C + \Psi$ is ANTI-Hermitian
[1.868 s] the Heisenberg equation on the Fock space
PASS FOCK (field level) [THE RESULT]: Heisenberg -- $[H_{\text{free}}, \Psi_a] = -(G^{-1} h \Psi)_a$ for all 16
 $\hookrightarrow$ components, so $i d_4 \Psi = i [H, \Psi]$ IS the field equation $i G d_4 \Psi = h \Psi$
PASS FOCK (field level) [control]: with the opposite sign of the anticommutator the Hamiltonian is
 $\hookrightarrow -H_{\text{free}}$ and $[H, \Psi_a] = +(G^{-1} h \Psi)_a$: the time-REVERSED equation
PASS FOCK (mode level) [THE RESULT]: $\{b_u, b_v^\dagger\} = \eta_u \delta_{uv}$ (the indefinite CAR demanded
 $\hookrightarrow$ by $\sum \eta_u u^\dagger u = G^{-1}$), and the J -involution $b^\dagger = \eta b^\dagger$ restores $\{b, b^\dagger\}$
 $\hookrightarrow = 1$
PASS FOCK (mode level) [has content]: the naive Krein 'norm' $\langle 0 | b_u b_u^\dagger | 0 \rangle$ equals η_u --
 $\hookrightarrow$ NEGATIVE for the four $\eta = -1$ modes of each frequency -- while the Fock norm $\langle 0 | b_u b_u^\dagger$
 $\hookrightarrow | 0 \rangle = 1$
PASS FOCK (mode level) [THE RESULT]: $\sum \omega_u \eta_u b_u^\dagger b_u = \sum \omega_u b_u^\dagger b_u$,
 $\hookrightarrow$ and $[H_{\text{am}}, b_u] = -\omega_u b_u$ for every u ($b_u(x_4) = b_u e^{-i \omega_u x_4}$)
PASS FOCK (mode level) [THE RESULT]: the spectrum -- a UNIQUE ground state at $-8 \omega = -40$, the
 $\hookrightarrow$ Dirac sea (modes 9..16 filled, 1..8 empty); $:H_{\text{am}} = H_{\text{am}} + 40 \geq 0$
PASS FOCK (mode level) [THE RESULT]: the first excited level, $\omega = 5$ above the sea, is 16-fold: 8
 $\hookrightarrow$ particle states ($N = +1$) and 8 antiparticle states ($N = -1$), with $N = \sum b^\dagger b - 8$
PASS FOCK (mode level) [THE RESULT]: in every Fock basis state tested (the sea and the 16 first
 $\hookrightarrow$ excited states) $\langle n | b_u^\dagger b_v | n \rangle = \eta_u \eta_v \delta_{uv}$ -- the expectation rule of Section 29
PASS CHARGE [THE RESULT]: $n^4 = -(i/H) \text{Psibar } \gamma^4 \Psi = (1/H) \Psi^\dagger G \Psi = (1/H) \Psi^\dagger$
 $\hookrightarrow \Psi$, whose mode form $\sum_u \eta_u b_u^\dagger b_u = \sum_u b_u^\dagger b_u$ is the particle number
 $\hookrightarrow$ (normal-ordered: particles - antiparticles)

CELL 191 $t=9.858$ s
[0.693 s] Hilbert tensor from first-order symmetric tetrad perturbations, five components
PASS EMT [THE RESULT]: the Hilbert (symmetric-tetrad) tensor of the symmetrized action IS That, OFF
 $\hookrightarrow$ shell -- components (x_1, x_1) , (x_4, x_4) , (x_1, x_4) , (x_0, x_5) , (x_5, x_4)
PASS EMT [has content]: the spin-connection variation contributes NOTHING -- dropping it leaves the
 $\hookrightarrow$ Hilbert tensor unchanged at (x_1, x_4) , (x_0, x_5) , (x_5, x_4)
PASS EMT [control]: the Hilbert tensor is NOT the partial-derivative tensor: at (x_1, x_4) they differ
 $\hookrightarrow$ -- witnessed on constant spinors ($\text{Psibar} = e_1$, $\Psi = e_{11}$: $(T_{16}[4] T_{16}[1] T_{16}[0])_{\{1,11\}} \neq 0$)
PASS EMT [definition]: That is symmetric
PASS EMT [THE RESULT]: That is Hermitian under the classical conjugation: real for $\Psi = u + i v$,
 $\hookrightarrow \text{Psibar} = (u - i v)^T \sigma_{16}$, all 64 components

CELL 192 $t=2.248$ s
PASS EMT CONSERVATION [definition]: F and $\bar{F}$ solve the field equation and the adjoint equation for
 $\hookrightarrow$ the x_4 -derivatives
[0.767 s] $\nabla_\mu \text{That}_{\{\mu\}}$, generic $\Psi(x_0, x_4)$, $\text{Psibar}(x_0, x_4)$
[2.599 s] the on-shell divergence, eight components
PASS EMT CONSERVATION [THE RESULT]: $\nabla_\mu \text{That}_{\{\mu\}} = 0$ ON SHELL in all eight components, for
 $\hookrightarrow$ generic $\Psi(x_0, x_4)$, $\text{Psibar}(x_0, x_4)$ and generic V
PASS EMT CONSERVATION [control]: OFF shell the divergence is not zero -- witnessed on $\Psi = (x_4 +$
 $\hookrightarrow x_0^2) e_1 + e_{13}$, $\text{Psibar} = e_1 + x_4 e_5$ with $V = s^2$, which is not a solution
PASS EMT [THE RESULT]: $\text{That} - T_{\text{par}} = -(1/(4H)) \text{Psibar} (\{\gamma_\mu, \Gamma_\mu\} + \{\gamma_\mu, \Gamma_\mu\})$
 $\hookrightarrow \Psi$: equal on the diagonal (every $\{\gamma_\mu, \Gamma_\mu\}$ vanishes), different off it (the (x_1, x_4)
 $\hookrightarrow$ witness above)

CELL 193 $t=4.431$ s
PASS EMT ON SHELL [THE RESULT]: the kinetic bilinear $= s V'(s)$ (not zero) and $L_{\text{hat}} = s V'(s) - V(s)$
PASS EMT ON SHELL [THE RESULT]: $\rho = \text{That}_{44} = V(s) + K_h$
PASS EMT ON SHELL [THE RESULT]: $T^i_i = T^h_h = s V'(s) - V(s)$ for the three observed and the three
 $\hookrightarrow$ hidden directions, and $T^0_0 = s V'(s) - V(s) + K_h$
PASS EMT [has content]: the trace $T^\mu_\mu = 7 L_{\text{kin}} - 8 V$ off shell, $= 7 s V'(s) - 8 V(s)$ on shell
PASS EMT [fidelity]: K_h in the real commuting limit IS Part VI's $\text{cf} K_h$, and $K_h = 0$ on the rescaled
 $\hookrightarrow$ solutions $\text{Sqrt}[\text{Sin}] \Psi'(x_4)$, $\text{Sqrt}[\text{Sin}] \text{Psibar}'(x_4)$, where $\rho = V(s)$

[3.561 s] the J-parity of all 64 components of That, admissible fields
PASS EMT HERMITICITY [THE RESULT]: J-parity table -- (mu, nu) in {0..4}: EVEN (Hermitian); (mu in
→ 0..4, h in 5..7): ODD (anti-Hermitian); (h, h'), h != h': ZERO identically; (h, h): even
PASS EMT HERMITICITY [has content]: for x5..x7-independent fields That_{hh} == g_hh Lhat exactly,
→ because {gamma_h, Gamma_h} == 0
CELL 194 t=4.080 s
PASS MODE SUMS [THE RESULT]: on a flat-frame plane wave (canonical normalization), T_44 + Lhat ==
→ omega ubar (-i T16[4]) u and T_j_j - Lhat == k_j ubar (-i T16[j]) u (j = 0..3): energy and
→ pressure are the frequency and k_j dh/dk_j bilinears
PASS MODE SUMS [has content]: with ubar = u^ddag sigma16 these are omega u^ddag G u and k_j u^ddag
→ (dh/dk_j) u, so by Section 28 each occupied mode contributes omega, k_j^2/omega and (to sigma)
→ m/omega
PASS VACUUM [THE RESULT]: the unrenormalized sea energy -eps_KS(Lambda) == -(g/(16 pi^2))(2 Lambda^4
→ + 2 m^2 Lambda^2 + m^4/4 - m^4 Log[2 Lambda/m]) + o(1) as Lambda -> infinity; its m-dependent
→ finite part contains -(g/(32 pi^2)) m^4 Log[m^2]
CELL 195 t=2.098 s
PASS KOHN-SHAM [definition]: the degeneracy g = 8 -- per momentum the positive-frequency eigenspace
→ has rank 8 (Section 28's P_{+,+} + P_{+,-}), four flavours times two spins
PASS KOHN-SHAM [THE RESULT]: each closed form has k_F-derivative (g/(2 pi^2)) k_F^2 f(k_F) with f =
→ 1, m/omega, omega, k^2/(3 omega), and vanishes at k_F = 0 -- so it IS the integral g Int d^3k/(2
→ pi)^3 f
PASS KOHN-SHAM [has content]: Integrate returns the same functions, with ArcTanh[k_F/w_F] in place of
→ L -- and ArcTanh[k_F/w_F] == L (equal derivatives, both 0 at k_F = 0)
PASS KOHN-SHAM [control]: numerical quadrature agrees at k_F = 3/2, m = 1, g = 8 to 1e-12 (a
→ cross-check; the proof is the symbolic one above)
PASS KOHN-SHAM [THE RESULT]: eps + P == w_F n (Gibbs), eps - 3 P == m sigma (trace), d eps/dm ==
→ sigma, d eps/dk_F == w_F dn/dk_F
PASS MEAN FIELD [THE RESULT]: rho = eps + W - sigma W', P_obs = P_KS + sigma W' - W give rho + P_obs
→ == w_F n exactly; and d rho/dn == w_F + (sigma_KS - sigma) W' sigma'(n), i.e. == w_F at the
→ self-consistent point sigma = sigma_KS (m = W'(sigma) throughout)
PASS AUTHOR'S MASS TERM [THE RESULT]: W = m sigma gives W - sigma W' == 0, so rho == eps_KS, P ==
→ P_KS, and w = P_KS/eps_KS depends only on x = k_F/m: w == I2/(3 I1), I_n the m = 1 integrals
PASS AUTHOR'S MASS TERM [THE RESULT]: 0 <= w <= 1/3 -- P_KS >= 0 (I2' = x^4/w_x > 0, I2(0) = 0) and
→ eps - 3P = m sigma = (g m^4/(2 pi^2)) I3 >= 0 (I3' = x^2/w_x > 0, I3(0) = 0)
PASS AUTHOR'S MASS TERM [THE RESULT]: w rises MONOTONICALLY -- dw/dx == x^2 Delta/(3 w_x I1^2) with
→ Delta = x^2 I1 - w_x^2 I2, Delta(0) = 0 and Delta' = 2 x I3 > 0 -- from w -> 0 (x -> 0, w = x^2/5
→ + 0(x^4): dust) to w -> 1/3 (x -> infinity: radiation)
PASS AUTHOR'S MASS TERM [has content]: the classical sign problem disappears -- sigma_KS = g Int
→ m/omega is ODD in m and m sigma_KS = (g m^4/(2 pi^2)) I3(k_F/|m|) >= 0 for either sign, so with m
→ = -2M: M sigma_KS <= 0, the branch M s < 0 that Part VI had to assume
PASS CLASSICAL LIMIT [THE RESULT]: at the self-consistent point rho == W + 3 P_KS and P_obs == (sigma
→ W' - W) + P_KS exactly (sigma = sigma_KS, m = W'); Part VI's rho = V(s), P = s V' - V is the limit
→ P_KS/eps_KS -> 0, i.e. k_F/m -> 0
CELL 196 t=4.945 s
PASS PRE-UNIVERSE (1) [THE RESULT]: the U(1) current is conserved on shell, (1/Sqrt[g]) d_mu(Sqrt[g]
→ j^mu) == 0, for generic Psi(x0, x4), Psi_bar(x0, x4)
PASS PRE-UNIVERSE (1) [THE RESULT]: for Psi = Sqrt[Sin] Psi'(x4), Sqrt[g] j^0 is x0-independent (Sec
→ Sin Cot == 1), so d_4(Sqrt[g] j^4) == 0 on shell; and Sqrt[g] and q^3 p^3 are x4-independent
PASS PRE-UNIVERSE (2) [THE RESULT]: Psi = Sqrt[Sin] Psi'(x4) satisfies the field equation iff d_x0
→ V'(Sin[6Hx0] s') == 0, and d_x0 V'(Sin s') == 6 H Cos[6 H x0] s' V''(Sin s'): V linear OR s' == 0
PASS PRE-UNIVERSE (2) [has content]: s' = Psi_bar Psi' is conserved by the x4-only equations (d_4
→ Psi' = -H V' T16[4] Psi', d_4 Psi_bar = H V' Psi_bar T16[4]) for ANY V; so null data s' = 0 are
→ solutions for any V, with rho == V(0); and for V = s^2 with s' != 0 the x0-dependence is real
→ (witness)
PASS PRE-UNIVERSE (3) [THE RESULT]: for Psi = Sqrt[Sin] u(x4) Exp[i k x1] the field equation (mass
→ term, V' = constant) is Sqrt[Sin] Exp[i k x1] (T16[4] u' + i (k/q) T16[1] u - H V' u): the
→ physical momentum is k/q
PASS PRE-UNIVERSE (3) [THE RESULT]: d Log[k/q]/dx4 == H a4'[H x4] == -H_obs (observed momenta
→ redshift as Exp[a4]), and E^2 == (k^2/q^2 + m^2) ID16 at the physical momentum
PASS PRE-UNIVERSE (4) [THE RESULT]: x = k_F,phys/m = x_c/(q m) obeys d Log[x]/dx4 == H a4'[H x4];
→ since dw/dx > 0, dw/dx4 = w'(x) x H a4' < 0 exactly when a4' < 0 (the observed sheet expanding):
→ the quantum gas COOLS
PASS PRE-UNIVERSE (4) [control]: the classical Part VI fable does NOT cool -- ds/dx4 == 0 on its
→ rescaled solutions (Part VI's result, re-asserted), and its w = s V'/V - 1 is a function of s
→ alone: frozen
PASS PART VII [control]: a4 is STILL UNDEFINED -- nothing in Part VII gave it a value
CELL 197 t=0.072 s
identities accepted on numerical evidence alone : 0 (stage 3 returned True)
non-vanishing witnesses : 38 (cfNonZeroWitnessQ found a probe point at which
→ every entry is a
number and one of them is non-zero: a complete
→ proof of non-vanishing)
PASS no identity in this notebook was accepted on numerical evidence alone
Assertions run: 528 passed: 528 failed: 0

CELL 198 t=0.000 s

10.3 The same assertions in the runs of Parts I–VIII

After Part VIII (the coupled system, notebook Sections 30–33) was added, the whole notebook was run twice more. **Part VII’s record is its own run** (section 10.1, section 10.2); the two later runs show that adding Part VIII changed nothing in it. With the CELL and [t s] timing lines removed, the lines printed by cells 173–198 are identical in all three logs — the Part VII run, the acceptance run of Part VIII and the final run — 174 lines each, 170 of them PASS lines [checked for this page with diff].

The acceptance run of Part VIII (claude-fable/run_fermion_fable_part8.log, committed in bc04ed1; a historical record). The Part VII cells took 29.706s in that run. Its summary:

```
===== RUN SUMMARY =====
cells evaluated : 217
total seconds   : 233.289
cells w/ msgs   : 0
---- slowest cells ----
cell 80  80.343 s
cell 88  23.494 s
cell 204 10.364 s
cell 86   9.821 s
cell 137   9.306 s
cell 191   9.225 s
cell 87   8.387 s
cell 121   6.069 s
cell 160   4.250 s
cell 196   4.185 s
cell 107   3.836 s
cell 193   3.140 s
===== ASSERTIONS =====
assertions run  : 642
passed         : 642
FAILED        : 0
=====
RUN-DONE
```

The last Part VII line of that log is still Assertions run: 528 passed: 528 failed: 0 (line 843), and the run ends with Assertions run: 642 passed: 642 failed: 0 and 46 non-vanishing witnesses. Two of Part VIII’s assertions did not run in it: its comparisons of the Rust solver with the Mathematica reference runs are made only if the solver’s CSVs exist, and they did not exist yet (two NOTE lines say Rust CSV not found; that comparison was skipped, not failed). And three of its labels, in Section 32, write the listed spin-connection components with both flat indices up — the notation that 3a5020d corrected:

- SPIN CONNECTION [THE RESULT]: the complete list of non-zero $\omega_{\mu}^{\{ab\}}$ of the WARPED frame is the thirteen stated components (and their antisymmetric partners) – nothing else
- SPIN CONNECTION [THE RESULT]: the complete list for the Bianchi-I frame is $\omega_a^{\{04\}} = C'$, $\omega_i^{\{i4\}} = A'$, $\omega_h^{\{4h\}} = B'$ – nothing else
- SPIN CONNECTION [fidelity]: on the canonical member (labelled substitution) the warped connection IS $\omega_{\text{Canonical}}$ of Section 16, and the new component $\omega_a^{\{04\}}$ vanishes there

The arrays these assertions compare (cfOmegaWarp, cfOmegaBI) carry both flat indices **down**, so the tests themselves were right and are unchanged. Raising both flat indices multiplies a component by $\eta_{aa}\eta_{bb}$, which is -1 exactly in the planes with one timelike index and one spacelike index $((0, 4), (i, 4), (0, h))$ and $+1$ in the $(0, i)$ and $(4, h)$ planes [derived here]. So, read

with the indices up as written, the second label states the wrong sign for ω_0^{04} and ω_i^{i4} , while $\omega_h^{4h} = B'$ holds in either position; and the list of values in the old Section 32 Text cell had the wrong sign for its $(0, 4)$, $(i, 4)$ and $(0, h)$ components. (The corrected Text cell of 3a5020d says that raising both indices “would flip the sign of every plane that contains x4 or a hidden timelike direction”; exactly, it flips the planes with one timelike and one spacelike index, and leaves the $(4, h)$ planes, whose two indices are both timelike, unchanged.)

The final run (claude-fable/run_fermion_fable_final.log, first recorded in 1671155 and re-recorded after the label correction in 3a5020d; the notebook in its committed state). The Part VII cells took 26.118s and the Part VIII cells 20.994s in that run. Its summary:

```
===== RUN SUMMARY =====
cells evaluated : 217
total seconds   : 198.352
cells w/ msgs   : 0
---- slowest cells ----
cell 80  68.159 s
cell 88  22.525 s
cell 204  9.360 s
cell 86   8.072 s
cell 191  7.887 s
cell 87   7.193 s
cell 137  6.985 s
cell 121  4.579 s
cell 160  3.822 s
cell 107  3.505 s
cell 196  3.334 s
cell 193  2.970 s
===== ASSERTIONS =====
assertions run  : 644
passed         : 644
FAILED         : 0
=====
```

The last Part VII line of that log is again Assertions run: 528 passed: 528 failed: 0 (line 843), and the run ends with Assertions run: 644 passed: 644 failed: 0 (line 1005), identities accepted on numerical evidence alone : 0 and 46 non-vanishing witnesses; no RUN-DONE line was appended to this log. Its PASS lines differ from those of the acceptance run in exactly five places [checked for this page with diff]. The two comparisons with the Rust solver now run and pass, each after a line Rust <path>: 71 rows compared; max difference per column ...:

- fable4d RUNS [fidelity]: the Rust solver’s mass30eV run agrees with this cell to 1e-6 in every column, at every one of the rows its grid covers
- fable4d RUNS [fidelity]: the Rust solver’s power run agrees with this cell to 1e-6 in every column, at every one of the rows its grid covers

and the three Section 32 labels write the components with both flat indices down, $\omega_{\mu ab}$, as the arrays they compare have them (the tests are unchanged):

- SPIN CONNECTION [THE RESULT]: the complete list of non-zero $\omega_{\mu ab}$ (both flat indices down) of the WARPED frame is the thirteen stated components (and their antisymmetric partners) - nothing else
- SPIN CONNECTION [THE RESULT]: the complete list for the Bianchi-I frame is $\omega_{0\ 04} = C'$, $\omega_{i\ i4} = A'$, $\omega_{h\ 4h} = B'$ (both flat indices down) - nothing else
- SPIN CONNECTION [fidelity]: on the canonical member (labelled substitution) the warped connection IS $\omega_{\mu ab}$ of Section 16, and the new component $\omega_{0\ 04}$ vanishes there

10.4 The design review, and every correction it made to Effort A

The design (revision 1) was reviewed by four adversarial reviewers — quantization (findings QK- $\ast$); field equations, energy–momentum tensor and spin connection (EMT- $\ast$); Einstein equations and cosmology (E- $\ast$); density-functional theory and numerics (DFT- $\ast$) — each followed by an independent skeptic who tried to refute each finding. **No finding was refuted.** Several were confirmed only in part, and then the corrected fix was adopted. The review records are working documents and are not part of the repository; every adopted correction that concerns Effort A is restated here, with where it now lives.

design item	findings	what revision 1 said, or lacked	what was adopted	where on this page (notebook)
refinement	QK-7	“fermion fable must be complex” without scope; F2(a) only on the flat frame	the invariant forms are exactly $\text{span}\{\sigma_{16}, \sigma_{16}T_{16}[8]\}$; Majorana has one quartic scalar and no commuting kinetic shadow; Weyl/Majorana–Weyl cannot propagate; the complex spinor is the minimal , not the unique, field; F2(a) holds on every frame	3.2, 3.4, 3.8 (REFINEMENT (a), (b))
Lagrangian	QK-4 (corrected), EMT-5	several Lagrangians used interchangeably; Part VI’s drop-out mechanism assumed to carry over	only the symmetrized covariant Lagrangian; the unsymmetrized partial one is inadmissible (its two equations are not conjugate); the drop-out is a property of the matrix $\{\gamma^\mu, \Gamma_\mu\}$, per μ , on diagonal frames only	4 (LAGRANGIAN (a)–(f))
field equations	EMT-9, EMT-11	8+8 form sketched; operator ordering “normal-ordered or Weyl”; x_4 -only solutions assumed	the 8+8 form with the conjugate rows; the axial phase is not a symmetry; the ordering rule with its scalar and γ^4 contraction parts; the x_4 -only family needs $V''s' = 0$, and s' is conserved	5.3, 5.8, 5.9, 5.11
quantization: truncation	QK-2	“physical states obey $P_5 = P_6 = P_7 = 0$ ” (a superselection rule) and Tomonaga–Schwinger integrability	a truncation with finite V_{hid} (closed timelike curves stated); anticommutator with $H/(V_{\text{hid}}\sqrt{ g })$ and δ^4 ; only $[P_4, P_h] = 0$ kept	6.4, 6.10
quantization: theorem	QK-1 (corrected)	admissibility asserted from one example	the theorem: positive quantization iff the momentum span is spacelike-definite, with items (1)–(7) (threshold Jordan block, G -neutral modes above threshold, traceless certificate, uniqueness)	6.10 (ADMISSIBILITY)
quantization: J	QK-5 (corrected), plus the missed basis point	Krein prescription $b^\dagger := \eta b^\ddagger$, basis unspecified; reflection positivity claimed	$J = -i\Omega_4 T_{16}[0]$, $\sigma_{16} = J\beta$, flavour metric (2, 2); the adjoint $\Psi^\dagger := \Psi^\ddagger J$ is mode-independent and requires a J-eigenmode basis (control: a Krein boost breaks Hermiticity of s); why it fails for bosons; reflection positivity dropped	6.5, 6.6, 6.13 (J, KREIN MODES [control])
quantization: other	QK-6, QK-8, QK-9, QK-10, QK-11, EMT-2	symmetry count, dynamical frames, wall condition, current sign and J -connection relation incomplete or wrong	13 kept / 15 lost generators, G invariant under $\text{Spin}(4,3)$; $\chi = (\sqrt{g}/H)^{1/2}\Psi$; one-particle space $L^2(dz)$, regular at the wall, limit-point at infinity, bag condition $T_{16}[0]\Psi' = \pm\Psi'$; $n^\mu = -\frac{i}{H}\bar{\Psi}\gamma^\mu\Psi$; the J table with $\Gamma_{5..7}$ anticommuting	6.9, 6.12, 6.15, 6.16, 6.8

design item	findings	what revision 1 said, or lacked	what was adopted	where on this page (notebook)
EMT	EMT-1, EMT-3, EMT-4, EMT-8, QK-3	EMT stated without derivation; conservation “for independent $\Psi, \bar{\Psi}$ ” without scope; “ $\langle T \rangle = 0$ in the vacuum”	Hilbert = T_{cov} off shell; spin-connection variation exactly zero; Hilbert $\neq T_{\text{par}}$ (witness at a non-vacuous pair); why the commuting proxy proves conservation; the J -parity table; the vacuum statement only for static backgrounds; do not run the symbolic divergence of T_{par}	7 (EMT, EMT CONSERVATION, EMT HERMITICITY), 8.3
expectation values	DFT-5, DFT-6, DFT-12, DFT-15, DFT-4	classical limit “ $k_F \rightarrow 0$ at fixed n ” (impossible) or a coherent zero mode; unstable closed forms; phantom crossing not examined; sea energy only for constant m	the classical limit is the non-relativistic limit, with the sign rule; the numerically stable forms; $w_f \geq -1$ (no phantom crossing); the pair-excitation thresholds ($\omega_F + m $ at $q = k_F$, $2\omega_F$ at $q = 0$; not asserted in Part VII); ΔE_{vac} for mass-varying W	8.5, 8.7, 8.9, 8.3
spin connection	EMT-6, EMT-10	“the connection enters the Hamiltonian”	it enters neither the anticommutator nor the Hamiltonian density; it re-emerges in the equation of motion as the term that makes $\hbar$ Hermitian; $\gamma^\mu \Gamma_\mu = -\frac{1}{2} T_\mu \gamma^\mu$ (the Weitzenböck torsion vector)	6.11, 9.12.3, 9.12.4

Two of these adopted corrections are also stated in Part VII’s own Text cells: the Hamiltonian-density correction (EMT-6; “This corrects the design’s ‘it enters the Hamiltonian’”, Section 28), and the identification of the classical limit with the non-relativistic one (DFT-5; Section 29). One numerical slip in the review’s vacuum-energy figures is corrected on this page (section 8.3).

10.5 The Fock check

The finite Fock space of section 6.14 is the independent check of the quantization: it realizes the Dirac-bracket anticommutator on a positive Hilbert space of dimension $2^{16} = 65536$, with exact integer sparse matrices, and confirms the sign of the anticommutator through the Heisenberg equation (the opposite sign gives the time-reversed equation). It shows the unique Dirac-sea ground state at -40 , the 16-fold first excited level (8 particles, 8 antiparticles), the negative Krein “norms” that the J -involution removes, and the expectation rule used for ρ , P and σ . Twelve assertions (FOCK, CHARGE), Input cell 191, 9.858 s.

10.6 The TeX export

`claude-fable/export_fermion_fable_tex.wls` evaluates the notebook’s Input cells (output suppressed), checks that the Part VII objects exist, and writes fourteen files into `provenance-latex/generated/`:

file (.tex and .txt)	content	taken from the notebook object
<code>fermion_fable_field_equations</code>	the 16 component field equations on the canonical frame	<code>cfFieldEqTerms</code>
<code>fermion_fable_adjoint_equations</code>	the 16 component adjoint equations	<code>cfAdjEqTerms</code>
<code>fermion_fable_field_equations_8plus8</code>	the split-octonion 8+8 form, compact and by component	<code>cfFieldOp16</code> , <code>cfUpper8</code> , <code>cfLower8</code> , <code>cfFieldEqTerms</code>

file (.tex and .txt)	content	taken from the notebook object
fermion_fable_field_ equations_rescaled	the 16 equations for $\Psi' = \Psi / \sqrt{\sin 6Hx_0}$	the rescaled operator of Section 27 (iv), asserted equal to cfE8 under the rescaling
fermion_fable_anticommutator	$\{\Psi_a, \Psi_b^\dagger\} = H \cos(6Hx_0) J_{ab} \delta^7$, with the matrix	cfAntiPsiPsiDd at $c = \sqrt{g}/H$
fermion_fable_krein_J	$J, \beta, \sigma_{16} = J\beta$	cfJK, cfBeta
fermion_fable_eos	the Kohn–Sham closed forms $n, \sigma, \varepsilon, P, w$ at $g = 8$	cfNKS, cfSigKS, cfEpsKS, cfPKS

Every expression comes from an object that Part VII asserts; the script does no mathematics of its own beyond formatting (its header says so, and its code only renders term lists and matrices). The .tex files are \input by this document in sections 5, 6 and 8; the .txt files hold the same content in plain text. The export was run on 2026-09-24 at 20:41 (PDT) on the 198-cell notebook, and it printed:

```
evaluating 198 Input cells of C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\claud-fable\claud-f
↪ fable_Einstein-Rosen-2-Planes.nb (output suppressed)
assertions: 528   failed: 0   cells with messages: 0
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\fermion_fable_f
↪ ield_equations.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\fermion_fable_a
↪ djoint_equations.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\fermion_fable_f
↪ ield_equations_8plus8.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\fermion_fable_f
↪ ield_equations_rescaled.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\fermion_fable_a
↪ nticommutator.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\fermion_fable_k
↪ rein_J.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\fermion_fable_e
↪ os.tex / .txt
done: 14 files in C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated
```

(That console output was captured outside the repository; the fourteen files it wrote were committed in 4fdffc40.) A test document that \inputs all seven .tex fragments (with amsmath) compiled with pdfLaTeX (MiKTeX) to 9 pages with no errors, warnings or overfull boxes (a scratch test on 2026-09-24, outside the repository).

11 Every command, in full

The commands run in Git Bash on Windows 11 (they are plain POSIX shell and run the same way on Linux and macOS). The toolchain recorded in the logs: Wolfram Language 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026), called as wolframscript (the fourth line of each of the three run logs); Python 3 for the builder (Python 3.14.5 when section 11.1’s output was recorded); and MiKTeX (pdfTeX 3.141592653-2.6-1.40.29, MiKTeX 26.5; Latexmk 4.88) for the LaTeX twin, installed on this machine at C:/ProgramFiles/MiKTeX/miktex/bin/x64.

11.1 Build the notebook from its manifests

The notebook is never edited by hand. build_tools.py reads every claud-fable/cells_part*.wl in order and writes both the notebook and run_all.wls (the same Input cells as a script).

```
cd "$(git rev-parse --show-toplevel)/claude-fable"
python build_tools.py          # use python3 where python is absent
```

It prints (recorded on 2026-09-24 for this page, with the eight committed manifests, in a scratch copy of the directory so that nothing in the repository was touched; it ran in 0.115s):

```
manifest files : ['cells_part1.wl', 'cells_part2.wl', 'cells_part3.wl', 'cells_part4.wl',
↪ 'cells_part5.wl', 'cells_part6.wl', 'cells_part7.wl', 'cells_part8.wl']
cells parsed   : 384 {'Title': 8, 'Subtitle': 1, 'Subsubtitle': 1, 'Text': 123, 'Section': 34, 'Input':
↪ 217}
run_all.wls    : 217 Input cells
notebook       : 384 cells -> <repo>\claude-fable\claude-fable_Einstein-Rosen-2-Planes.nb
```

(the last line prints the absolute path of the directory it ran in). The rebuilt `claude-fable_Einstein-Rosen-2-Planes.nb` and `run_all.wls` were **byte-identical** to the committed ones (cmp reported no difference). The check was repeated on 2026-09-25 on the manifests of 3a5020d, whose `cells_part8.wl` carries the corrected Section 32 labels: the same four lines, and the rebuilt notebook and `run_all.wls` again byte-identical to the committed ones. With only the seven manifests of Parts I–VII present — the state in which Part VII was run — it prints

```
cells parsed   : 340 {'Title': 7, 'Subtitle': 1, 'Subsubtitle': 1, 'Text': 103, 'Section': 30, 'Input':
↪ 198}
```

and `run_all.wls` : 198 Input cells, which is the style tally that the checker reported for the Part VII notebook (section 11.3).

11.2 Run the notebook end to end, straight out of the .nb

```
cd "$(git rev-parse --show-toplevel)/claude-fable"
wolframscript -file run_from_nb.wls > run_fermion_fable_final.log 2>&1
git checkout -- claude-fable_Einstein-Rosen-2-Planes-eLa.mx
↪ claude-fable_Einstein-Rosen-2-Planes-eLazt.mx
```

- **What it does.** `run_from_nb.wls` imports the `.nb`, takes its Input cells in order, and evaluates each one as a single unit (a multi-line cell becomes one `CompoundExpression`, exactly as the front end evaluates it), through the harness `runner_header.wl`. The harness prints `CELL n t=... s` after each cell, with `MESSAGES: ...` appended if the cell raised any message, and ends with the run summary and the assertion counts.
- **Why from inside `claude-fable/`.** Run headless, the notebook cannot ask for its own file name, so Section 10 looks for the author's helper packages `ConvertMapleToMathematicaV2.wl` and `EtoExp.wl` in the working directory (and in `Pre-Universe_14SEP26-77/` below it, and one level up). They sit in `claude-fable/`. The reference CSVs of Parts VI and VIII are found as `claude-fable\..\fable-cosmology\reference\...` (the log prints these paths). The script itself is self-locating (`$InputFileName`) and finds the notebook from any directory.
- **The `.mx` restore.** Section 12 of the notebook writes `claude-fable_Einstein-Rosen-2-Planes-eLa.mx` and `...-eLazt.mx` with `DumpSave`, whose bytes are not reproducible from run to run. The two files are restored to the committed version after every run (commits 4fdcf40 and bc04ed1 record that this was done).
- **How long.** 198.352s for the 217 cells of the final run; 232.146s for the 198 cells of the Part VII run and 233.289s for the 217 cells of the acceptance run of Part VIII (the **total** seconds lines of the three logs).
- **What it prints.** The first line is evaluating 217 Input cells straight out of the `.nb` (evaluating 198 ... for the Part VII state). Then the provenance banner of the original notebook and every cell's line, `PASS` labels and notes. The Part VII portion is quoted in full in section 10.2. The end of the run is quoted in sections 10.1 and 10.3.

The final log was produced by this command on the committed notebook of 3a5020d. The Part VII log was produced by the same command with the output redirected to `run_fermion_fable_part7.log`, when the notebook had Parts I–VII, and the acceptance log of Part VIII with the output redirected to `run_fermion_fable_part8.log`, before the Rust solver’s CSVs existed. The last line of those two, RUN-DONE, was appended by the wrapper that ran the command; the final log has no such line.

Checking a log (the numbers are the results on the three committed logs):

```
cd "$(git rev-parse --show-toplevel)/claude-fable"
grep -c '^ PASS' run_fermion_fable_part7.log      # 528
grep -c '^ PASS' run_fermion_fable_part8.log      # 642
grep -c '^ PASS' run_fermion_fable_final.log      # 644
grep -n '^ FAIL' run_fermion_fable_final.log      # no output: no assertion
↪ failed
grep -c 'MESSAGES: ' run_fermion_fable_final.log  # 0: no cell raised a message
grep -E 'identities accepted|non-vanishing witnesses' run_fermion_fable_final.log |
↪ tail -2
# identities accepted on numerical evidence alone : 0    (stage 3 returned True)
# non-vanishing witnesses                        : 46    (cfNonZeroWitnessQ found)
↪ a probe point at which every entry is a
sed -n '/RUN SUMMARY/, $p' run_fermion_fable_final.log # the summary quoted in
↪ section 10.3
diff <(grep '^ PASS' run_fermion_fable_part8.log) <(grep '^ PASS'
↪ run_fermion_fable_final.log)
# five differences: the three Section 32 labels, and the two Rust comparisons
↪ added (section 10.3)
```

A plain `grep FAIL` also finds the summary line `FAILED : 0` and the two `PASS` labels whose text contains the word “FAILS” (control assertions that show a wrong sign failing, lines 298 and 921 of the final log, and of the Part VIII log). The run prints four `EXPECTED MESSAGE(S)` lines (three in Part II, one in Part V: messages that the original notebook’s time budgets or its underdetermined `Solve` calls may raise, announced in advance) and one `NOTE` line in Part III (that a_4 is undefined); in Part VIII it prints the two lines `Rust <path>: 71 rows compared; ... of the comparisons with the Rust solver’s CSVs fable-cosmology/results/nb06_fable4d_mass30eV.csv and nb06_fable4d_power.csv` (committed in aca5102). In the acceptance run of Part VIII those CSVs were not yet present, and two `NOTE` lines said so instead (`Rust CSV not found; that comparison was skipped, not failed`). None of these lines is a message raised by a cell.

11.3 Check the generated notebook without evaluating it

```
wolframscript -file "$(git rev-parse --show-toplevel)/claude-fable/verify_nb.wls"
```

It imports the `.nb`, counts its cells by style, extracts the Input cells, and checks that every one of them parses. It takes about 4s (4.118s when timed for the draft of this page). For the committed notebook it prints (`claude-fable/verify_nb_part8.log`, re-recorded in 3a5020d after the Section 32 labels were corrected; before that correction it printed `file bytes: 545408` and `total input characters: 347187`, the rest unchanged):

```
file bytes: 545713
Head: Notebook
cells: 384
style tally: {{Title, 8}, {Subtitle, 1}, {Subsubtitle, 1}, {Text, 123}, {Section, 34}, {Input, 217}}
input cells with plain-string BoxData: 217
total input characters: 347237
input cells that FAIL to parse: {}
```

```

first cell: InputForm[(* --- provenance banner, as in the original notebook
↪ -----]
last cell : InputForm[cfAssert["PART VIII [control]: a4 is STILL UNDEFINED -- nothing in Part VIII gave
↪ it a val]

```

For the Part VII notebook it printed (claude-fable/verify_nb_part7.log):

```

file bytes: 440320
Head: Notebook
cells: 340
style tally: {{Title, 7}, {Subtitle, 1}, {Subsubtitle, 1}, {Text, 103}, {Section, 30}, {Input, 198}}
input cells with plain-string BoxData: 198
total input characters: 279955
input cells that FAIL to parse: {}
first cell: InputForm[(* --- provenance banner, as in the original notebook
↪ -----]
last cell : InputForm[(* --- final tally of every assertion made in this notebook, Parts I-VII
↪ -----]

```

11.4 Print the section-to-cell map

```

wolframscript -file "$(git rev-parse --show-toplevel)/claude-fable/nb_section_map.wls" >
↪ "$(git rev-parse --show-toplevel)/claude-fable/nb_section_map.log"

```

It reads the .nb without evaluating anything and prints, in document order, each Part title and each Section with the range of its Input cells (3.340s when timed for this page). Its output, claude-fable/nb_section_map.log:

```

== claude-fable_Einstein-Rosen-2-Planes
0. What this notebook does, and how it is organized | Input cells none
1. Session setup | Input cells 1-6
2. Coordinates and the flat 4+4 Minkowski metric eta | Input cells 7-12
3. SO(4): self-dual and anti-self-dual antisymmetric 4x4 matrices | Input cells 13-15
4. The real 8x8 Clifford generators tau, their conjugates, and the spinor metric sigma | Input cells
↪ 16-21
5. The 16x16 Dirac matrices T16, the chirality operator, sigma16 and the so(4,4) generators | Input
↪ cells 22-29
6. A complete basis of the 16x16 matrix algebra | Input cells 30-35
7. A complete basis of the 8x8 matrix algebra, and the induced flat metric | Input cells 36-38
8. The 4x4 Dirac matrices, and a complete basis of the 4x4 matrix algebra | Input cells 39-44
9. Cartan triality: the constant bridge between the vector and spinor realizations,
and the split-octonion structure constants | Input cells 45-52
== PART II -- The wave function of the un-universe, its field equations, and the 3 generations
10. Housekeeping: where this notebook lives, and the two helper packages | Input cells 53-55
11. The wave function of the un-universe, and its Lagrangian | Input cells 56-60
12. The Euler-Lagrange equations, the coupling pattern, and the (z,t) chart | Input cells 61-66
13. Decoupling into four blocks of four, and the closed-form solutions | Input cells 67-78
14. Bilinears, the two solution branches, and M6 = 3 generations of Einstein-Rosen 2-planes | Input
↪ cells 79-90
== PART III -- The curved 4+4 spacetime, its frame field, and the canonical spin connection
15. The local flat 4+4 Minkowski coordinate system at every spacetime point | Input cells 91-99
16. The generalized Christoffel symbols, and the canonical spin connection | Input cells 100-110
17. The gauge-covariant derivative of the 16-component spinor | Input cells 111-117
== PART IV -- Three bridges between curved and flat indices: two gauge transformations, and one new
↪ connection
18. Bridge 1 -- a locally boosted frame: the same connection in a local gauge | Input cells 118-125
19. Bridge 2 -- the null (light-cone) frame: the same connection in a constant gauge, and the spinor
↪ metric sigma | Input cells 126-130
20. Bridge 3 -- the triality frame with totally antisymmetric split-octonion torsion | Input cells
↪ 131-139
== PART V -- The fable-5.1 bridge: the connection in which the frame itself is parallel
21. The fable-5.1 bridge -- the Weitzenboeck (teleparallel) connection of the canonical frame | Input
↪ cells 140-148
22. Master comparison of the five spin connections | Input cells 149-152
== PART VI -- fableScalar and fable: two fields on the pre-universe, and the equation of state of dark
↪ energy
23. fableScalar -- a scalar field on the pre-universe, its energy-momentum tensor, and wScalar |
↪ Input cells 153-160
24. fable -- the 16-component spinor field, its energy-momentum tensor, and w | Input cells 161-169
25. Reading the two mechanisms side by side | Input cells 170-172
== PART VII -- fermion fable: the complex 16-spinor, its canonical quantization in 4+4 dimensions, its
↪ field equations in the primordial gravitational field, and its energy-momentum tensor operator

```

```

26. The refinement: why fermion fable is a complex 16-spinor | Input cells 173-179
27. The Lagrangian, the field equations in the primordial gravitational field, and the canonical spin
   ↪ connection | Input cells 180-184
28. Canonical quantization in 4+4 dimensions: the Dirac bracket, the Krein structure J, and the
   ↪ admissible sector | Input cells 185-191
29. The energy-momentum tensor operator of fermion fable: definition, conservation, and rho, P and w |
   ↪ Input cells 192-198
== PART VIII -- fable as a source of the Einstein equations of the primordial gravitational field
30. The Einstein tensor of the canonical metric, and what it demands of its source | Input cells
   ↪ 199-203
31. The generalized warped frame, its asymptotic region, and the 8-dimensional Bianchi-I limit |
   ↪ Input cells 204-207
32. The canonical spin connection of the interacting system | Input cells 208-210
33. The coupled equations of fable and the primordial field, and their solution from the radiation era
   ↪ to today | Input cells 211-217
total Input cells: 217

```

11.5 Export the field equations, the anticommutator, J and the equations of state to TeX and text

```

wolframscript -file "$(git rev-parse
↪ --show-toplevel)/claude-fable/export_fermion_fable_tex.wls"

```

It is self-locating and sets its own working directory to `claude-fable/`, so it runs from anywhere. It evaluates every Input cell of the notebook with the output suppressed (so it takes about as long as a run; its duration was not recorded), prints the assertion count, stops with exit code 1 if the Part VII objects are missing, and writes the fourteen files listed in section 10.6. Its recorded output is quoted there. On today's 217-cell notebook the first two lines would read `evaluating 217 Input cells ... and assertions: 644 ...` (644 because the Rust solver's CSVs are committed, so Part VIII's two comparisons run, as in the final run of section 10.3); the Part VII objects it exports are the same. The `.tex` files carry the date of the run in their header comment, so a re-run on another day changes that one line; the `.txt` files carry no date. Like every evaluation of the notebook, it rewrites the two `.mx` files of Section 12, which are then restored as in section 11.2.

11.6 Build the LaTeX twin and its PDF

```

bash "$(git rev-parse --show-toplevel)/provenance-latex/build_all.sh"

```

or, in PowerShell,

```

powershell -ExecutionPolicy Bypass -File provenance-latex\build_all.ps1

```

For each of the three documents (`PROVENANCE-14-FERMION-FABLE-CANONICAL-QUANTIZATION`, `...-15-...`, `...-16-...`) the script runs `latexmk -pdf -interaction=nonstopmode -halt-on-error <doc>.tex`, with the output in `provenance-latex/<doc>.latexmk.log`; it stops on the first failure, printing `FAILED`; see `provenance-latex/<doc>.latexmk.log` and `<doc>.log`; it warns if the `.log` contains “undefined” or “Rerun to get”; and it lists each PDF. So it prints `== <doc>` and an `ls -la` line per document, and finally `== all three PDFs built`. Every document begins with `\input{preamble}` (`provenance-latex/preamble.tex`: pdfLaTeX, Latin Modern, A4, 2.5 cm margins, and macros for the notebook's objects, with separate symbols for the classical conjugation `\conjug` and the Hilbert adjoint `\hadj`). MiKTeX's `latexmk` needs Perl on the PATH; Git for Windows ships one (`$env:PATH += ";C:\Program Files\Git\usr\bin"` in PowerShell, as `build_all.ps1` notes).

The preamble was test-compiled when it was added: commit `0a789e2` records that a minimal test document compiles with `latexmk` with no warnings. The LaTeX twin of this page, in its final state, was compiled on its own on 2026-09-25, inside `provenance-latex/`, with the command of its header, `latexmk -pdf -interaction=nonstopmode -halt-on-error`

PROVENANCE-14-FERMION-FABLE-CANONICAL-QUANTIZATION.tex: it ran with no errors, no warnings and no undefined references, and wrote a PDF of 114 pages whose only two overfull lines are 5.5pt and 0.6pt too wide. The run of build_all.sh over the three pages belongs to the fresh-clone verification of section 12.

12 Push and repository verification

The repository. origin is https://github.com/once-ere/Pre-Universe_with_Claude.git, and the work is pushed to its `main` branch. (The remote named `upstream`, the author's other repository, is never pushed to.) The repository carries a `.gitattributes` with `* -text`, so that git never rewrites line endings and every file checks out with the bytes that were committed.

The commits of the work, 2026-09-24 and 2026-09-25 (`git log -oneline -18`, times from `git log -date=iso, PDT`; the work ran past midnight, so the last four commits are dated 2026-09-25):

commit	time	what it contains
e34d454	18:52:36	the 2026-09-16 work, as it stood: Part VI of the notebook (Sections 23–25; 172 Input cells, 358/358), the <code>fable-cosmology/</code> solver, notebooks, results and references; with its known gaps listed in the commit message
028b379	19:24:32	the reference generator <code>make_reference.wls</code> no longer prints harmless underflow messages (the values it writes are unchanged)
2f6d322	19:57:29	the 2026-09-16 gaps closed: the student instructions and the paper of the fable-cosmology work, its provenance page completed, seven corrections to its design; a fresh local clone ran its <code>setup.sh</code> and <code>run_all.sh</code> end to end
8459c57	20:02:04	a work-in-progress snapshot of the fermion-fable solver crate <code>fable-cosmology/rust/fable_fermion</code>
e130337	20:14:25	a second work-in-progress snapshot of that crate
0a789e2	20:17:18	the LaTeX build tree of the three new provenance pages: <code>provenance-latex/preamble.tex</code> , <code>build_all.sh</code> , <code>build_all.ps1</code>
d0f2a99	20:18:53	<code>build_all.sh</code> tracked without the executable bit, like the repository's other scripts
c5892bd	20:38:49	the finished fermion-fable solver: the Kohn–Sham fermion gas (stable integrals, per-7-volume mean field, gap equation), the stabilized and 8D models, <code>cargo test</code> 21 + 8 passing, and the independent scipy cross-check
4fd4c40	20:47:28	Part VII (this effort): <code>claude-fable/cells_part7.wl</code> , the rebuilt notebook (198 Input cells, 528/528, 0 cells with messages), <code>run_fermion_fable_part7.log</code> , <code>verify_nb_part7.log</code> , <code>export_fermion_fable_tex.wls</code> and the fourteen generated TeX/text files
b7ea18d	20:48:42	a wording correction, in an earlier provenance page, about the canonical connection's planes: for $k = 5, 6, 7$ the $(0, k)$ plane is a boost and the $(4, k)$ plane a rotation (the fact is stated in section 9.6 of this page)
bc04ed1	21:24:04	Part VIII (the coupled system, Sections 30–33; 217 Input cells, 642/642, with its two comparisons of the Rust solver skipped because the solver's CSVs did not exist yet), its acceptance run log <code>run_fermion_fable_part8.log</code> , and the Mathematica fermion references
18a2501	22:13:42	the wall-state solver of the coupled system (<code>fable-cosmology/fermion/waveguide*</code>): its derivation from the notebook's own T16 (94/94 checks pass), its Python solver (validation 64/64), its independent Mathematica cross-check, and its report
fee5603	22:31:20	the draft of this page, its LaTeX twin and its PDF (110 pages), with section 1 still a placeholder
2bc936d	23:12:51	six defects of the solver <code>fable_fermion</code> fixed, among them the stable evaluation of the Dirac-sea energy <code>vac_over_rho</code> (section 8.3); <code>cargo test</code> 34/34 (23 unit + 11 integration, previously 21 + 8)

commit	time	what it contains
aca5102	2026-09-25 00:23:46	notebooks 05–07 of the fermion fable executed and committed with their results (all seven notebooks pass the checker, <code>nbcheck</code> 7/7; <code>fable-cosmology/run_all.sh</code> passes end to end); the wall-state solver’s input <code>fable-cosmology/fermion/waveguide_T16.json</code> committed (its derivation log still reports 94 checks, 94 PASS, 0 FAIL); <code>waveguide.py validate</code> exits 1 whenever a check fails
1671155	00:28:54	the final full run of the notebook recorded, <code>claude-fable/run_fermion_fable_final.log</code> (217 Input cells, 644/644, 0 cells with messages, 208.715s, 46 witnesses): with the solver’s CSVs now committed, Part VIII’s two comparisons of the Rust solver with the Mathematica reference runs run and pass ($642 + 2 = 644$)
38701ef	05:40:19	the drafts of the pages of Efforts B and C (markdown, LaTeX, PDF), the notebook figures their PDFs use (<code>provenance-latex/figures/</code>), and the repository’s front page updated for Parts VII–VIII and the three pages
3a5020d	05:54:25	Part VIII, Section 32: its Text cell and three assertion labels now write the listed spin-connection components with both flat indices down — $\omega_{\mu ab}$, $\omega_{0 04}$, $\omega_{i i4}$, $\omega_{h 4h}$ — as the arrays they compare have them (raising both indices flips the sign of every component in a plane with one timelike and one spacelike index; section 10.3); no computation changed. The final run re-recorded in <code>run_fermion_fable_final.log</code> (217 Input cells, 644/644, 0 cells with messages, 198.352s, 46 witnesses, 0 identities accepted on numerical evidence alone, both Rust-vs-Mathematica comparisons passing); <code>verify_nb_part8.log</code> re-recorded (section 11.3); a comment in the solver’s <code>models.rs</code>

Checking that the remote has what was pushed:

```
cd "$(git rev-parse --show-toplevel)"
git fetch origin
git rev-parse HEAD origin/main          # both must print the same hash
git ls-remote origin refs/heads/main    # the hash GitHub serves for main
git status --short                      # what is not yet committed
```

When the draft of this page was written (2026-09-24, about 21:28 PDT), `git ls-remote origin refs/heads/main` printed

```
bc04ed16096e134857111ae16df5220d4ff18435      refs/heads/main
```

which was the local `main` (bc04ed1, “Add Part VIII: fable as a source of the Einstein equations of the primordial field”). Everything of this effort that exists as code, log or generated file — `cells_part7.w1`, the rebuilt notebook, the Part VII run log, the checker’s logs, the export script and the generated TeX/text files — was in that pushed commit.

What was not yet committed when the draft was written, and where it went. The draft of this page, its LaTeX twin and its PDF were committed in fee5603. The working-tree changes of the other efforts that were pending at that moment are all committed since: the changes to the solver’s `constants.rs` and to `fable-cosmology/fermion/crosscheck.py` in 2bc936d, and the new notebook `05_fermion_fable_quantum_eos.ipynb`, the edits to the other notebooks and their results in aca5102. None of the numbers on this page is taken from a file that is not committed. The two `.mx` files of Section 12 were also showing as modified in the working tree then, rewritten by a notebook evaluation running at the time; they are restored as in section 11.2 before any commit. This final version of the page, its LaTeX twin and its PDF are committed together with the final versions of the pages of Efforts B and C.

The final push and the fresh-clone verification. The final state is verified from a fresh clone of the pushed main, with these commands (<scratch> is any empty directory outside the repository):

```
cd <scratch>
git clone https://github.com/once-ere/Pre-Universe_with_Claude.git fresh
cd fresh
git log --oneline -1                # the hash of the pushed main
cd claude-fable
cp claude-fable_Einstein-Rosen-2-Planes.nb committed.nb
python build_tools.py                # prints the four lines of section 11.1
cmp claude-fable_Einstein-Rosen-2-Planes.nb committed.nb && echo "notebook rebuilds byte for
↪ byte"
wolframscript -file verify_nb.wls    # section 11.3
wolframscript -file run_from_nb.wls > fresh_run.log 2>&1
sed -n '/RUN SUMMARY/, $p' fresh_run.log # 217 cells, 644/644, cells w/ msgs 0
diff <(grep '^ PASS' fresh_run.log) <(grep '^ PASS' run_fermion_fable_final.log) && echo
↪ "every label identical"
cd .. && bash provenance-latex/build_all.sh # section 11.6
```

The Rust solver's two CSVs that Part VIII compares with its Mathematica reference runs, `fable-cosmology/results/nb06_fable4d_mass30eV.csv` and `nb06_fable4d_power.csv`, are committed (since `aca5102`), so a fresh clone runs both comparisons and its summary is compared with the final run, 644/644. Before the final one, the most recent completed fresh-clone verification in the repository's history is the one recorded in `2f6d322` (for the fable-cosmology work). The delivered copy of the notebook outside the repository, `C:/Users/nsh/Documents/8-dim/claude-fable_Einstein-Rosen-2-Planes.nb`, is byte-identical to the committed notebook of `3a5020d` (545713 bytes, the file bytes of section 11.3; compared with `cmp` when this page was brought to its final state on 2026-09-25).

The final commit hashes, the `git ls-remote` line of the final push, and the results of the fresh-clone verification of that pushed state (the rebuild, the run summary, the comparison of every label, the LaTeX build) are recorded here: `{{FINAL-PUSH}}`